Configuration Summary

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DHW ANALYSIS - CONFIGURATION SUMMARY
Project: oisst_regression
Generated: 2026-10-09 13:49:57
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-- Data Preparation --

  input_dir.............................. /g/data/ia39/aus-ref-clim-data-nci/oisst/data/yearly/
  file_pattern........................... oisst-avhrr-v02r01_{year}.nc
  year_range............................. (1981, 2024)
  variable............................... sst
  detrend_method......................... global_linear
  standardise_method..................... global
  fill_method............................ diffusion
  fill_diffusion_iterations.............. 500
  fill_fallback_value.................... 0
  downscale.............................. True
  target_lat_size........................ 64
  target_lon_size........................ 128
  temporal_average....................... none
  subset_region.......................... False
  lat_bounds............................. None
  lon_bounds............................. None
  diagnostic_year........................ 2020
  diagnostic_pdf......................... diagnostics/data_preparation_diagnostics_64x128.pdf
  output_dir............................. data/processed_64x128/
  save_intermediates..................... True
  downscaled_file........................ sst_downscaled.nc
  temporally_averaged_file............... sst_temporally_averaged.nc
  detrended_file......................... sst_detrended.nc
  standardised_file...................... sst_standardised.nc
  filled_file............................ sst_filled.nc
  final_file............................. sst_final.nc
  subset_file............................ sst_subset.nc
  trend_params_file...................... trend_params.nc
  standardise_params_file................ std_params.nc
  resume_after_stage..................... 0
  chunks................................. {'time': 366, 'lat': 720, 'lon': 720}
  dask_scheduler......................... threads
  dask_n_workers......................... 8
  dask_threads_per_worker................ 2
  dask_memory_limit...................... 12GB
  dask_num_threads....................... 16

-- Model Training --

  model_name............................. sfno
  input_history.......................... 10
  output_channels........................ 1
  embed_dim.............................. 80
  num_layers............................. 5
  num_modes_lat.......................... 16
  num_modes_lon.......................... 32
  mlp_ratio.............................. 2.5
  activation............................. gelu
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norm_type.............................. instance_norm
dropout_rate........................... 0.05
use_residual........................... True
spectral_bias.......................... True
add_grid_features...................... True
add_seasonal_encoding.................. False
forecast_horizon....................... 90
initial_rollout_steps.................. 1
max_rollout_steps...................... 8
rollout_epoch_start.................... 5
rollout_epoch_end...................... 50
rollout_loss_decay..................... 0.95
train_years............................ (1982, 2015)
val_years.............................. (2016, 2019)
test_years............................. (2020, 2024)
learning_rate.......................... 0.0002
min_learning_rate...................... 1e-06
batch_size............................. 16
num_epochs............................. 120
weight_decay........................... 1e-05
optimizer.............................. adamw
scheduler.............................. cosine
scheduler_patience..................... 10
scheduler_factor....................... 0.5
warmup_epochs.......................... 5
gradient_clip_norm..................... 1
loss_function.......................... huber
early_stopping_patience................ 20
early_stopping_metric.................. val_loss
input_data_path........................ data/processed_64x128/sst_final.nc
checkpoint_dir......................... checkpoints_64x128/
best_model_path........................ checkpoints_64x128/best_model.pt
training_log_path...................... results_64x128/training_log.csv
random_seed............................ 42
num_workers............................ 4
distributed............................ True
dist_backend........................... nccl
find_unused_parameters................. False
sync_batchnorm......................... False
master_port............................ 29500

-- Model Inference --

  model_path............................. checkpoints_64x128/best_model.pt
  input_data_path........................ data/processed_64x128/sst_final.nc
  run_validation......................... True
  run_test............................... True
  forecast_lead_steps.................... 90
  chain_stride_steps..................... 90
  output_dir............................. results_64x128/
  val_predictions_file................... predictions_val.nc
  test_predictions_file.................. predictions_test.nc

-- Concat Chains --
```

```
  predictions_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predictions_file_template..... predictions_{split}.nc
  std_params_path..... data/processed_64x128/std_params.nc
  trend_params_path..... data/processed_64x128/trend_params.nc
  output_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  actual_file_template..... sst_actual_{split}_continuous.nc
  predicted_file_template..... sst_predicted_{split}_chains.nc
  splits..... ['test', 'val']
  lon_correction_threshold..... 0.6

-- Calc DHW --

  mmm_path..... /home/272/xl5758/ct5km_climatology_v3.1.nc
  results_dir..... /scratch/um09/xl5758/DHW-project/results_64x128
  predicted_file_template..... sst_predicted_{split}_chains.nc
  actual_file_template..... sst_actual_{split}_continuous.nc
  dhw_output_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_file_template..... dhw_products_{split}.nc
  run_test..... True
  run_validation..... True
  window_days..... 84
  hotspot_threshold..... 1

-- DHW Plots --

  dhw_dir..... /scratch/um09/xl5758/DHW-project/dhw_output_64x128
  output_pdf..... dhw_analysis_combined.pdf
  output_csv..... dhw_metrics_combined.csv
  spatial_lead_steps..... [1, 7, 14, 30, 60, 90]
  n_sample_chains..... 6
  plot_init_dates..... ['2016-05-20', '2016-10-15', '2017-02-10', '2017-07-09', '20
  dhw_vmax..... 20
  dhw_diff_vmax..... 8
  tropic_lat..... 30
  dhw_cmap..... ['#CCe6F0', '#441C94', '#6F47BF', '#815CCC', '#A383E6', '#ED
  error_cmap..... ['#053061', '#2166AC', '#92C5DE', '#F7F7F7', '#F4A582', '#D6
  rmse_cmap..... YlOrRd
  bias_cmap..... RdBu_r
  bleaching_thresholds..... [(0, 4, 'No Stress', '#27AE60'), (4, 8, 'Warning', '#F39C12'
  gbr_lat_bounds..... (-25.0, -10.0)
  gbr_lon_bounds..... (142.0, 157.0)
  reef_boxes..... [{'name': 'Great Barrier Reef', 'lat': (-25.0, -10.0), 'lon'
```

DHW Analysis - Summary

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DEGREE HEATING WEEKS (DHW) - COMBINED ANALYSIS
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METHODOLOGY  (NOAA Coral Reef Watch v3.1)
  SSTA      = SST - DC              (DC = daily climatology)
  HotSpot   = max(SST - MMM, 0)     (MMM = Maximum Monthly Mean)
  DHW_i     = (1/7) x sum(HotSpot_j) for j=i-83..i, HotSpot_j >= 1.0C

SLIDING WINDOW CONSTRUCTION  (always 84 days ending on valid date)
  Lead L:  (84-L) actual + L predicted SST days
  Lead 1:  83 actual +  1 predicted
  Lead 30: 54 actual + 30 predicted
  Lead 84:  0 actual + 84 predicted
  Lead 90: previous 84 predicted days (days 7-90)

VERIFICATION
  Actual DHW  : 84 observed days ending on the forecast valid date
  Persistence : observed DHW on the day before lead 1, held constant
  Sample      : chains defined at every lead step (same chains for
                model and persistence at all leads)

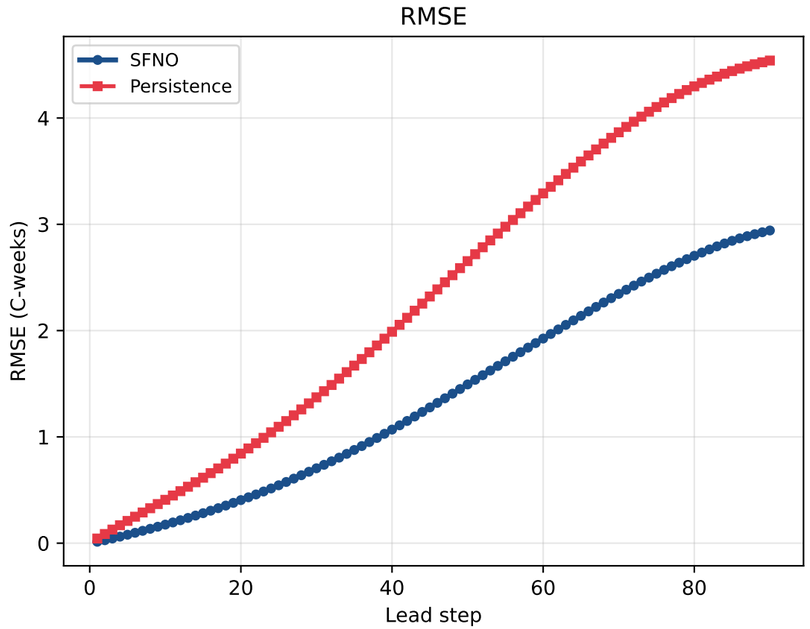
BLEACHING ALERT THRESHOLDS
  DHW  0-4  : No Stress
  DHW  4-8  : Warning
  DHW  8-12 : Alert Level 1
  DHW 12-16 : Alert Level 2
  DHW 16-20 : Alert Level 3
  DHW 20->= : Alert Level 4-5

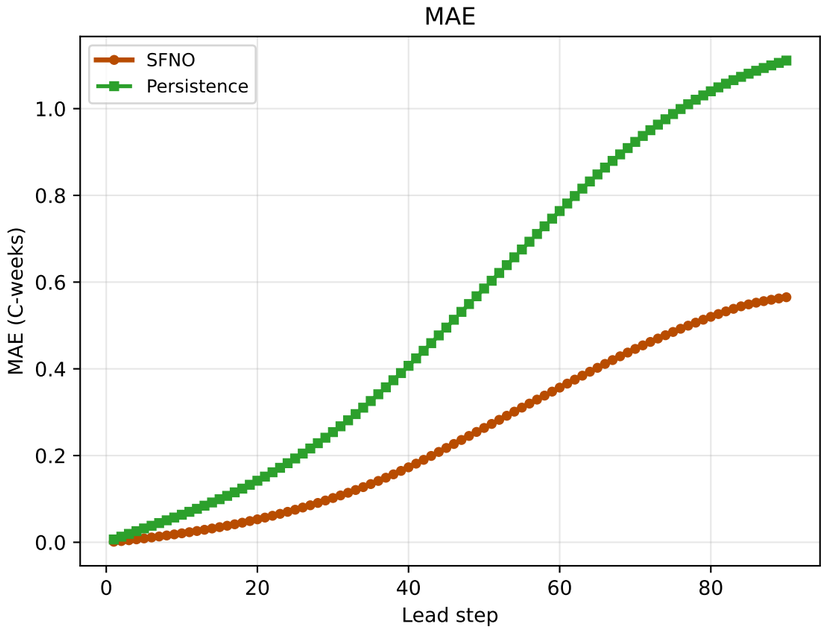
DATASETS
  Test          19 chains x 90 leads x 64 lat x 128 lon
  Val           16 chains x 90 leads x 64 lat x 128 lon
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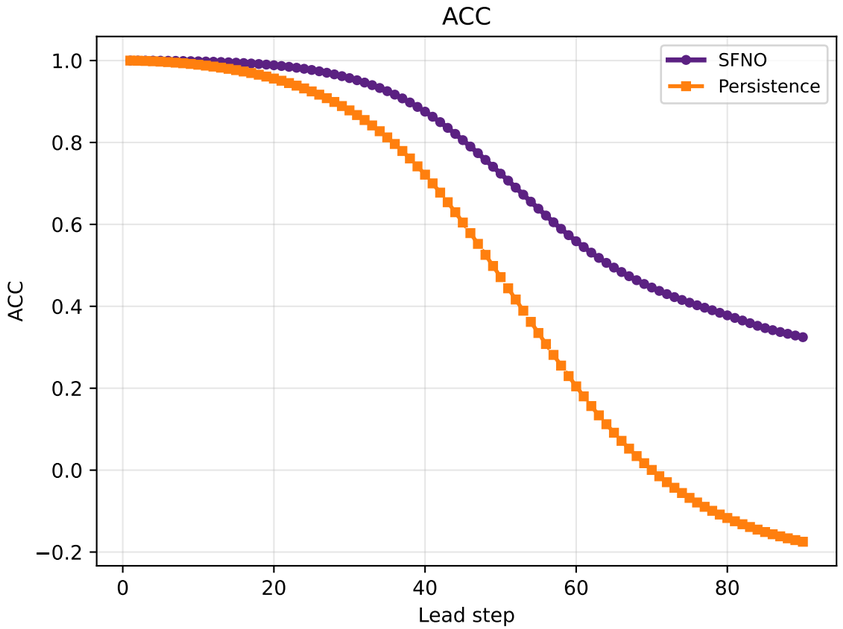
DHW Metrics by Lead Step - Test (18 chains scored at every lead)

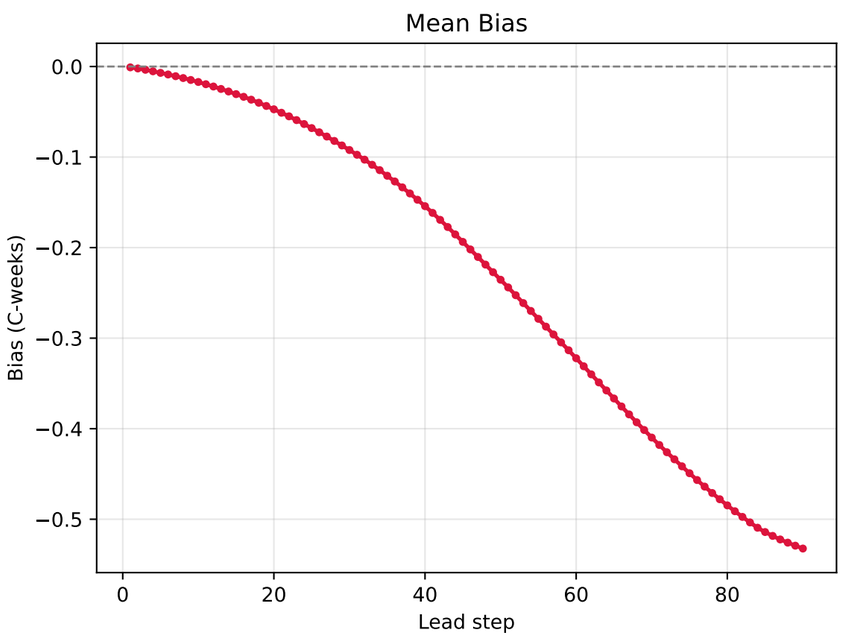
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0134	0.0437	69.3%	0.0013	0.0065	-0.0009	1.0000	0.9999	1.0000
2	0.0277	0.0862	67.8%	0.0028	0.0130	-0.0022	1.0000	0.9996	0.9999
3	0.0439	0.1278	65.6%	0.0047	0.0193	-0.0036	0.9999	0.9990	0.9998
4	0.0613	0.1687	63.6%	0.0068	0.0257	-0.0053	0.9998	0.9983	0.9996
5	0.0795	0.2091	62.0%	0.0089	0.0319	-0.0070	0.9996	0.9973	0.9994
6	0.0976	0.2491	60.8%	0.0111	0.0381	-0.0088	0.9994	0.9962	0.9990
7	0.1161	0.2888	59.8%	0.0134	0.0443	-0.0107	0.9992	0.9949	0.9986
8	0.1351	0.3285	58.9%	0.0158	0.0507	-0.0127	0.9989	0.9934	0.9981
9	0.1547	0.3680	58.0%	0.0182	0.0571	-0.0149	0.9985	0.9917	0.9975
10	0.1745	0.4079	57.2%	0.0207	0.0635	-0.0171	0.9981	0.9898	0.9968
11	0.1948	0.4484	56.5%	0.0233	0.0702	-0.0195	0.9977	0.9877	0.9960
12	0.2157	0.4893	55.9%	0.0260	0.0772	-0.0221	0.9971	0.9853	0.9951
13	0.2374	0.5310	55.3%	0.0288	0.0844	-0.0248	0.9964	0.9827	0.9940
14	0.2598	0.5733	54.7%	0.0318	0.0918	-0.0276	0.9957	0.9798	0.9927
15	0.2822	0.6158	54.2%	0.0349	0.0993	-0.0305	0.9948	0.9767	0.9913
16	0.3054	0.6590	53.7%	0.0381	0.1071	-0.0335	0.9939	0.9733	0.9897
17	0.3292	0.7031	53.2%	0.0415	0.1152	-0.0366	0.9928	0.9695	0.9879
18	0.3538	0.7484	52.7%	0.0451	0.1237	-0.0400	0.9915	0.9654	0.9858
19	0.3793	0.7951	52.3%	0.0490	0.1326	-0.0436	0.9901	0.9609	0.9835
20	0.4053	0.8429	51.9%	0.0529	0.1419	-0.0473	0.9885	0.9560	0.9809
21	0.4316	0.8913	51.6%	0.0569	0.1515	-0.0511	0.9867	0.9507	0.9780
22	0.4583	0.9406	51.3%	0.0611	0.1614	-0.0550	0.9847	0.9450	0.9749
23	0.4861	0.9912	51.0%	0.0656	0.1717	-0.0592	0.9824	0.9387	0.9713
24	0.5148	1.0427	50.6%	0.0702	0.1824	-0.0635	0.9799	0.9319	0.9673
25	0.5447	1.0951	50.3%	0.0751	0.1933	-0.0680	0.9770	0.9246	0.9628
26	0.5758	1.1486	49.9%	0.0802	0.2046	-0.0726	0.9738	0.9167	0.9578
27	0.6073	1.2029	49.5%	0.0853	0.2163	-0.0773	0.9702	0.9081	0.9523
28	0.6392	1.2585	49.2%	0.0907	0.2285	-0.0822	0.9662	0.8989	0.9463
29	0.6718	1.3152	48.9%	0.0963	0.2411	-0.0872	0.9619	0.8889	0.9397
30	0.7044	1.3721	48.7%	0.1021	0.2541	-0.0922	0.9571	0.8783	0.9326
31	0.7372	1.4298	48.4%	0.1080	0.2675	-0.0975	0.9519	0.8669	0.9249
32	0.7705	1.4883	48.2%	0.1142	0.2813	-0.1029	0.9462	0.8547	0.9166
33	0.8047	1.5482	48.0%	0.1205	0.2956	-0.1085	0.9399	0.8416	0.9075
34	0.8402	1.6090	47.8%	0.1272	0.3104	-0.1145	0.9330	0.8274	0.8975
35	0.8768	1.6707	47.5%	0.1342	0.3257	-0.1206	0.9252	0.8123	0.8867
36	0.9138	1.7328	47.3%	0.1414	0.3414	-0.1270	0.9168	0.7963	0.8750
37	0.9514	1.7960	47.0%	0.1489	0.3573	-0.1335	0.9076	0.7792	0.8624
38	0.9898	1.8599	46.8%	0.1566	0.3737	-0.1402	0.8976	0.7609	0.8490
39	1.0291	1.9244	46.5%	0.1646	0.3902	-0.1471	0.8868	0.7417	0.8346
40	1.0686	1.9890	46.3%	0.1728	0.4070	-0.1543	0.8752	0.7214	0.8194
41	1.1088	2.0544	46.0%	0.1814	0.4241	-0.1616	0.8629	0.7001	0.8035
42	1.1498	2.1205	45.8%	0.1901	0.4416	-0.1694	0.8497	0.6776	0.7868
43	1.1919	2.1870	45.5%	0.1991	0.4592	-0.1773	0.8356	0.6541	0.7693
44	1.2346	2.2540	45.2%	0.2083	0.4771	-0.1854	0.8209	0.6297	0.7514
45	1.2773	2.3209	45.0%	0.2175	0.4951	-0.1937	0.8057	0.6045	0.7331
46	1.3200	2.3878	44.7%	0.2267	0.5132	-0.2020	0.7900	0.5787	0.7146
47	1.3632	2.4546	44.5%	0.2359	0.5313	-0.2104	0.7739	0.5524	0.6959
48	1.4068	2.5213	44.2%	0.2452	0.5494	-0.2188	0.7574	0.5255	0.6769
49	1.4501	2.5876	44.0%	0.2545	0.5673	-0.2271	0.7407	0.4985	0.6581
50	1.4936	2.6535	43.7%	0.2637	0.5852	-0.2355	0.7238	0.4712	0.6392
51	1.5367	2.7188	43.5%	0.2729	0.6031	-0.2440	0.7069	0.4439	0.6207
52	1.5804	2.7840	43.2%	0.2823	0.6211	-0.2526	0.6898	0.4166	0.6021
53	1.6240	2.8487	43.0%	0.2918	0.6391	-0.2613	0.6726	0.3892	0.5838
54	1.6675	2.9131	42.8%	0.3012	0.6572	-0.2700	0.6554	0.3620	0.5658
55	1.7111	2.9772	42.5%	0.3106	0.6752	-0.2786	0.6383	0.3348	0.5481
56	1.7543	3.0409	42.3%	0.3198	0.6931	-0.2873	0.6216	0.3079	0.5309
57	1.7971	3.1042	42.1%	0.3290	0.7110	-0.2959	0.6052	0.2813	0.5142
58	1.8399	3.1668	41.9%	0.3383	0.7287	-0.3046	0.5891	0.2552	0.4980
59	1.8826	3.2292	41.7%	0.3474	0.7463	-0.3134	0.5737	0.2296	0.4825
60	1.9252	3.2912	41.5%	0.3567	0.7638	-0.3222	0.5588	0.2045	0.4676
61	1.9681	3.3528	41.3%	0.3659	0.7813	-0.3311	0.5446	0.1801	0.4534
62	2.0110	3.4139	41.1%	0.3751	0.7985	-0.3400	0.5311	0.1565	0.4399
63	2.0538	3.4741	40.9%	0.3843	0.8155	-0.3489	0.5182	0.1338	0.4271
64	2.0962	3.5329	40.7%	0.3933	0.8320	-0.3578	0.5060	0.1121	0.4149
65	2.1386	3.5910	40.4%	0.4024	0.8483	-0.3666	0.4945	0.0913	0.4035
66	2.1808	3.6481	40.2%	0.4113	0.8641	-0.3755	0.4837	0.0715	0.3927
67	2.2227	3.7045	40.0%	0.4202	0.8795	-0.3842	0.4735	0.0524	0.3824
68	2.2643	3.7596	39.8%	0.4289	0.8945	-0.3929	0.4637	0.0343	0.3727
69	2.3052	3.8136	39.6%	0.4375	0.9091	-0.4015	0.4545	0.0169	0.3634
70	2.3452	3.8659	39.3%	0.4459	0.9233	-0.4098	0.4458	0.0005	0.3546
71	2.3846	3.9166	39.1%	0.4541	0.9370	-0.4180	0.4376	-0.0149	0.3464
72	2.4233	3.9656	38.9%	0.4620	0.9503	-0.4260	0.4298	-0.0294	0.3386
73	2.4615	4.0132	38.7%	0.4698	0.9631	-0.4338	0.4225	-0.0431	0.3312
74	2.4986	4.0591	38.4%	0.4775	0.9755	-0.4416	0.4155	-0.0558	0.3243
75	2.5354	4.1036	38.2%	0.4850	0.9875	-0.4492	0.4088	-0.0678	0.3177
76	2.5713	4.1466	38.0%	0.4924	0.9991	-0.4567	0.4025	-0.0790	0.3114
77	2.6058	4.1876	37.8%	0.4995	1.0101	-0.4640	0.3965	-0.0895	0.3054
78	2.6393	4.2264	37.6%	0.5065	1.0206	-0.4710	0.3904	-0.0993	0.2993
79	2.6720	4.2630	37.3%	0.5133	1.0305	-0.4779	0.3841	-0.1084	0.2933
80	2.7038	4.2978	37.1%	0.5199	1.0400	-0.4847	0.3779	-0.1168	0.2873
81	2.7343	4.3305	36.9%	0.5263	1.0491	-0.4912	0.3716	-0.1248	0.2814
82	2.7639	4.3612	36.6%	0.5325	1.0576	-0.4974	0.3653	-0.1321	0.2755
83	2.7924	4.3898	36.4%	0.5384	1.0657	-0.5035	0.3588	-0.1388	0.2695
84	2.8198	4.4167	36.2%	0.5441	1.0733	-0.5094	0.3524	-0.1452	0.2636
85	2.8447	4.4418	36.0%	0.5486	1.0805	-0.5141	0.3468	-0.1511	0.2586
86	2.8674	4.4649	35.8%	0.5525	1.0873	-0.5184	0.3418	-0.1565	0.2541
87	2.8883	4.4865	35.6%	0.5561	1.0937	-0.5223	0.3372	-0.1616	0.2501
88	2.9074	4.5062	35.5%	0.5593	1.0998	-0.5259	0.3329	-0.1664	0.2464
89	2.9253	4.5244	35.3%	0.5623	1.1055	-0.5292	0.3287	-0.1708	0.2428
90	2.9419	4.5408	35.2%	0.5651	1.1109	-0.5324	0.3246	-0.1748	0.2395

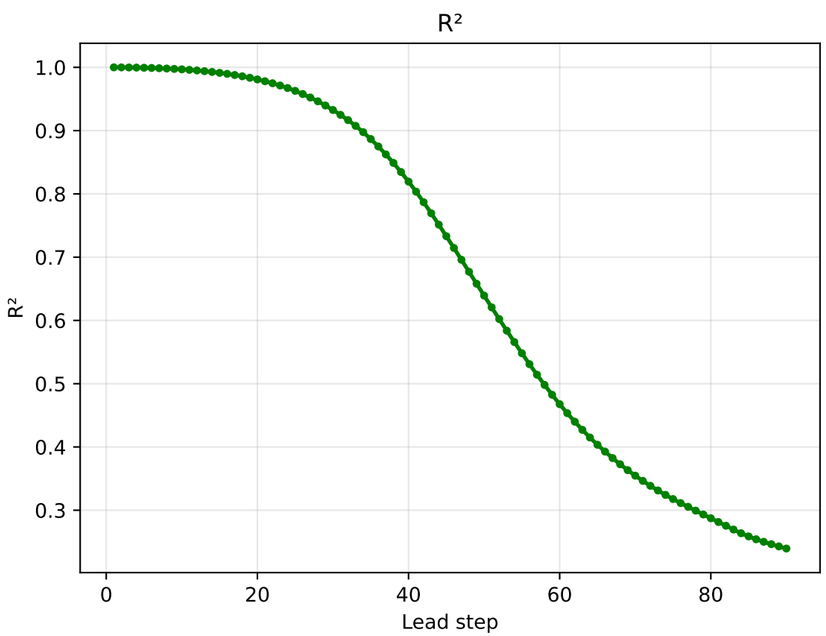
DHW Forecast Skill vs Lead Step - Test

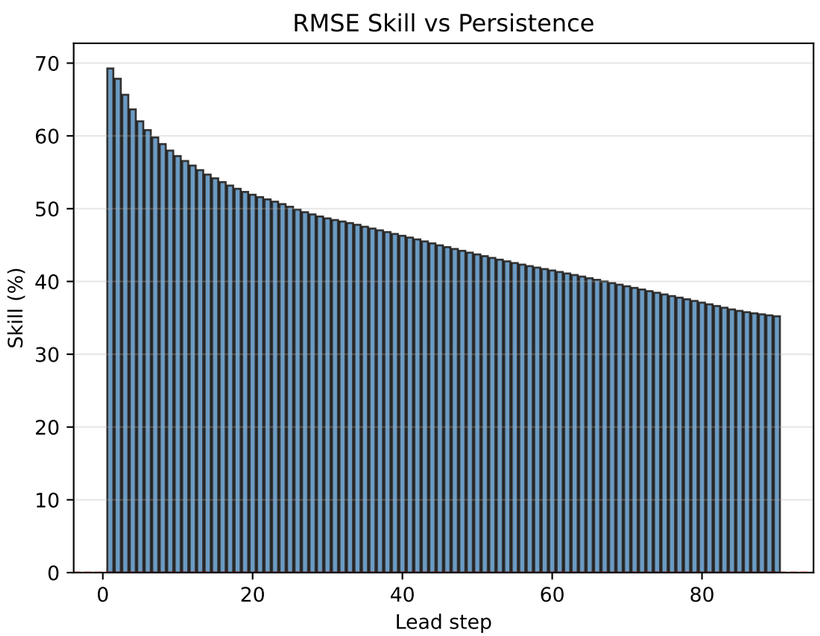


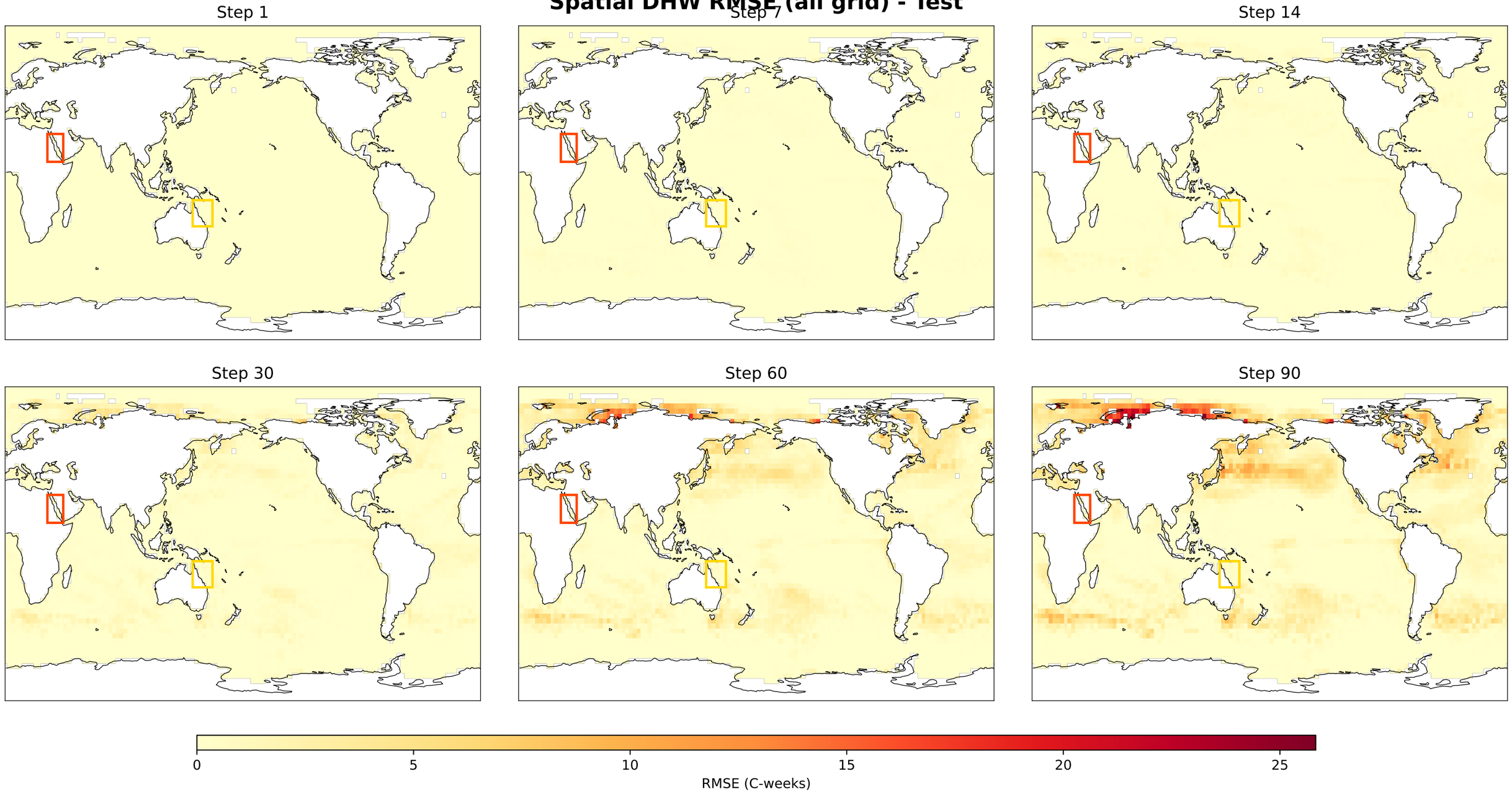


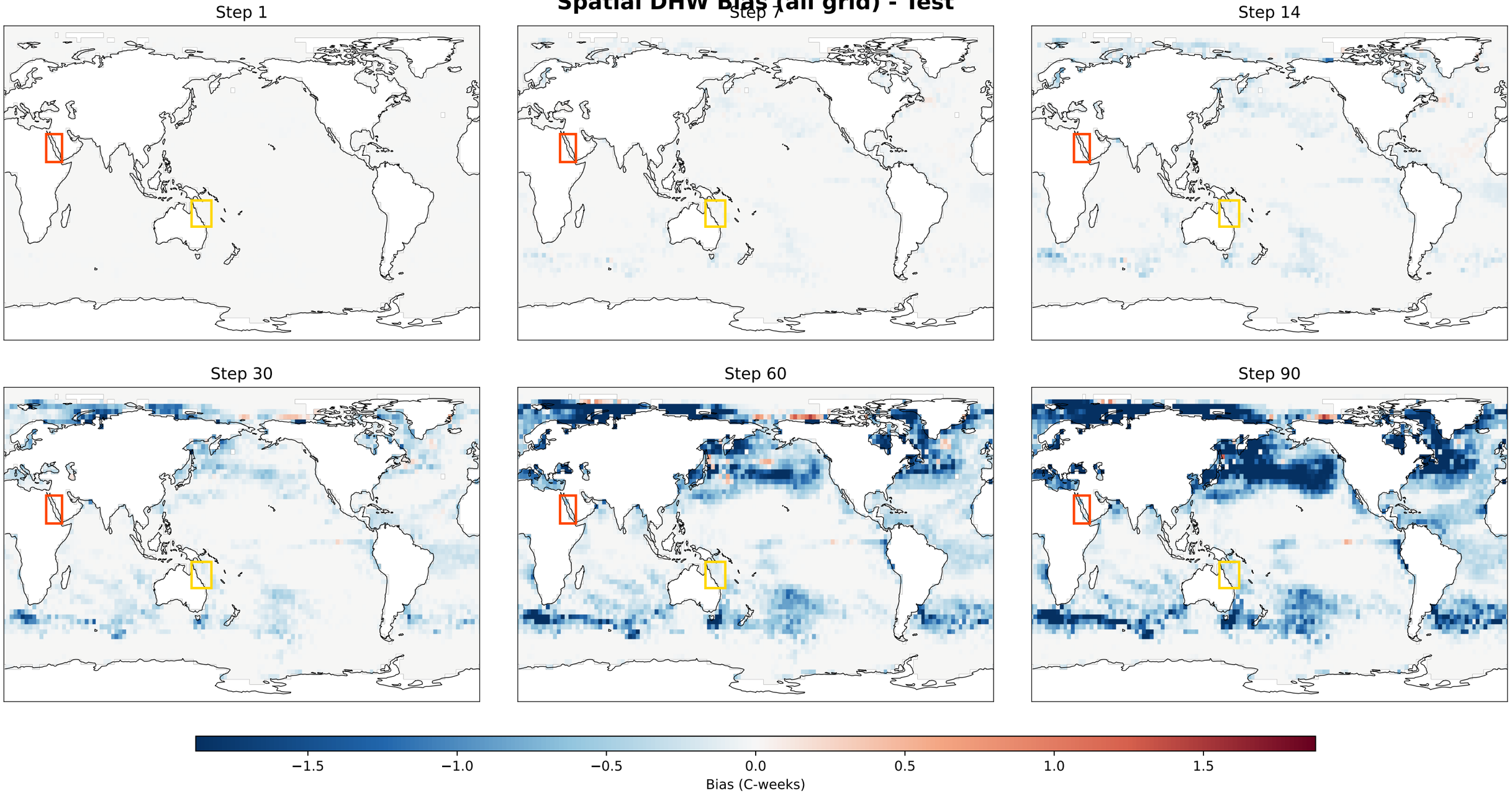


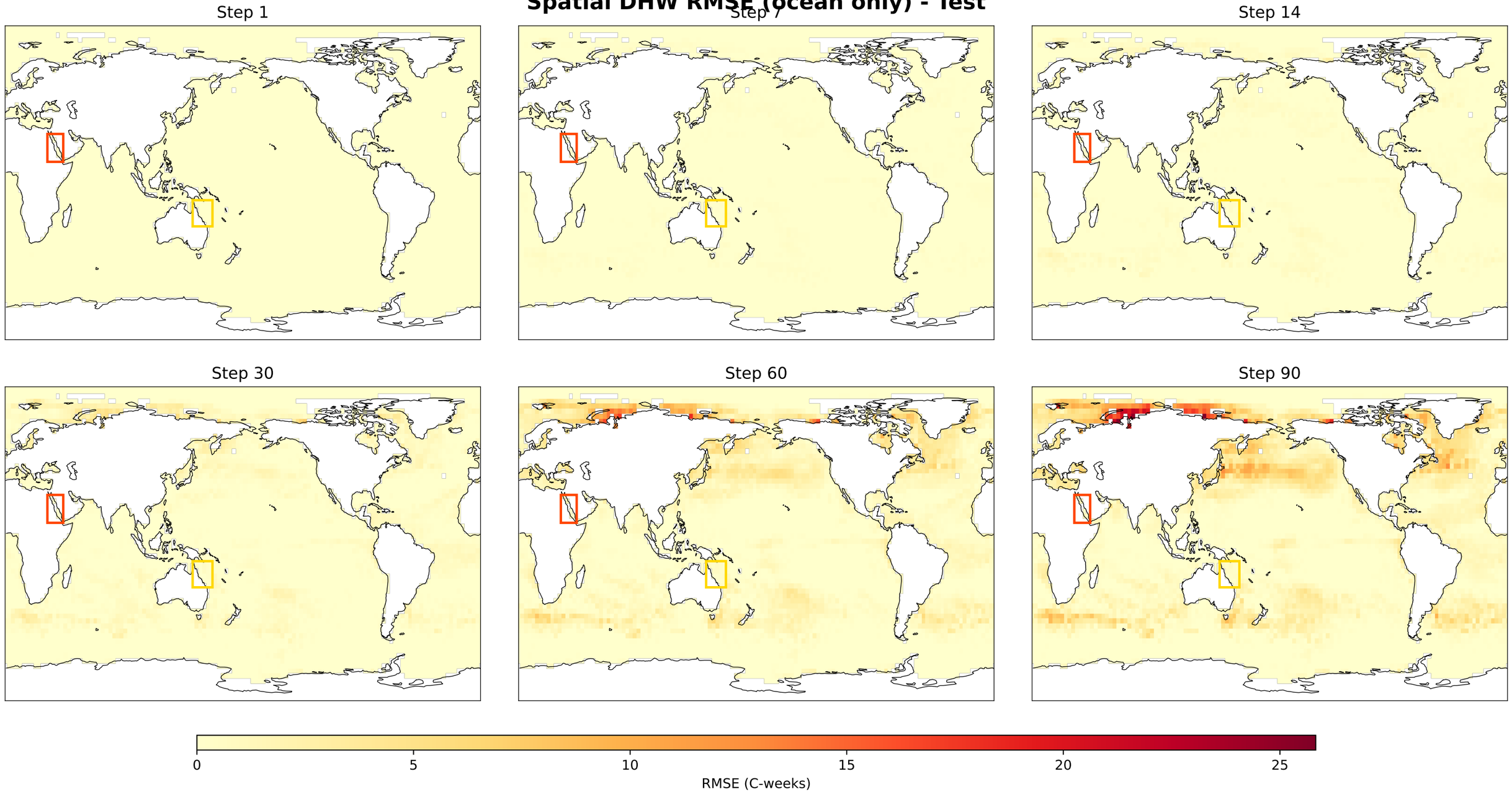


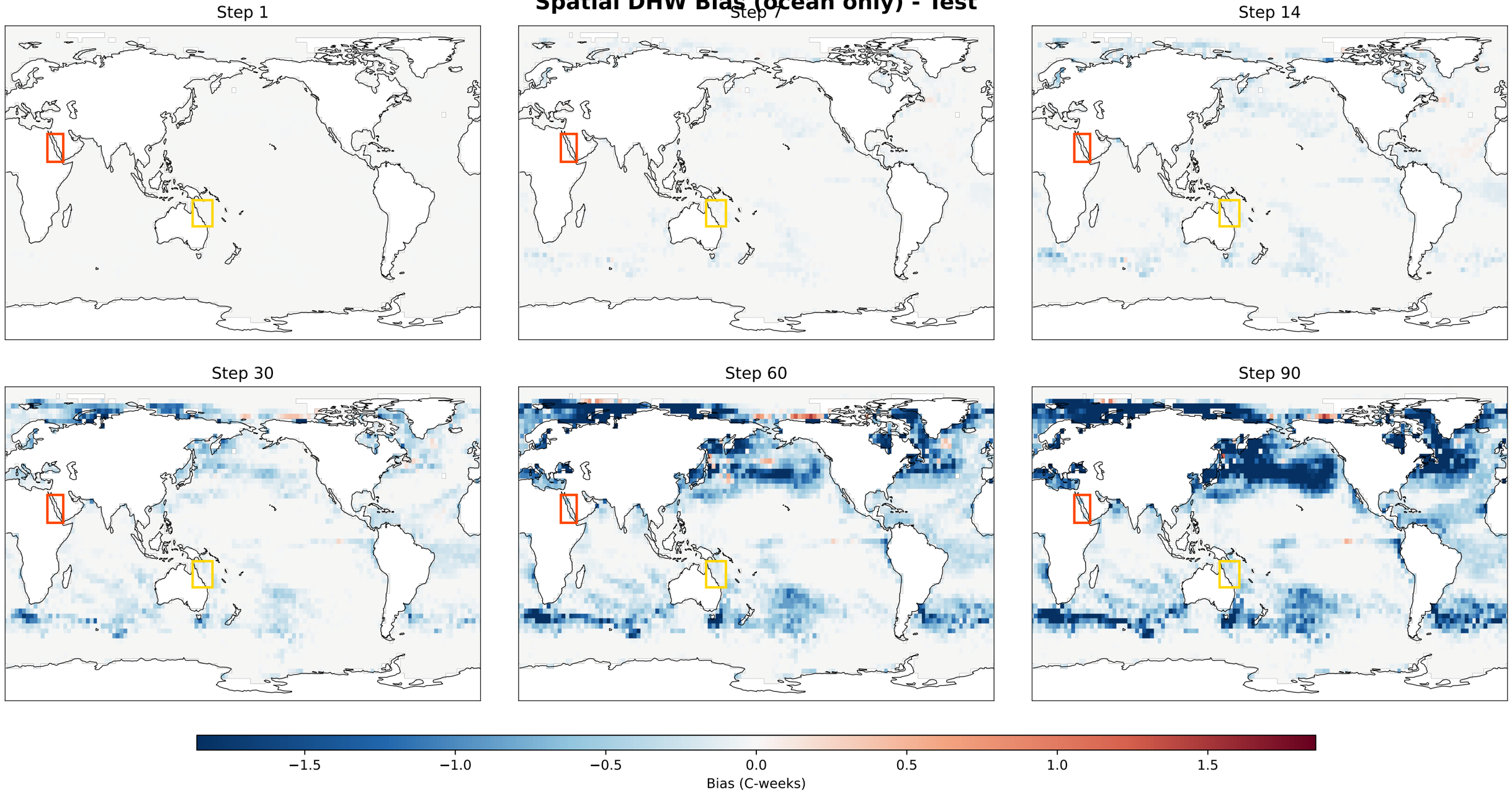





Spatial DHW RMSE (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

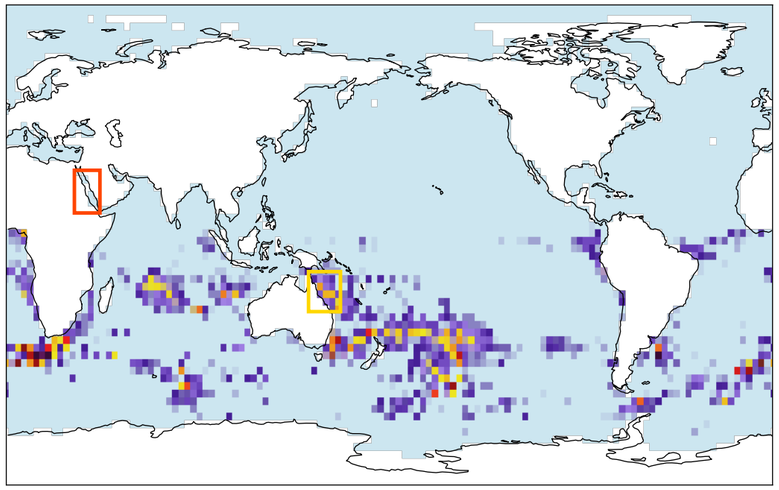

Spatial DHW Bias (all grid) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
5
10
15
20
25
RMSE (C-weeks)

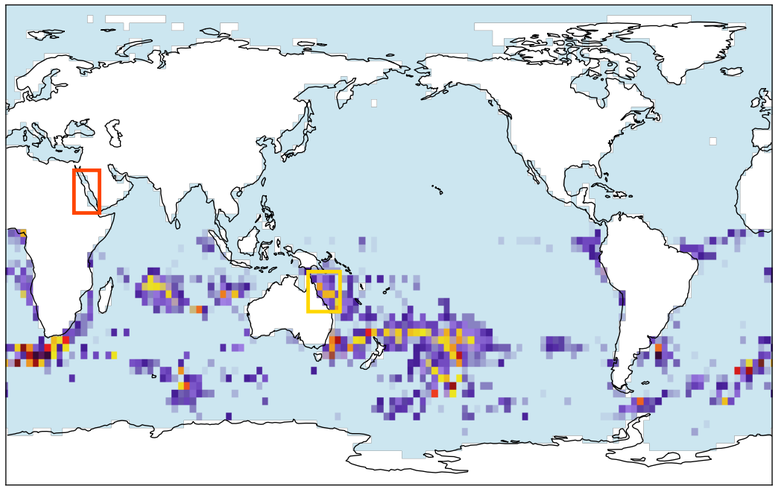

Spatial DHW Bias (ocean only) - Test
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test (C-weeks)

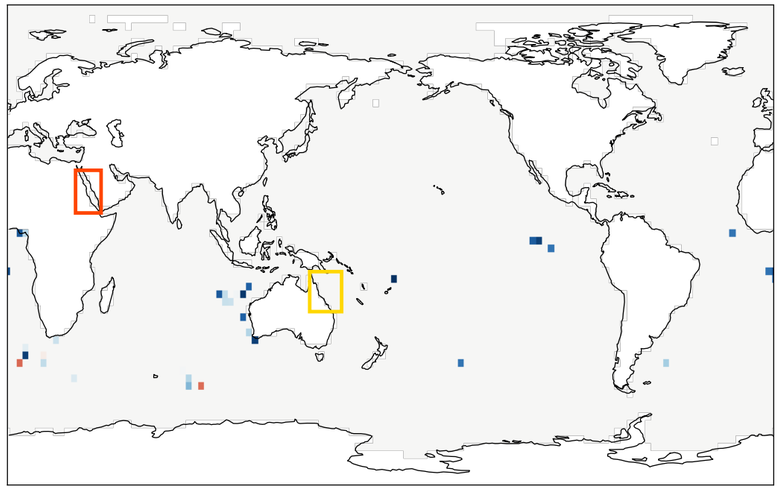
Actual DHW - Step 1



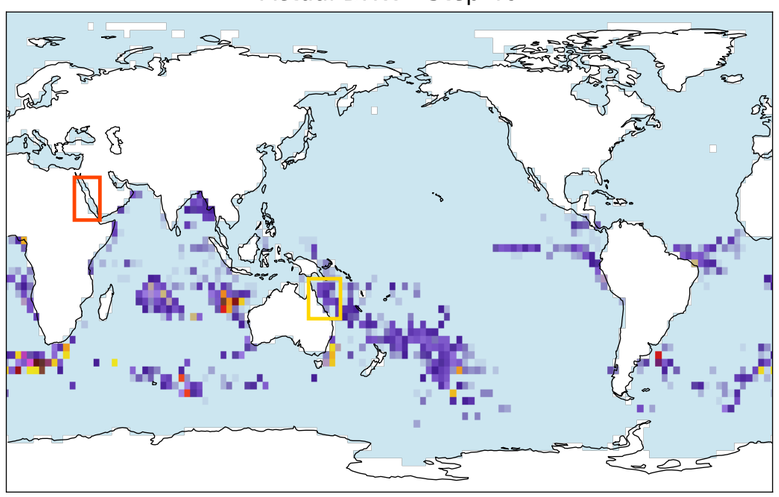
Predicted DHW - Step 1



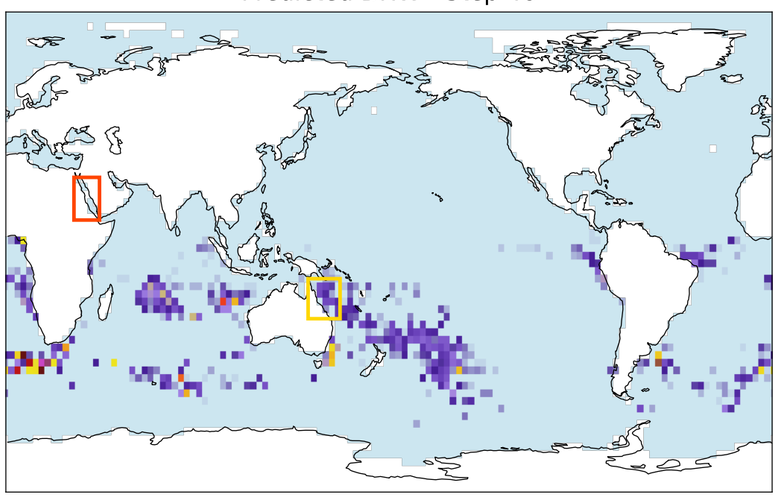
Error (Pred - Actual) - Step 1 [scale ±0.24]



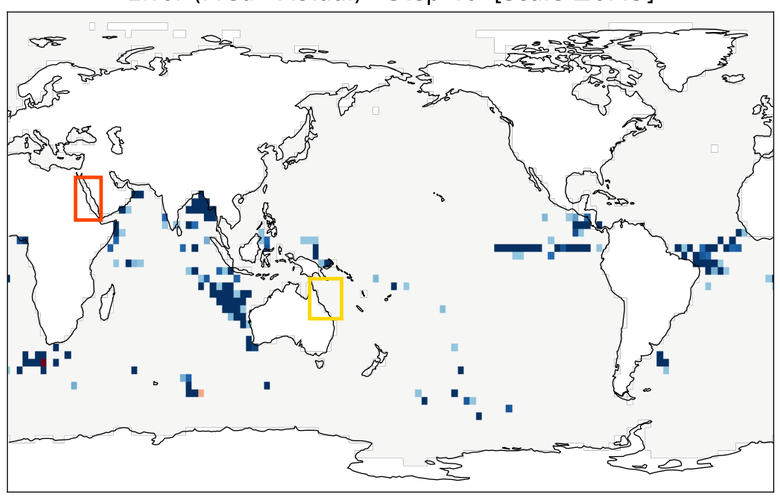
Actual DHW - Step 46



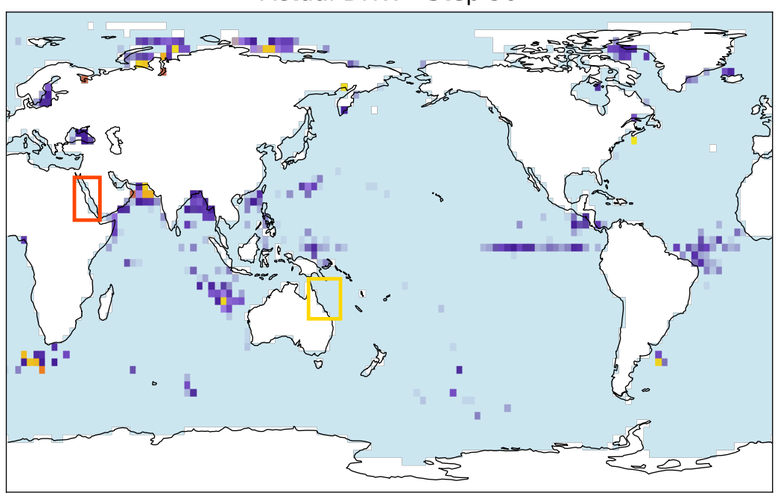
Predicted DHW - Step 46



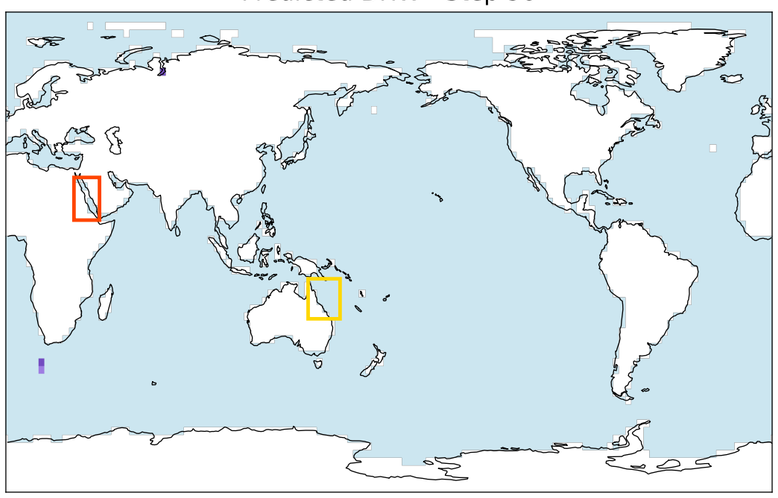
Error (Pred - Actual) - Step 46 [scale ±0.45]



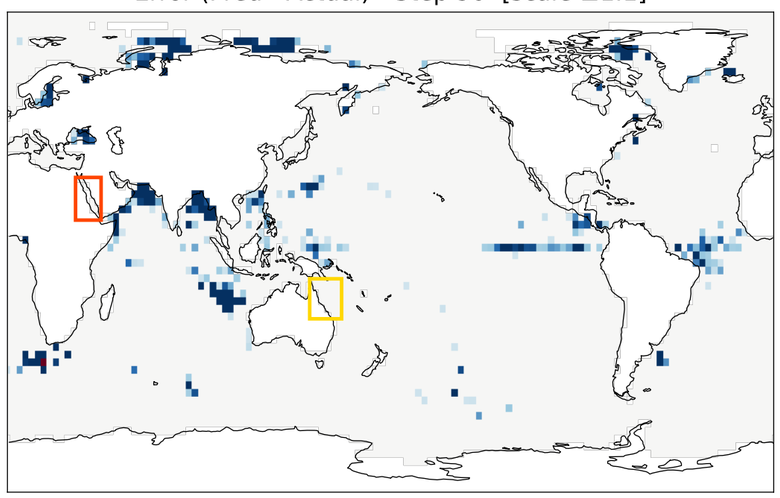
Actual DHW - Step 90

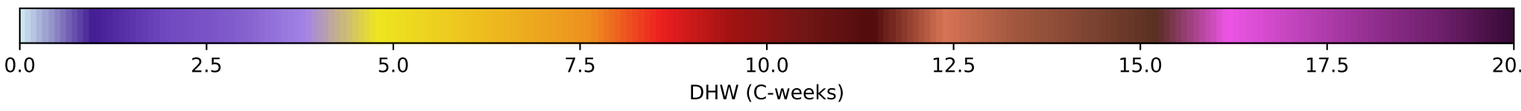


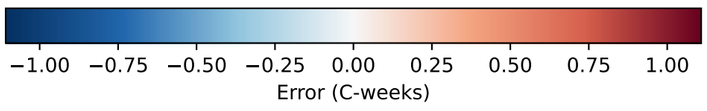
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±1.1]

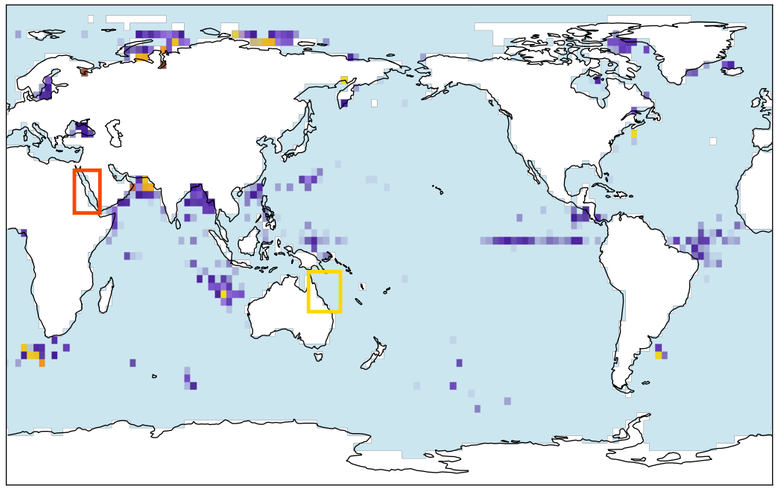




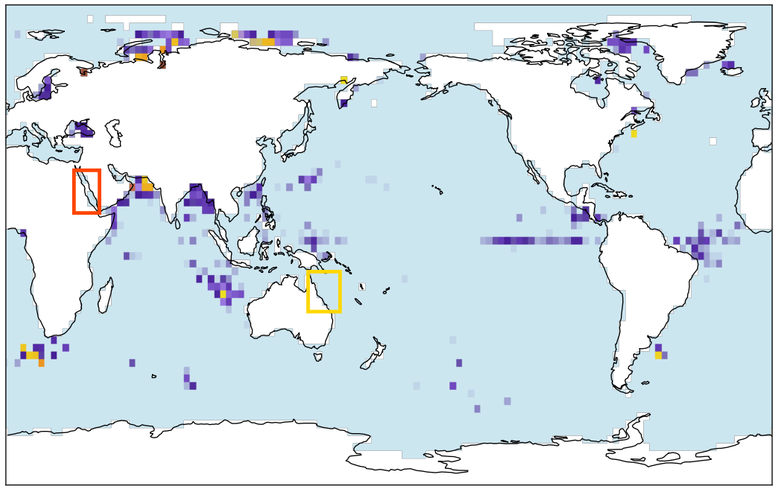


DHW Chain init: 2020-07-09 - Test (C-weeks)

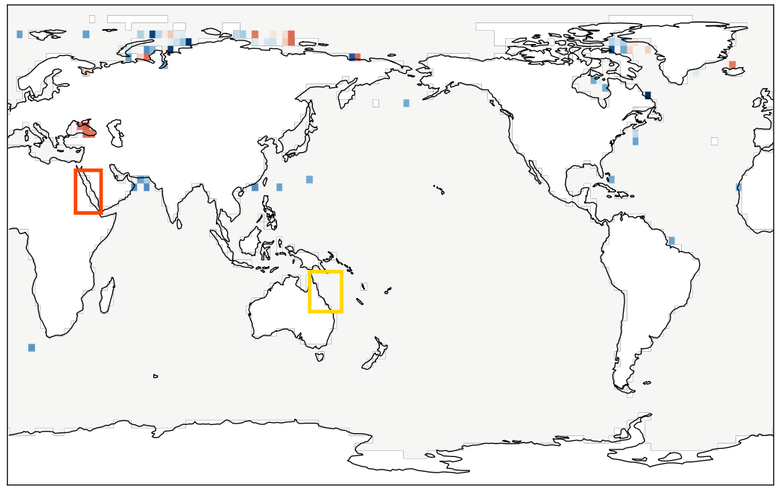
Actual DHW - Step 1



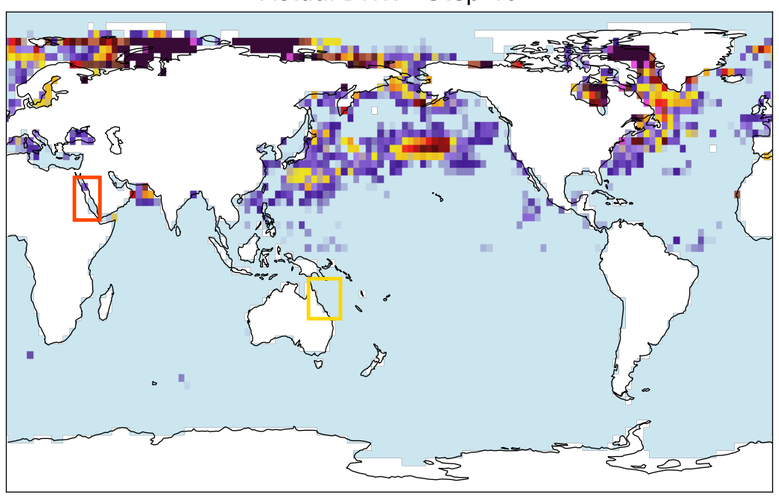
Predicted DHW - Step 1



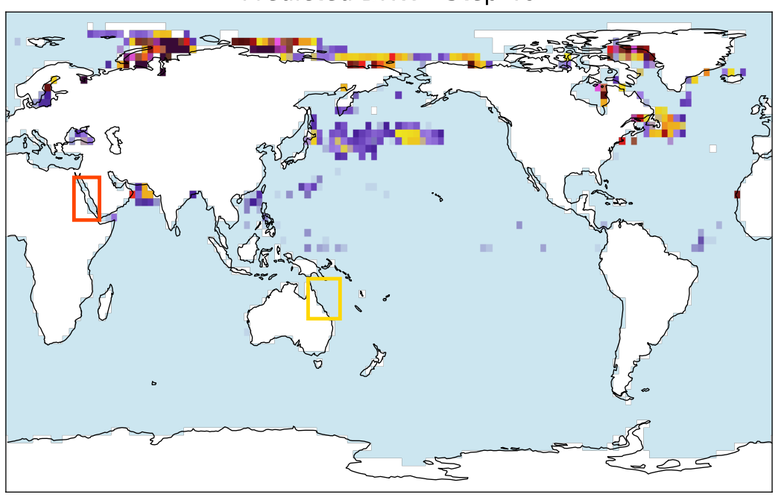
Error (Pred - Actual) - Step 1 [scale ±0.36]



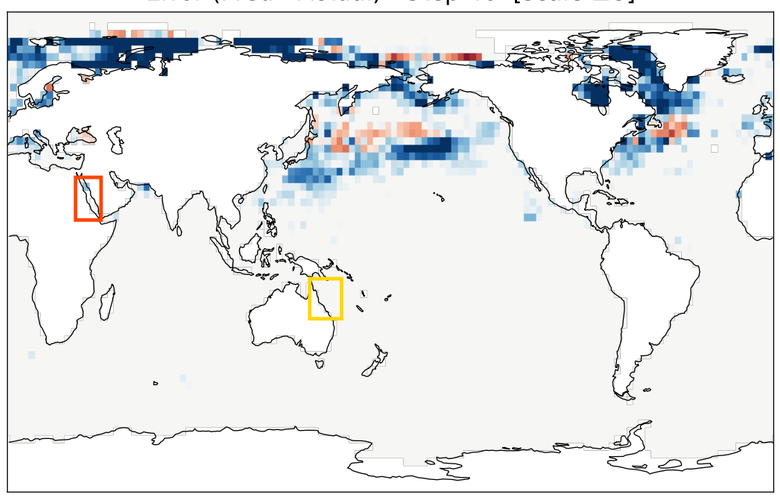
Actual DHW - Step 46



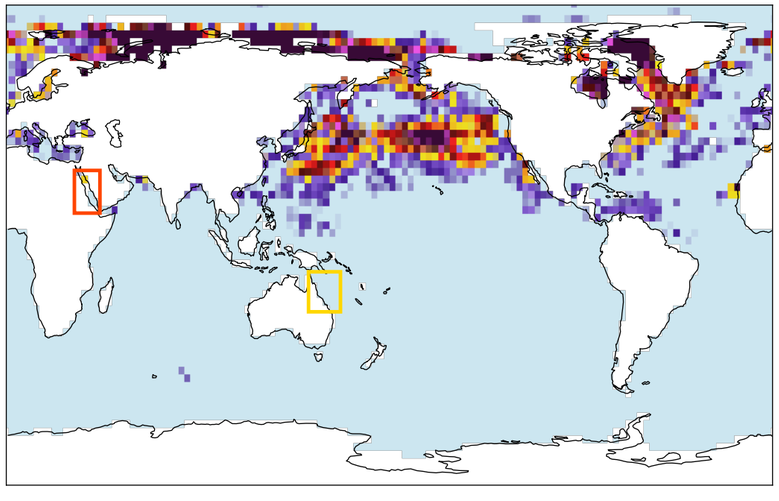
Predicted DHW - Step 46



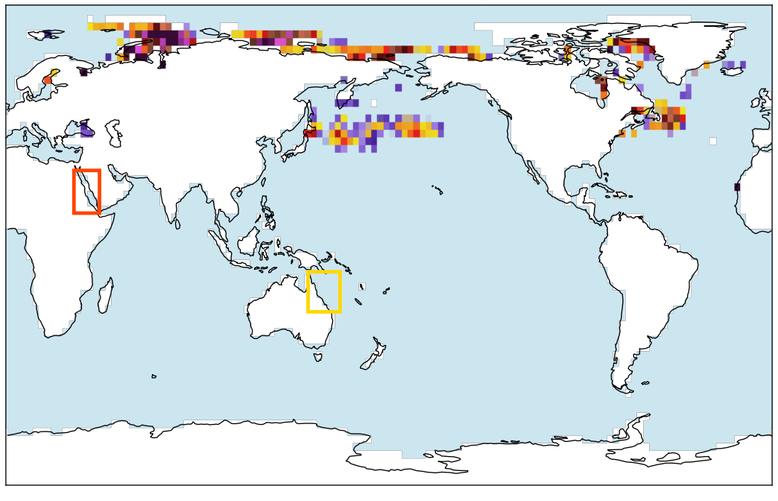
Error (Pred - Actual) - Step 46 [scale ±8]



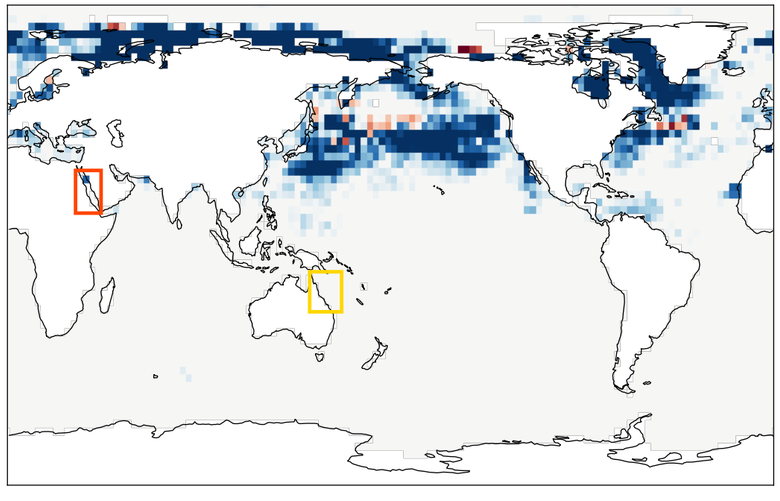
Actual DHW - Step 90

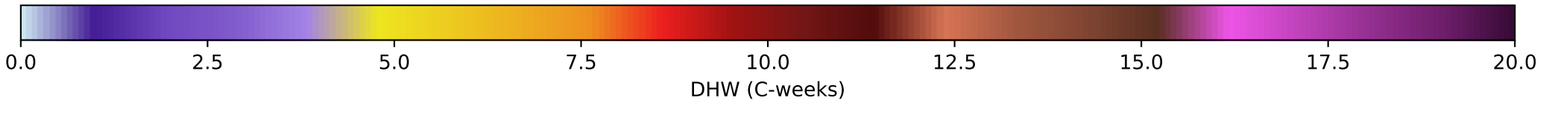


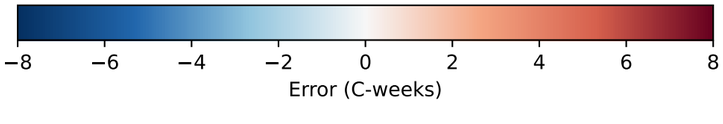
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

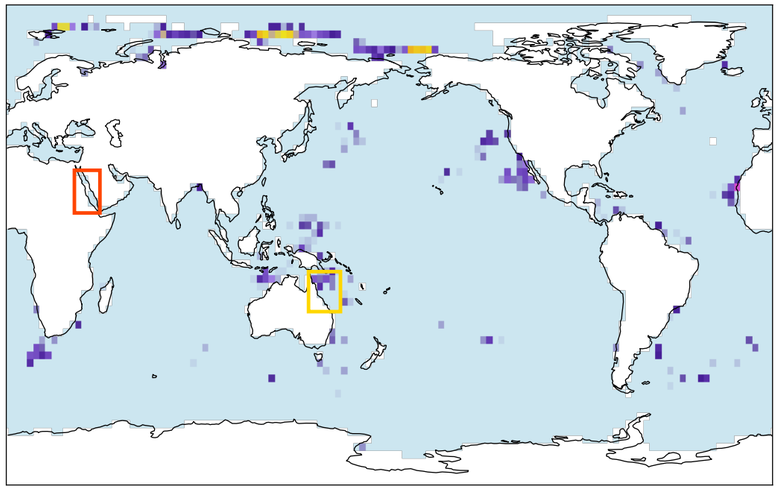




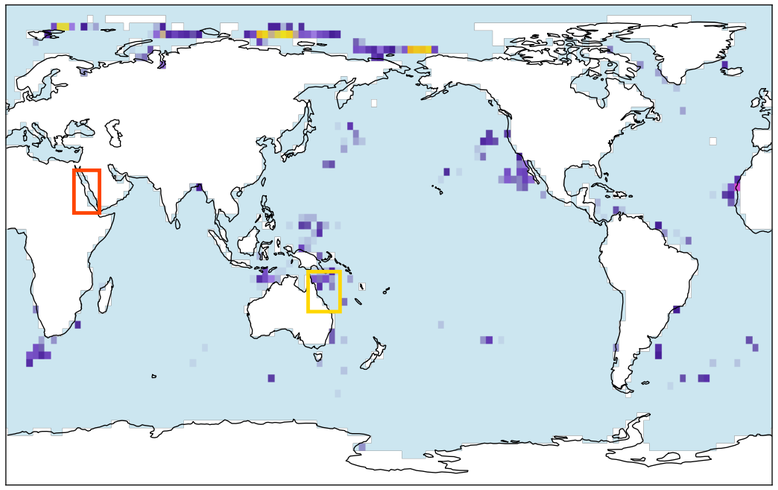


DHW Chain init: 2021-01-06 - Test (C-weeks)

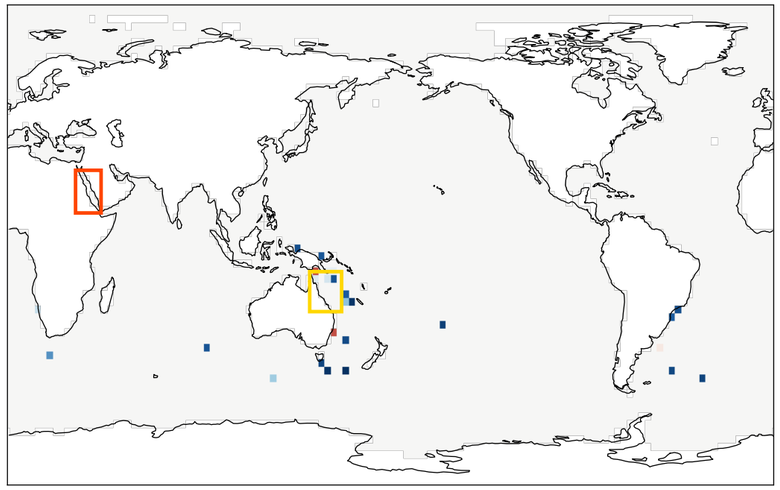
Actual DHW - Step 1



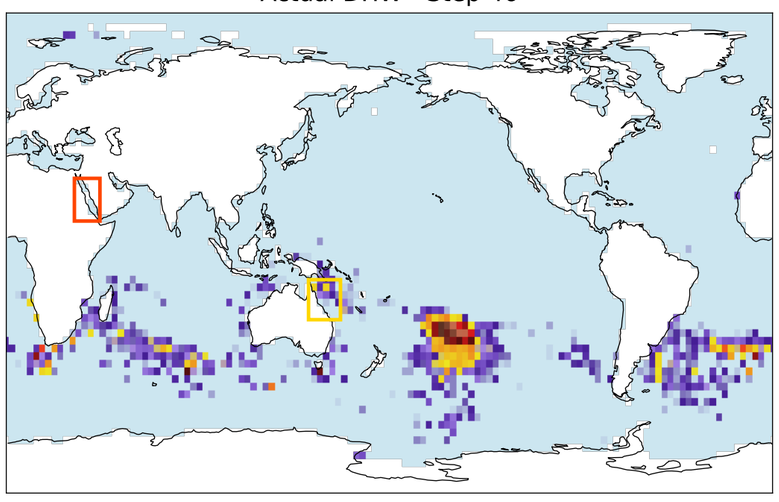
Predicted DHW - Step 1



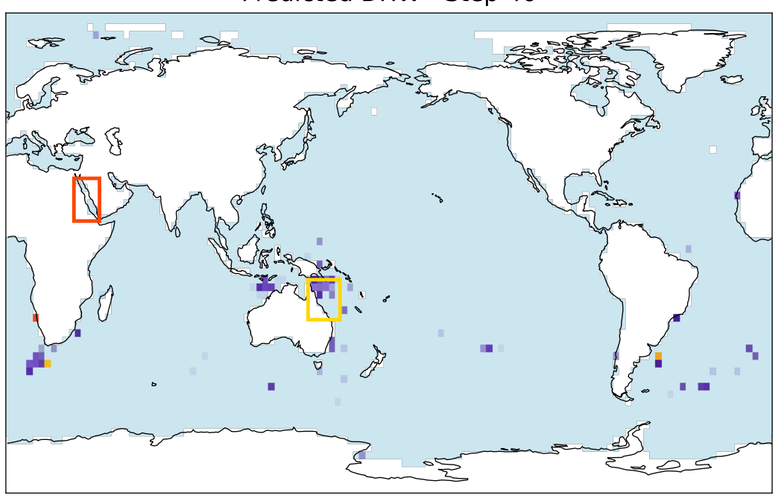
Error (Pred - Actual) - Step 1 [scale ±0.2]



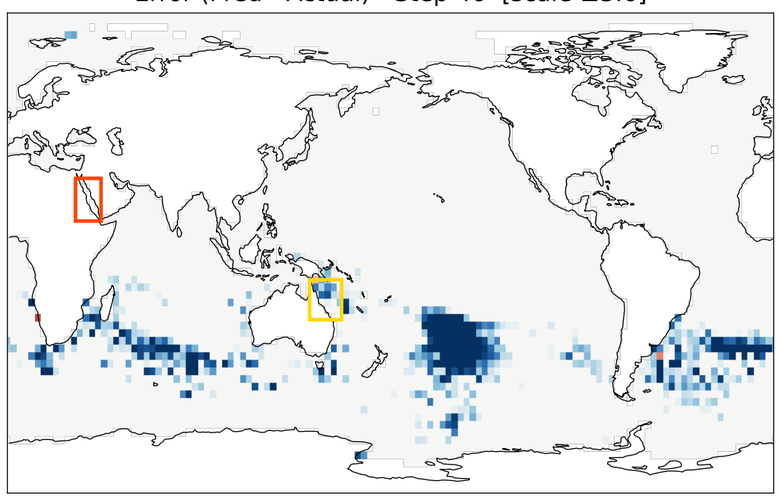
Actual DHW - Step 46



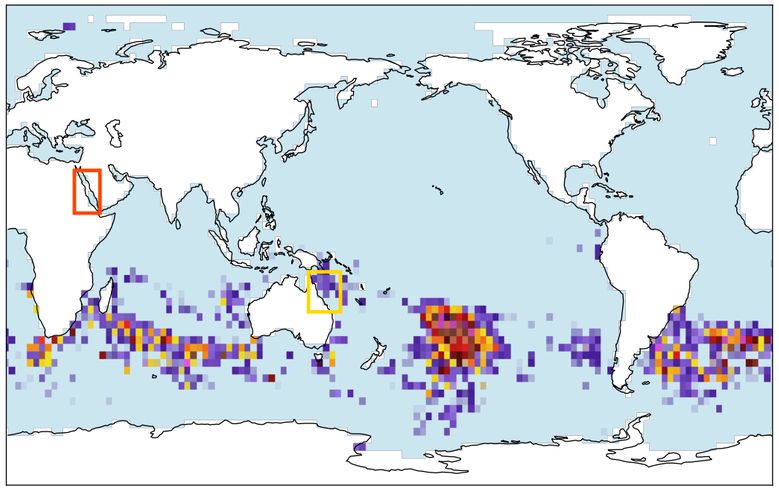
Predicted DHW - Step 46



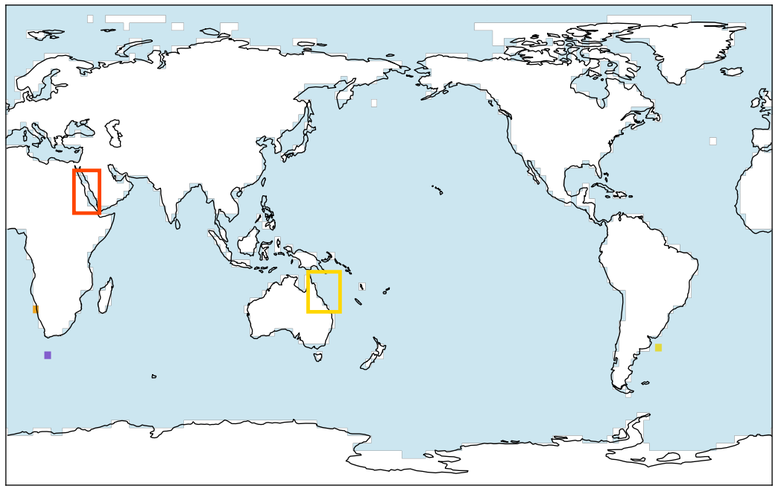
Error (Pred - Actual) - Step 46 [scale ±3.6]



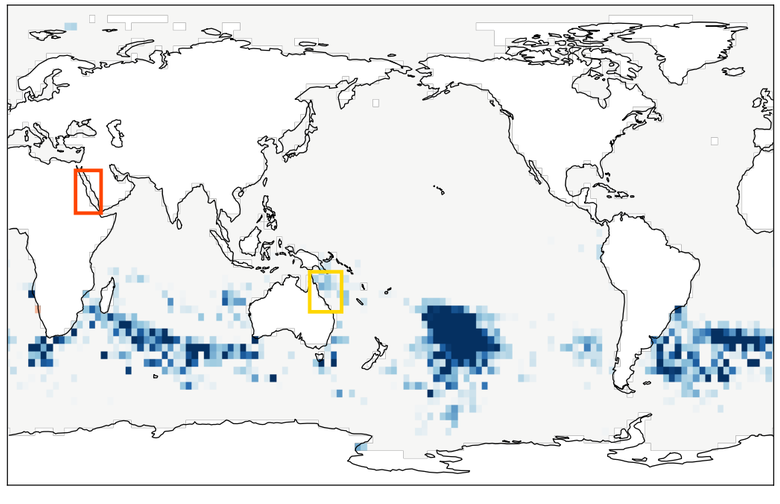
Actual DHW - Step 90

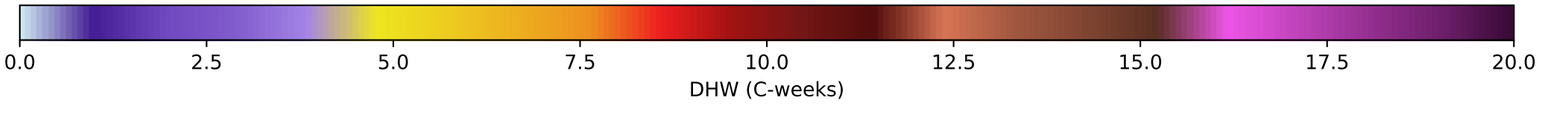


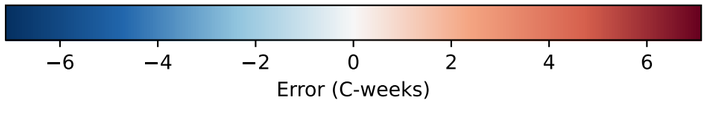
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±7.1]

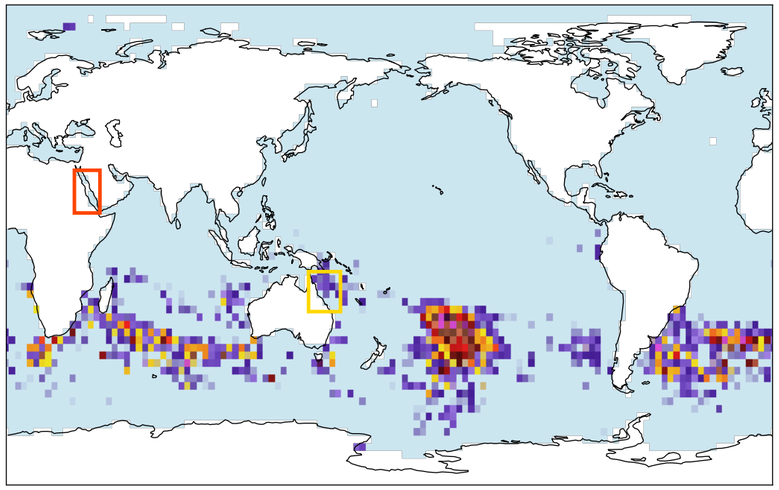




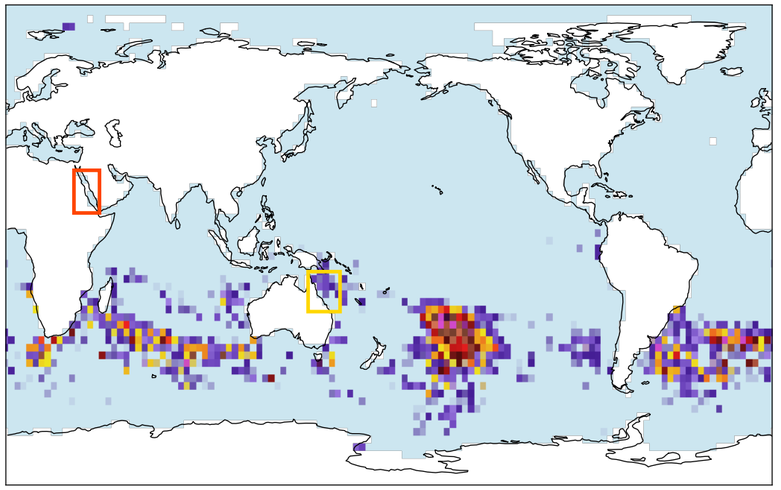


DHW Chain init: 2021-04-06 - Test (C-weeks)

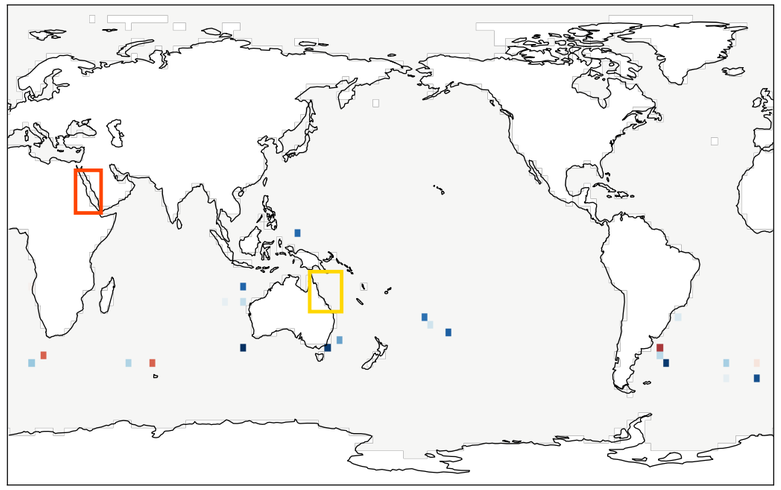
Actual DHW - Step 1



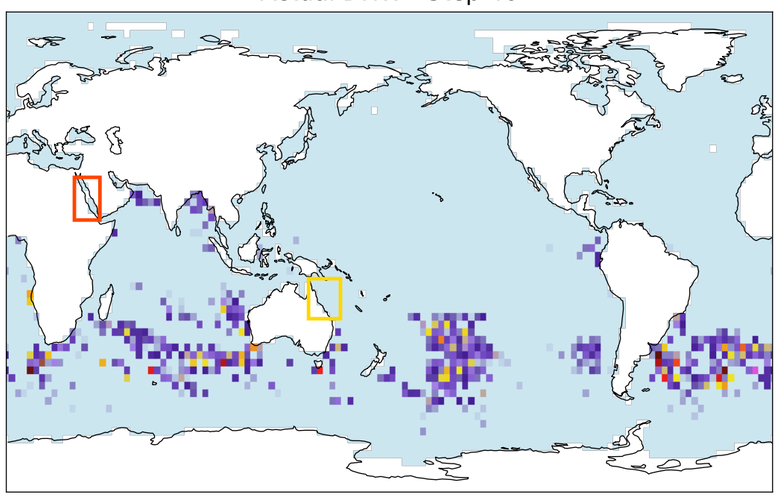
Predicted DHW - Step 1



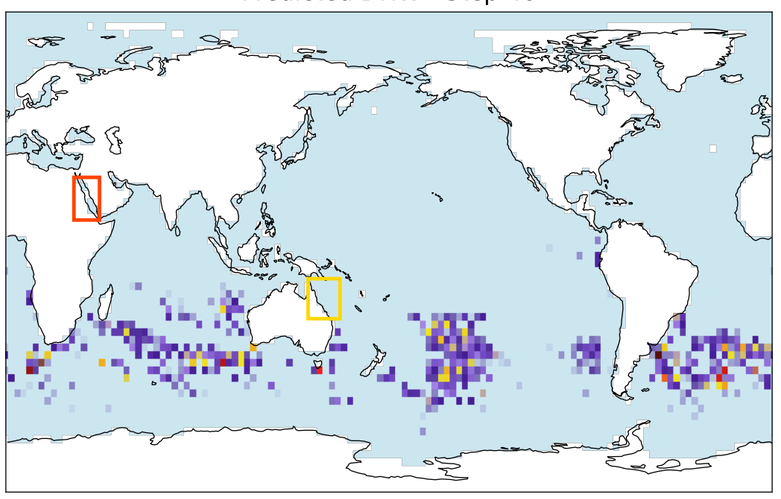
Error (Pred - Actual) - Step 1 [scale ±0.23]



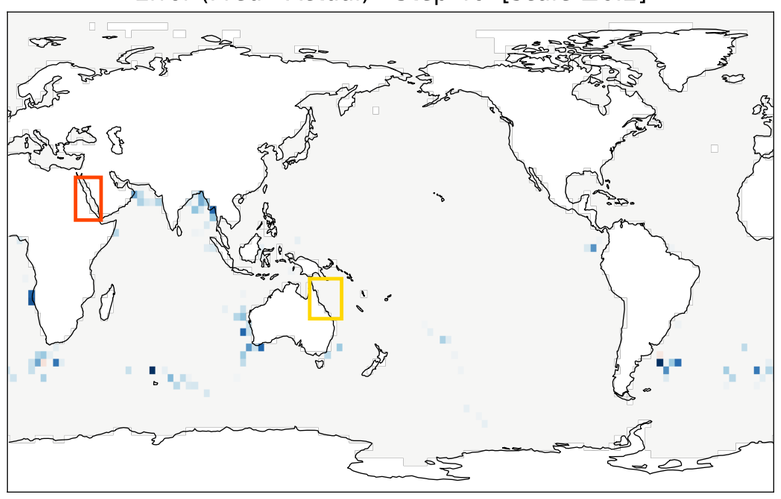
Actual DHW - Step 46



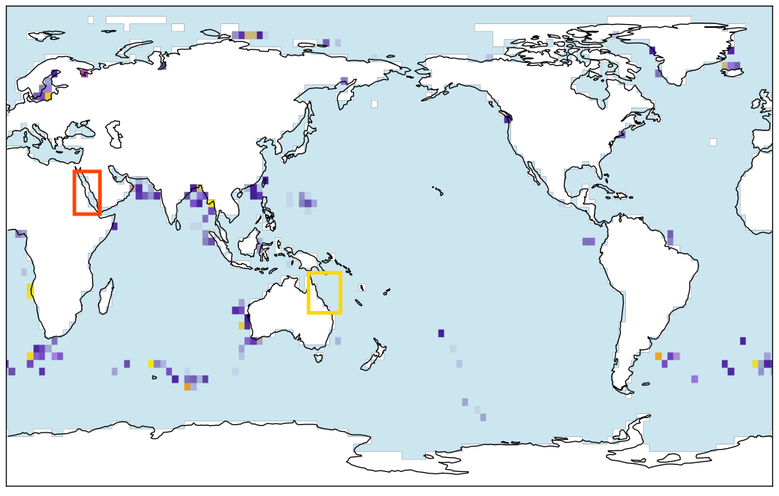
Predicted DHW - Step 46



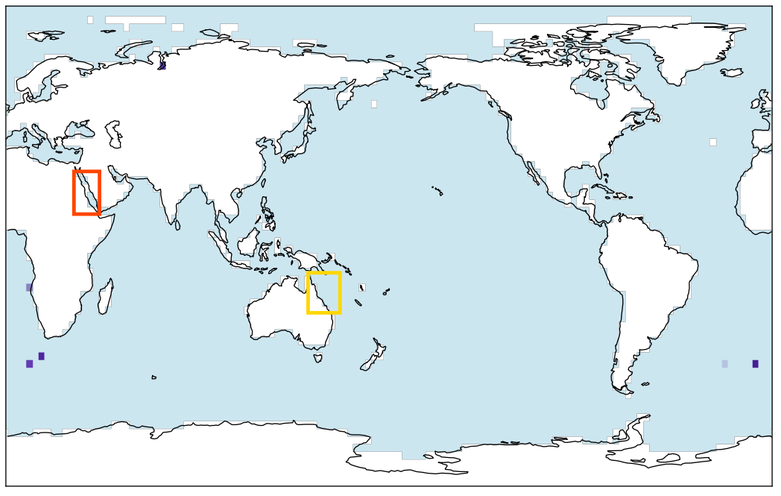
Error (Pred - Actual) - Step 46 [scale ±6.2]



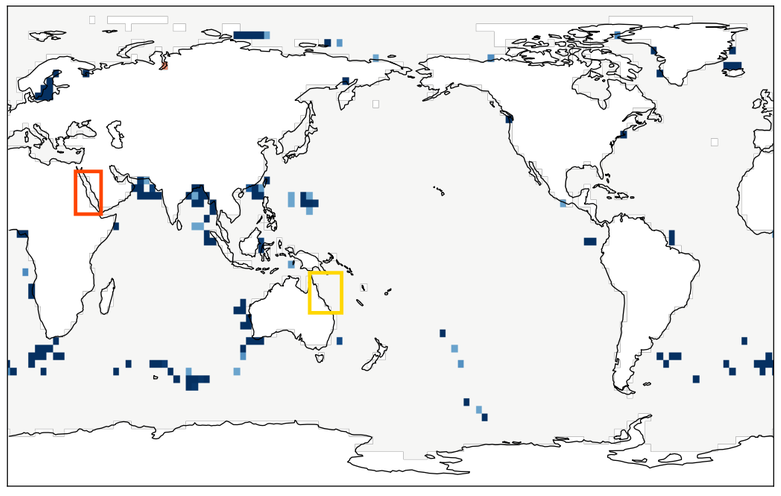
Actual DHW - Step 90

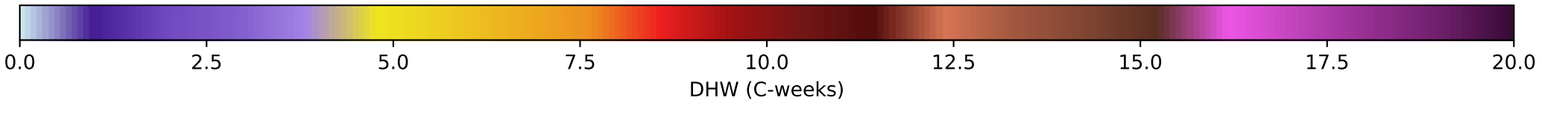


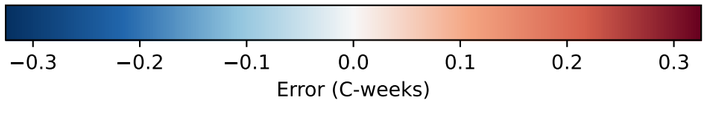
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

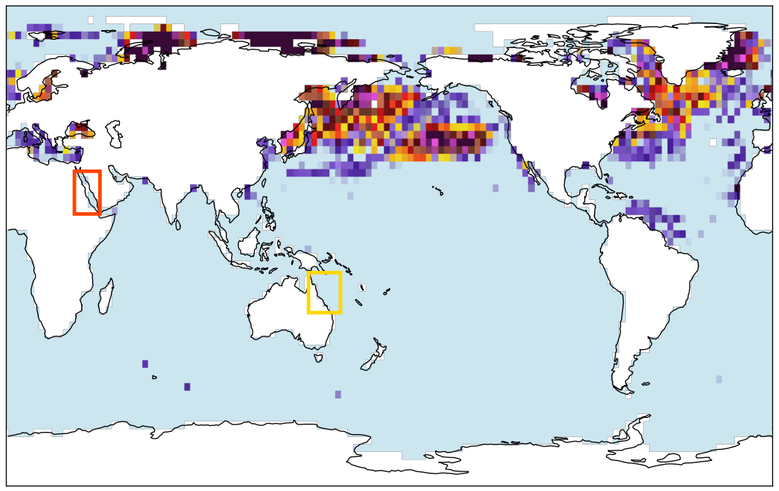




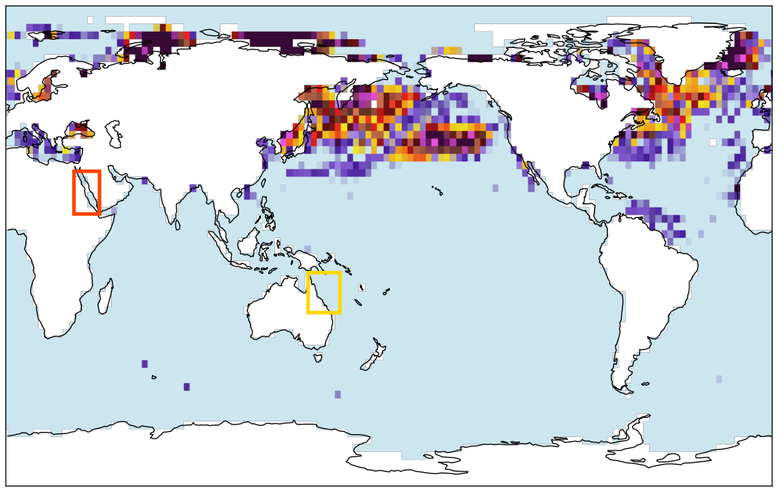


DHW Chain init: 2021-10-03 - Test (C-weeks)

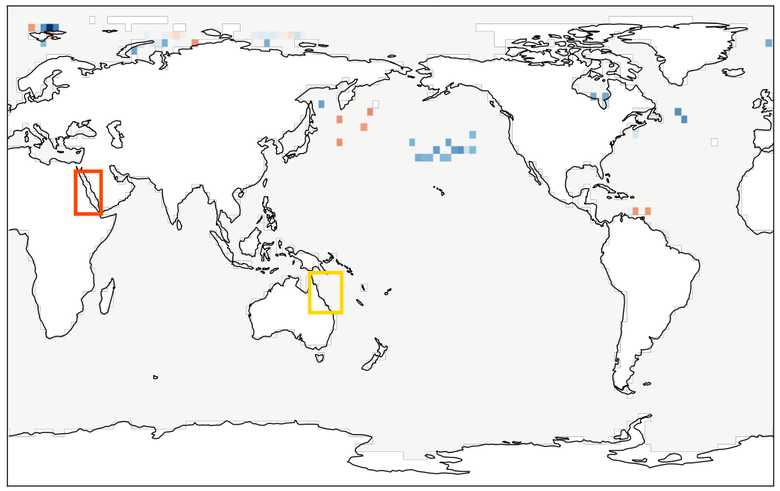
Actual DHW - Step 1



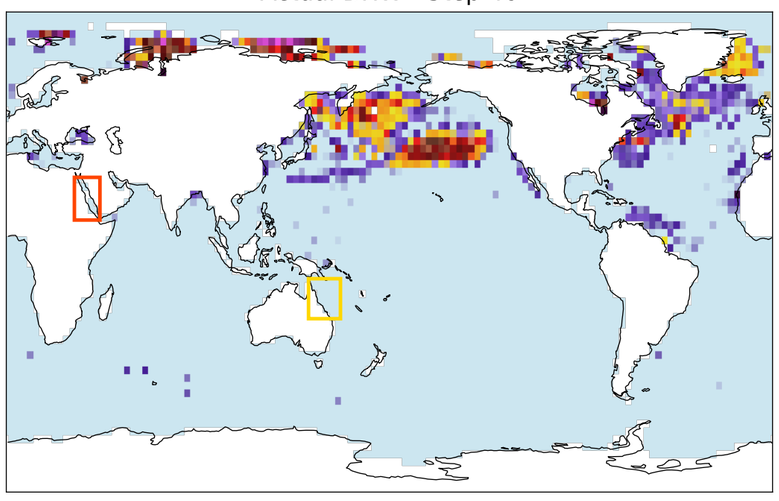
Predicted DHW - Step 1



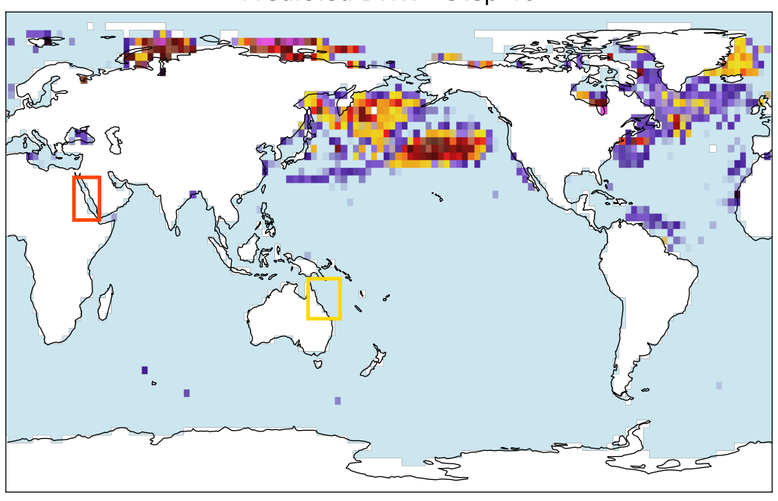
Error (Pred - Actual) - Step 1 [scale ±0.39]



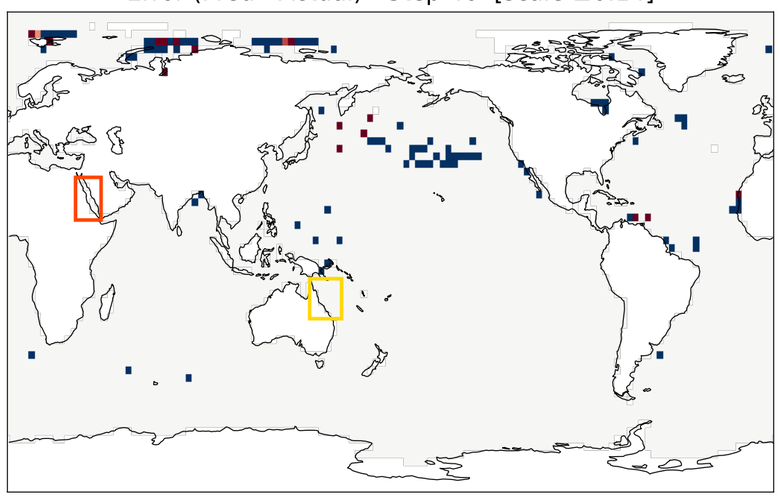
Actual DHW - Step 46



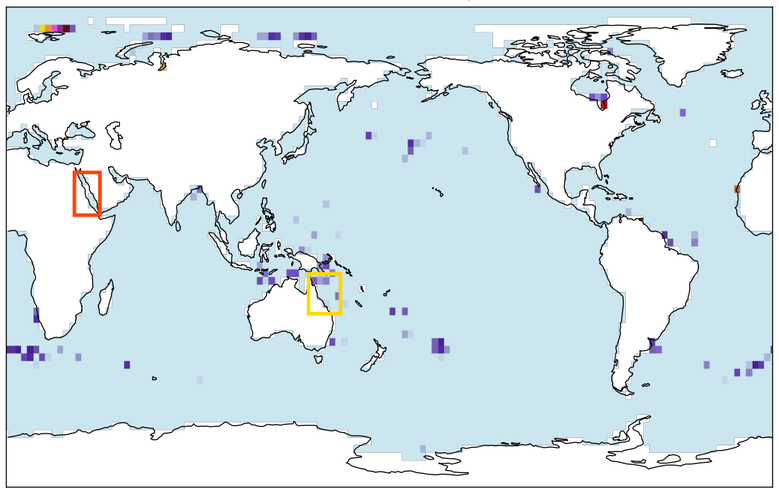
Predicted DHW - Step 46



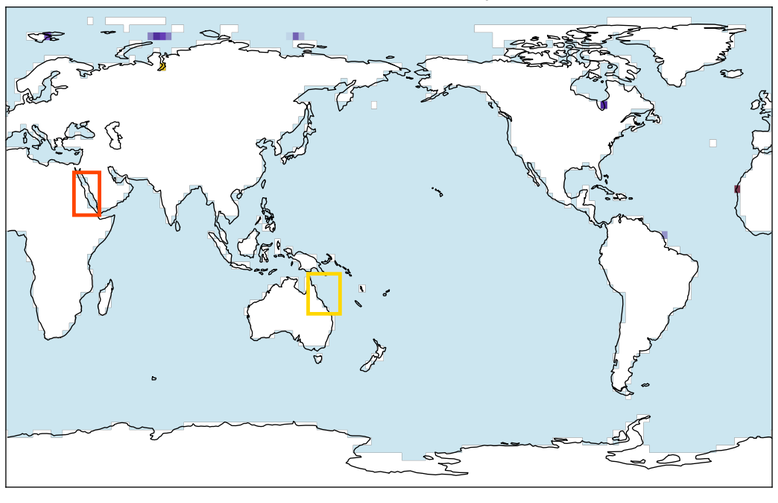
Error (Pred - Actual) - Step 46 [scale ±0.14]



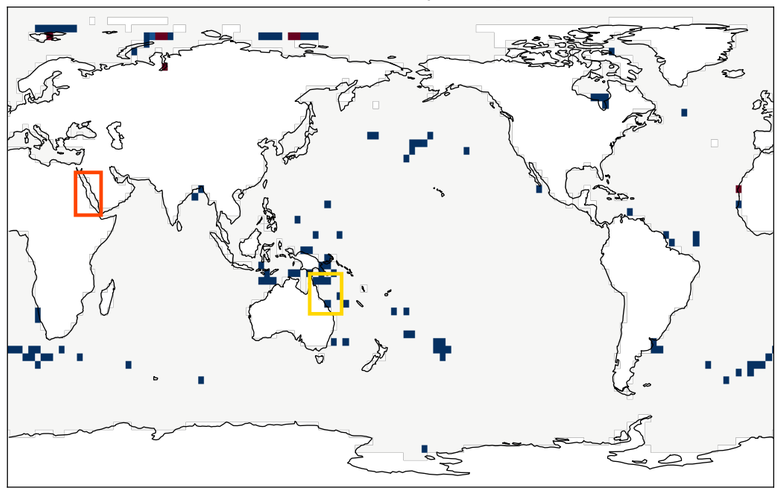
Actual DHW - Step 90

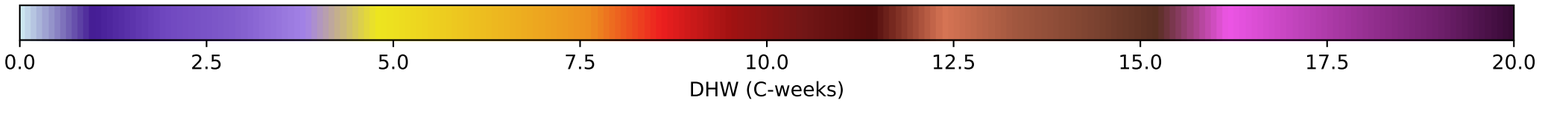


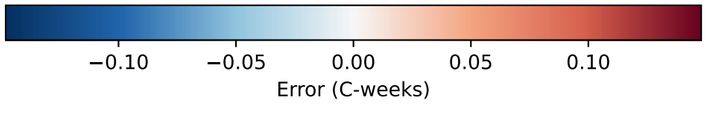
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.15]

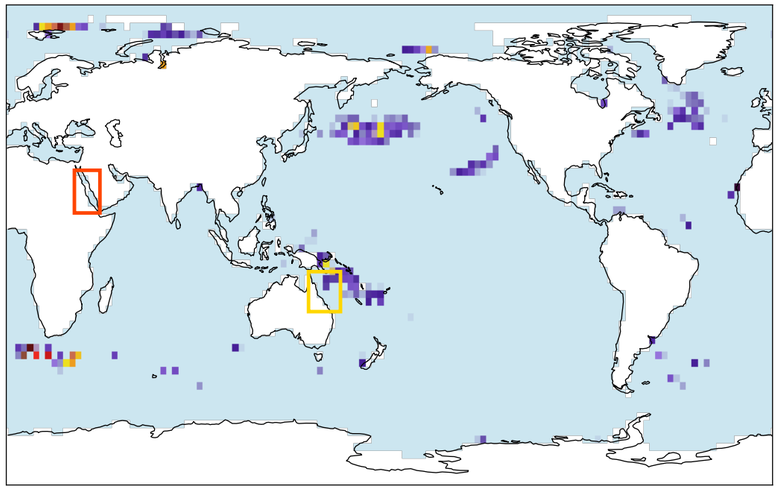




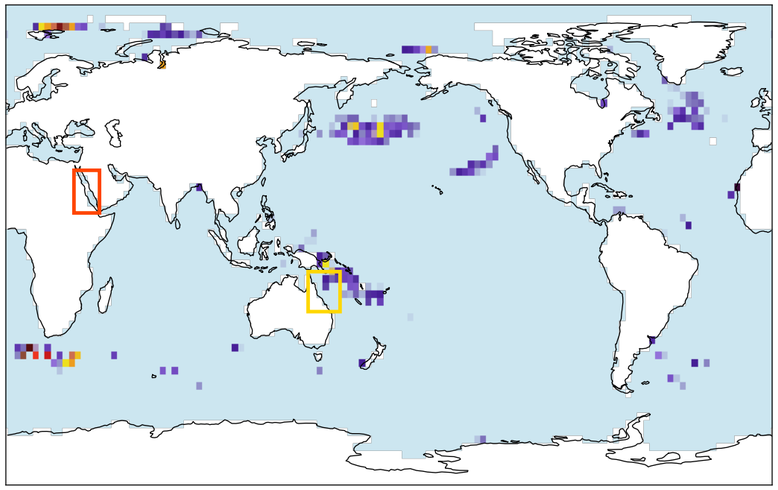


DHW Chain init: 2022-12-27 - Test (C-weeks)

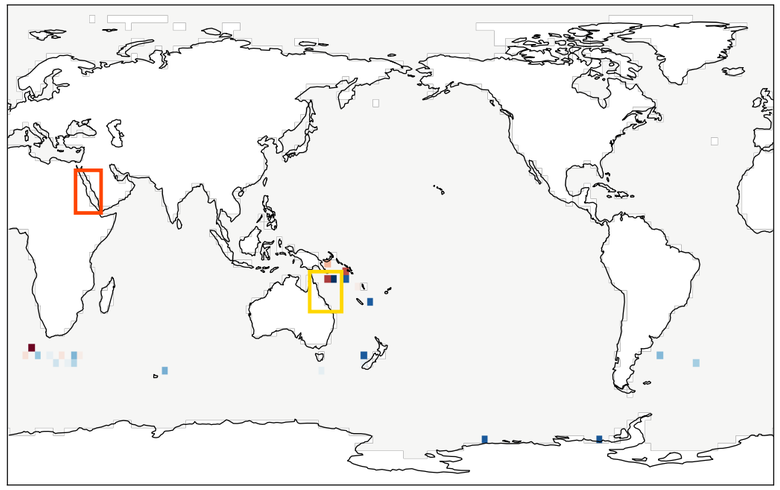
Actual DHW - Step 1



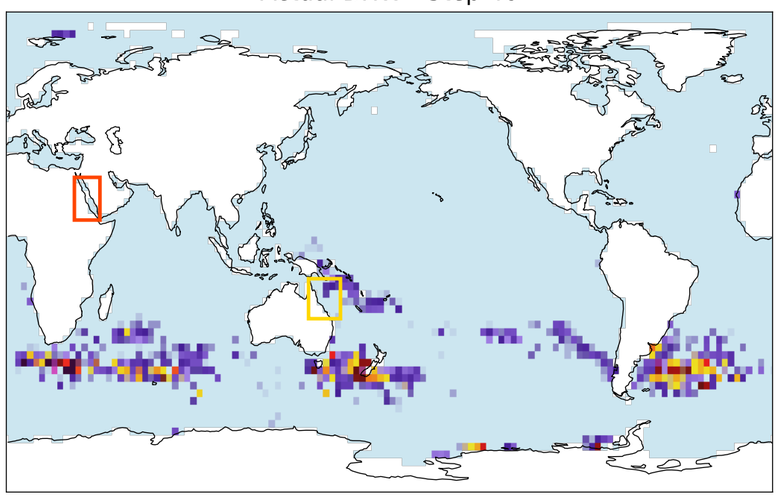
Predicted DHW - Step 1



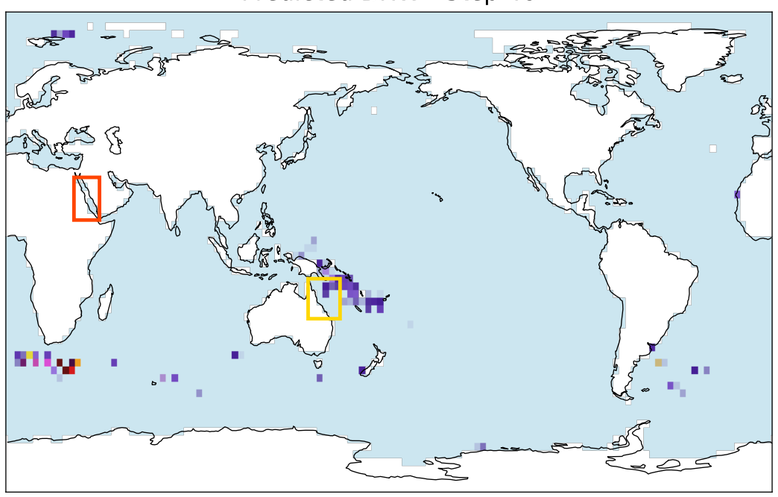
Error (Pred - Actual) - Step 1 [scale ±0.2]



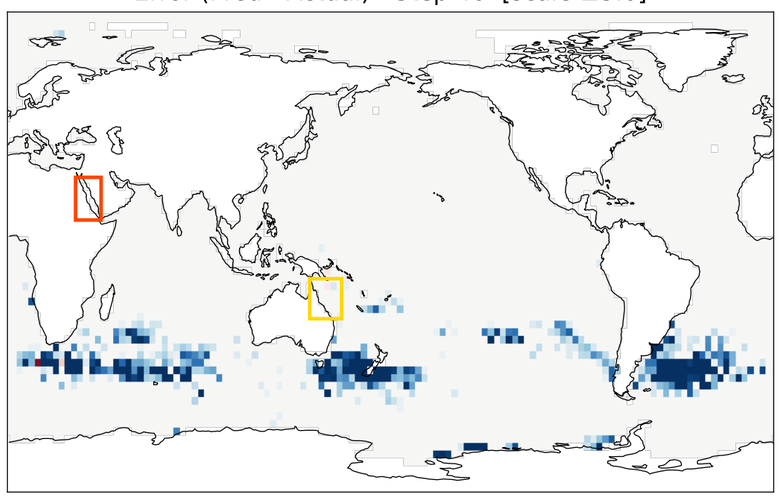
Actual DHW - Step 46



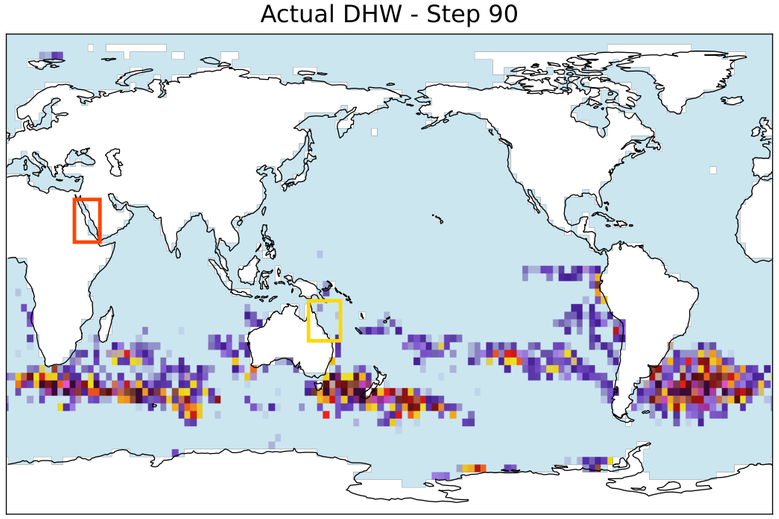
Predicted DHW - Step 46



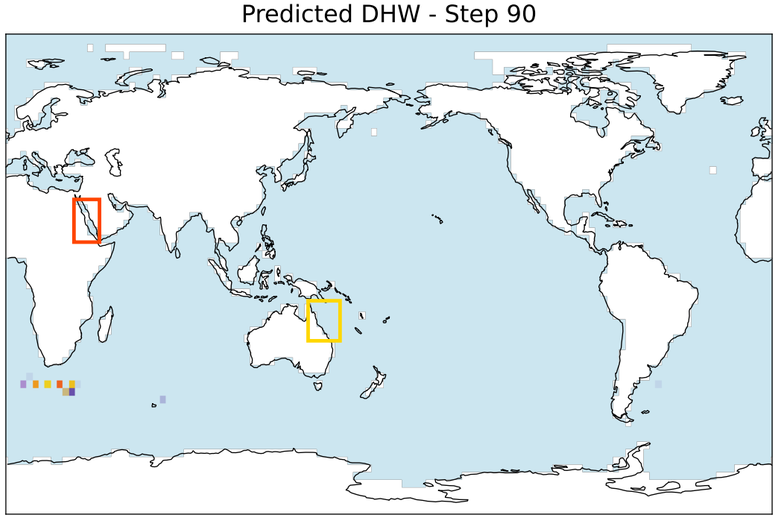
Error (Pred - Actual) - Step 46 [scale ±3.6]



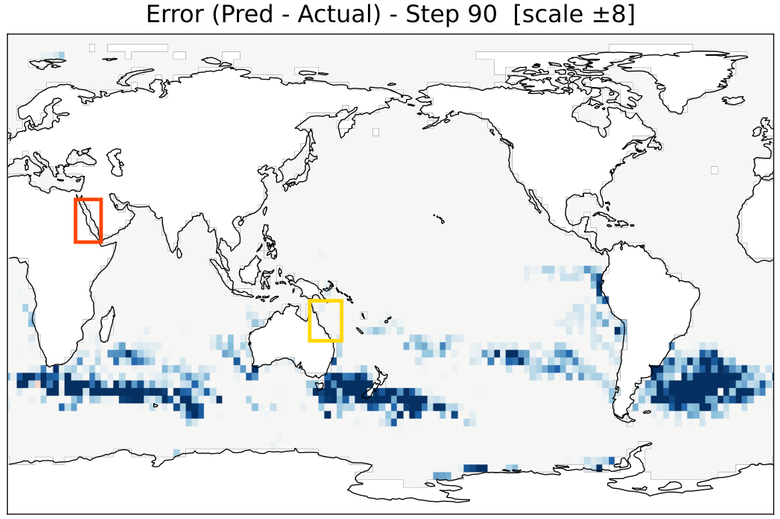
Actual DHW - Step 90

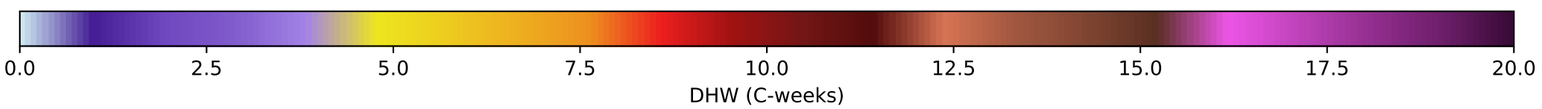


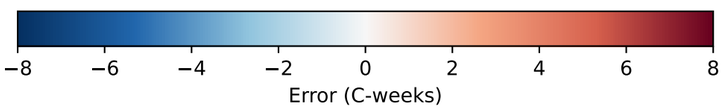
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

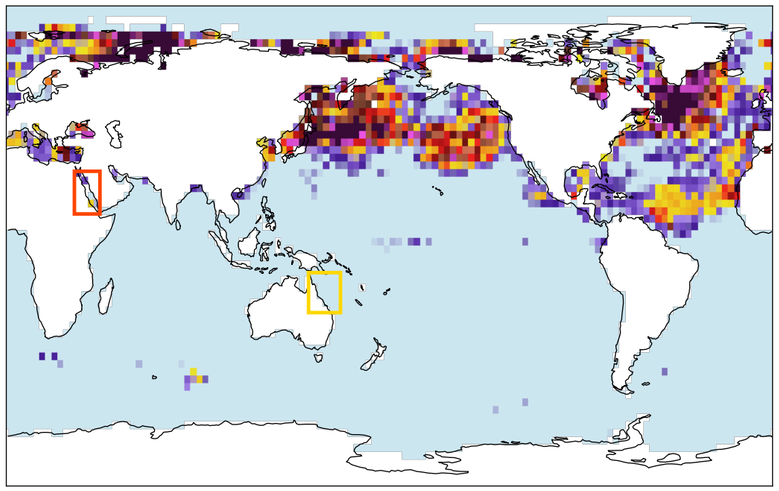




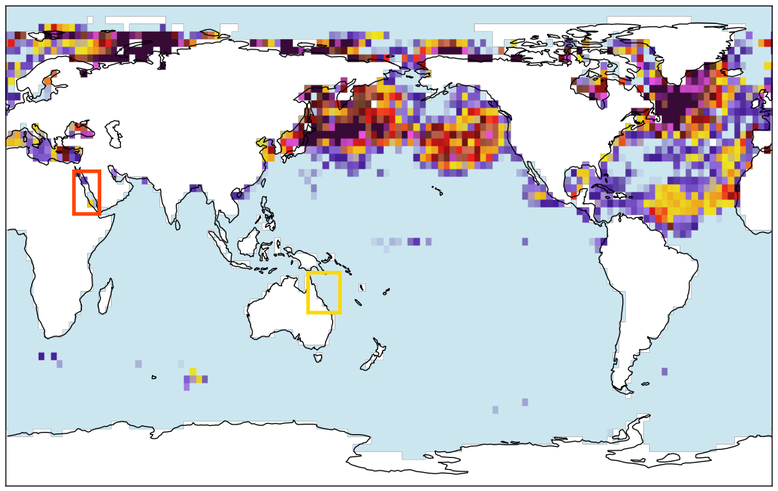


DHW Chain init: 2023-09-23 - Test (C-weeks)

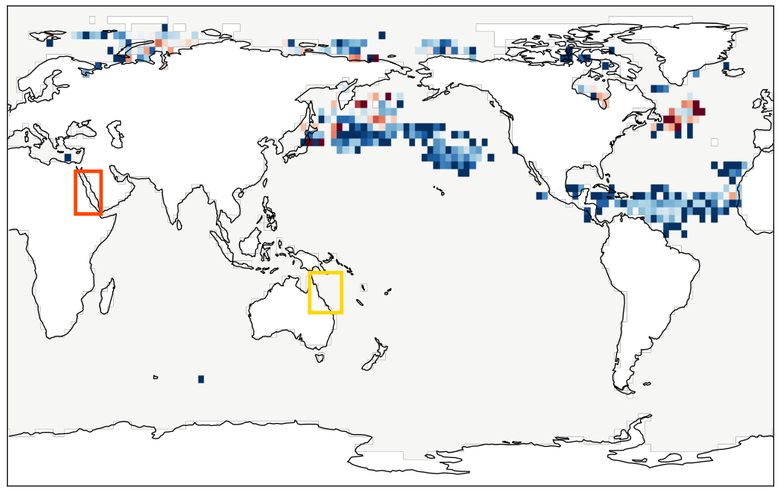
Actual DHW - Step 1



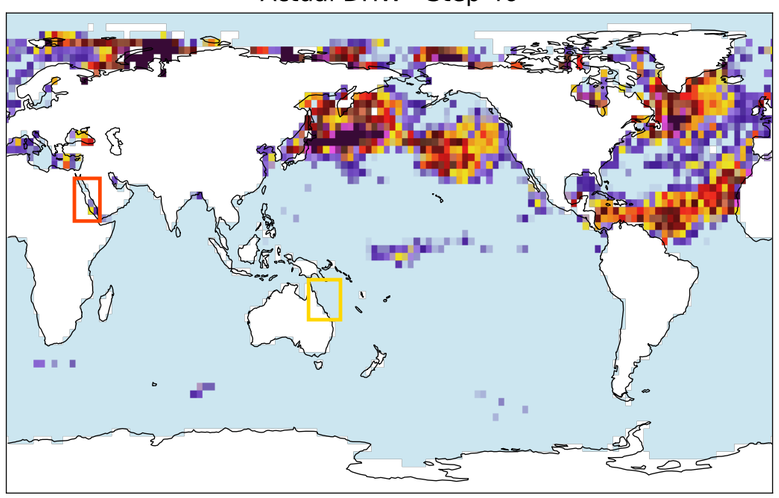
Predicted DHW - Step 1



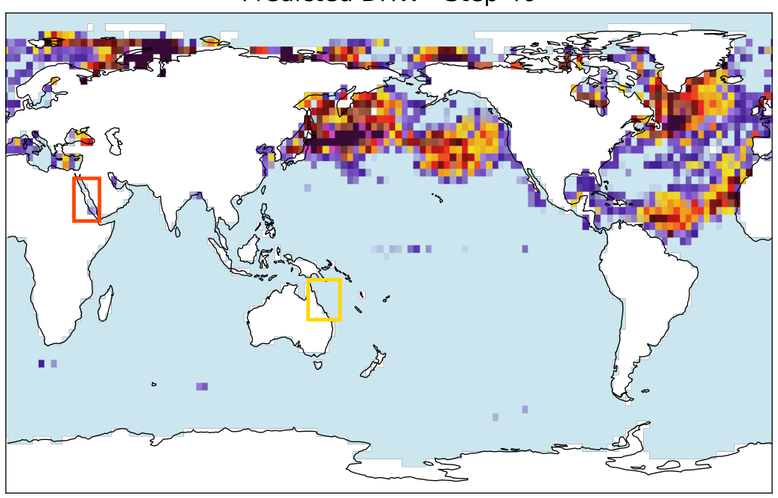
Error (Pred - Actual) - Step 1 [scale ±0.063]



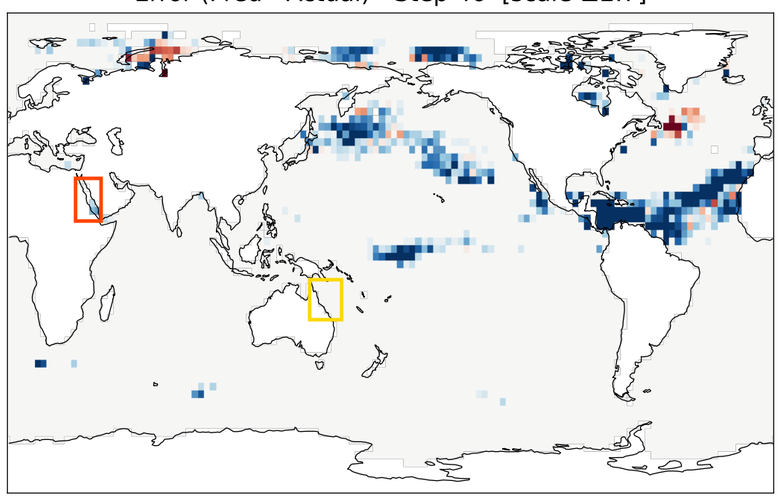
Actual DHW - Step 46



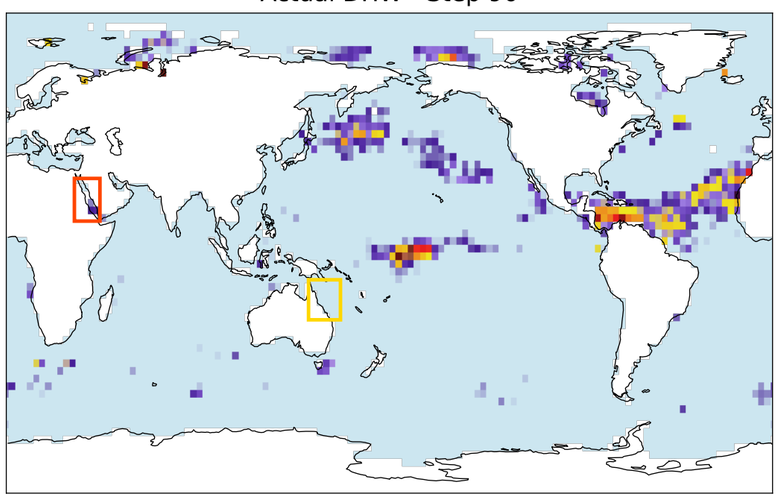
Predicted DHW - Step 46



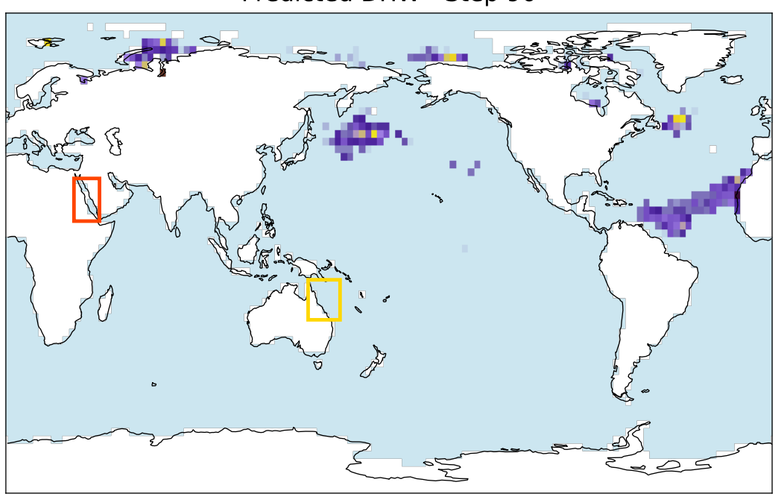
Error (Pred - Actual) - Step 46 [scale ±2.7]



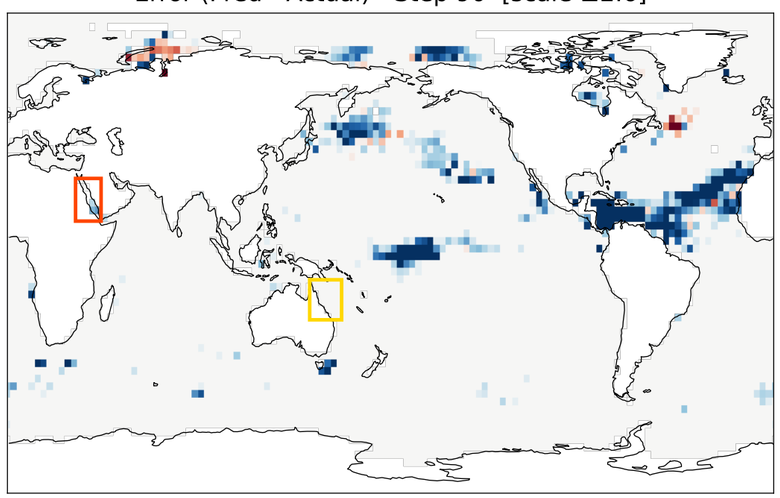
Actual DHW - Step 90

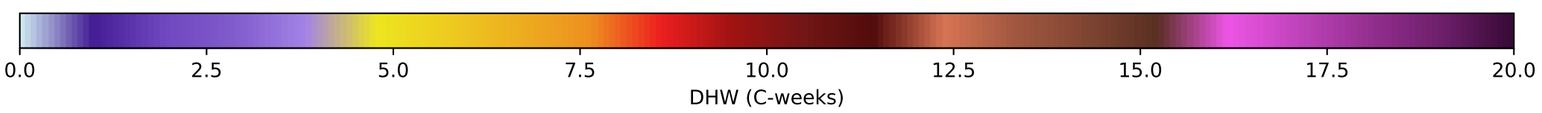


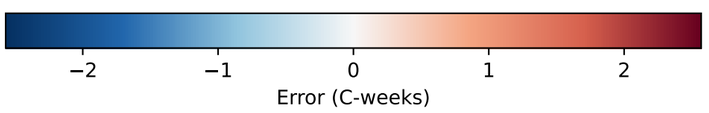
Predicted DHW - Step 90



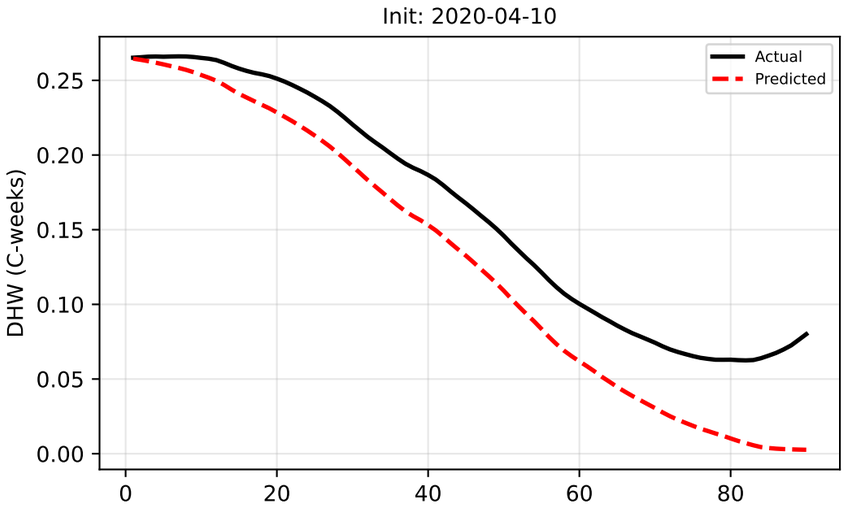
Error (Pred - Actual) - Step 90 [scale ±2.6]

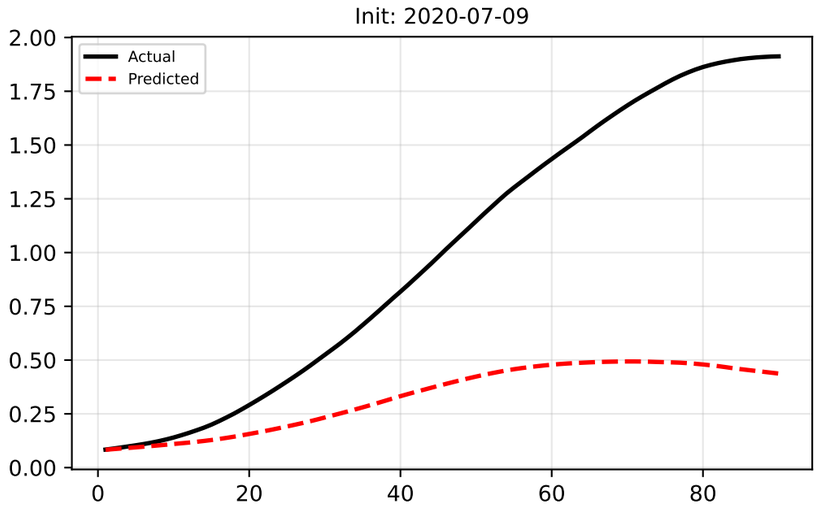


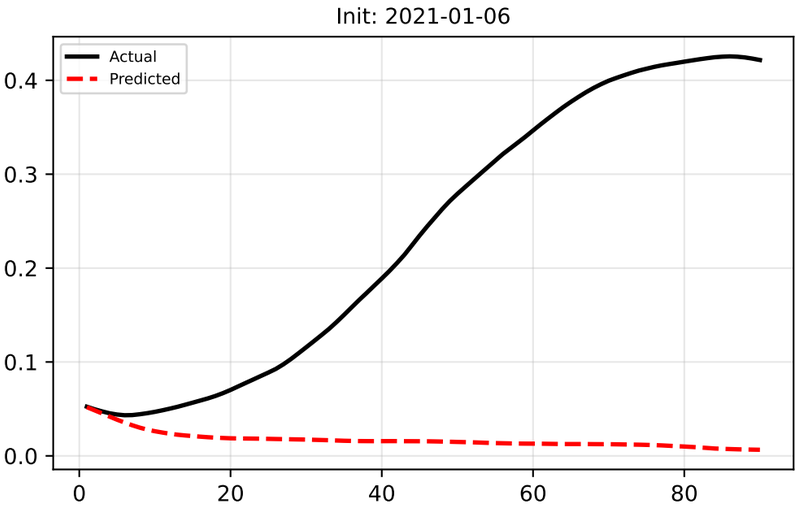


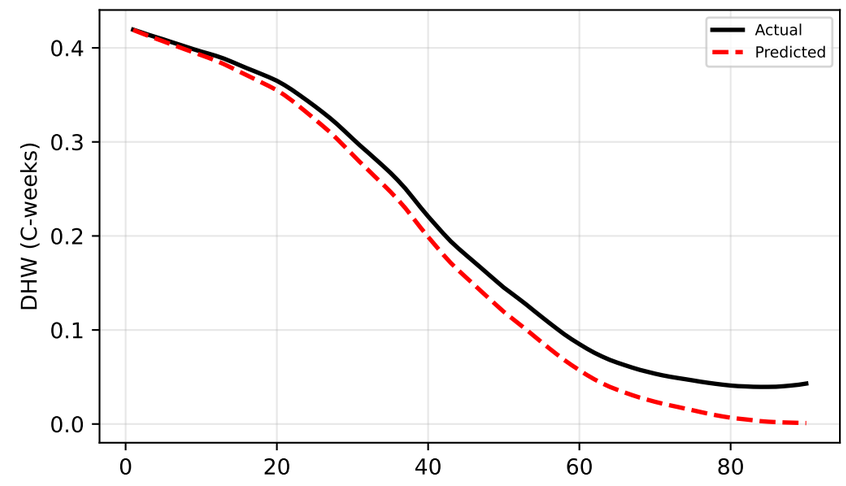


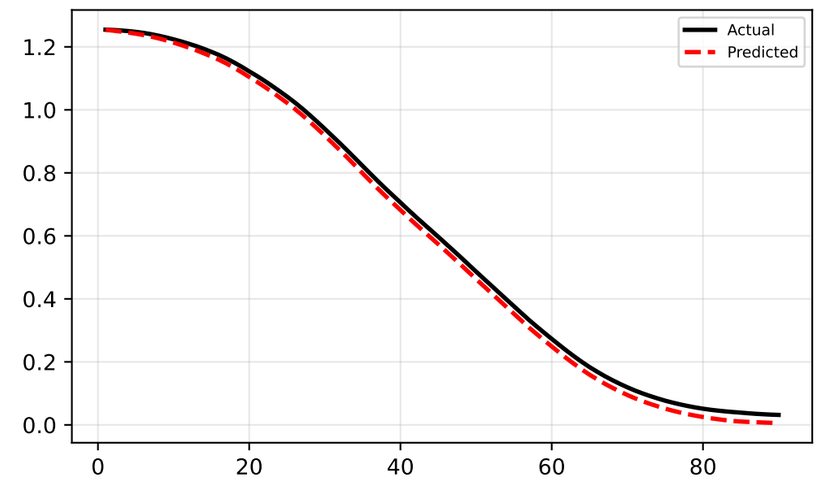
Global-Mean DHW - Test (C-weeks)

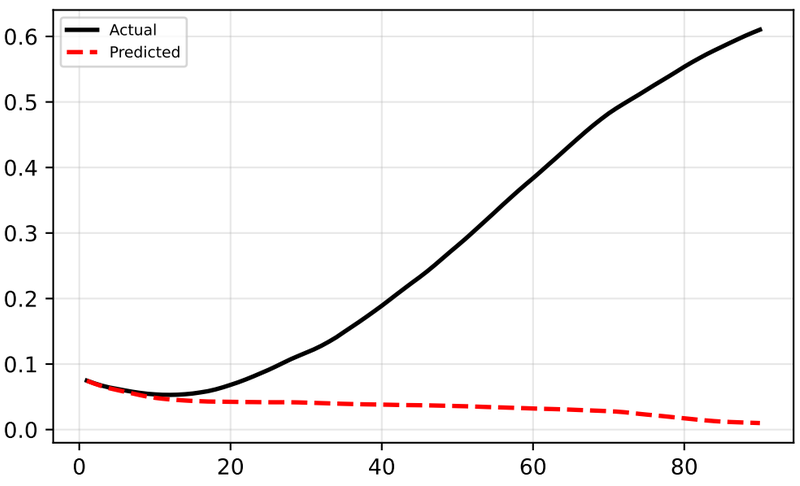


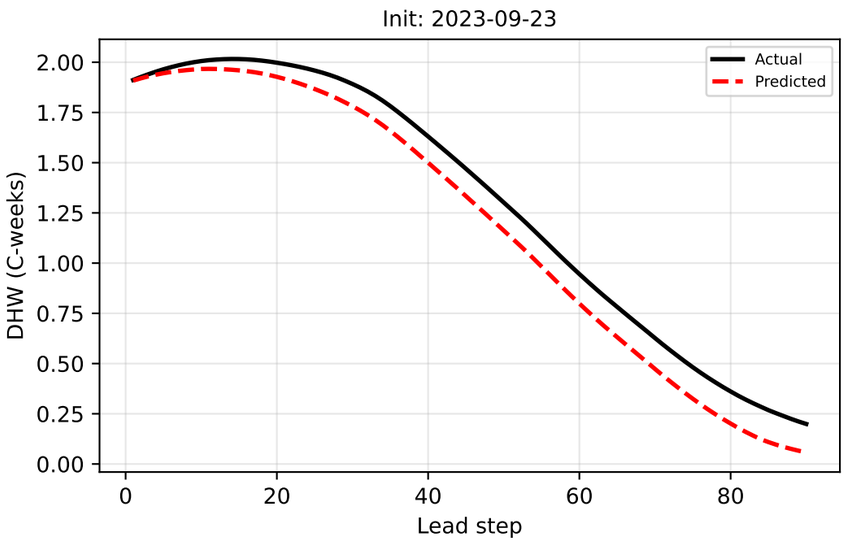




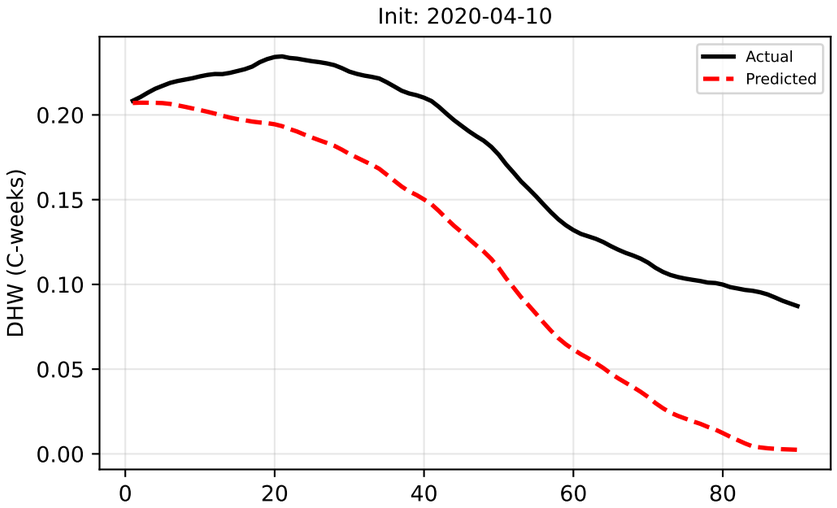


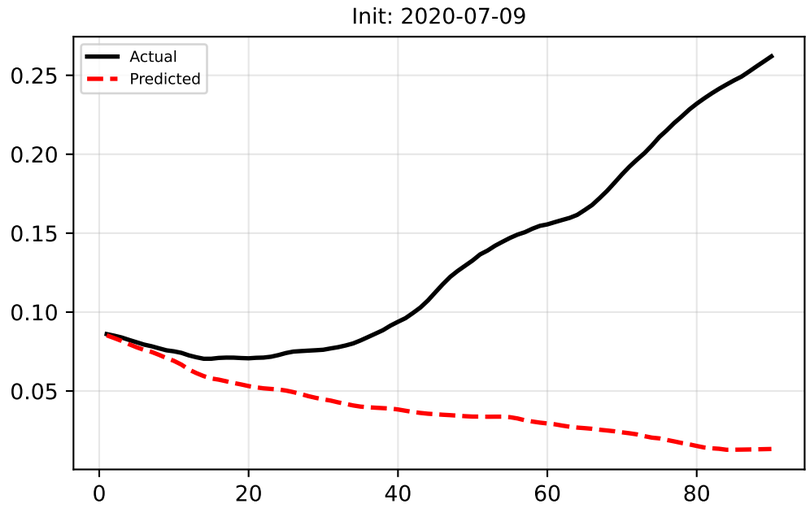


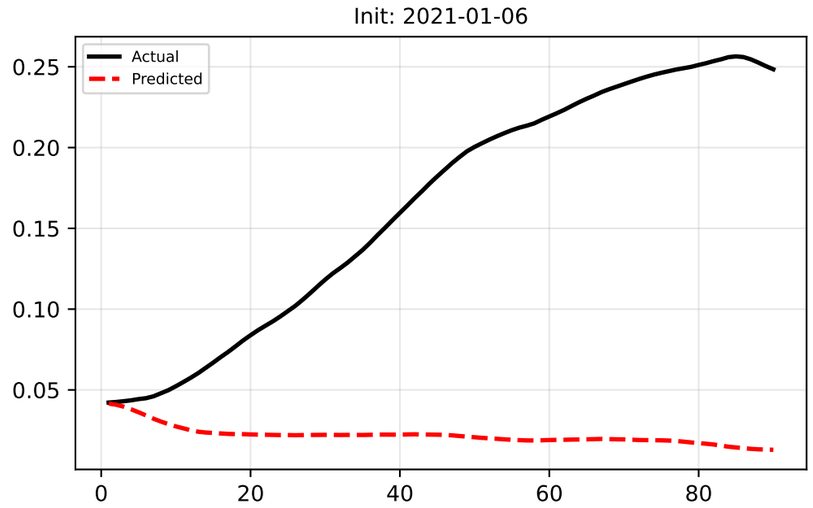


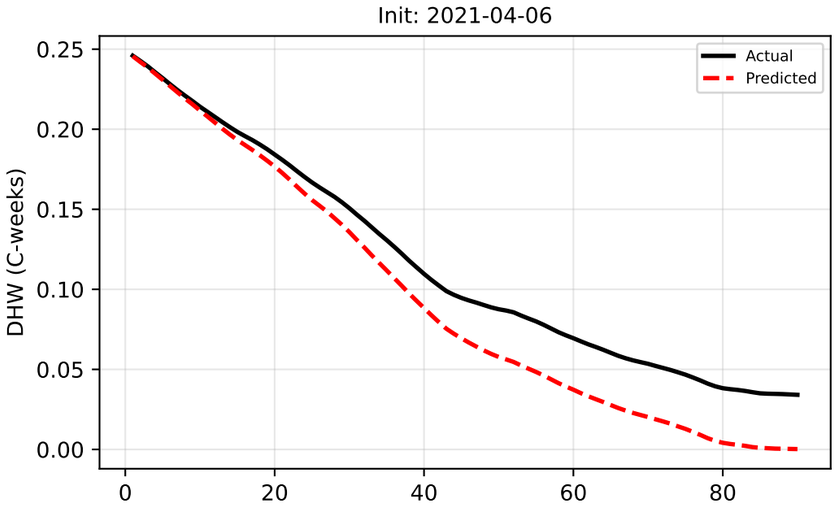


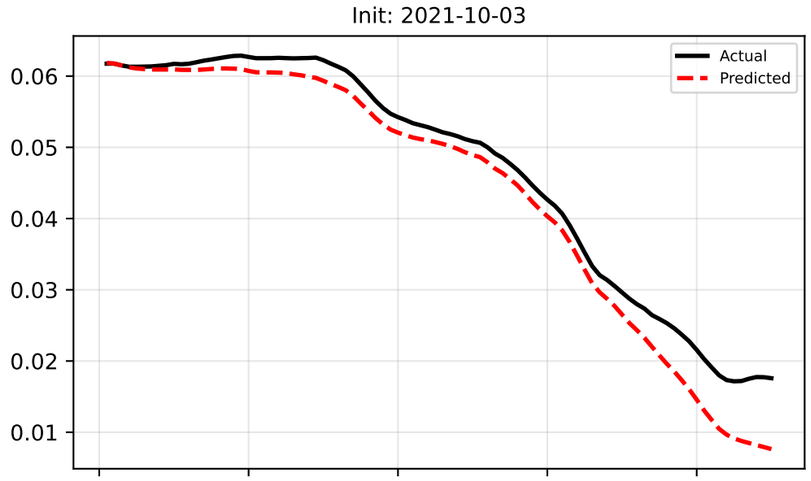
Tropical-Mean DHW (|lat|<=30) - Test (C-weeks)

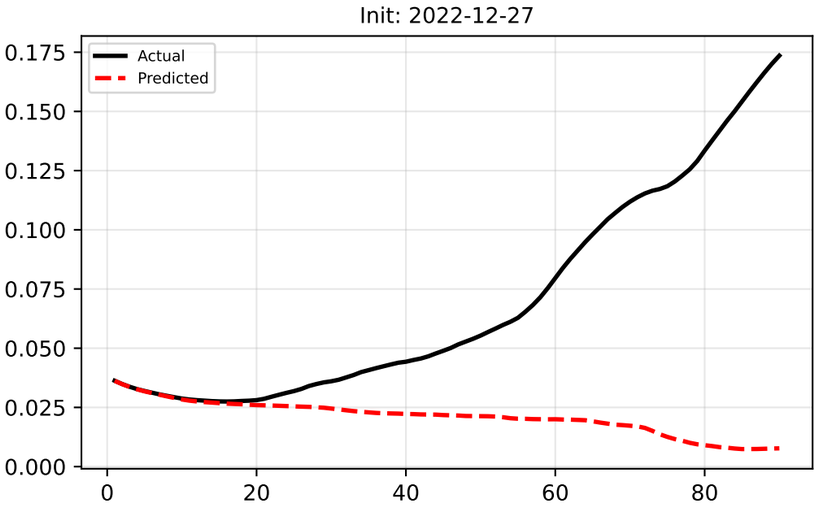


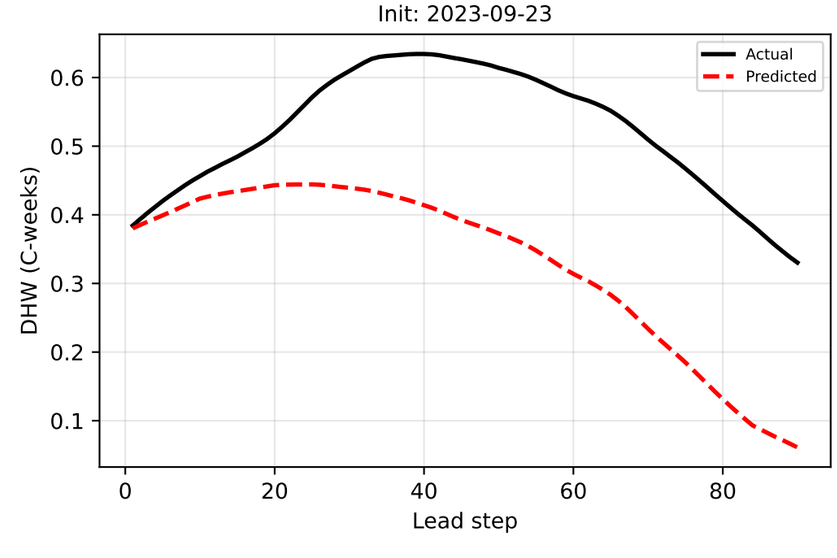


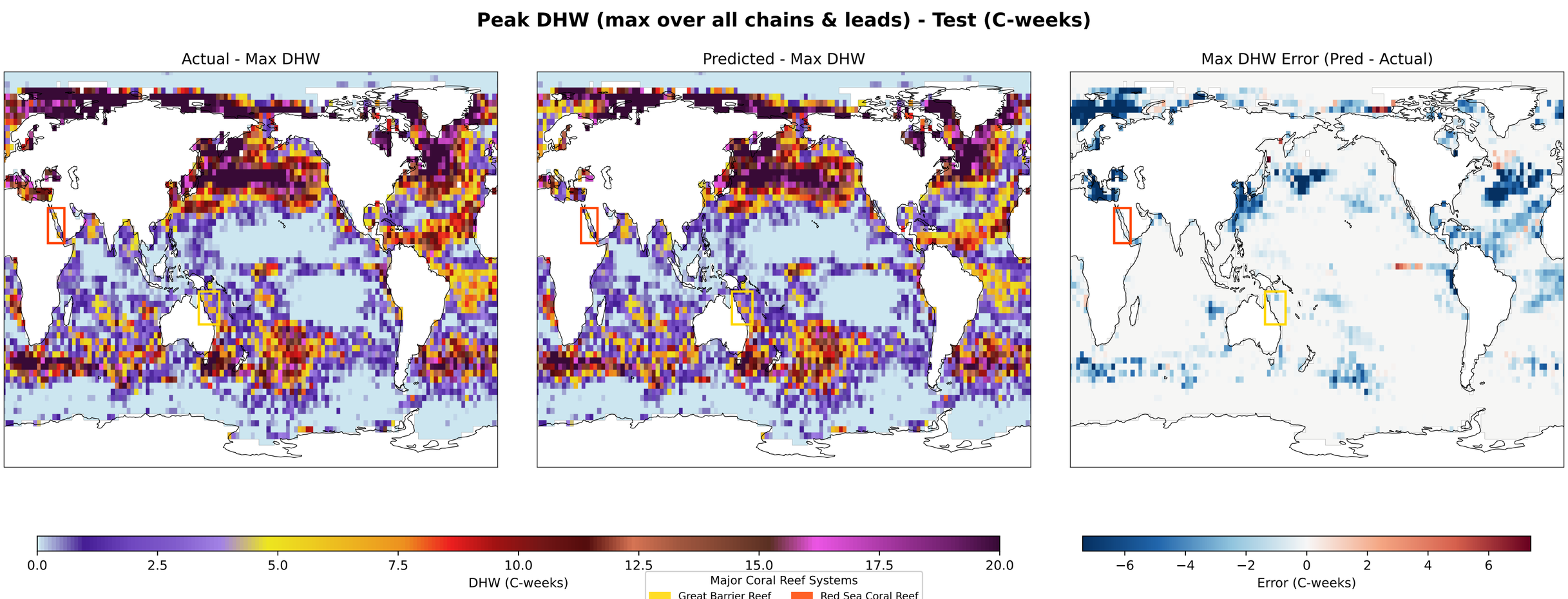


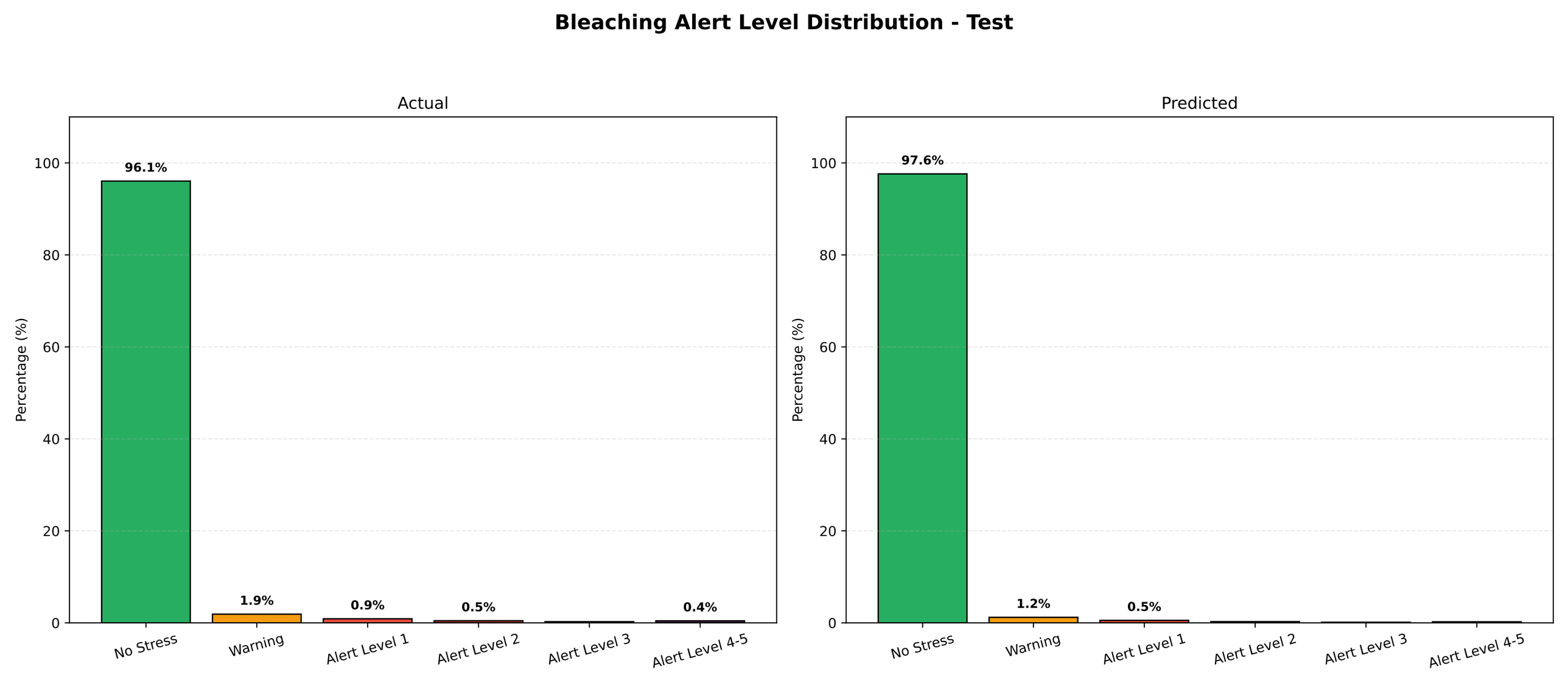






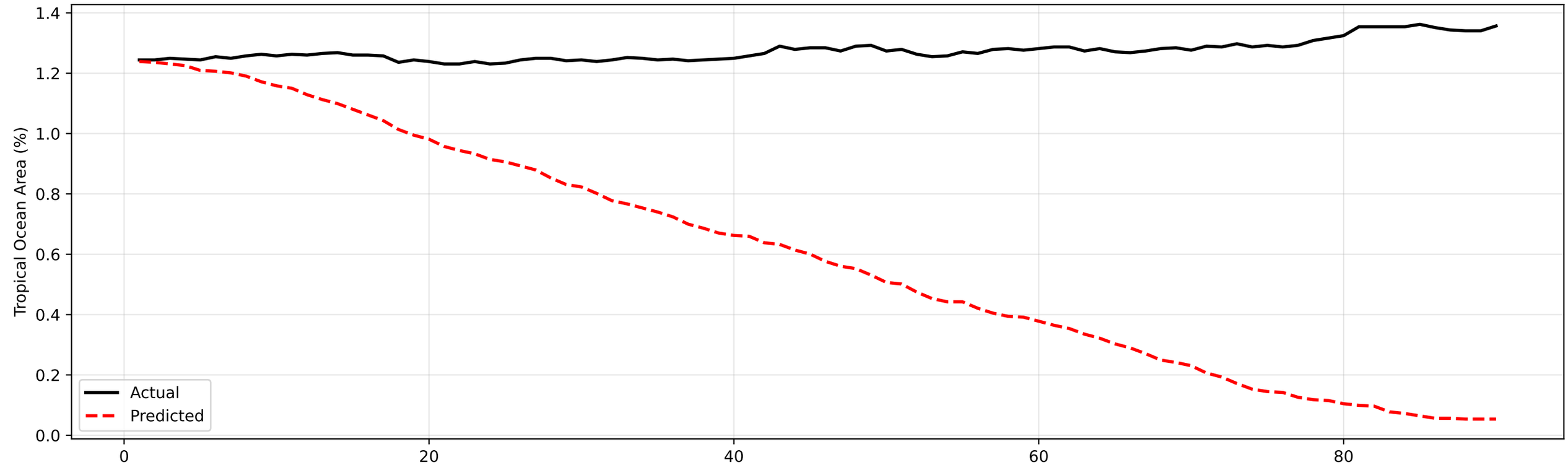



Peak DHW (max over all chains & leads) - Test (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−6
−4
−2
0
2
4
6
Error (C-weeks)

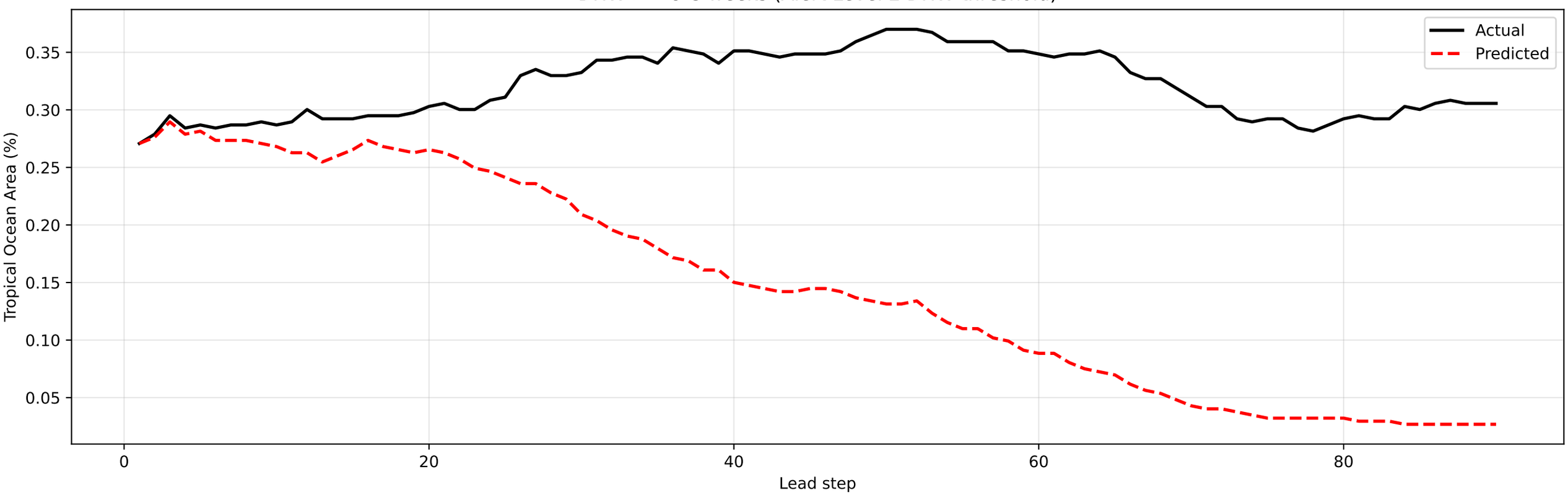

Bleaching Alert Level Distribution - Test
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.1%
1.9%
0.9%
0.5%
0.4%
97.6%
1.2%
0.5%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



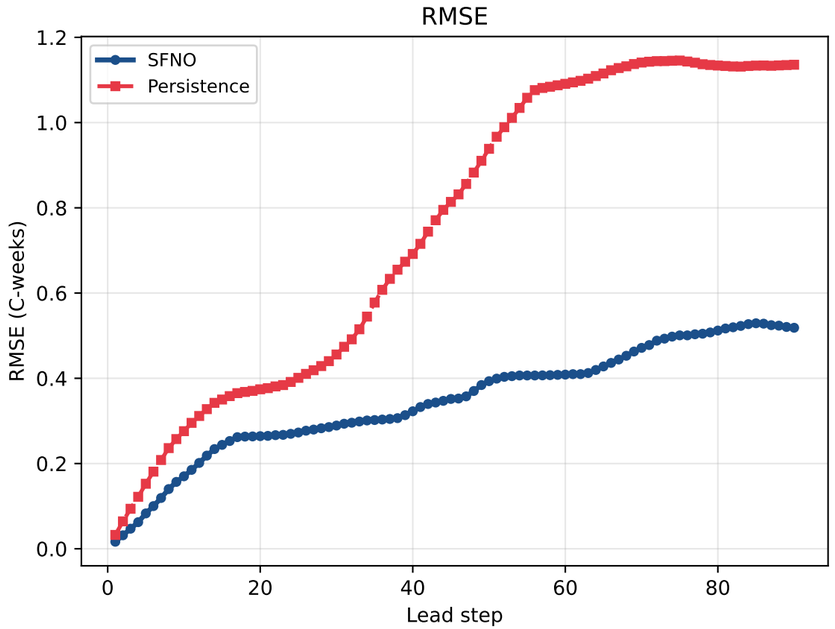
Great Barrier Reef Analysis - Test

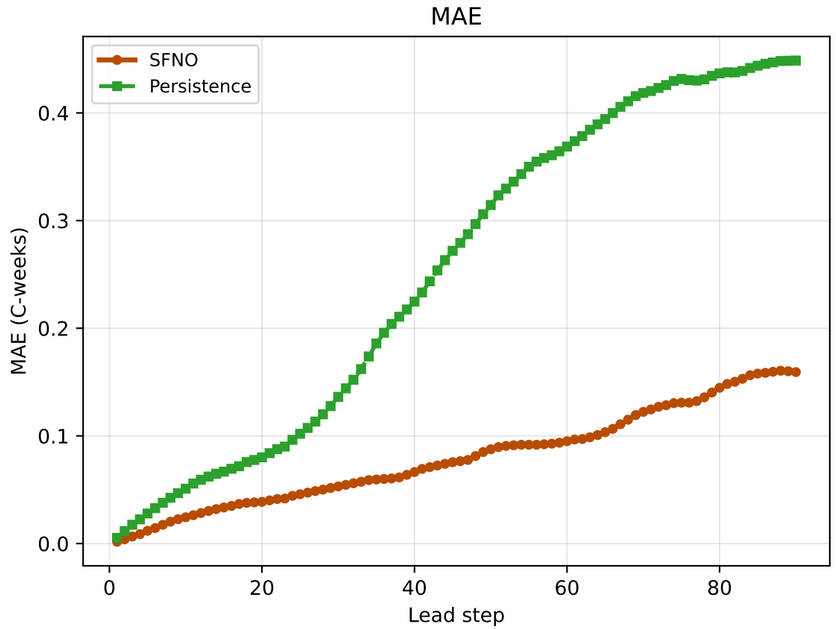
```
                    GREAT BARRIER REEF - TEST
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                      Grid: 5 lat x 5 lon points
```

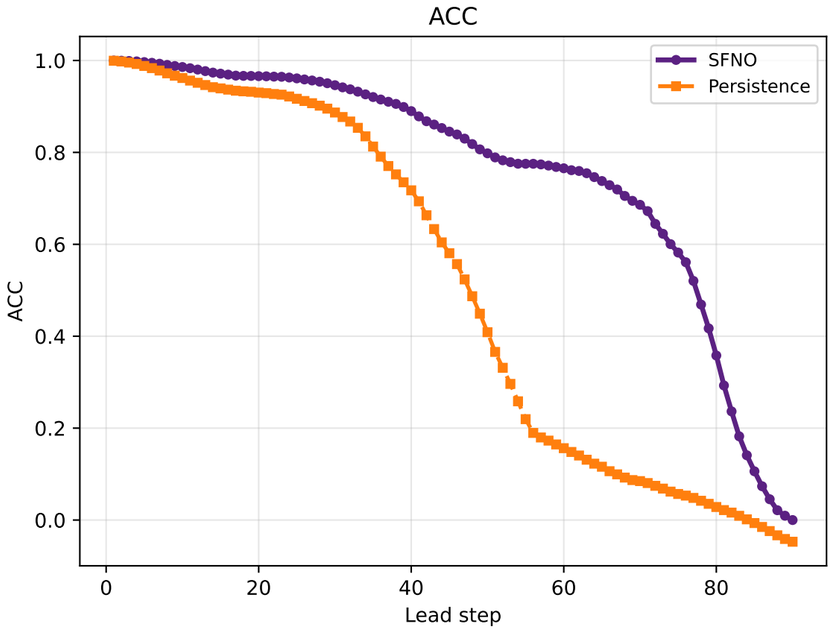
DHW Metrics by Lead Step - Test - GBR (18 chains scored at every lead)

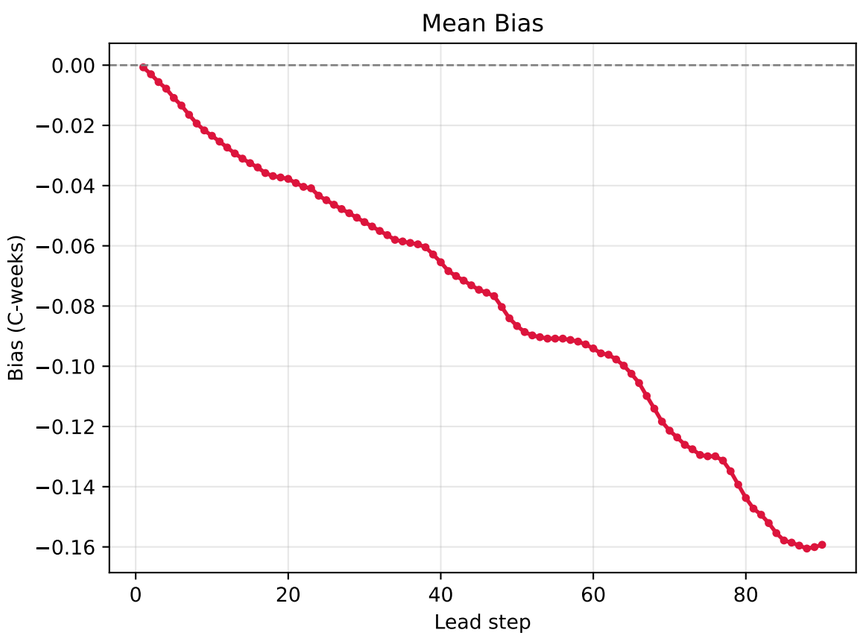
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0165	0.0325	49.2%	0.0017	0.0054	-0.0007	0.9999	0.9995	0.9997
2	0.0316	0.0643	50.8%	0.0040	0.0116	-0.0030	0.9995	0.9979	0.9990
3	0.0471	0.0939	49.8%	0.0066	0.0175	-0.0056	0.9989	0.9955	0.9978
4	0.0624	0.1219	48.8%	0.0088	0.0224	-0.0078	0.9980	0.9924	0.9962
5	0.0830	0.1525	45.6%	0.0118	0.0279	-0.0109	0.9965	0.9881	0.9933
6	0.1002	0.1810	44.7%	0.0144	0.0329	-0.0134	0.9950	0.9834	0.9903
7	0.1192	0.2083	42.8%	0.0174	0.0379	-0.0165	0.9929	0.9781	0.9863
8	0.1401	0.2362	40.7%	0.0203	0.0425	-0.0194	0.9903	0.9721	0.9813
9	0.1569	0.2574	39.0%	0.0226	0.0468	-0.0217	0.9879	0.9669	0.9766
10	0.1701	0.2759	38.4%	0.0244	0.0510	-0.0235	0.9858	0.9619	0.9725
11	0.1852	0.2955	37.3%	0.0263	0.0558	-0.0254	0.9832	0.9562	0.9674
12	0.2014	0.3116	35.4%	0.0283	0.0593	-0.0274	0.9802	0.9515	0.9616
13	0.2186	0.3277	33.3%	0.0303	0.0622	-0.0293	0.9769	0.9466	0.9551
14	0.2340	0.3424	31.7%	0.0320	0.0649	-0.0310	0.9736	0.9419	0.9488
15	0.2438	0.3502	30.4%	0.0335	0.0668	-0.0325	0.9714	0.9393	0.9445
16	0.2529	0.3582	29.4%	0.0349	0.0694	-0.0340	0.9693	0.9366	0.9405
17	0.2619	0.3652	28.3%	0.0368	0.0718	-0.0358	0.9671	0.9343	0.9363
18	0.2632	0.3680	28.5%	0.0378	0.0757	-0.0368	0.9665	0.9329	0.9353
19	0.2636	0.3705	28.9%	0.0383	0.0777	-0.0373	0.9664	0.9319	0.9350
20	0.2643	0.3742	29.4%	0.0387	0.0801	-0.0378	0.9659	0.9302	0.9341
21	0.2649	0.3768	29.7%	0.0401	0.0840	-0.0392	0.9656	0.9289	0.9335
22	0.2666	0.3810	30.0%	0.0414	0.0876	-0.0404	0.9649	0.9271	0.9323
23	0.2673	0.3841	30.4%	0.0418	0.0901	-0.0409	0.9645	0.9256	0.9316
24	0.2697	0.3914	31.1%	0.0443	0.0963	-0.0434	0.9629	0.9217	0.9287
25	0.2728	0.4011	32.0%	0.0458	0.1020	-0.0448	0.9611	0.9167	0.9252
26	0.2771	0.4104	32.5%	0.0473	0.1074	-0.0463	0.9588	0.9118	0.9210
27	0.2797	0.4189	33.2%	0.0487	0.1133	-0.0478	0.9565	0.9070	0.9168
28	0.2828	0.4287	34.0%	0.0501	0.1201	-0.0492	0.9538	0.9017	0.9120
29	0.2856	0.4402	35.1%	0.0516	0.1277	-0.0506	0.9505	0.8952	0.9059
30	0.2891	0.4558	36.6%	0.0531	0.1363	-0.0521	0.9462	0.8868	0.8983
31	0.2934	0.4739	38.1%	0.0545	0.1442	-0.0536	0.9415	0.8770	0.8898
32	0.2958	0.4915	39.8%	0.0560	0.1522	-0.0551	0.9373	0.8672	0.8822
33	0.2986	0.5150	42.0%	0.0574	0.1621	-0.0564	0.9322	0.8535	0.8731
34	0.3011	0.5448	44.7%	0.0590	0.1738	-0.0580	0.9263	0.8349	0.8626
35	0.3021	0.5777	47.7%	0.0595	0.1858	-0.0585	0.9204	0.8127	0.8522
36	0.3033	0.6076	50.1%	0.0600	0.1958	-0.0590	0.9151	0.7908	0.8428
37	0.3046	0.6333	51.9%	0.0605	0.2040	-0.0595	0.9102	0.7703	0.8344
38	0.3065	0.6548	53.2%	0.0614	0.2107	-0.0605	0.9055	0.7521	0.8264
39	0.3135	0.6735	53.4%	0.0638	0.2175	-0.0629	0.8988	0.7349	0.8150
40	0.3226	0.6918	53.4%	0.0664	0.2248	-0.0655	0.8898	0.7175	0.7997
41	0.3326	0.7155	53.5%	0.0694	0.2332	-0.0684	0.8784	0.6935	0.7803
42	0.3397	0.7441	54.3%	0.0710	0.2436	-0.0700	0.8678	0.6632	0.7622
43	0.3431	0.7709	55.5%	0.0725	0.2537	-0.0715	0.8606	0.6331	0.7491
44	0.3471	0.7952	56.3%	0.0741	0.2632	-0.0731	0.8530	0.6042	0.7354
45	0.3516	0.8139	56.8%	0.0756	0.2720	-0.0746	0.8453	0.5806	0.7209
46	0.3525	0.8315	57.6%	0.0765	0.2793	-0.0756	0.8390	0.5570	0.7091
47	0.3576	0.8560	58.2%	0.0777	0.2874	-0.0767	0.8299	0.5236	0.6920
48	0.3702	0.8827	58.1%	0.0813	0.2967	-0.0803	0.8181	0.4870	0.6700
49	0.3842	0.9103	57.8%	0.0850	0.3059	-0.0841	0.8066	0.4490	0.6484
50	0.3933	0.9384	58.1%	0.0876	0.3145	-0.0866	0.7980	0.4087	0.6316
51	0.3992	0.9668	58.7%	0.0896	0.3234	-0.0886	0.7891	0.3658	0.6160
52	0.4034	0.9892	59.2%	0.0907	0.3297	-0.0897	0.7828	0.3313	0.6053
53	0.4050	1.0111	59.9%	0.0913	0.3362	-0.0903	0.7788	0.2963	0.5975
54	0.4066	1.0344	60.7%	0.0918	0.3432	-0.0908	0.7753	0.2584	0.5902
55	0.4066	1.0582	61.6%	0.0918	0.3501	-0.0908	0.7752	0.2195	0.5878
56	0.4066	1.0761	62.2%	0.0918	0.3548	-0.0908	0.7752	0.1896	0.5861
57	0.4069	1.0812	62.4%	0.0922	0.3581	-0.0913	0.7738	0.1797	0.5817
58	0.4074	1.0841	62.4%	0.0928	0.3607	-0.0918	0.7713	0.1728	0.5762
59	0.4081	1.0875	62.5%	0.0937	0.3643	-0.0927	0.7683	0.1645	0.5694
60	0.4086	1.0909	62.5%	0.0951	0.3688	-0.0941	0.7654	0.1564	0.5630
61	0.4097	1.0943	62.6%	0.0967	0.3738	-0.0957	0.7613	0.1481	0.5538
62	0.4098	1.0980	62.7%	0.0971	0.3785	-0.0962	0.7595	0.1404	0.5478
63	0.4124	1.1030	62.6%	0.0987	0.3843	-0.0977	0.7547	0.1313	0.5400
64	0.4195	1.1092	62.2%	0.1008	0.3894	-0.0998	0.7462	0.1227	0.5288
65	0.4278	1.1152	61.6%	0.1035	0.3943	-0.1025	0.7377	0.1160	0.5186
66	0.4361	1.1228	61.2%	0.1065	0.3999	-0.1056	0.7287	0.1063	0.5078
67	0.4441	1.1280	60.6%	0.1108	0.4056	-0.1099	0.7193	0.0994	0.4971
68	0.4529	1.1321	60.0%	0.1150	0.4108	-0.1141	0.7052	0.0924	0.4800
69	0.4628	1.1374	59.3%	0.1193	0.4156	-0.1184	0.6946	0.0870	0.4686
70	0.4714	1.1409	58.7%	0.1224	0.4186	-0.1214	0.6859	0.0847	0.4592
71	0.4780	1.1428	58.2%	0.1246	0.4203	-0.1236	0.6722	0.0805	0.4435
72	0.4883	1.1442	57.3%	0.1271	0.4232	-0.1261	0.6445	0.0744	0.4122
73	0.4930	1.1443	56.9%	0.1285	0.4259	-0.1276	0.6229	0.0684	0.3863
74	0.4978	1.1451	56.5%	0.1304	0.4297	-0.1295	0.6003	0.0619	0.3619
75	0.5007	1.1456	56.3%	0.1309	0.4316	-0.1299	0.5820	0.0567	0.3421
76	0.5007	1.1433	56.2%	0.1309	0.4305	-0.1299	0.5610	0.0532	0.3220
77	0.5033	1.1405	55.9%	0.1323	0.4300	-0.1313	0.5203	0.0482	0.2868
78	0.5046	1.1370	55.6%	0.1358	0.4312	-0.1349	0.4689	0.0419	0.2462
79	0.5076	1.1351	55.3%	0.1403	0.4344	-0.1393	0.4171	0.0353	0.2080
80	0.5120	1.1339	54.8%	0.1447	0.4368	-0.1437	0.3580	0.0284	0.1655
81	0.5169	1.1328	54.4%	0.1482	0.4377	-0.1473	0.2928	0.0216	0.1191
82	0.5197	1.1317	54.1%	0.1503	0.4376	-0.1493	0.2364	0.0162	0.0787
83	0.5229	1.1313	53.8%	0.1531	0.4390	-0.1521	0.1822	0.0092	0.0398
84	0.5271	1.1328	53.5%	0.1564	0.4417	-0.1554	0.1409	0.0015	0.0103
85	0.5291	1.1337	53.3%	0.1579	0.4438	-0.1579	0.1061	-0.0065	-0.0154
86	0.5282	1.1340	53.4%	0.1586	0.4457	-0.1586	0.0736	-0.0150	-0.0396
87	0.5247	1.1333	53.7%	0.1596	0.4469	-0.1596	0.0453	-0.0240	-0.0634
88	0.5235	1.1341	53.8%	0.1605	0.4482	-0.1605	0.0214	-0.0332	-0.0849
89	0.5205	1.1349	54.1%	0.1601	0.4484	-0.1601	0.0094	-0.0412	-0.0956
90	0.5186	1.1357	54.3%	0.1593	0.4487	-0.1593	0.0000	-0.0472	-0.1042

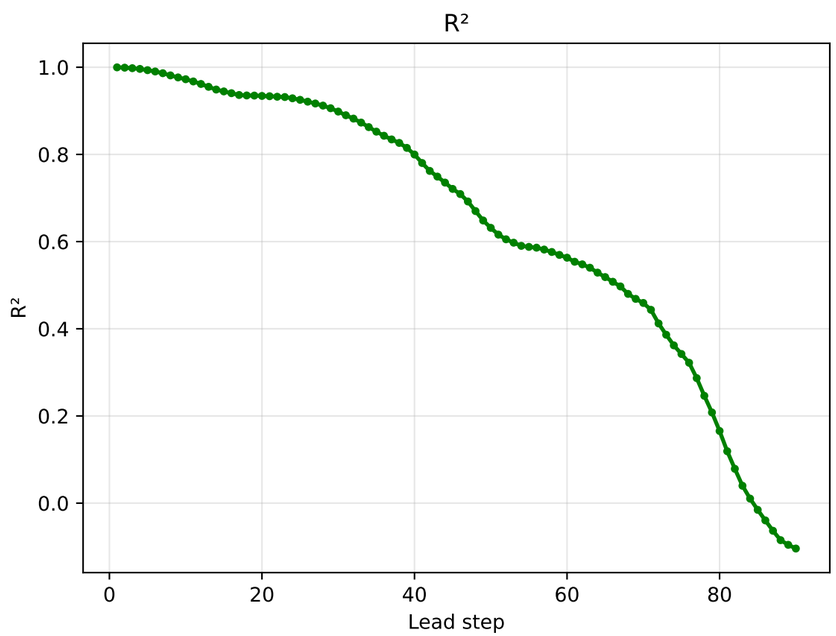
DHW Forecast Skill vs Lead Step - Test - GBR

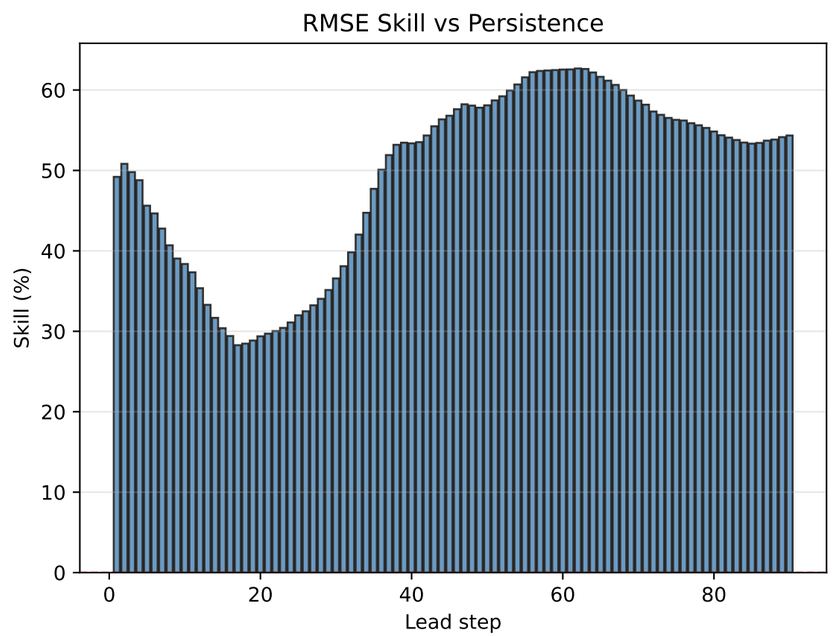


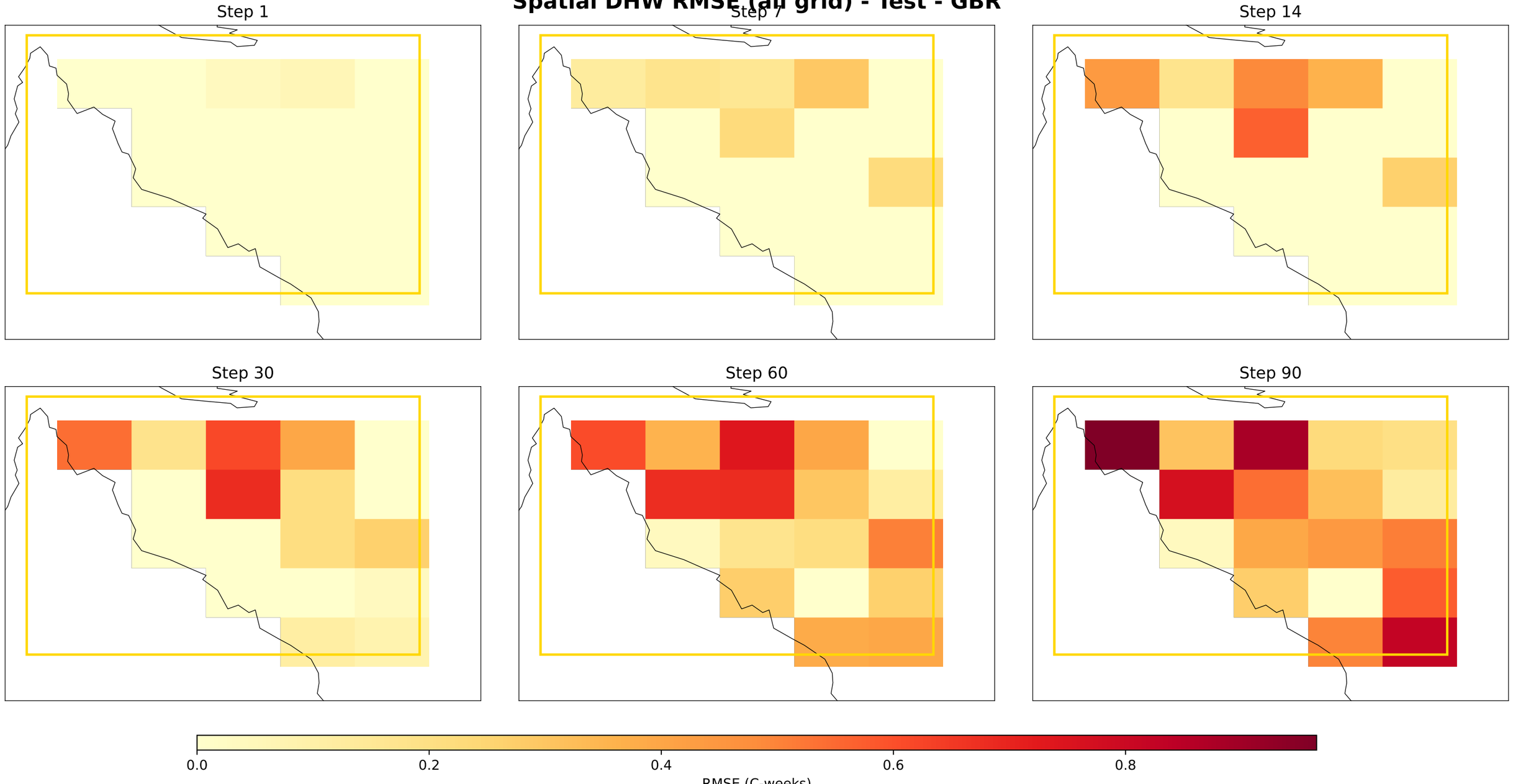


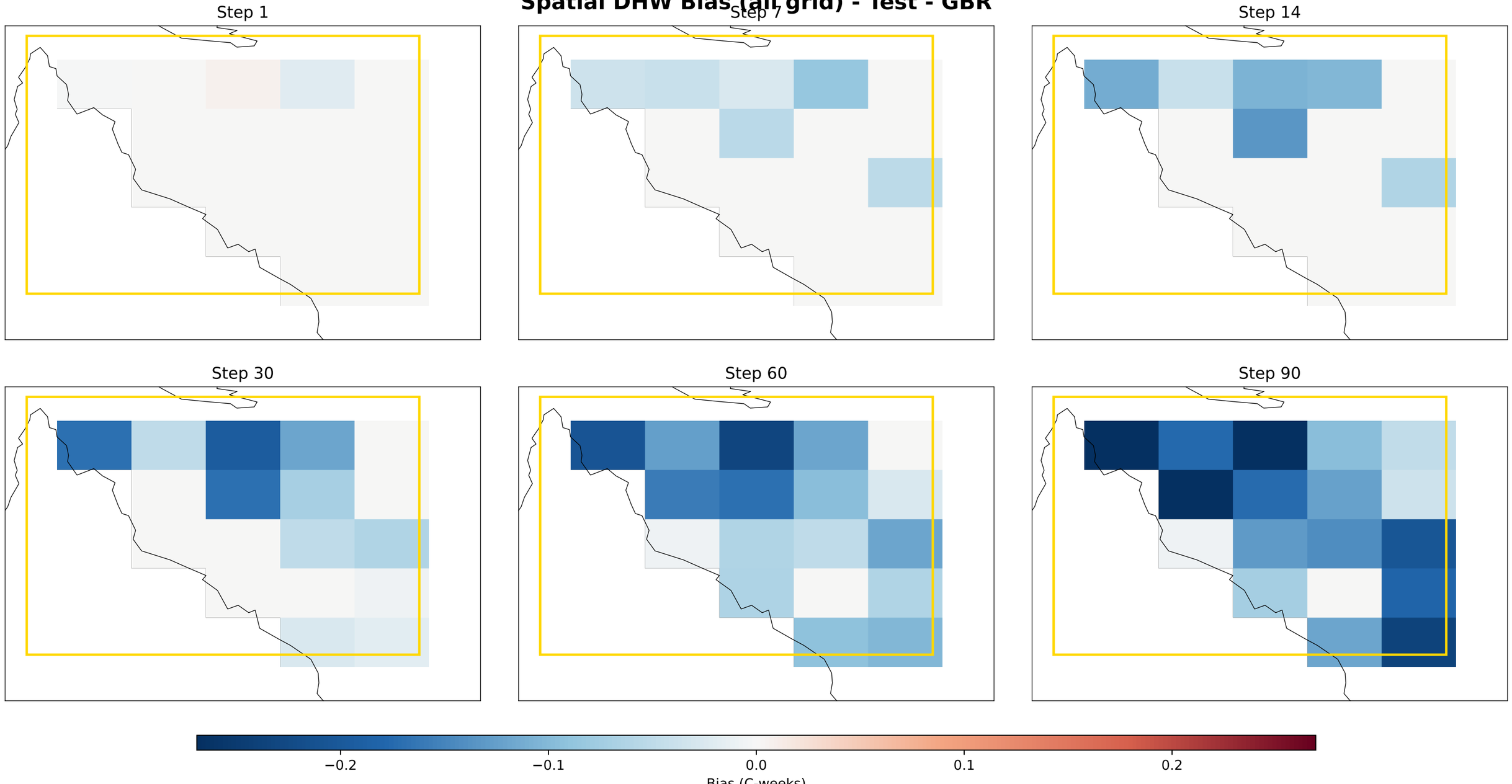


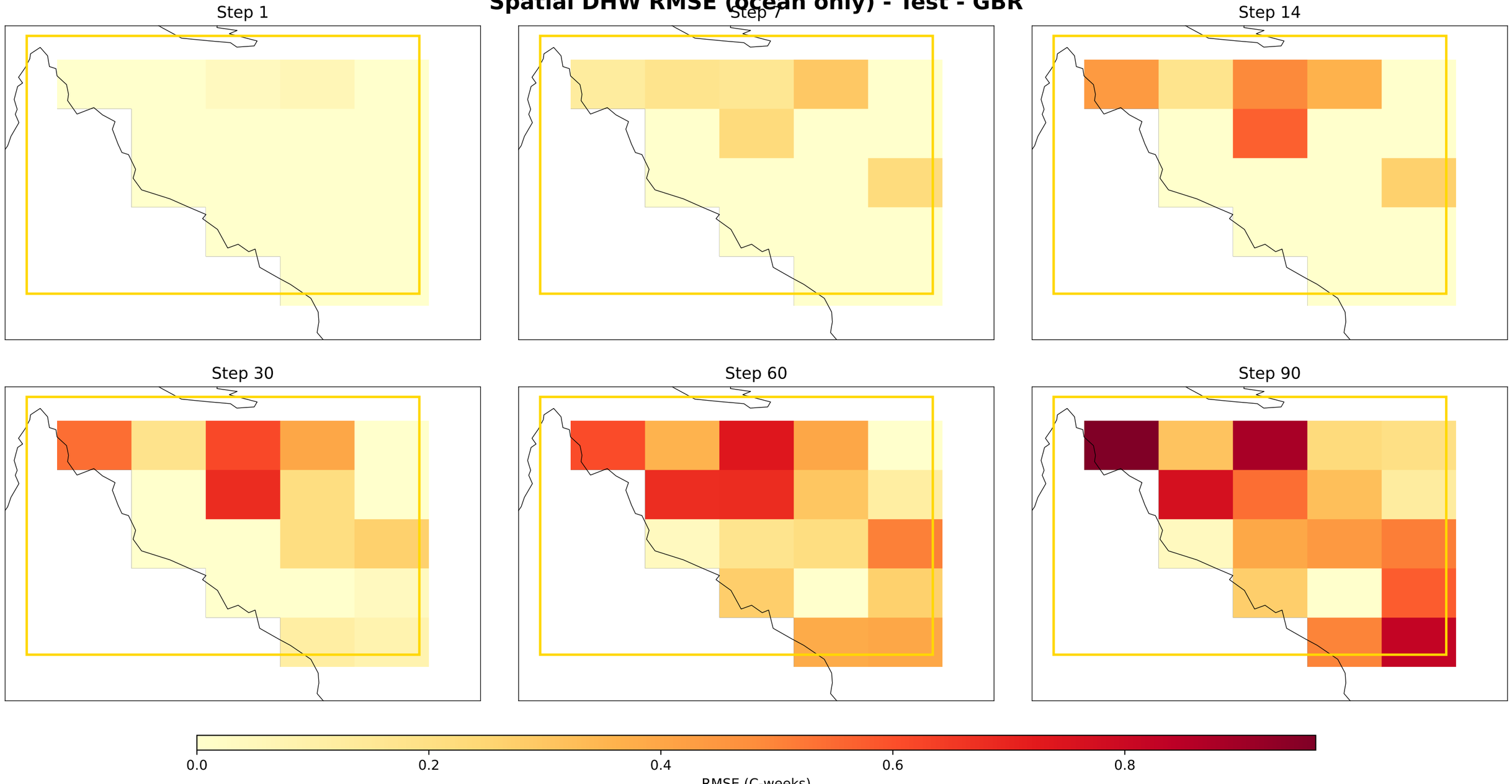


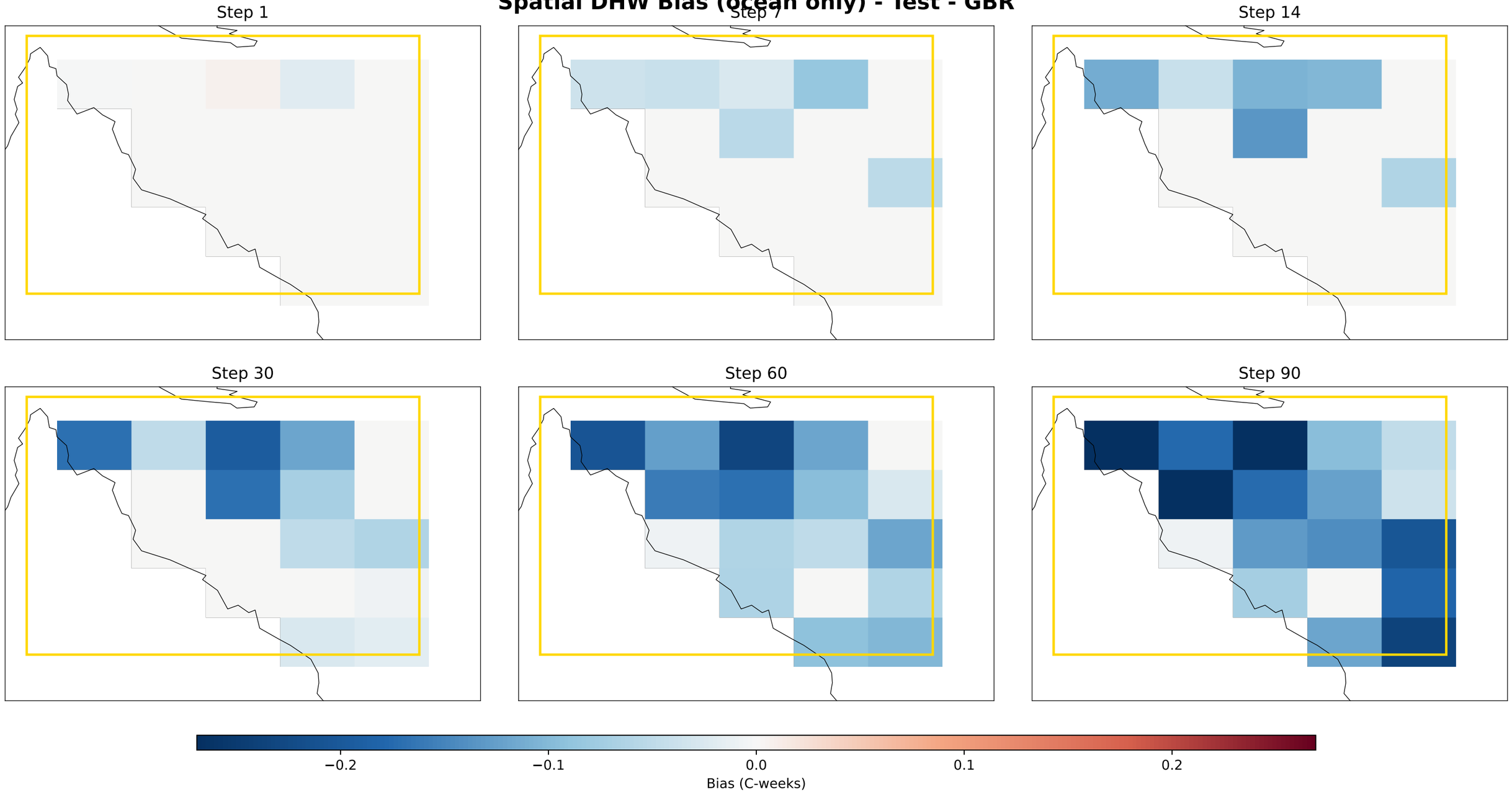





Spatial DHW RMSE (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

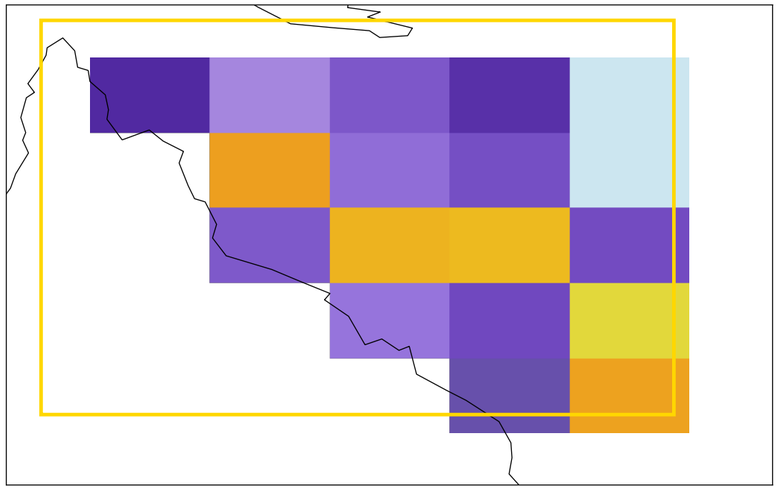

Spatial DHW Bias (all grid) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.2
0.4
0.6
0.8
RMSE (C-weeks)

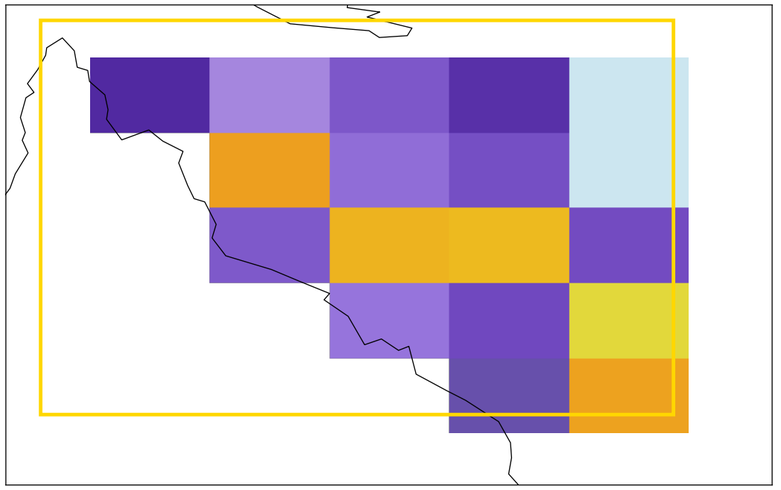

Spatial DHW Bias (ocean only) - Test - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.2
−0.1
0.0
0.1
0.2
Bias (C-weeks)

DHW Chain init: 2020-04-10 - Test - GBR (C-weeks)

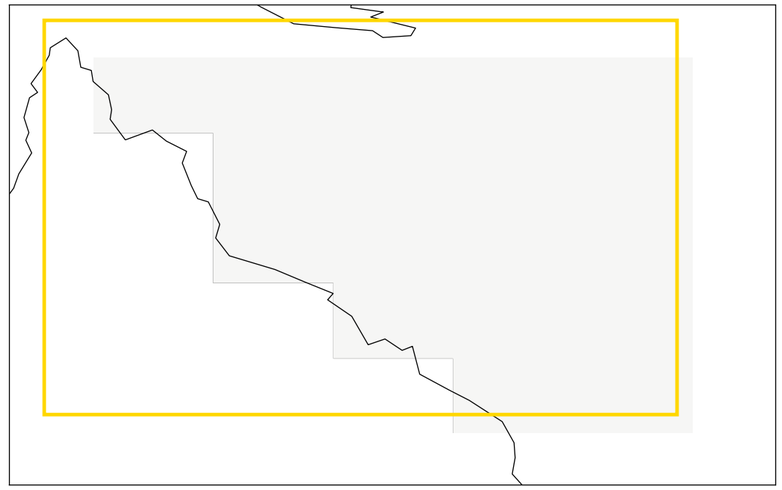
Actual DHW - Step 1



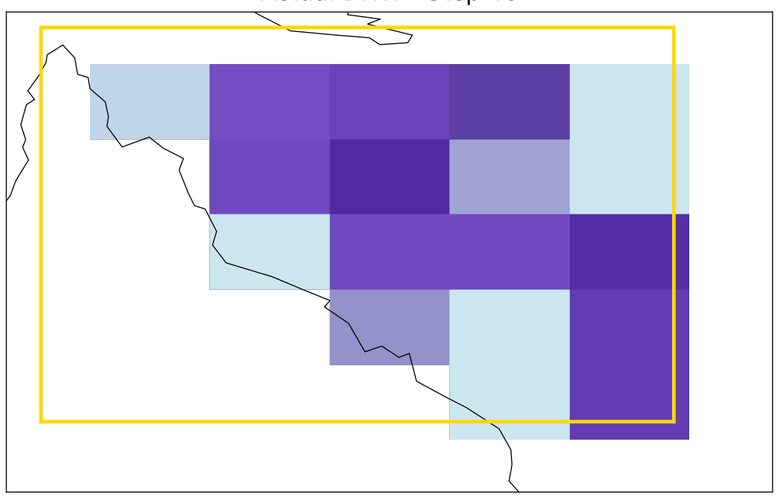
Predicted DHW - Step 1



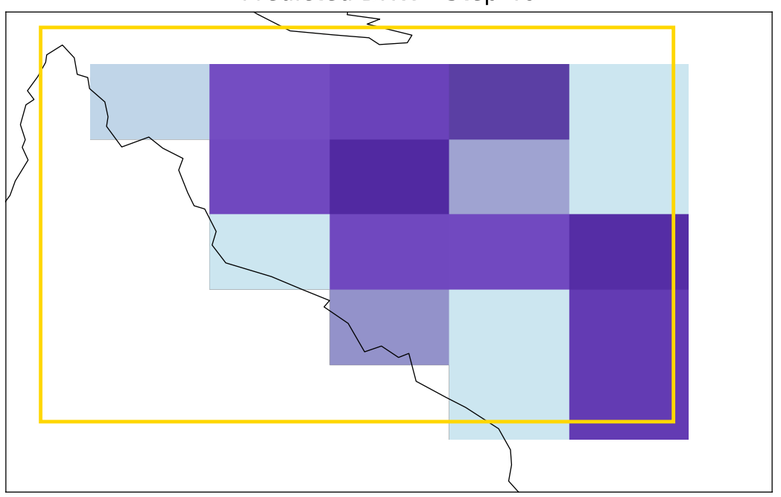
Error (Pred - Actual) - Step 1 [scale ±8]



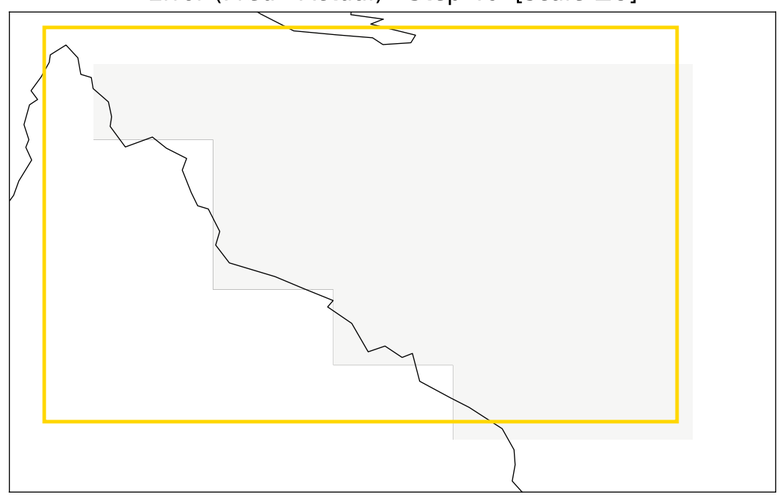
Actual DHW - Step 46



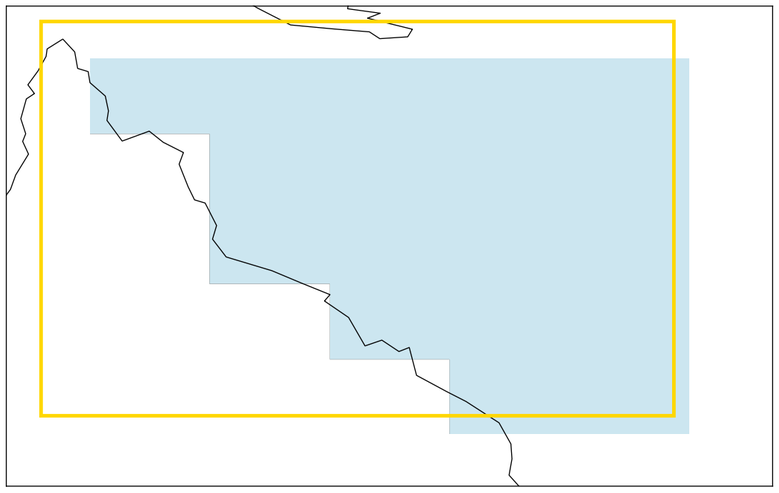
Predicted DHW - Step 46



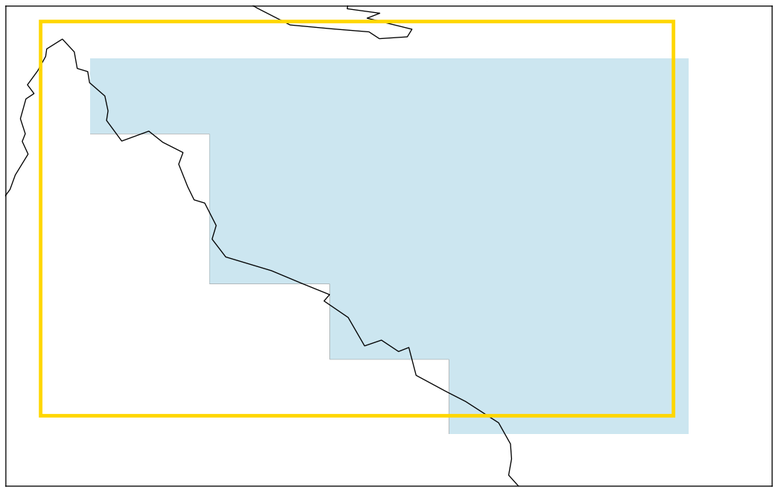
Error (Pred - Actual) - Step 46 [scale ±8]



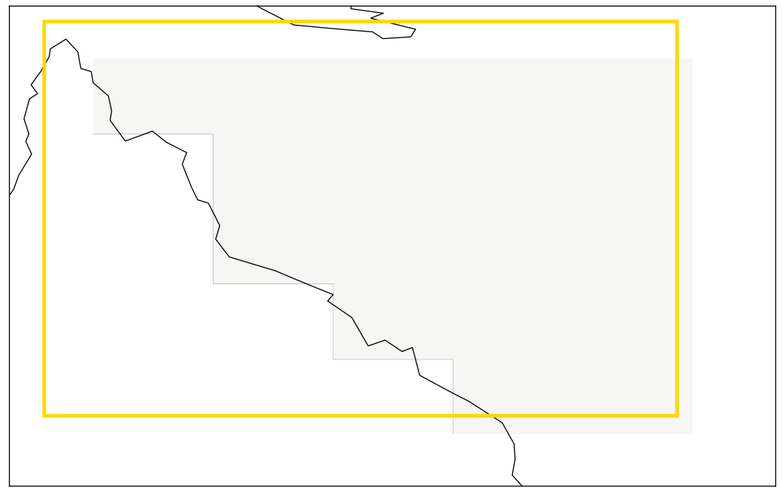
Actual DHW - Step 90

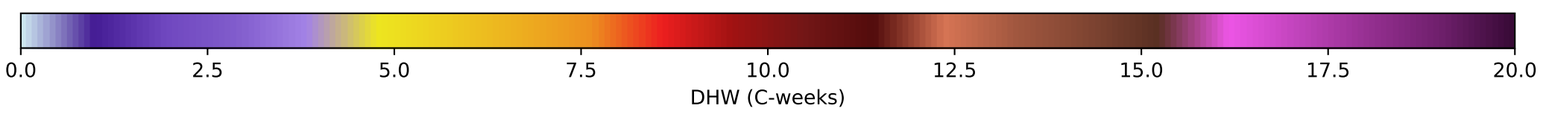


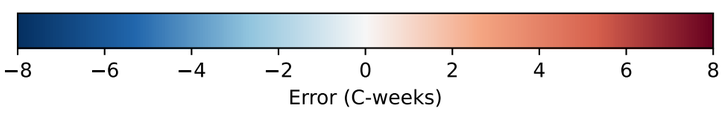
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

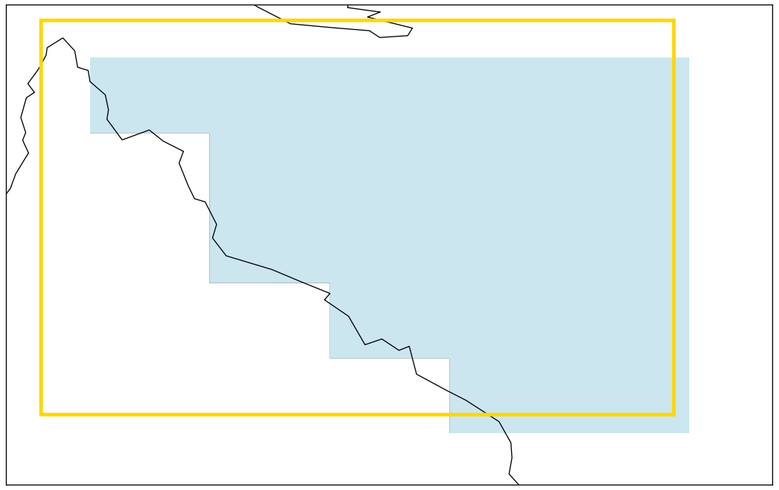




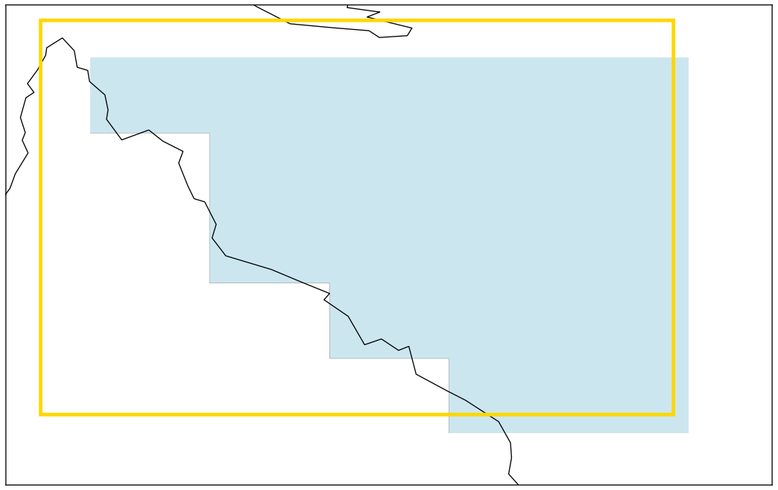


DHW Chain init: 2020-07-09 - Test - GBR (C-weeks)

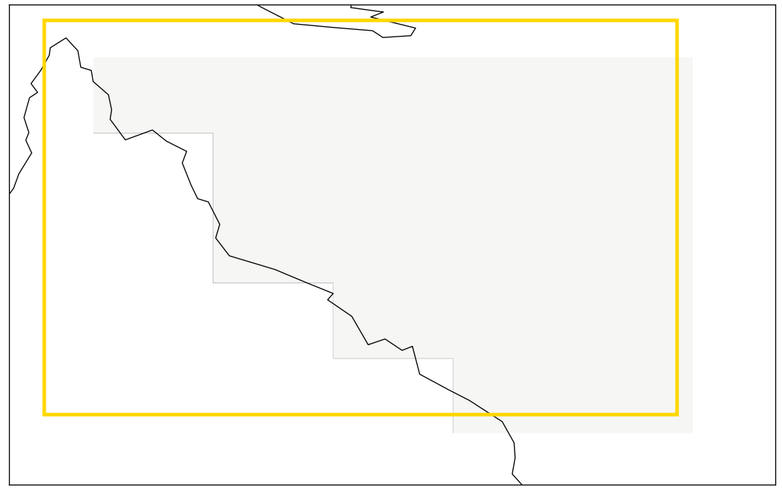
Actual DHW - Step 1



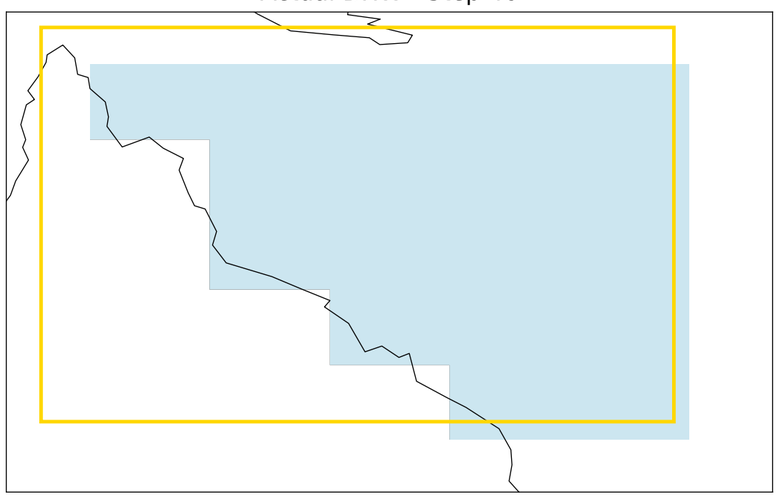
Predicted DHW - Step 1



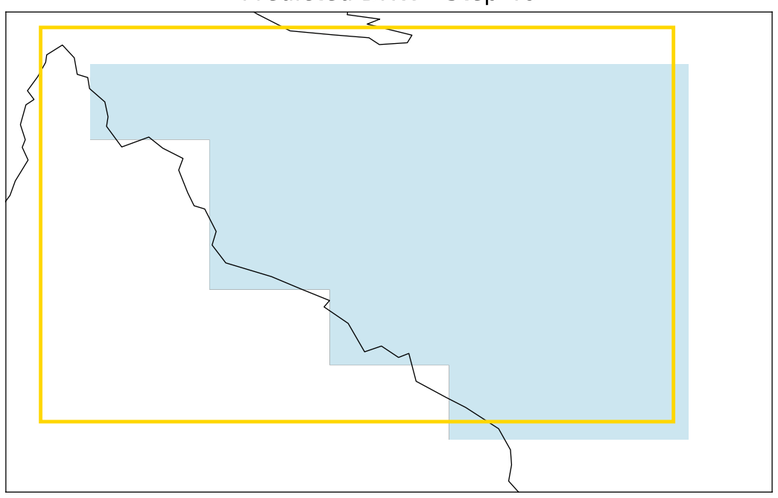
Error (Pred - Actual) - Step 1 [scale ±8]



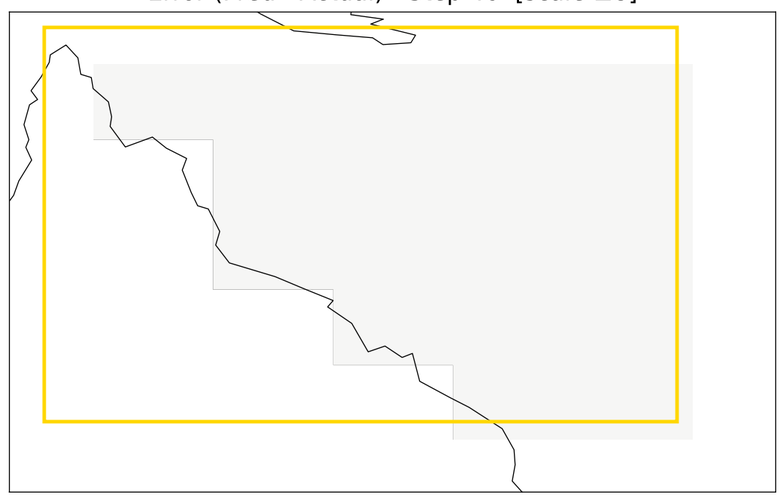
Actual DHW - Step 46



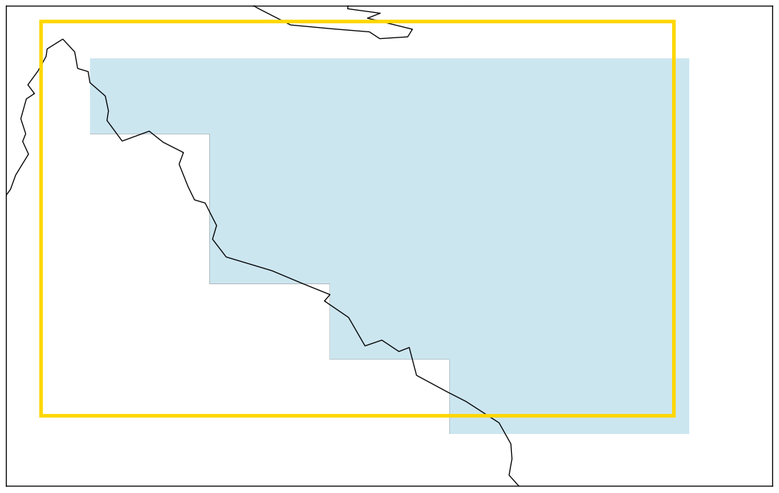
Predicted DHW - Step 46



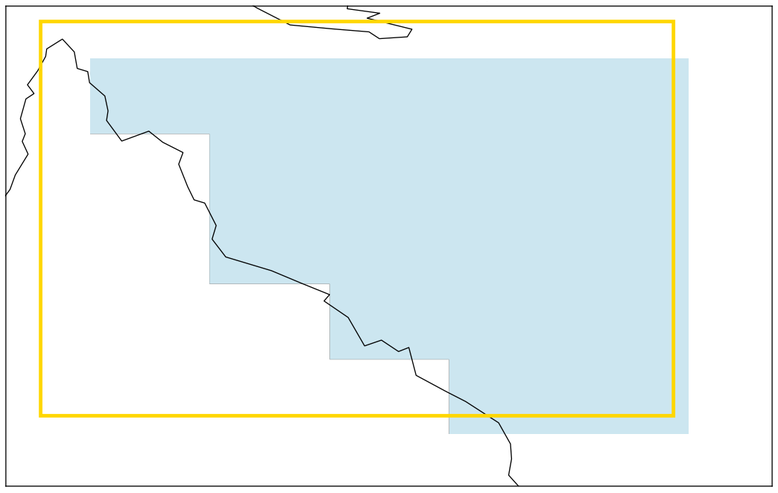
Error (Pred - Actual) - Step 46 [scale ±8]



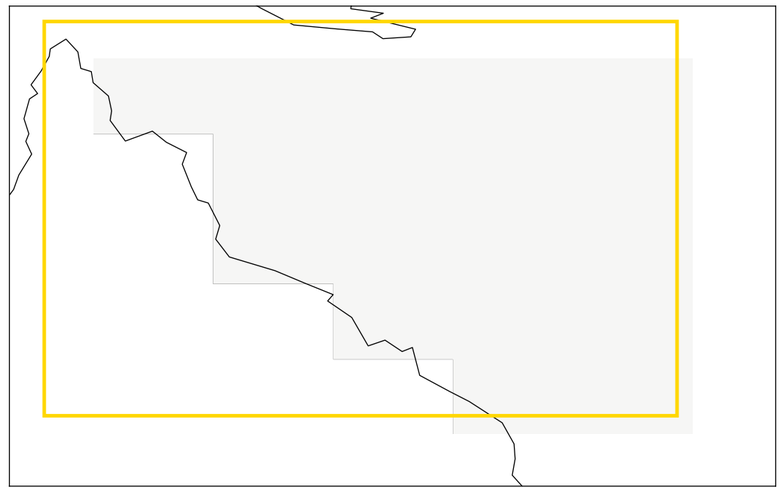
Actual DHW - Step 90

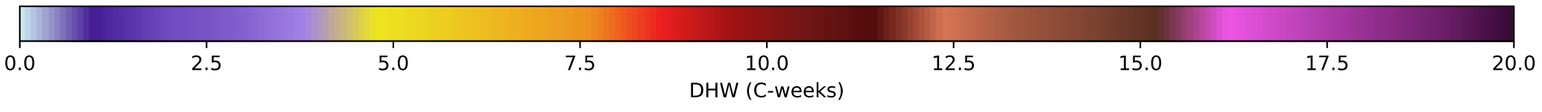


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

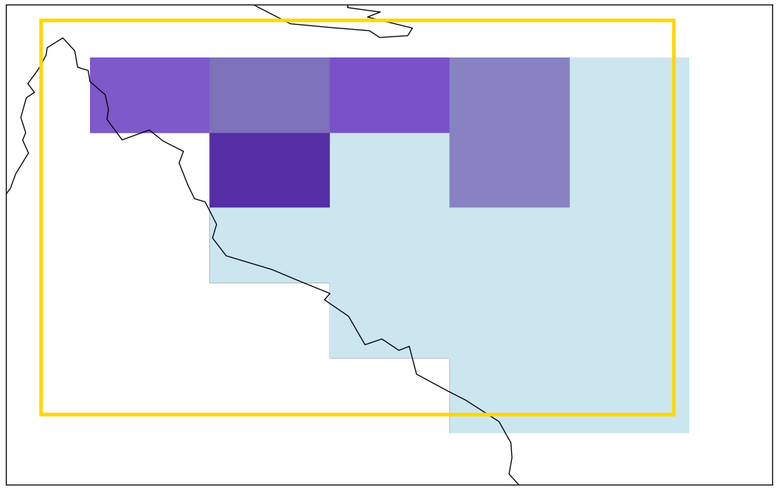




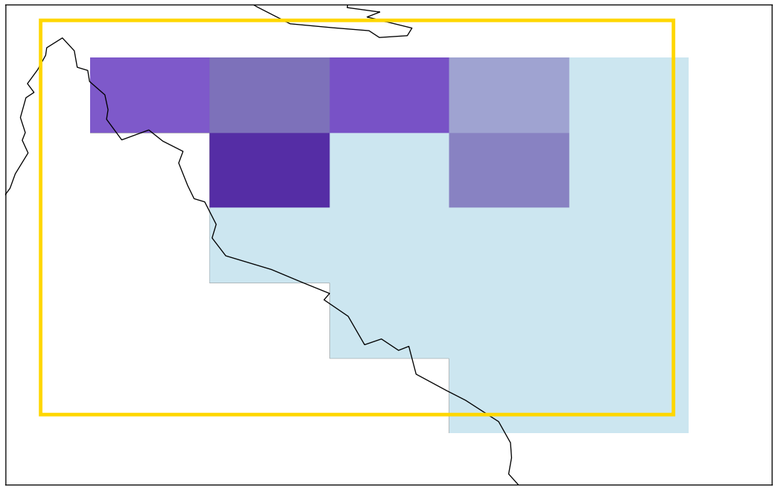


DHW Chain init: 2021-01-06 - Test - GBR (C-weeks)

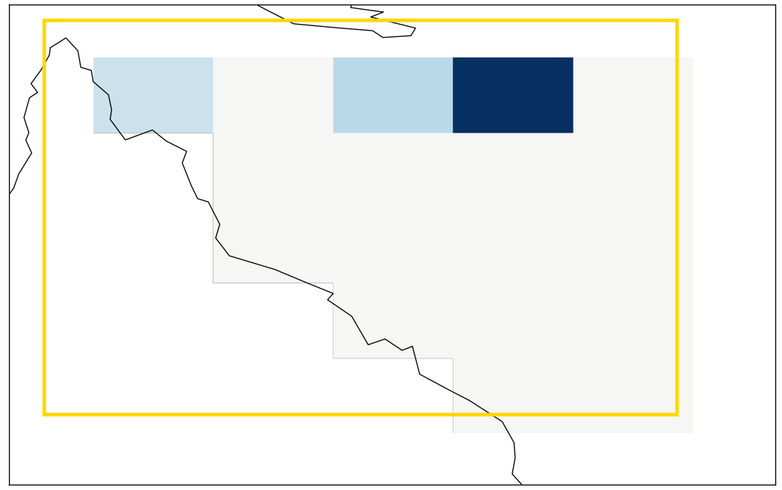
Actual DHW - Step 1



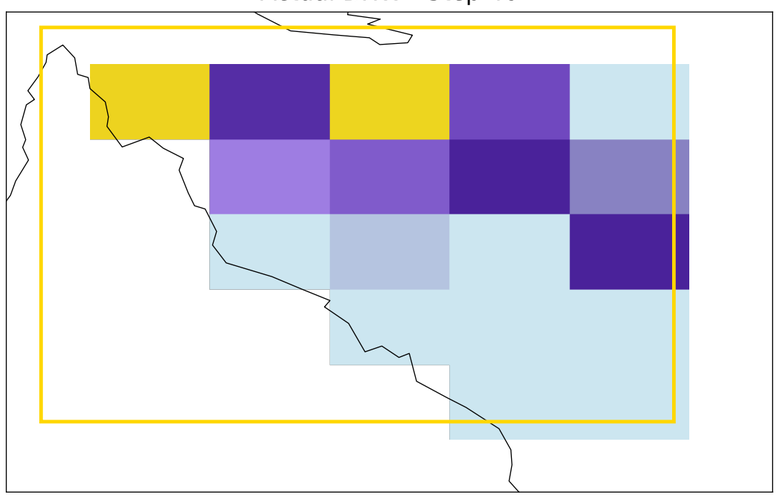
Predicted DHW - Step 1



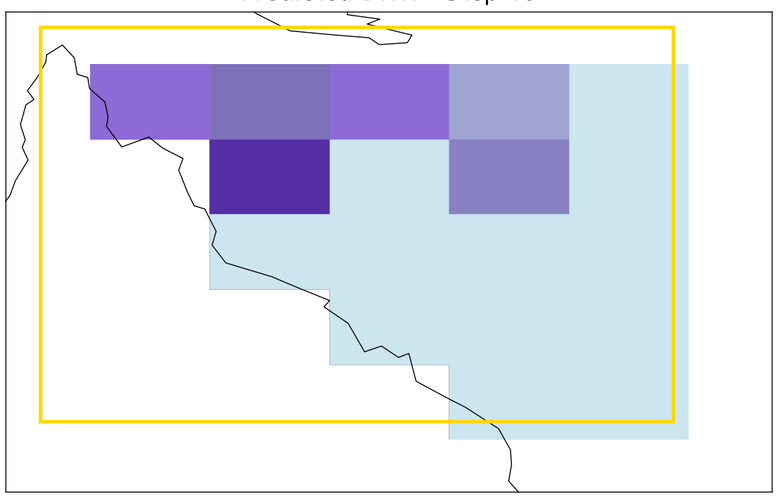
Error (Pred - Actual) - Step 1 [scale ±0.11]



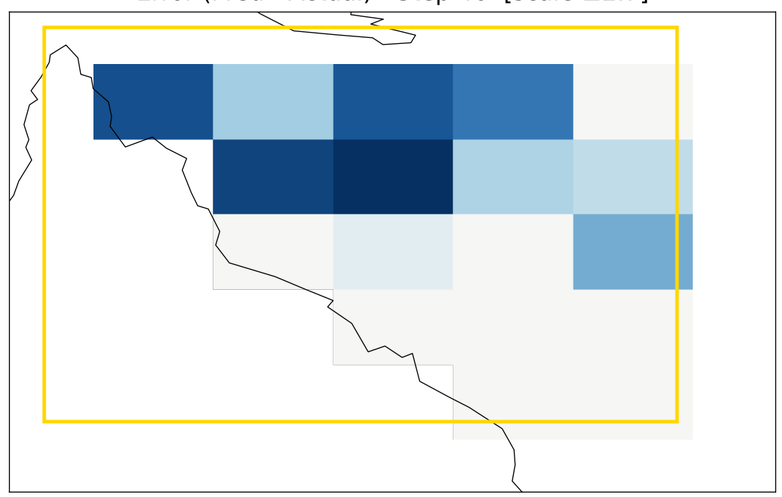
Actual DHW - Step 46



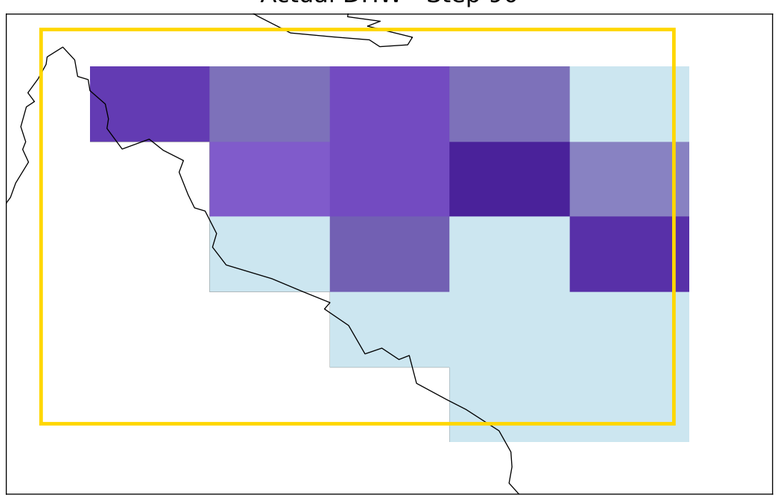
Predicted DHW - Step 46



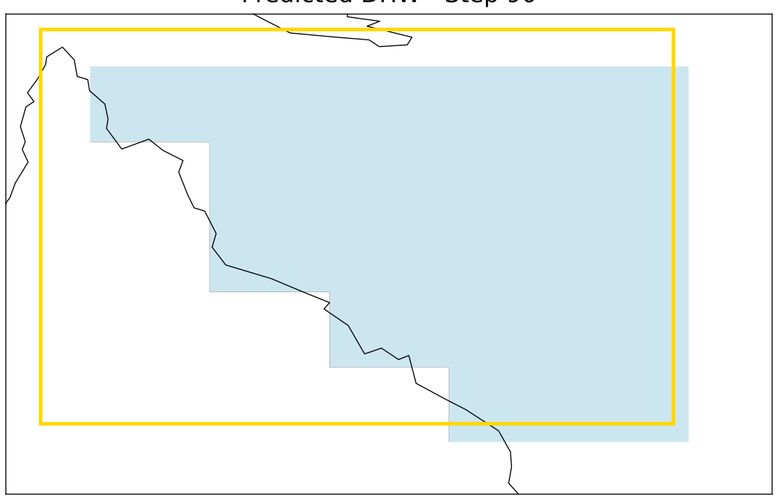
Error (Pred - Actual) - Step 46 [scale ±2.7]



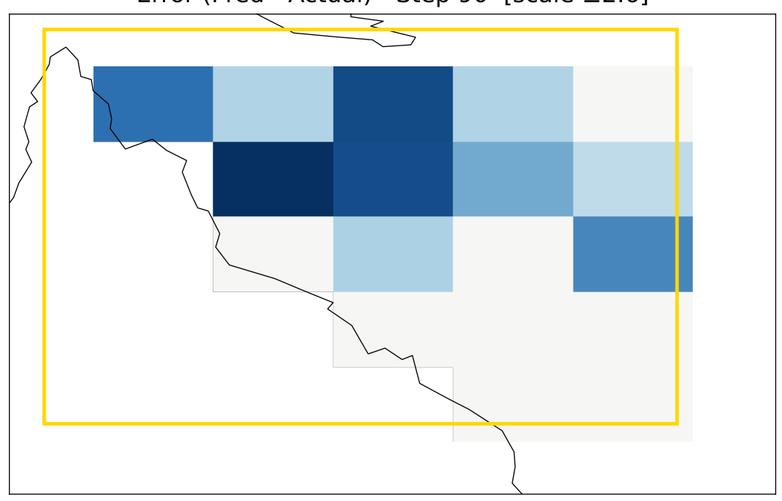
Actual DHW - Step 90

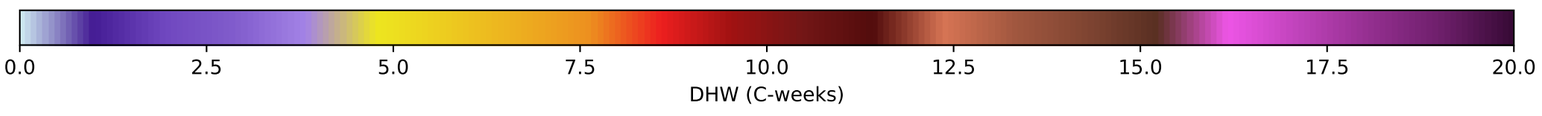


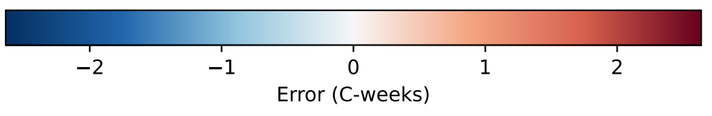
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.6]

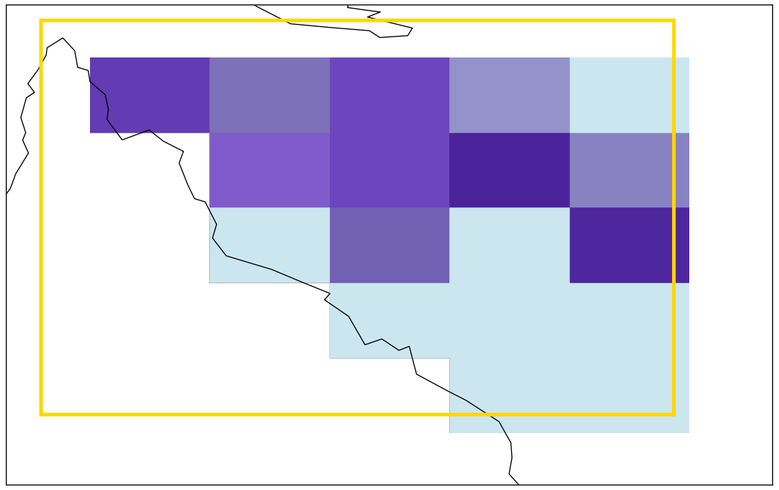




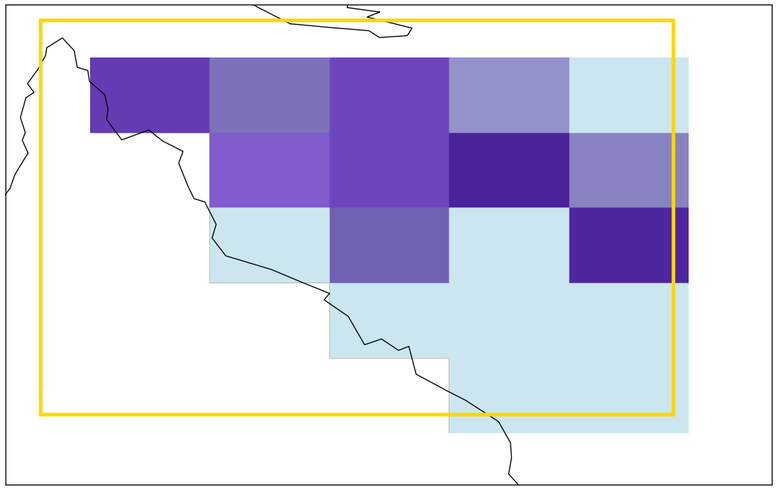


DHW Chain init: 2021-04-06 - Test - GBR (C-weeks)

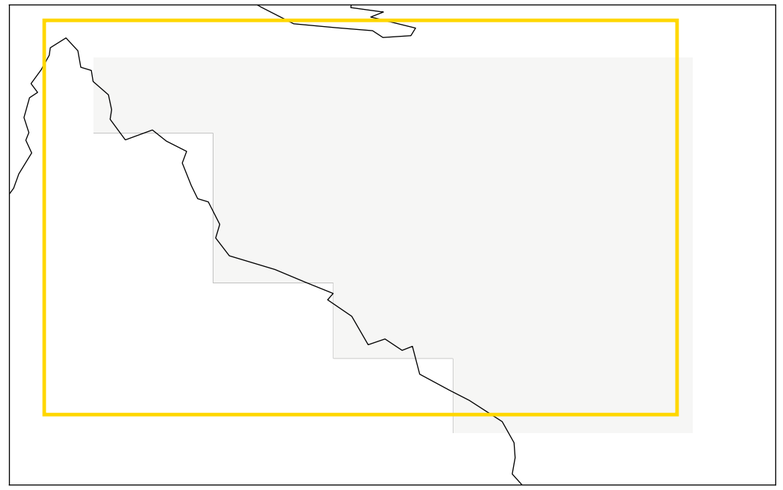
Actual DHW - Step 1



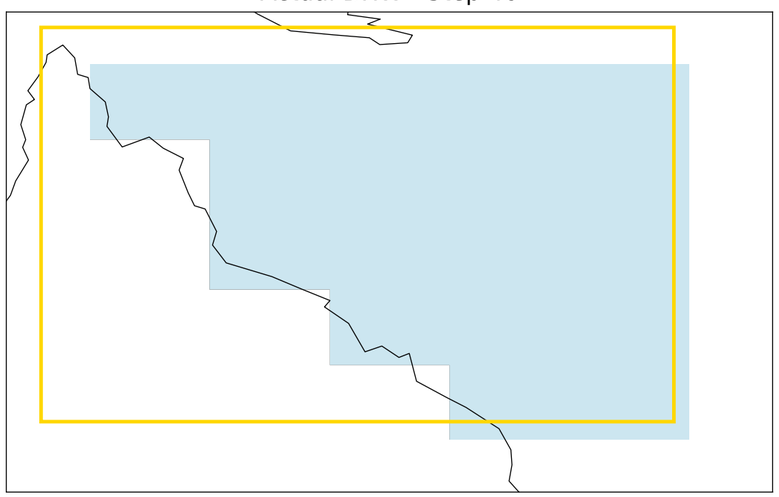
Predicted DHW - Step 1



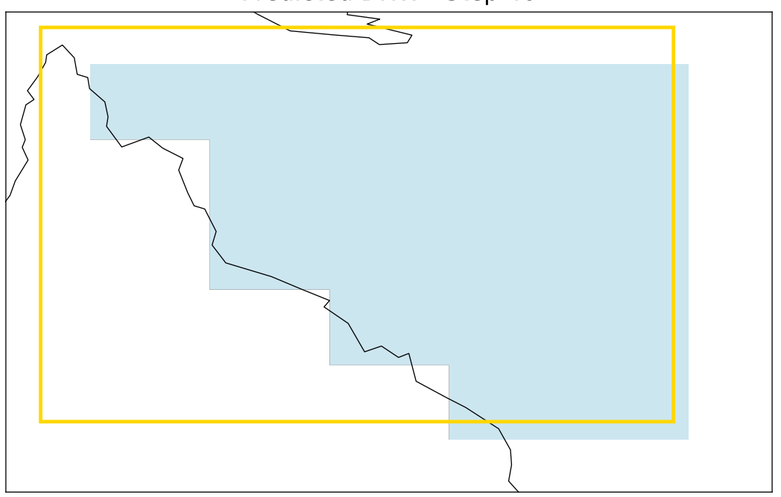
Error (Pred - Actual) - Step 1 [scale ±8]



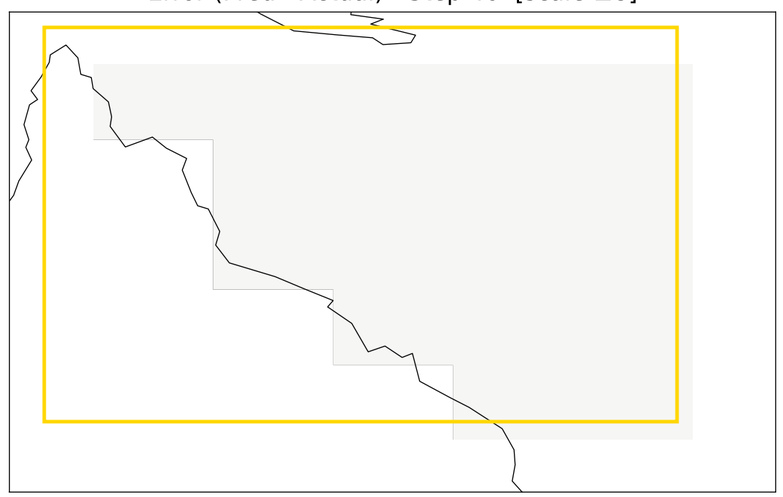
Actual DHW - Step 46



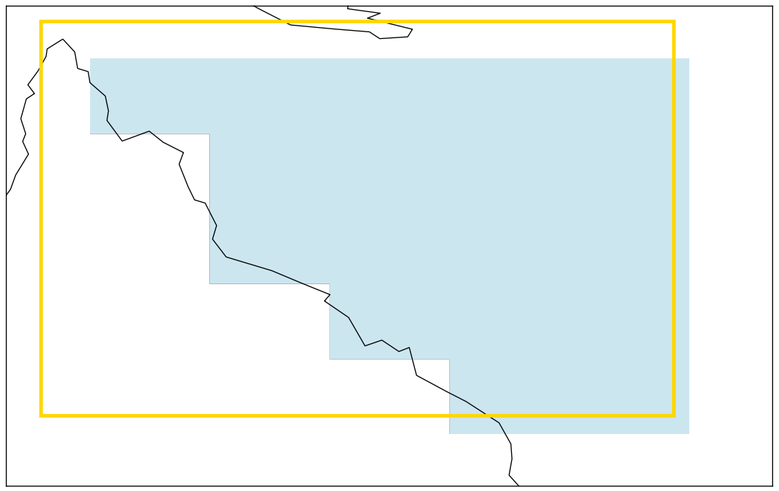
Predicted DHW - Step 46



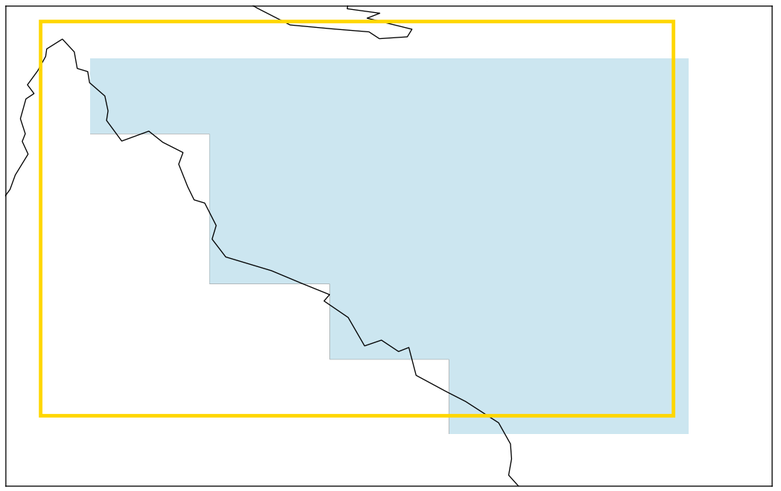
Error (Pred - Actual) - Step 46 [scale ±8]



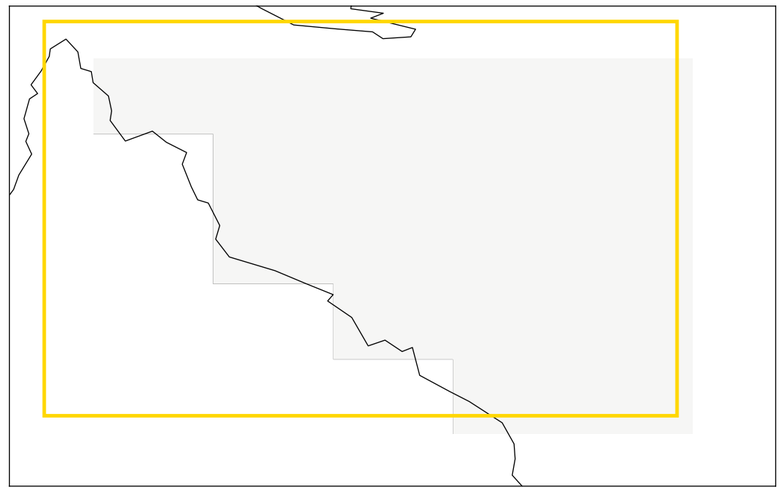
Actual DHW - Step 90

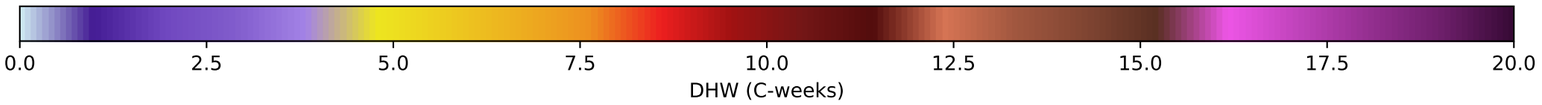


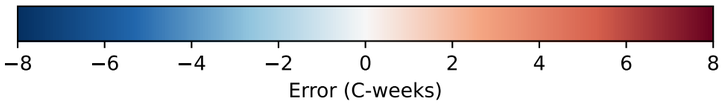
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

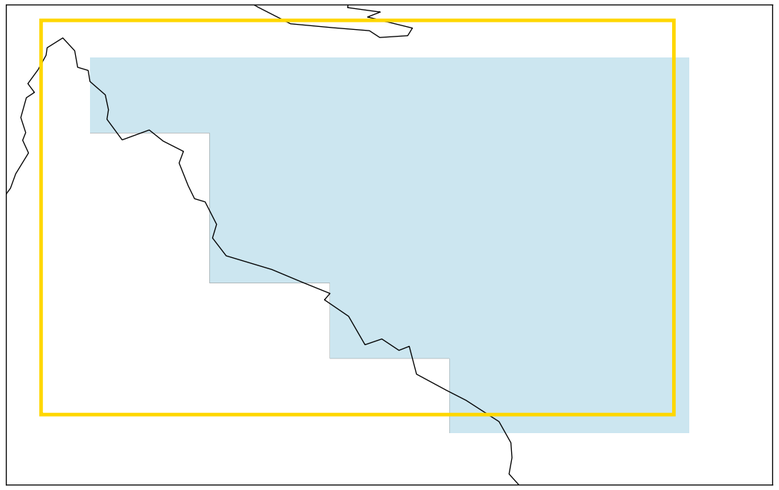




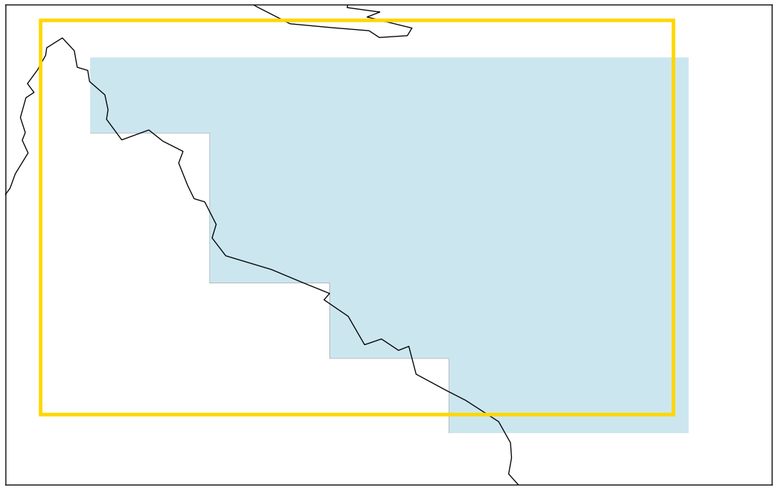


DHW Chain init: 2021-10-03 - Test - GBR (C-weeks)

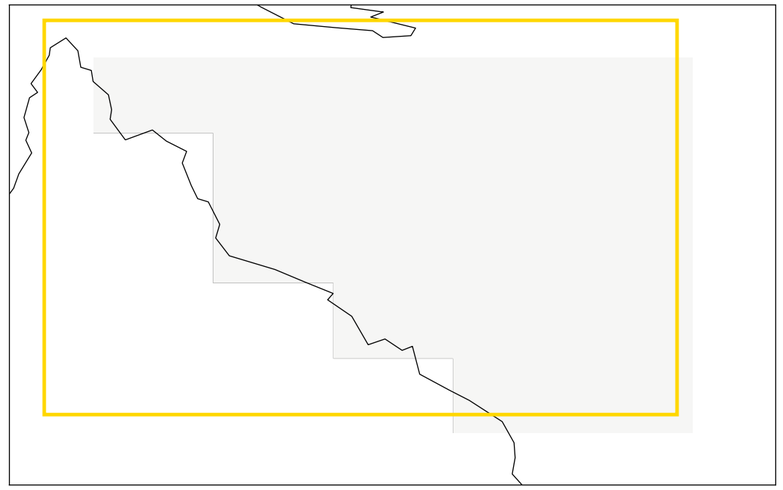
Actual DHW - Step 1



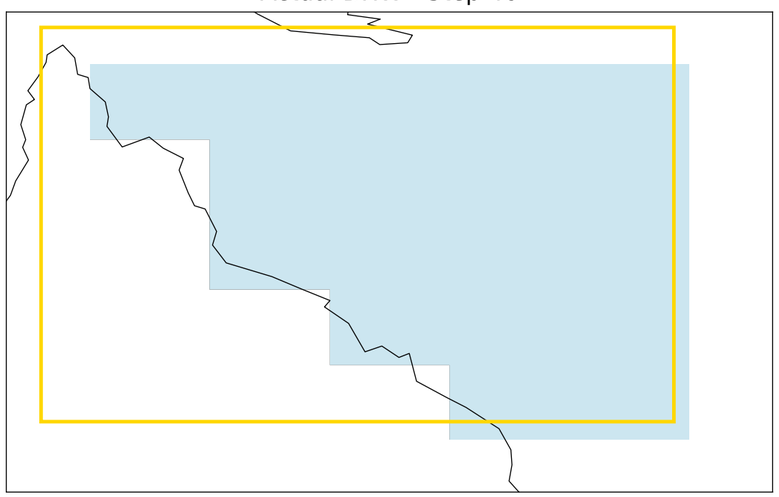
Predicted DHW - Step 1



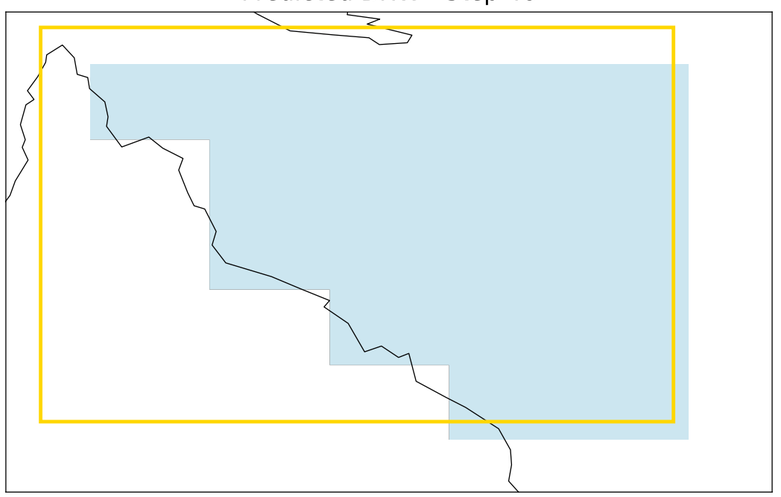
Error (Pred - Actual) - Step 1 [scale ±8]



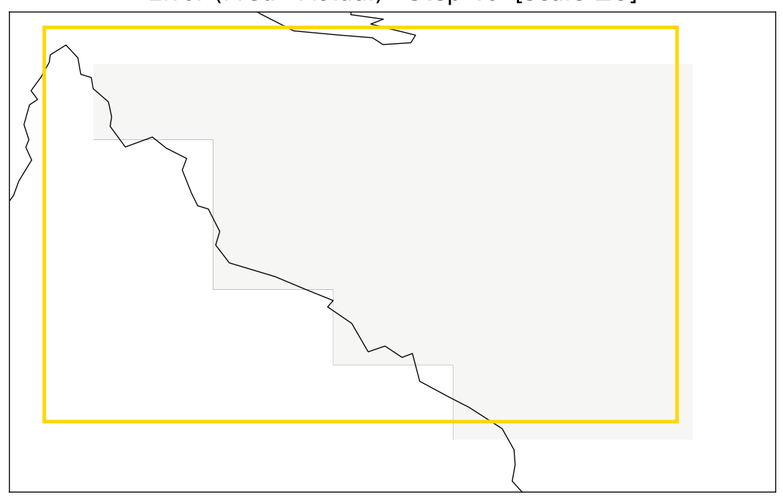
Actual DHW - Step 46



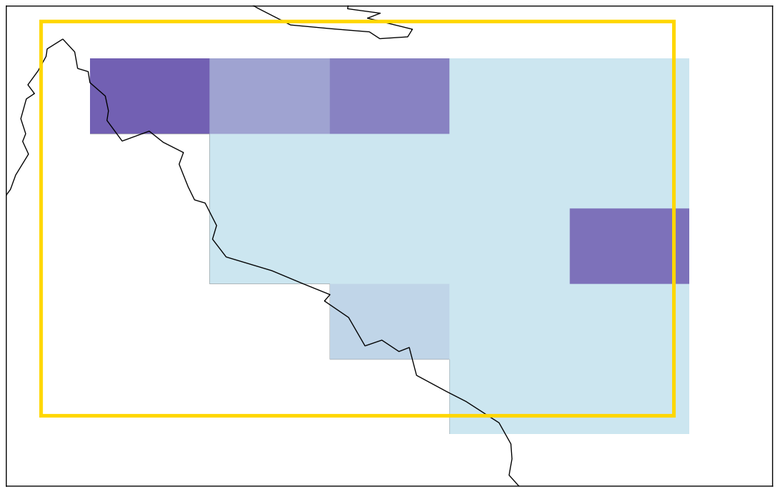
Predicted DHW - Step 46



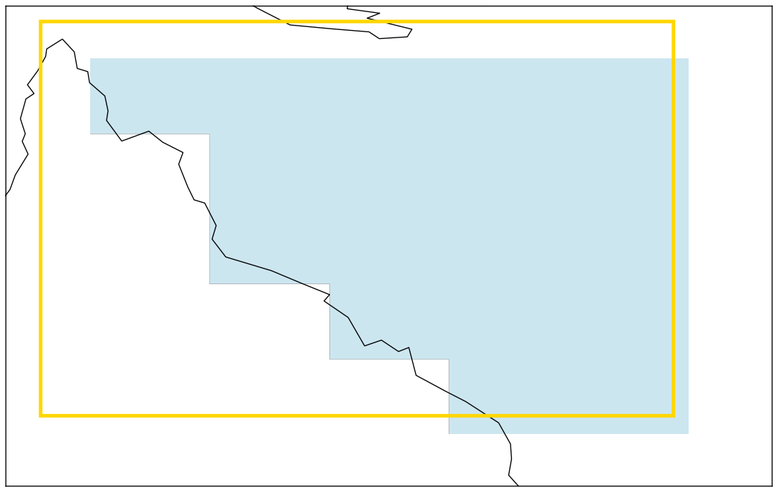
Error (Pred - Actual) - Step 46 [scale ±8]



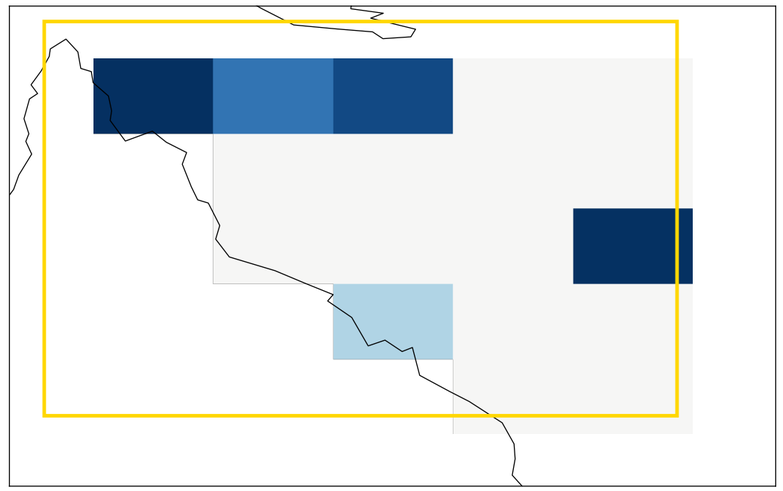
Actual DHW - Step 90

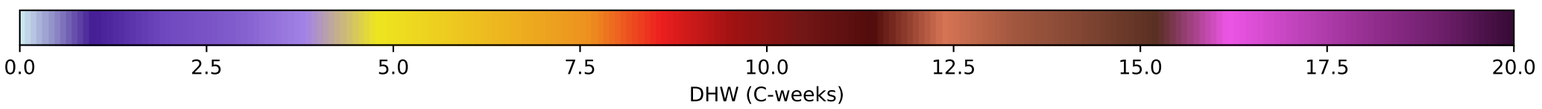


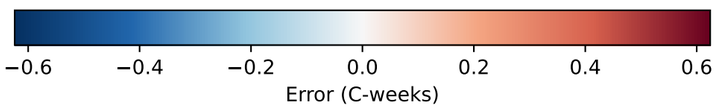
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.62]

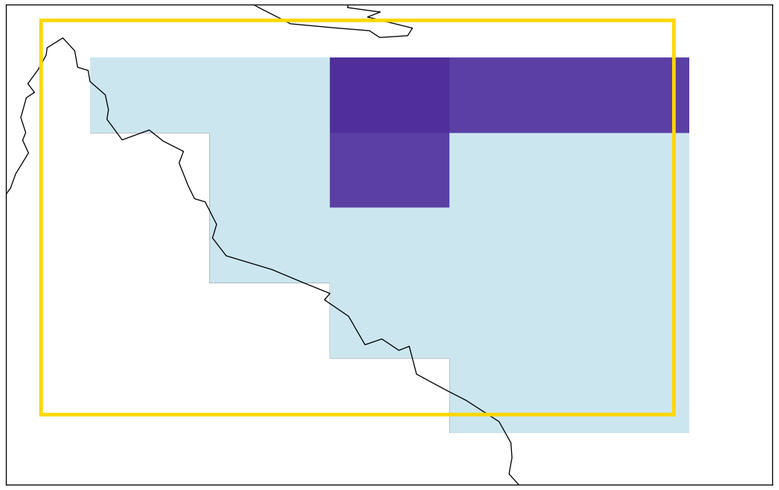




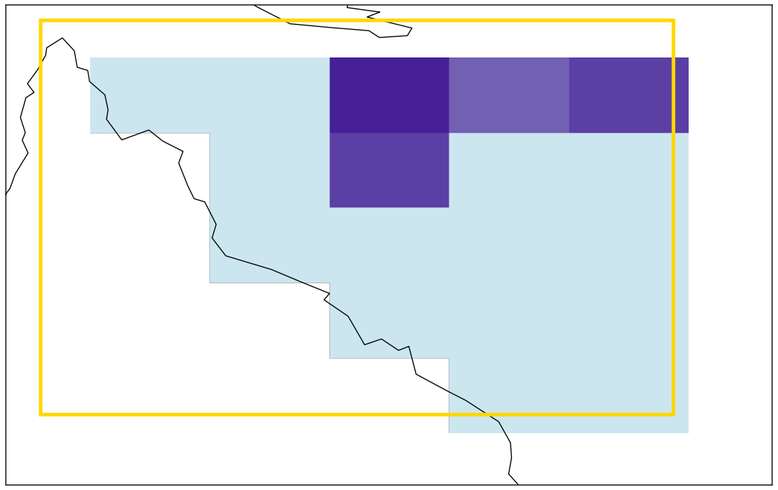


DHW Chain init: 2022-12-27 - Test - GBR (C-weeks)

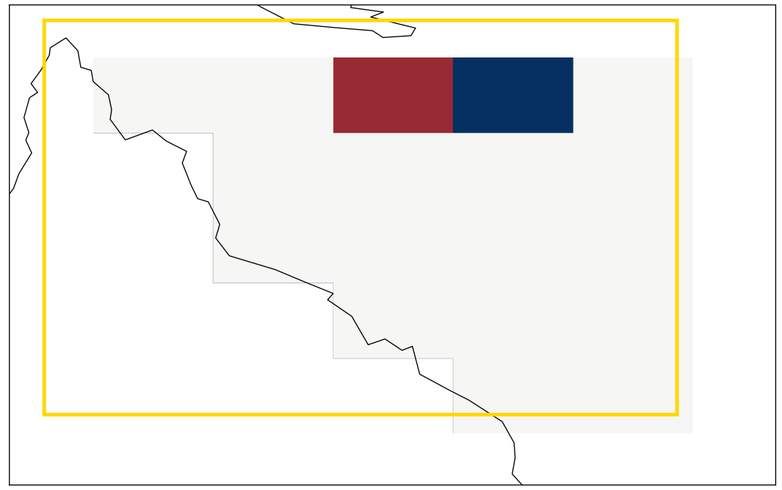
Actual DHW - Step 1



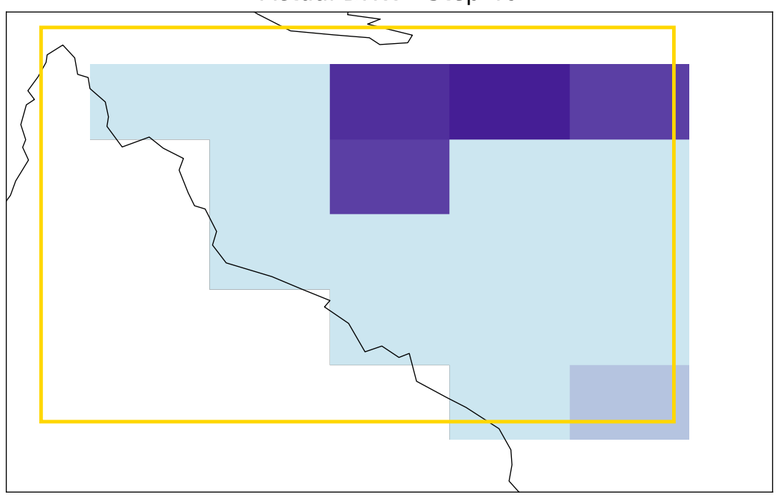
Predicted DHW - Step 1



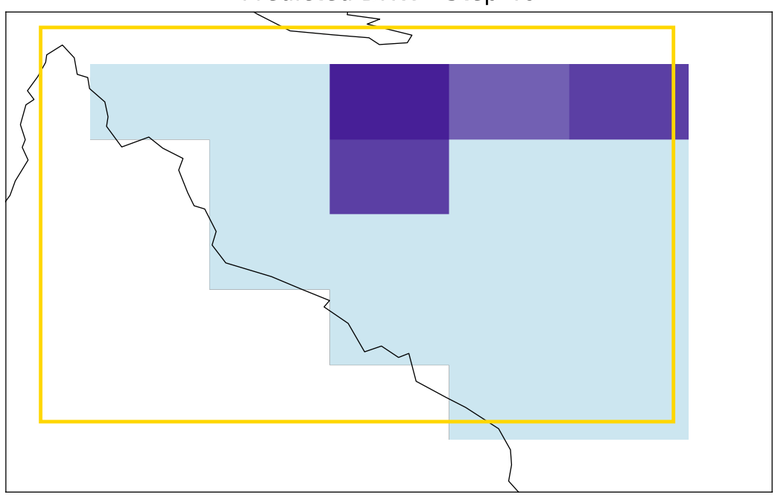
Error (Pred - Actual) - Step 1 [scale ±0.18]



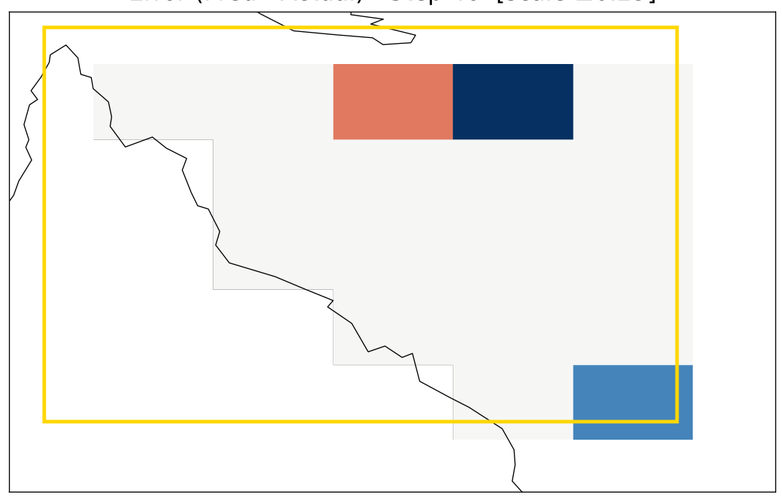
Actual DHW - Step 46



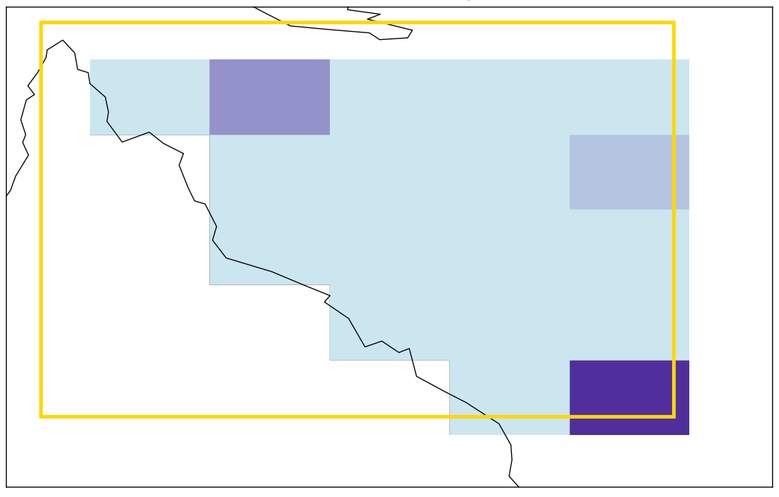
Predicted DHW - Step 46



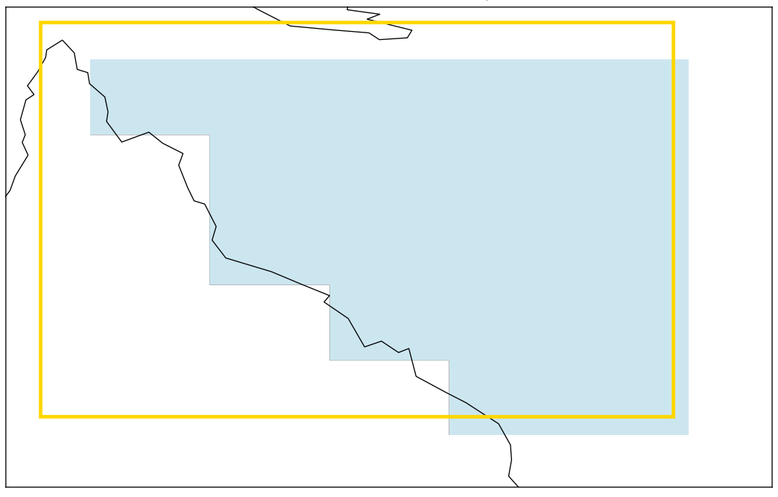
Error (Pred - Actual) - Step 46 [scale ±0.29]



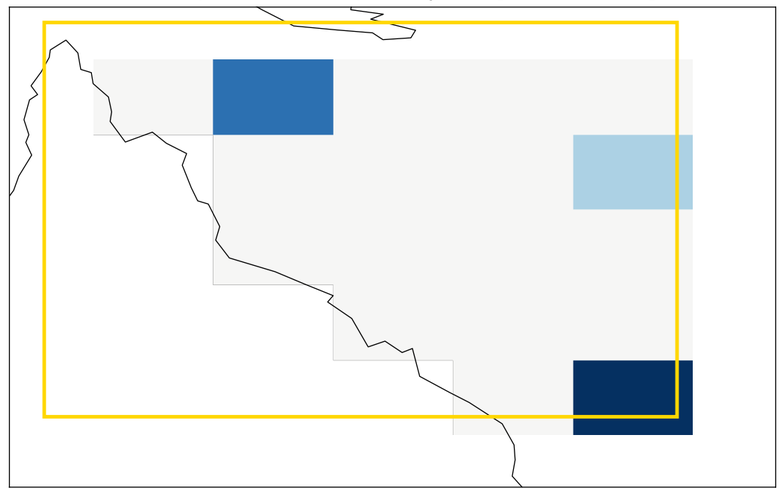
Actual DHW - Step 90

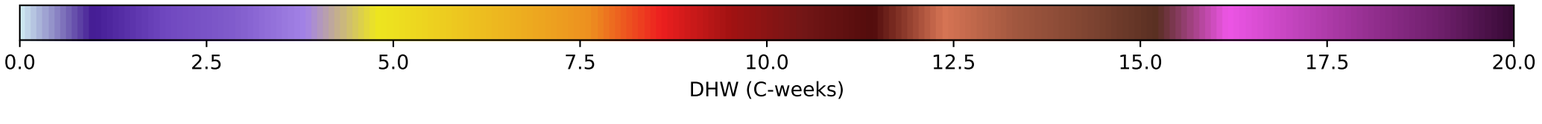


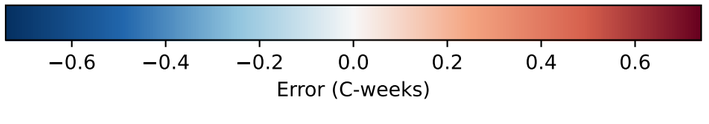
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.74]

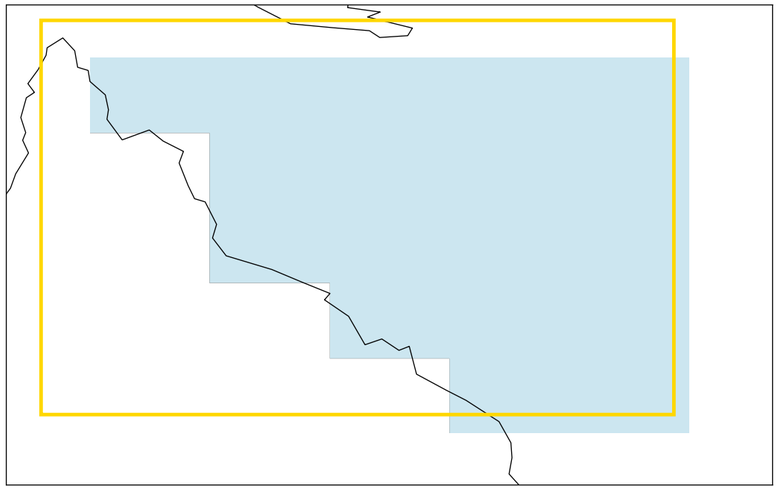




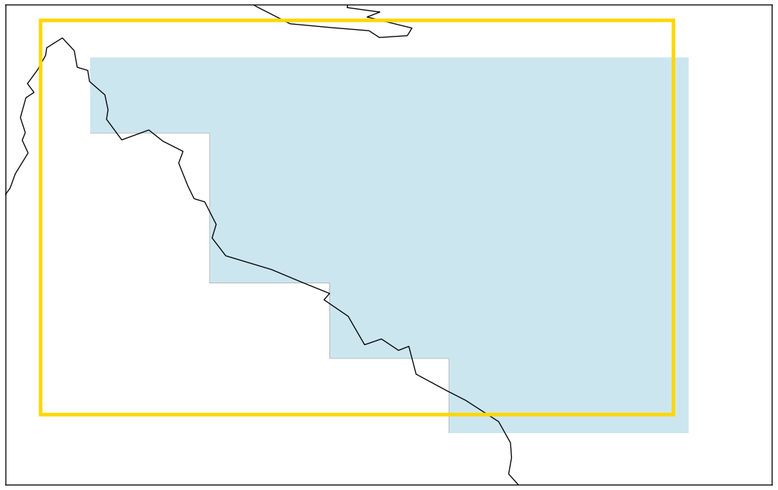


DHW Chain init: 2023-09-23 - Test - GBR (C-weeks)

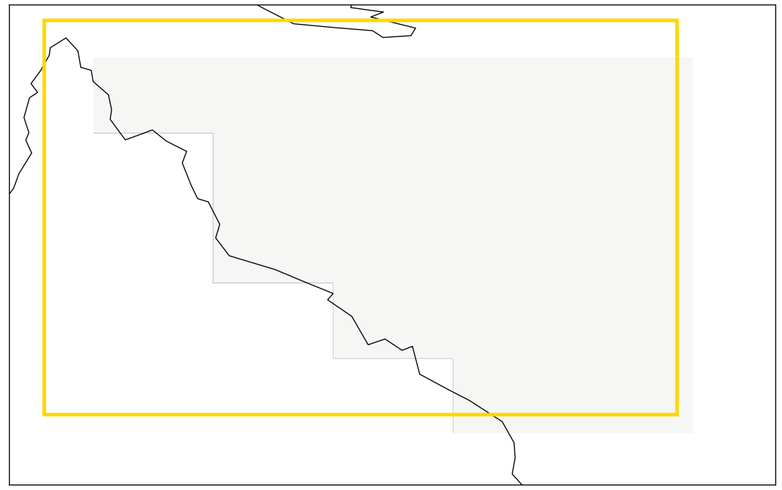
Actual DHW - Step 1



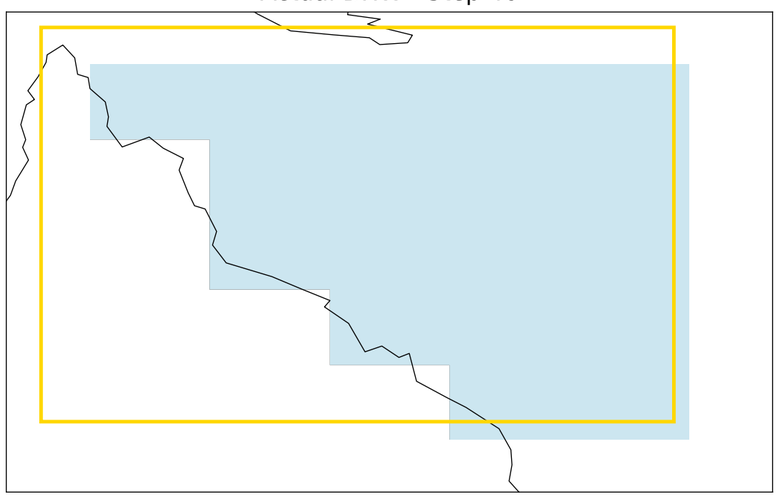
Predicted DHW - Step 1



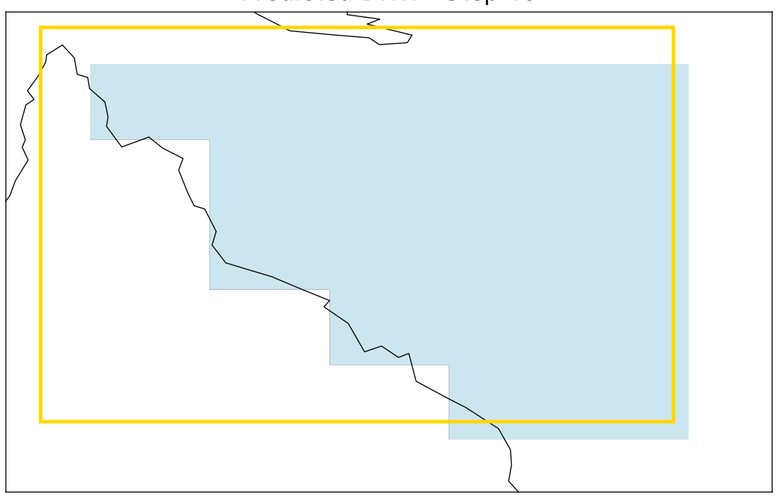
Error (Pred - Actual) - Step 1 [scale ±8]



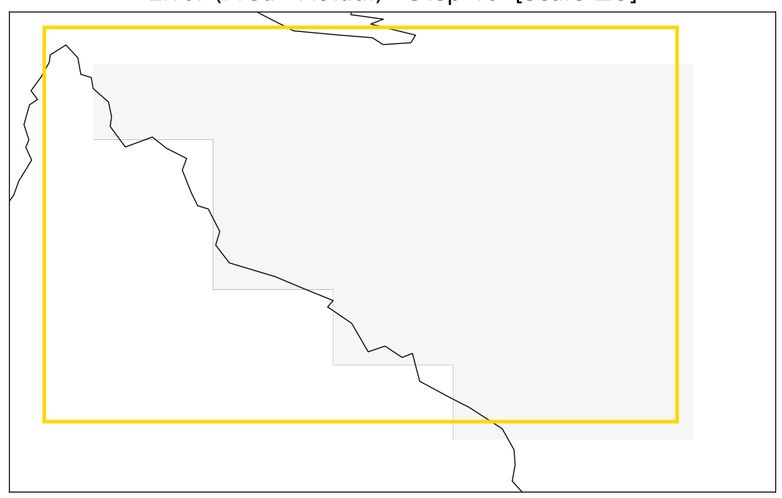
Actual DHW - Step 46



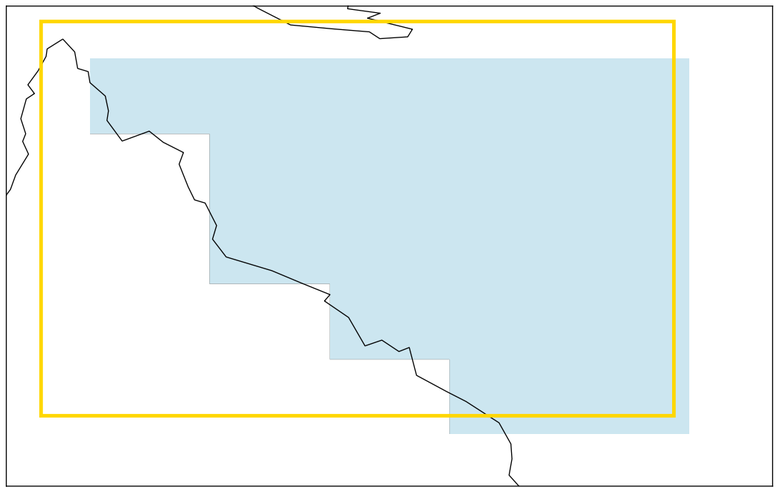
Predicted DHW - Step 46



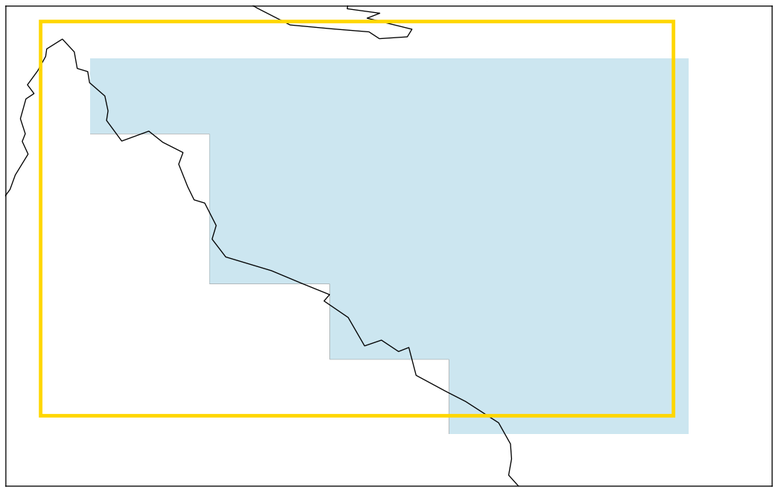
Error (Pred - Actual) - Step 46 [scale ±8]



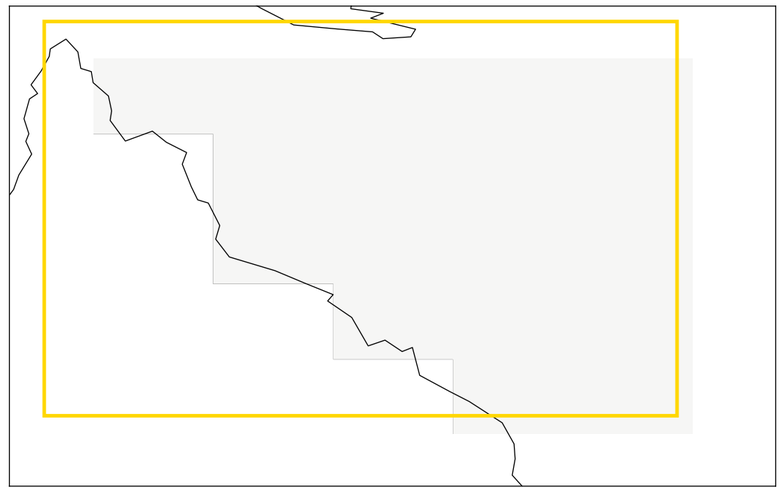
Actual DHW - Step 90

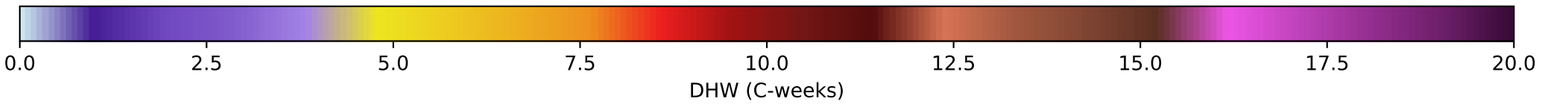


Predicted DHW - Step 90



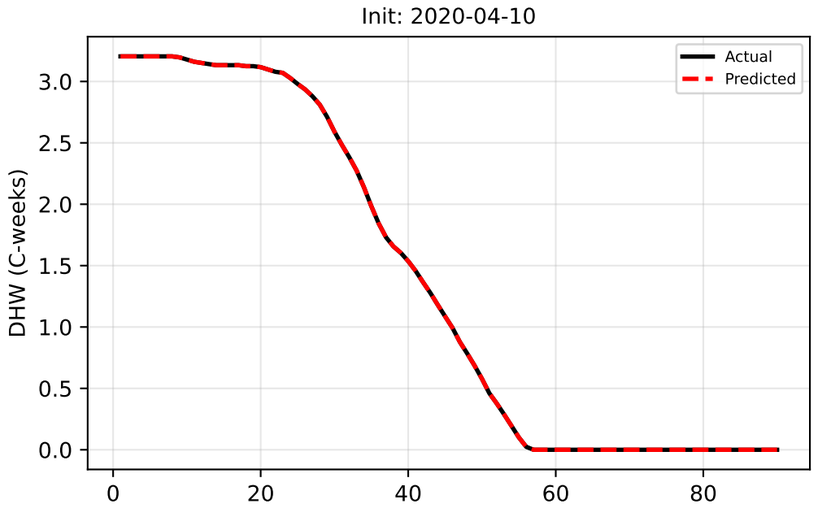
Error (Pred - Actual) - Step 90 [scale ±8]

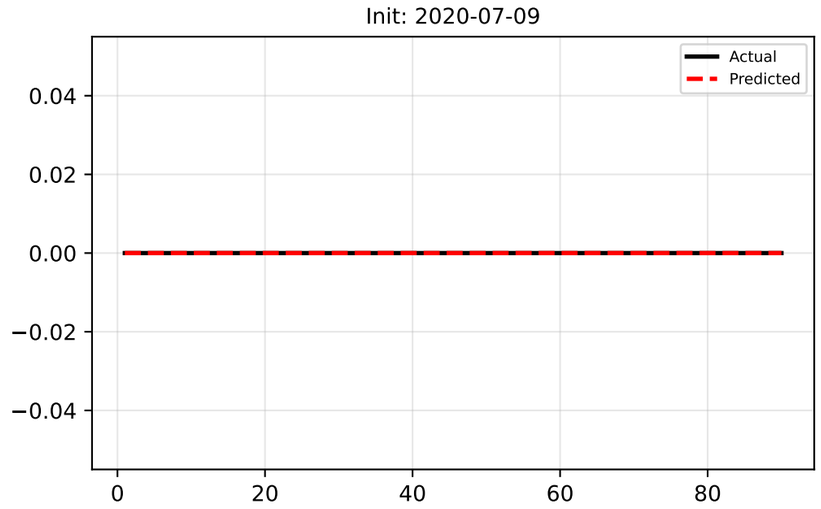


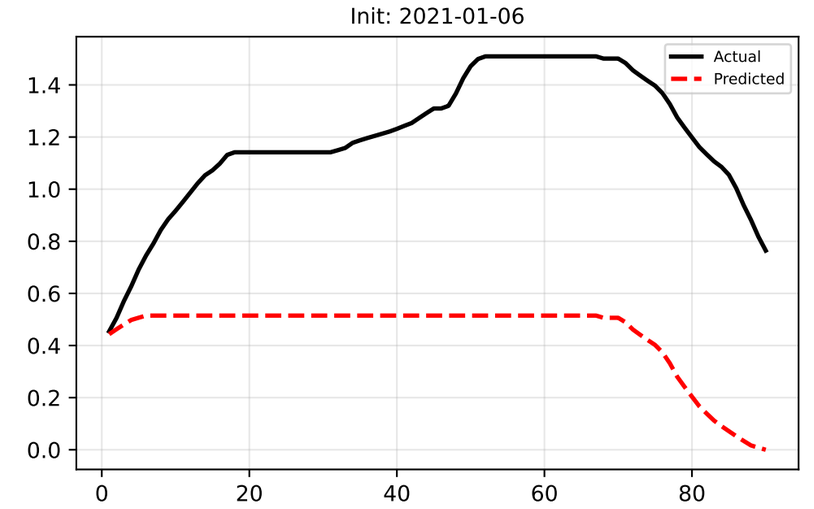


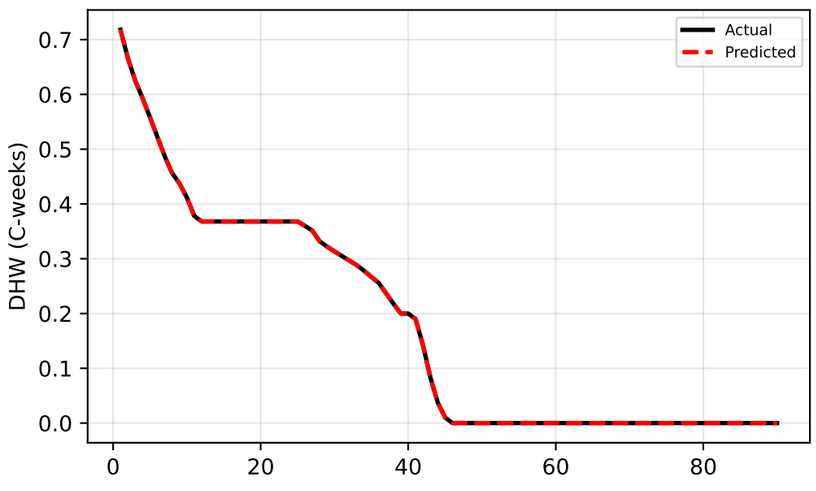


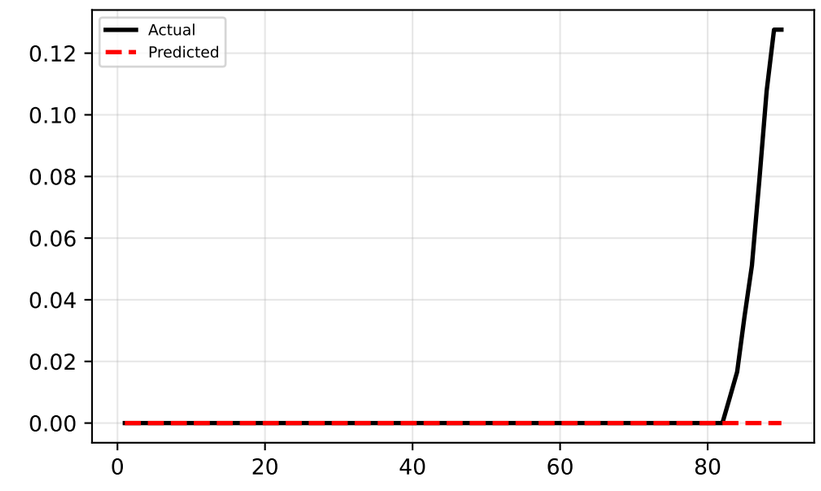
Global-Mean DHW - Test - GBR (C-weeks)

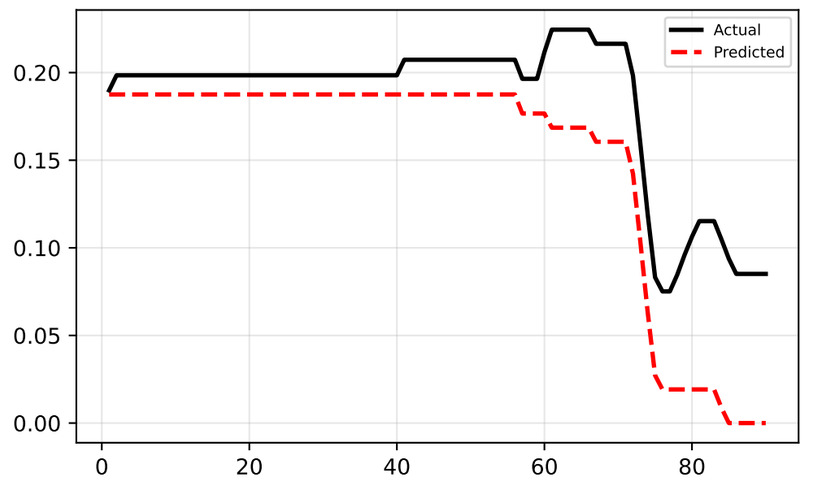


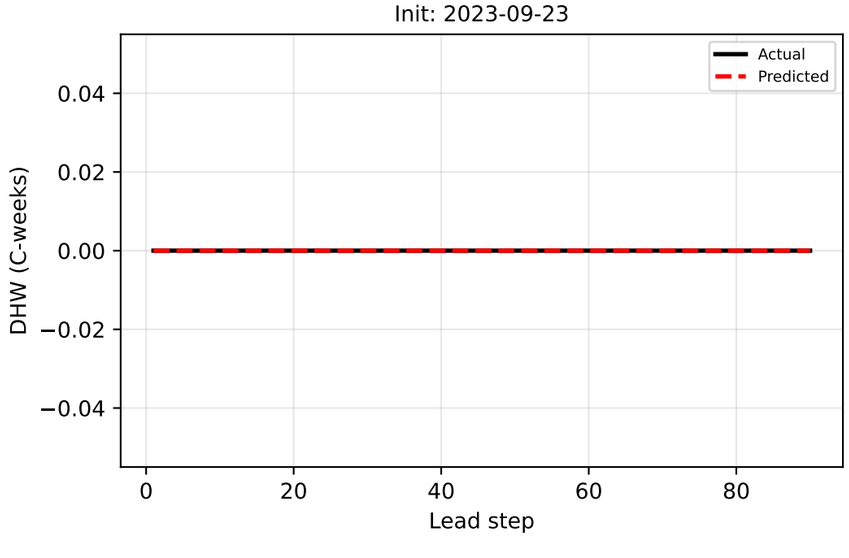


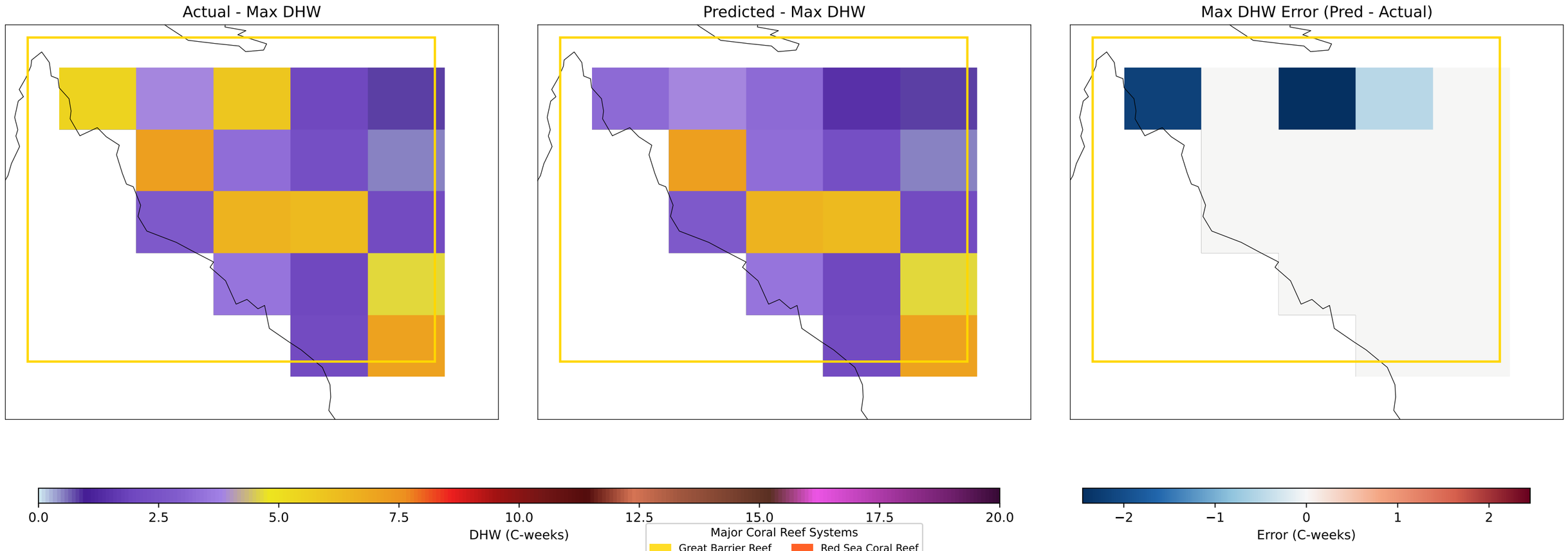


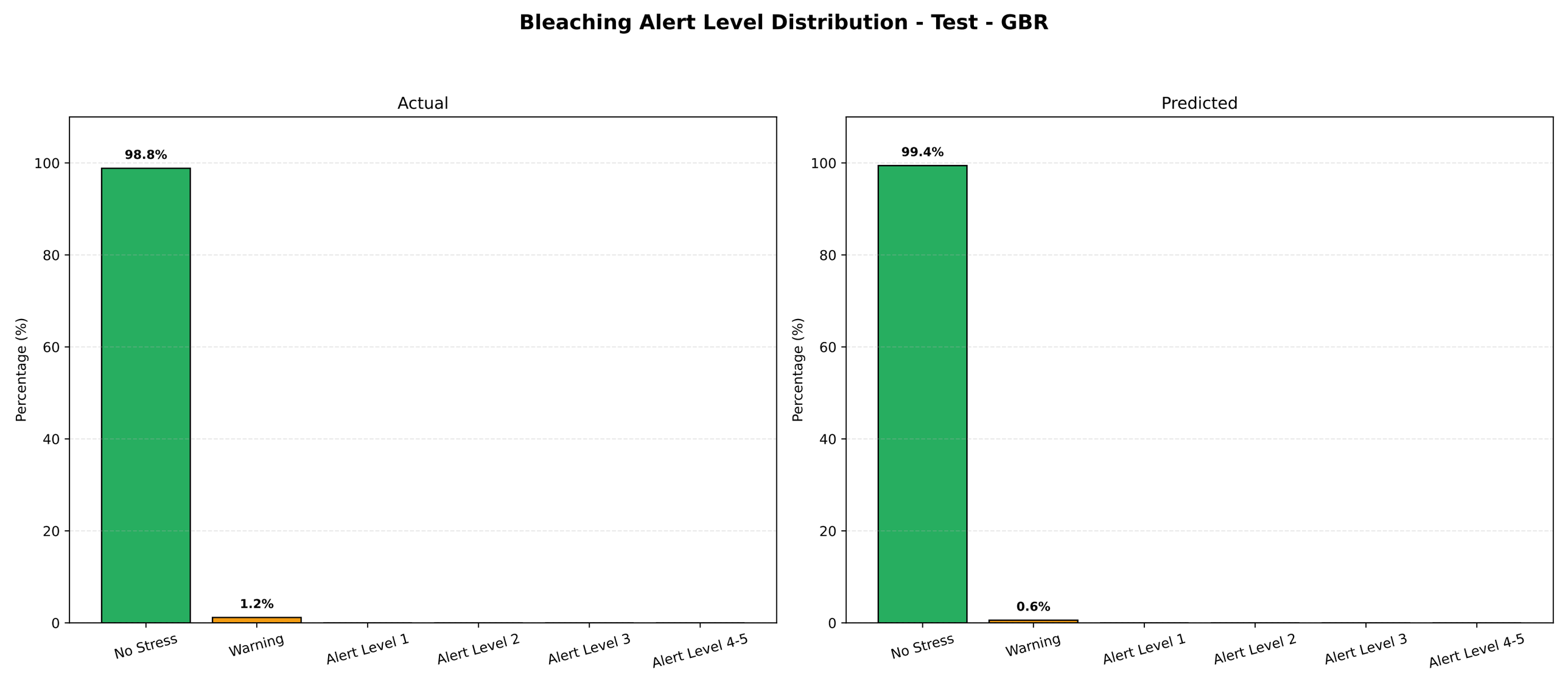






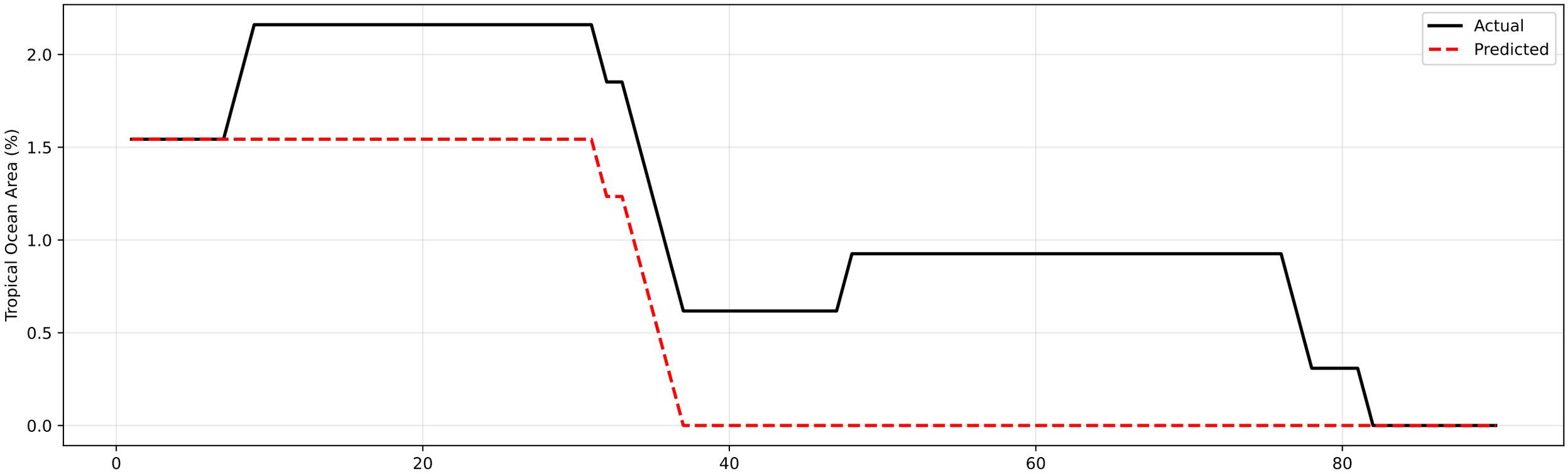



Peak DHW (max over all chains & leads) - Test - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−2
−1
0
1
2
Error (C-weeks)

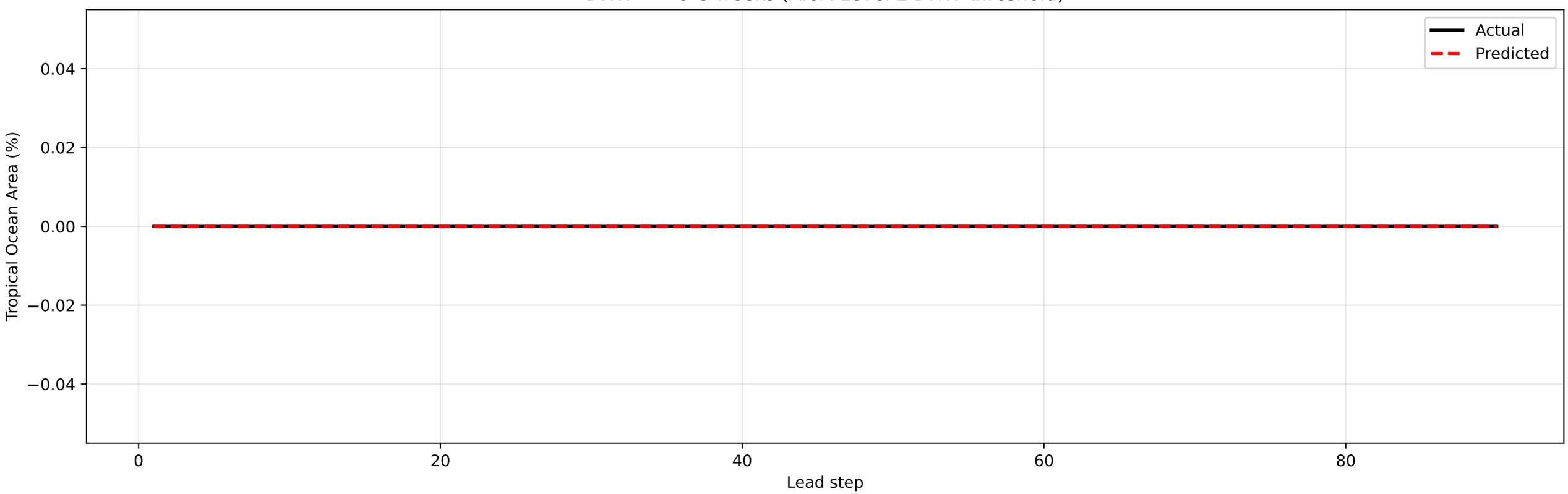

Bleaching Alert Level Distribution - Test - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
98.8%
1.2%
99.4%
0.6%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Test - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



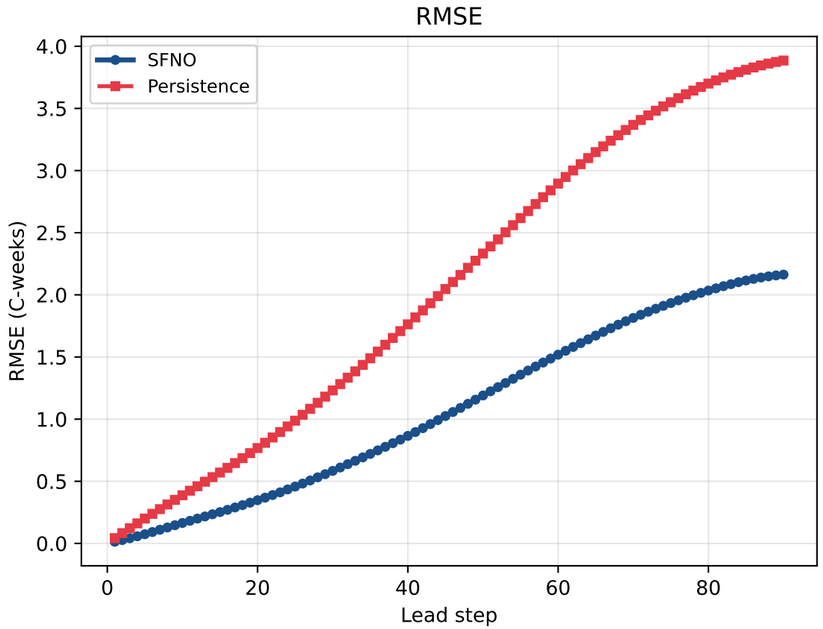
DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

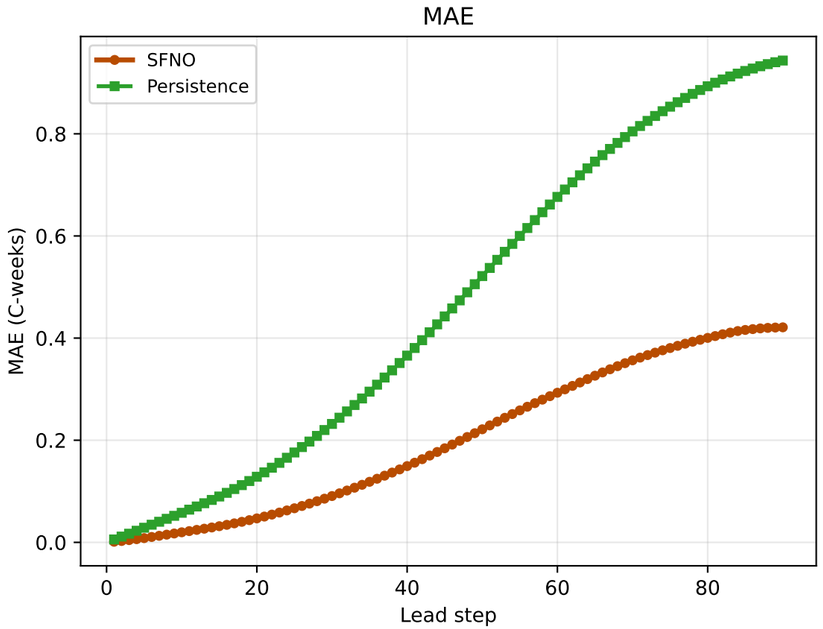


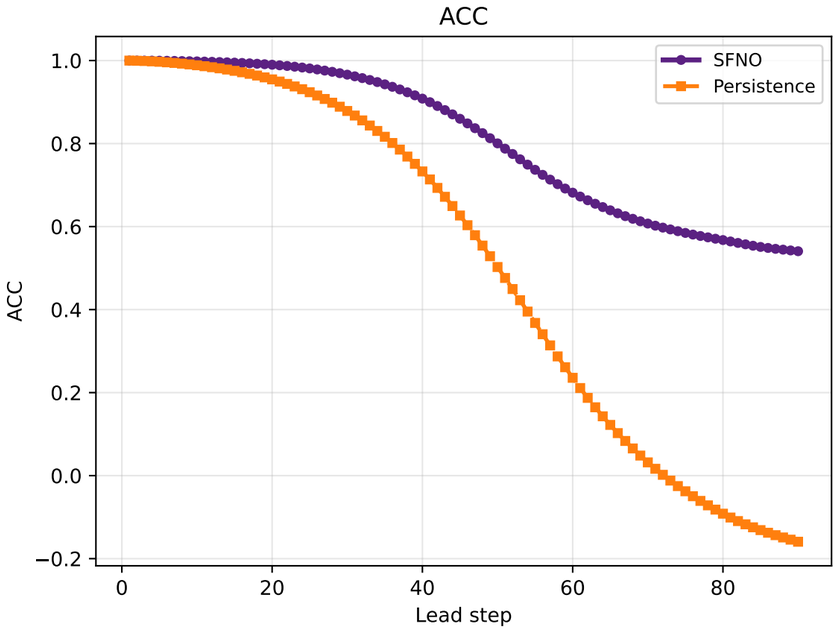
DHW Metrics by Lead Step - Val (15 chains scored at every lead)

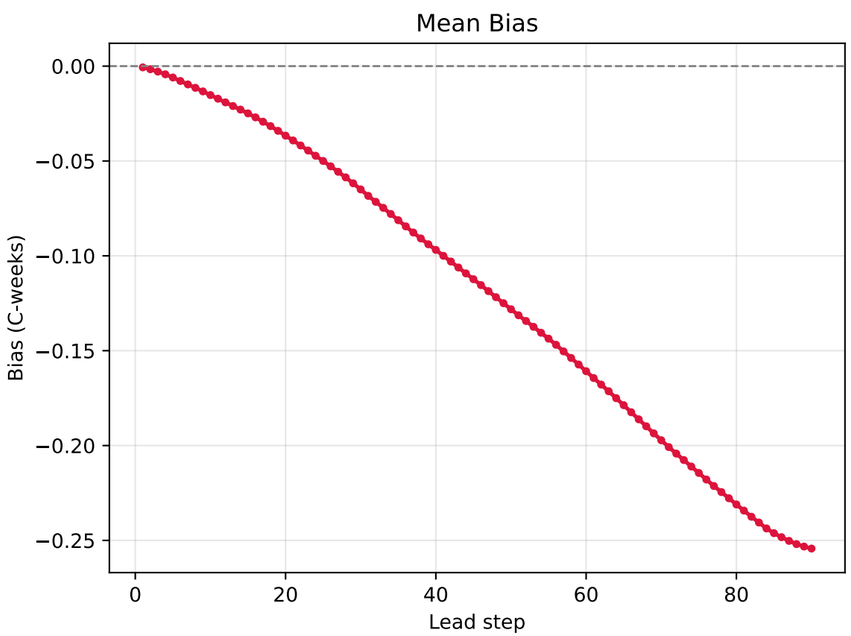
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0135	0.0425	68.1%	0.0012	0.0057	-0.0006	1.0000	0.9999	1.0000
2	0.0271	0.0829	67.3%	0.0027	0.0114	-0.0016	0.9999	0.9995	0.9999
3	0.0421	0.1222	65.5%	0.0044	0.0170	-0.0029	0.9999	0.9989	0.9998
4	0.0575	0.1613	64.3%	0.0063	0.0227	-0.0043	0.9998	0.9980	0.9996
5	0.0741	0.2000	62.9%	0.0083	0.0286	-0.0060	0.9996	0.9970	0.9993
6	0.0918	0.2381	61.5%	0.0106	0.0346	-0.0078	0.9994	0.9957	0.9990
7	0.1101	0.2758	60.1%	0.0129	0.0404	-0.0096	0.9991	0.9943	0.9985
8	0.1283	0.3133	59.0%	0.0151	0.0461	-0.0114	0.9988	0.9926	0.9980
9	0.1464	0.3504	58.2%	0.0174	0.0520	-0.0133	0.9984	0.9907	0.9973
10	0.1644	0.3874	57.6%	0.0197	0.0579	-0.0153	0.9980	0.9886	0.9966
11	0.1820	0.4241	57.1%	0.0221	0.0641	-0.0172	0.9975	0.9863	0.9958
12	0.1996	0.4601	56.6%	0.0244	0.0702	-0.0191	0.9970	0.9839	0.9950
13	0.2170	0.4965	56.3%	0.0268	0.0766	-0.0210	0.9964	0.9812	0.9940
14	0.2346	0.5332	56.0%	0.0292	0.0831	-0.0229	0.9957	0.9783	0.9930
15	0.2522	0.5701	55.8%	0.0317	0.0898	-0.0249	0.9950	0.9752	0.9918
16	0.2704	0.6080	55.5%	0.0343	0.0970	-0.0270	0.9942	0.9717	0.9905
17	0.2893	0.6467	55.3%	0.0372	0.1044	-0.0293	0.9933	0.9680	0.9890
18	0.3081	0.6861	55.1%	0.0402	0.1121	-0.0316	0.9923	0.9639	0.9874
19	0.3279	0.7264	54.9%	0.0435	0.1201	-0.0341	0.9911	0.9594	0.9856
20	0.3479	0.7676	54.7%	0.0469	0.1285	-0.0366	0.9899	0.9546	0.9837
21	0.3683	0.8096	54.5%	0.0505	0.1371	-0.0392	0.9885	0.9494	0.9815
22	0.3893	0.8524	54.3%	0.0544	0.1462	-0.0418	0.9869	0.9437	0.9791
23	0.4111	0.8961	54.1%	0.0584	0.1557	-0.0445	0.9852	0.9376	0.9763
24	0.4342	0.9412	53.9%	0.0625	0.1657	-0.0472	0.9832	0.9309	0.9732
25	0.4575	0.9875	53.7%	0.0668	0.1760	-0.0500	0.9810	0.9237	0.9699
26	0.4815	1.0350	53.5%	0.0712	0.1866	-0.0528	0.9785	0.9158	0.9662
27	0.5065	1.0830	53.2%	0.0759	0.1974	-0.0557	0.9758	0.9074	0.9621
28	0.5317	1.1316	53.0%	0.0806	0.2084	-0.0586	0.9728	0.8984	0.9576
29	0.5573	1.1811	52.8%	0.0855	0.2200	-0.0618	0.9696	0.8888	0.9527
30	0.5836	1.2311	52.6%	0.0907	0.2319	-0.0650	0.9660	0.8785	0.9474
31	0.6108	1.2818	52.4%	0.0961	0.2441	-0.0684	0.9620	0.8675	0.9416
32	0.6379	1.3332	52.2%	0.1015	0.2564	-0.0715	0.9577	0.8558	0.9354
33	0.6647	1.3846	52.0%	0.1070	0.2690	-0.0747	0.9531	0.8434	0.9289
34	0.6921	1.4364	51.8%	0.1127	0.2818	-0.0779	0.9481	0.8304	0.9218
35	0.7203	1.4894	51.6%	0.1186	0.2951	-0.0812	0.9427	0.8164	0.9140
36	0.7487	1.5434	51.5%	0.1246	0.3088	-0.0845	0.9367	0.8014	0.9058
37	0.7773	1.5981	51.4%	0.1306	0.3227	-0.0877	0.9304	0.7854	0.8970
38	0.8063	1.6531	51.2%	0.1367	0.3368	-0.0908	0.9235	0.7687	0.8876
39	0.8354	1.7081	51.1%	0.1429	0.3512	-0.0939	0.9162	0.7511	0.8778
40	0.8652	1.7629	50.9%	0.1493	0.3657	-0.0969	0.9083	0.7329	0.8672
41	0.8960	1.8185	50.7%	0.1560	0.3806	-0.1000	0.8997	0.7136	0.8559
42	0.9276	1.8750	50.5%	0.1629	0.3959	-0.1030	0.8905	0.6932	0.8439
43	0.9602	1.9323	50.3%	0.1699	0.4113	-0.1061	0.8806	0.6717	0.8312
44	0.9926	1.9898	50.1%	0.1770	0.4269	-0.1093	0.8704	0.6495	0.8181
45	1.0251	2.0466	49.9%	0.1842	0.4424	-0.1123	0.8597	0.6267	0.8046
46	1.0575	2.1032	49.7%	0.1915	0.4581	-0.1154	0.8486	0.6033	0.7907
47	1.0903	2.1599	49.5%	0.1989	0.4738	-0.1186	0.8370	0.5792	0.7764
48	1.1233	2.2173	49.3%	0.2063	0.4897	-0.1218	0.8251	0.5542	0.7617
49	1.1567	2.2749	49.2%	0.2138	0.5055	-0.1250	0.8129	0.5286	0.7467
50	1.1904	2.3325	49.0%	0.2214	0.5215	-0.1281	0.8005	0.5026	0.7314
51	1.2240	2.3896	48.8%	0.2290	0.5374	-0.1313	0.7877	0.4762	0.7158
52	1.2574	2.4464	48.6%	0.2365	0.5532	-0.1343	0.7749	0.4495	0.7002
53	1.2908	2.5032	48.4%	0.2439	0.5689	-0.1374	0.7621	0.4226	0.6847
54	1.3244	2.5604	48.3%	0.2512	0.5845	-0.1405	0.7492	0.3952	0.6691
55	1.3578	2.6175	48.1%	0.2585	0.6001	-0.1437	0.7368	0.3679	0.6540
56	1.3908	2.6741	48.0%	0.2656	0.6155	-0.1469	0.7246	0.3407	0.6394
57	1.4234	2.7302	47.9%	0.2726	0.6310	-0.1504	0.7131	0.3138	0.6254
58	1.4554	2.7859	47.8%	0.2795	0.6464	-0.1538	0.7022	0.2872	0.6121
59	1.4871	2.8409	47.7%	0.2863	0.6616	-0.1573	0.6916	0.2611	0.5993
60	1.5187	2.8953	47.5%	0.2930	0.6764	-0.1608	0.6817	0.2357	0.5871
61	1.5500	2.9487	47.4%	0.2996	0.6909	-0.1644	0.6723	0.2110	0.5755
62	1.5810	3.0007	47.3%	0.3064	0.7050	-0.1679	0.6634	0.1873	0.5643
63	1.6117	3.0513	47.2%	0.3131	0.7188	-0.1714	0.6550	0.1647	0.5536
64	1.6421	3.1006	47.0%	0.3197	0.7323	-0.1750	0.6470	0.1429	0.5434
65	1.6723	3.1484	46.9%	0.3263	0.7456	-0.1787	0.6392	0.1222	0.5335
66	1.7019	3.1948	46.7%	0.3327	0.7582	-0.1824	0.6320	0.1024	0.5241
67	1.7312	3.2402	46.6%	0.3389	0.7704	-0.1862	0.6252	0.0835	0.5151
68	1.7599	3.2842	46.4%	0.3450	0.7822	-0.1899	0.6188	0.0655	0.5065
69	1.7876	3.3266	46.3%	0.3509	0.7936	-0.1936	0.6127	0.0483	0.4983
70	1.8142	3.3677	46.1%	0.3565	0.8047	-0.1972	0.6073	0.0319	0.4908
71	1.8397	3.4069	46.0%	0.3618	0.8153	-0.2008	0.6022	0.0165	0.4836
72	1.8644	3.4445	45.9%	0.3667	0.8253	-0.2042	0.5975	0.0020	0.4771
73	1.8884	3.4811	45.8%	0.3715	0.8349	-0.2076	0.5931	-0.0119	0.4710
74	1.9116	3.5164	45.6%	0.3761	0.8443	-0.2111	0.5889	-0.0252	0.4651
75	1.9341	3.5499	45.5%	0.3805	0.8532	-0.2144	0.5847	-0.0377	0.4594
76	1.9561	3.5823	45.4%	0.3848	0.8619	-0.2179	0.5808	-0.0496	0.4540
77	1.9770	3.6139	45.3%	0.3890	0.8703	-0.2213	0.5773	-0.0609	0.4493
78	1.9968	3.6437	45.2%	0.3929	0.8782	-0.2245	0.5740	-0.0716	0.4448
79	2.0160	3.6724	45.1%	0.3967	0.8858	-0.2278	0.5707	-0.0819	0.4403
80	2.0345	3.6998	45.0%	0.4004	0.8931	-0.2311	0.5675	-0.0917	0.4360
81	2.0523	3.7255	44.9%	0.4040	0.9000	-0.2343	0.5641	-0.1009	0.4316
82	2.0697	3.7494	44.8%	0.4075	0.9064	-0.2375	0.5606	-0.1095	0.4272
83	2.0861	3.7716	44.7%	0.4107	0.9123	-0.2406	0.5573	-0.1173	0.4230
84	2.1018	3.7922	44.6%	0.4138	0.9179	-0.2437	0.5539	-0.1245	0.4189
85	2.1158	3.8114	44.5%	0.4159	0.9231	-0.2462	0.5510	-0.1313	0.4155
86	2.1283	3.8295	44.4%	0.4176	0.9278	-0.2483	0.5485	-0.1376	0.4126
87	2.1395	3.8461	44.4%	0.4190	0.9323	-0.2502	0.5464	-0.1434	0.4102
88	2.1488	3.8609	44.3%	0.4200	0.9365	-0.2520	0.5444	-0.1489	0.4080
89	2.1565	3.8738	44.3%	0.4206	0.9402	-0.2532	0.5425	-0.1542	0.4058
90	2.1630	3.8852	44.3%	0.4211	0.9435	-0.2543	0.5406	-0.1593	0.4037

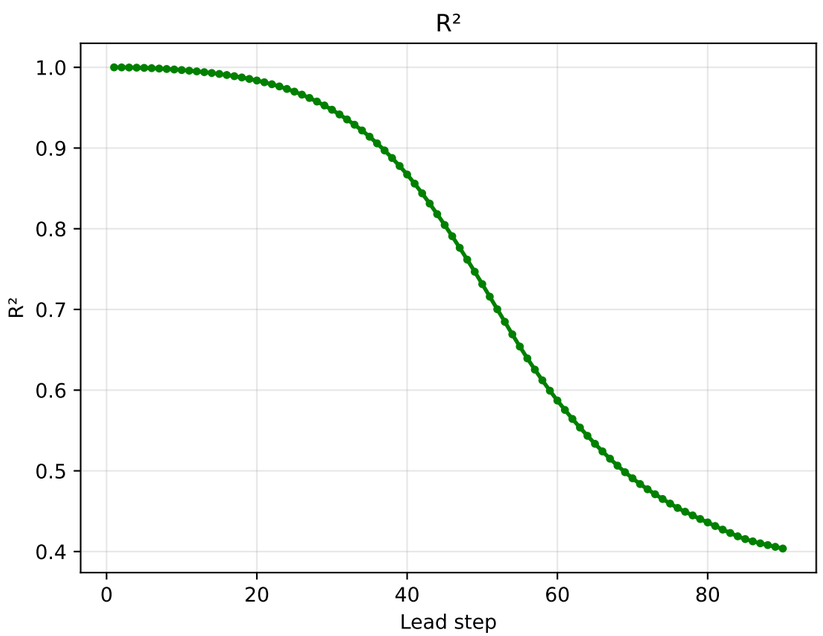
DHW Forecast Skill vs Lead Step - Val

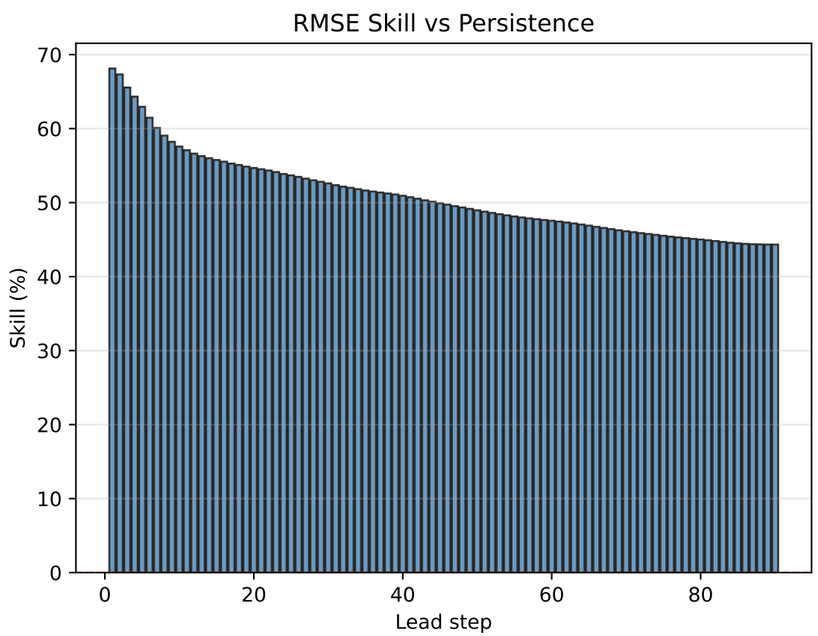


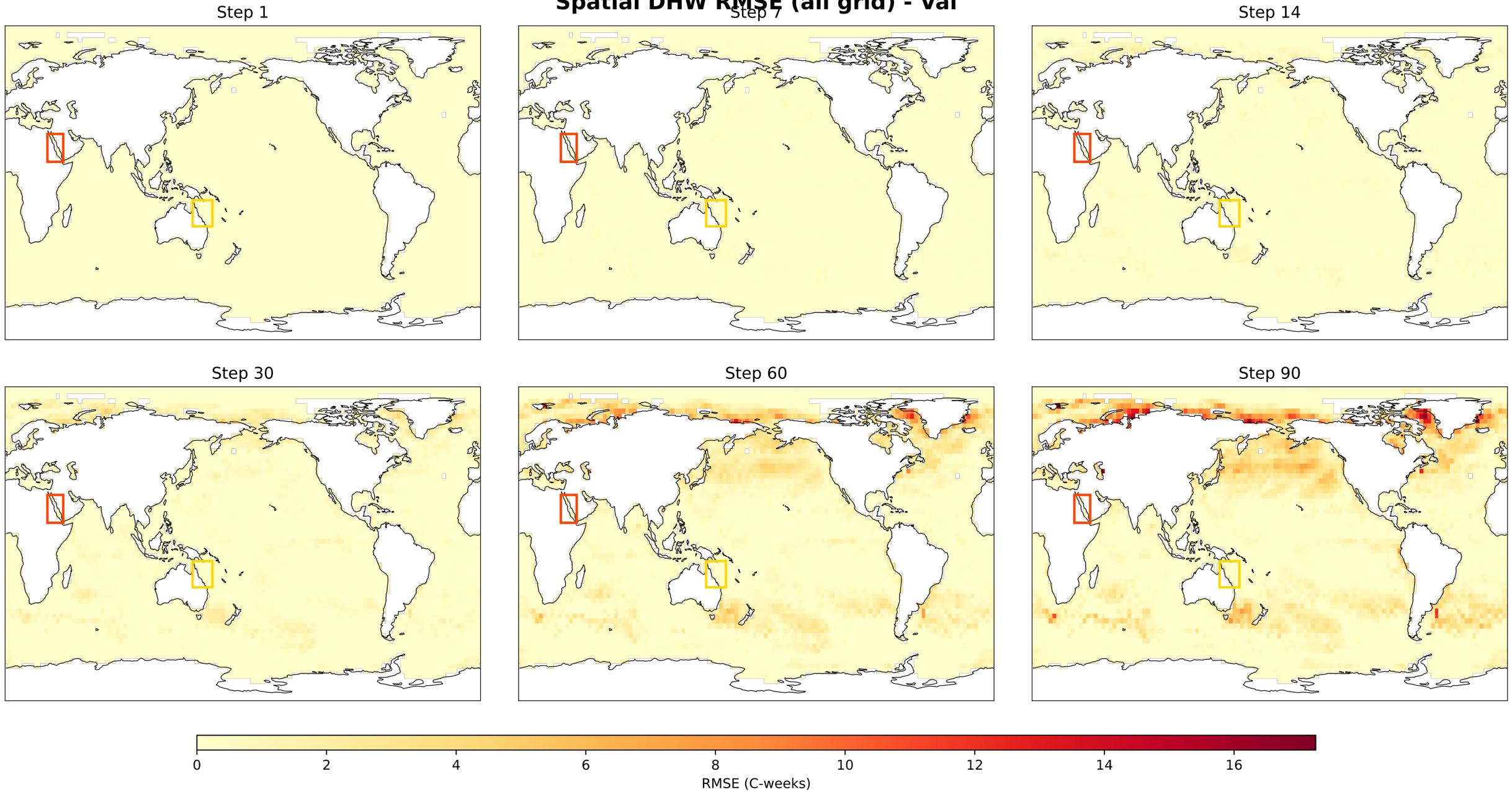


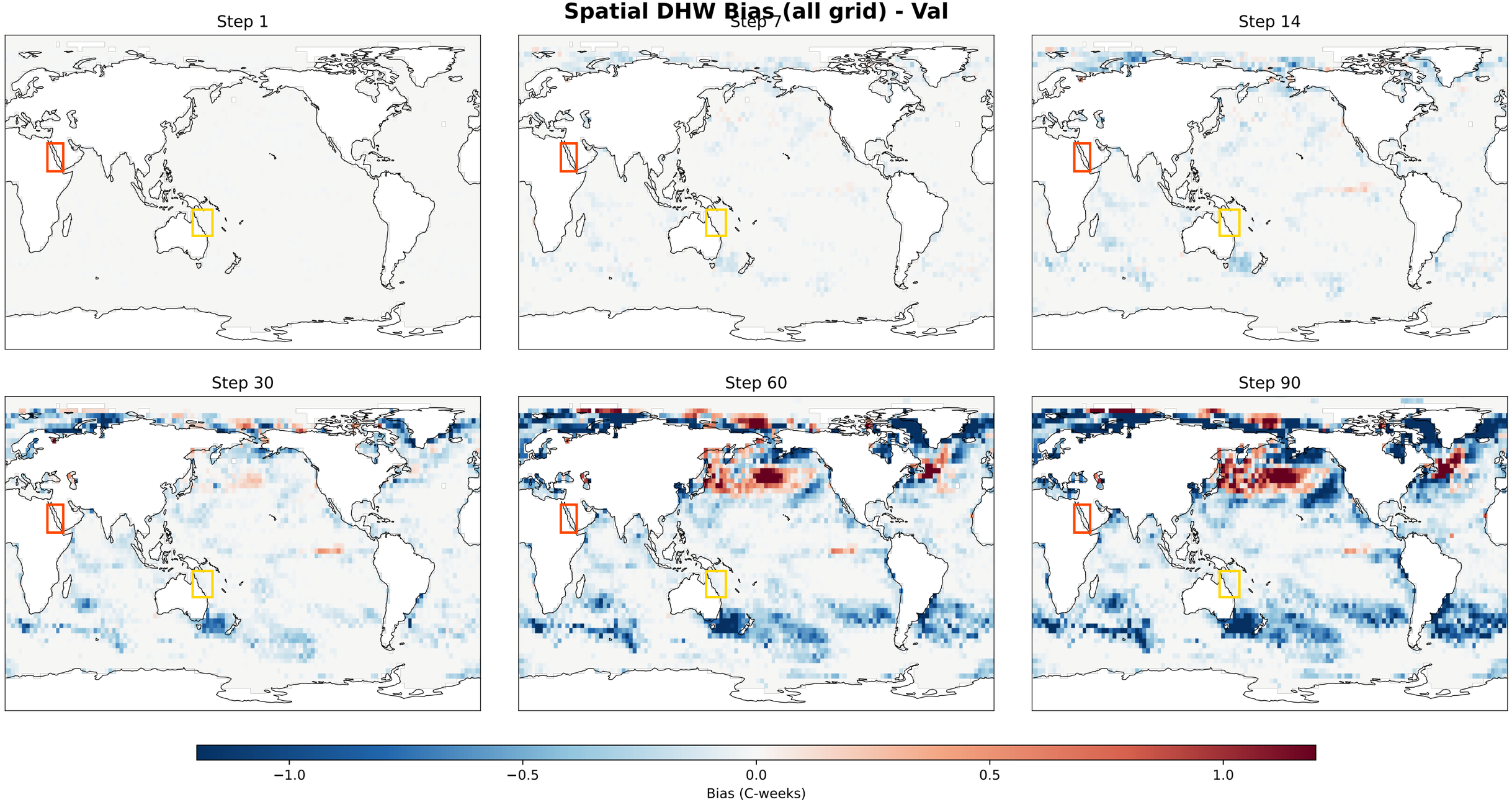


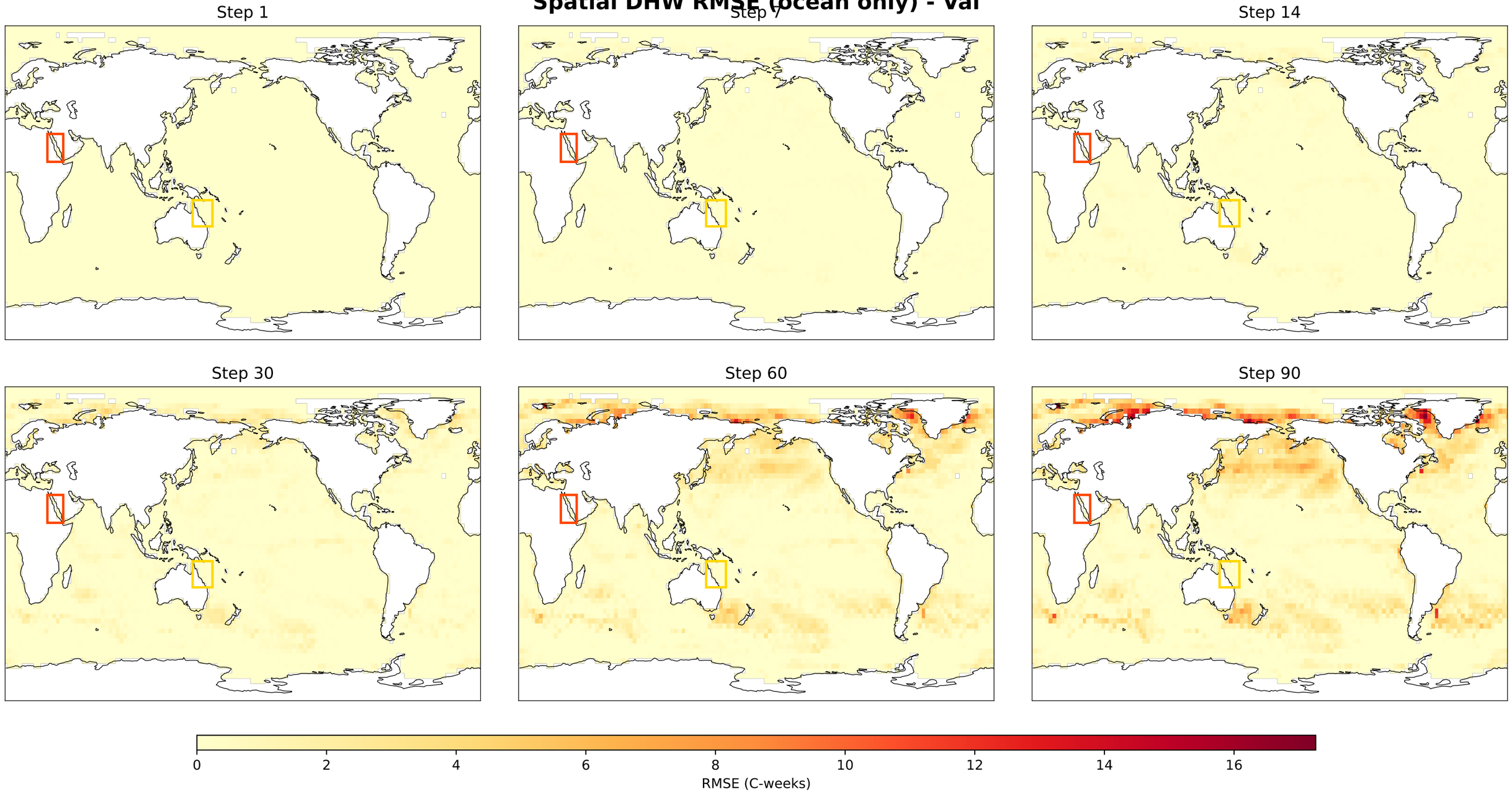


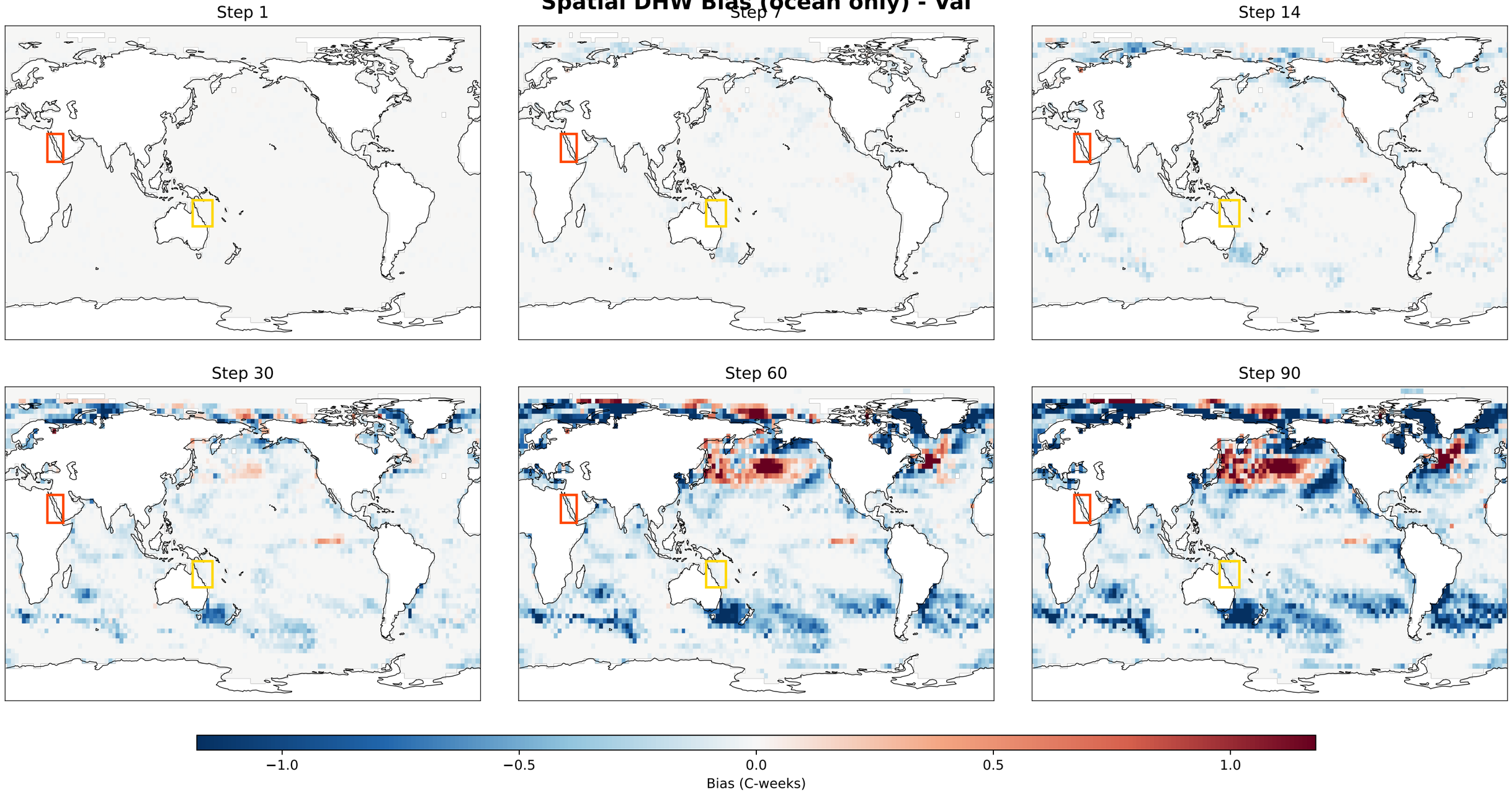





Spatial DHW RMSE (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

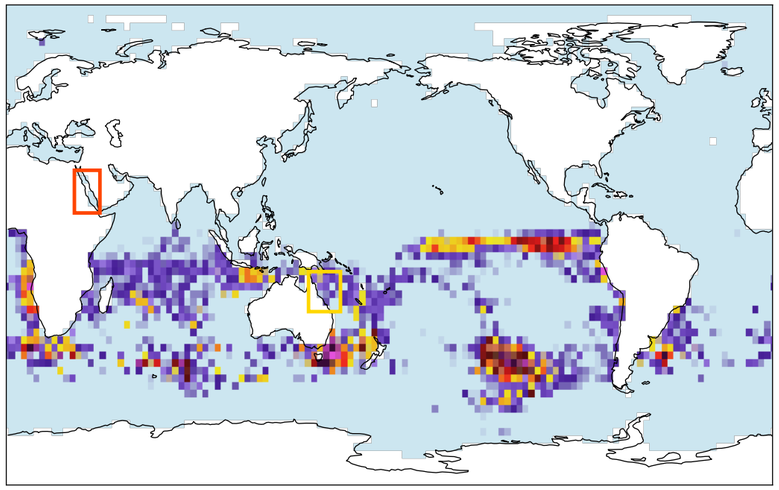

Spatial DHW Bias (all grid) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0
2
4
6
8
10
12
14
16
RMSE (C-weeks)

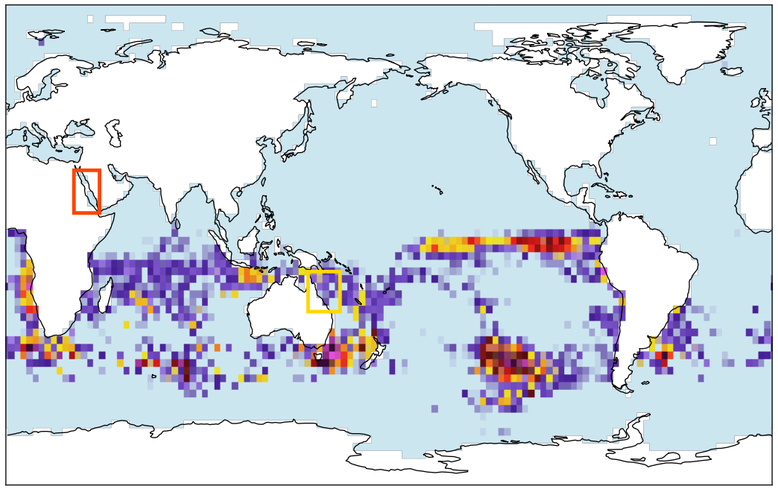

Spatial DHW Bias (ocean only) - Val
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−1.0
−0.5
0.0
0.5
1.0
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val (C-weeks)

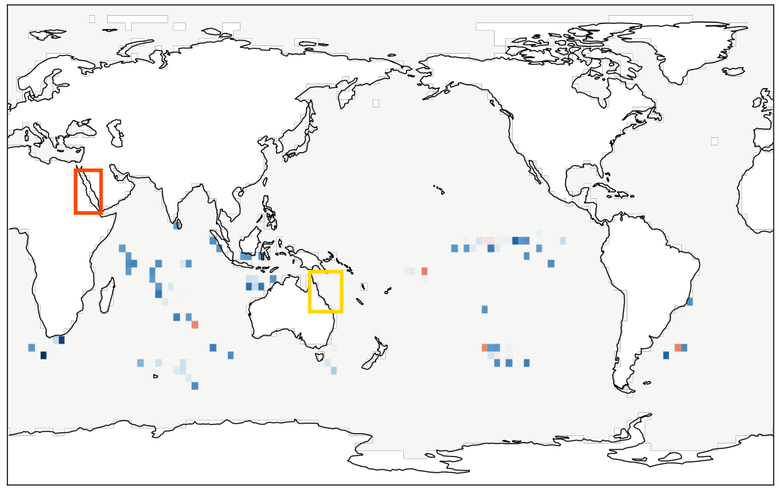
Actual DHW - Step 1



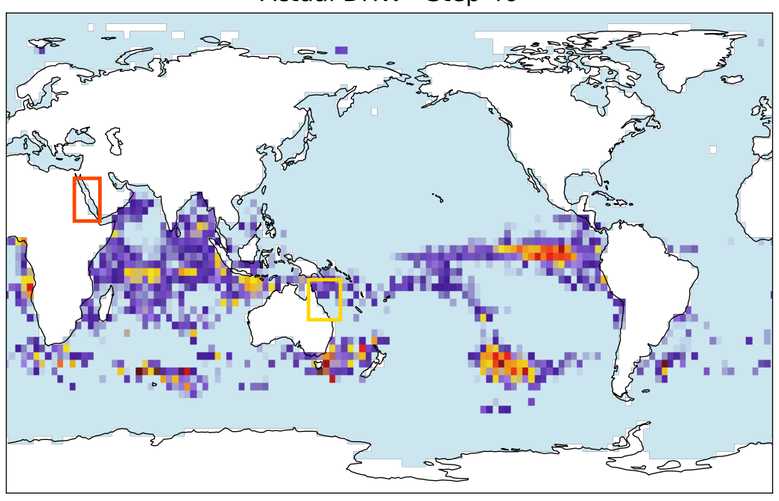
Predicted DHW - Step 1



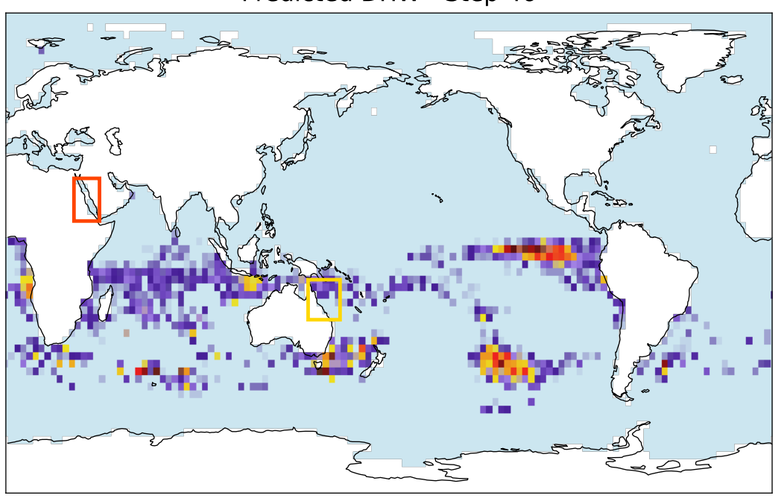
Error (Pred - Actual) - Step 1 [scale ±0.31]



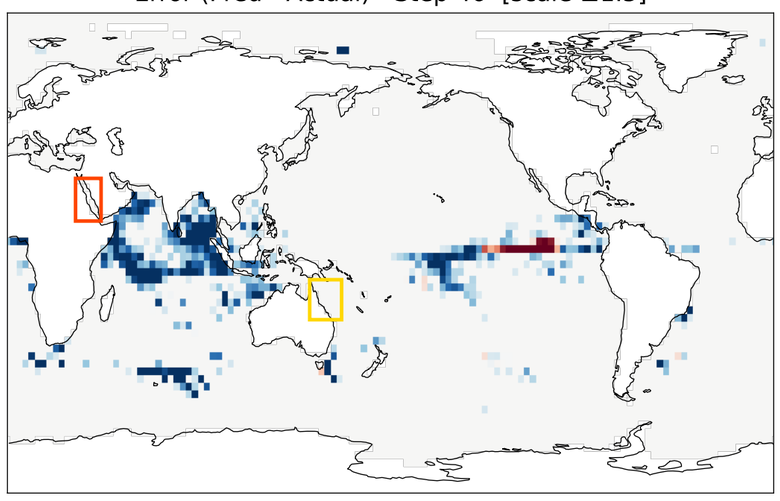
Actual DHW - Step 46



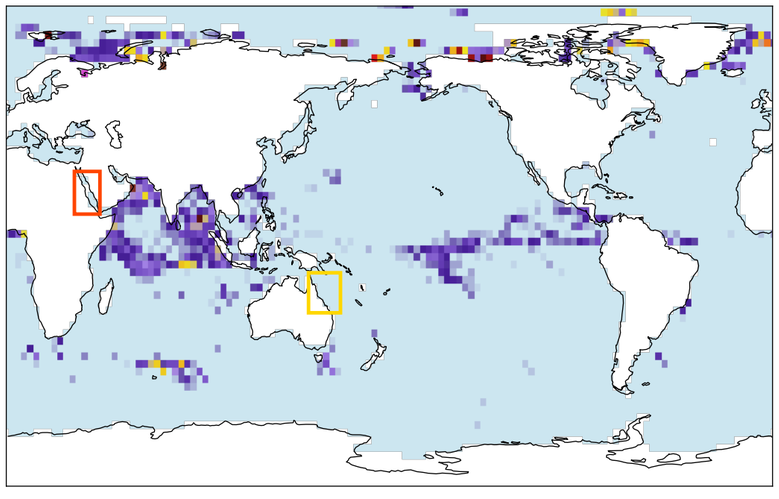
Predicted DHW - Step 46



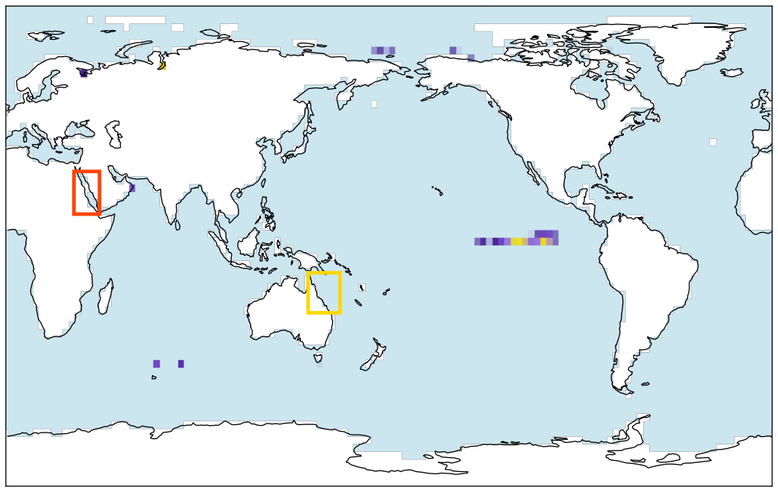
Error (Pred - Actual) - Step 46 [scale ±1.5]



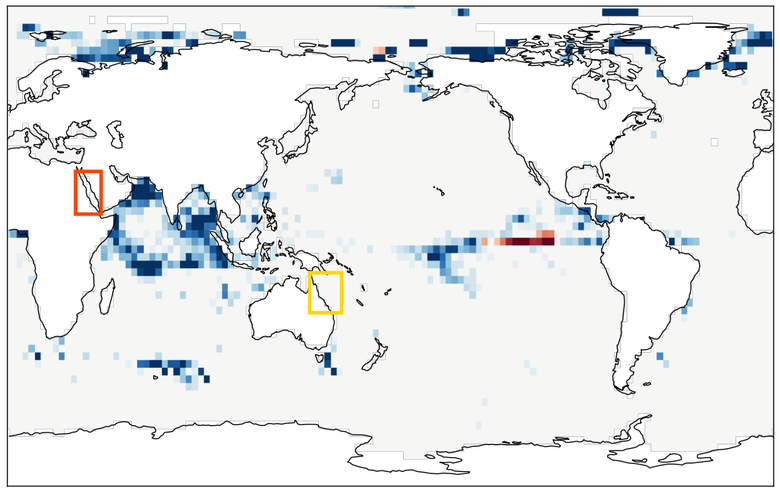
Actual DHW - Step 90

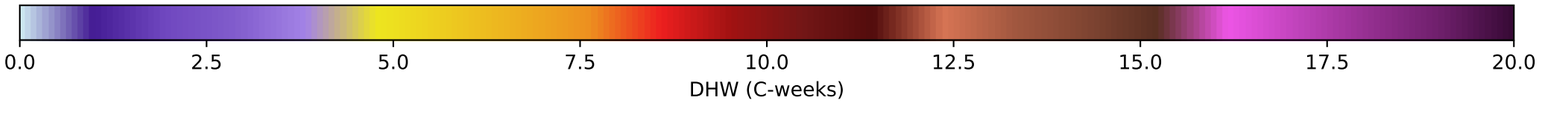


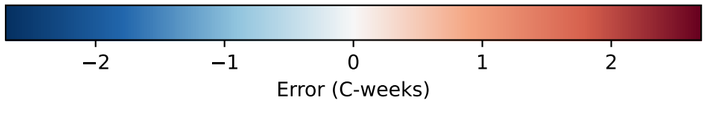
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.7]

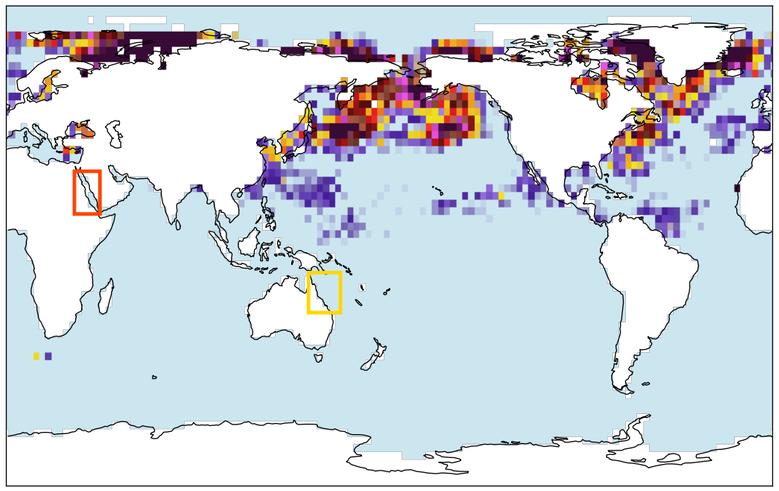




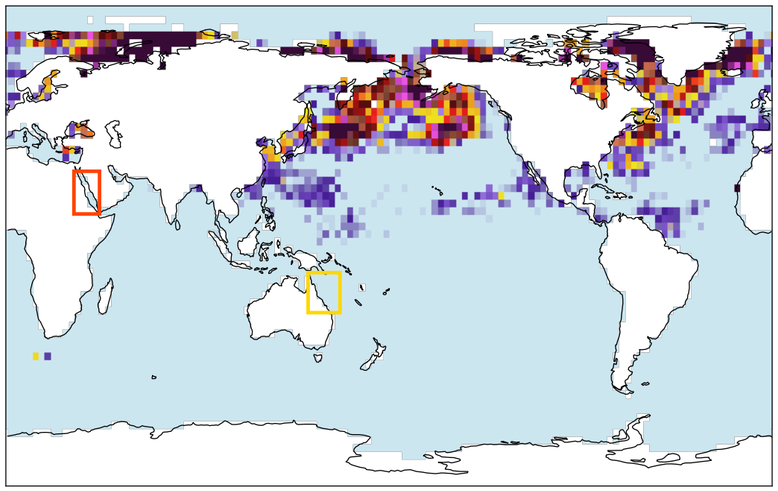


DHW Chain init: 2016-10-07 - Val (C-weeks)

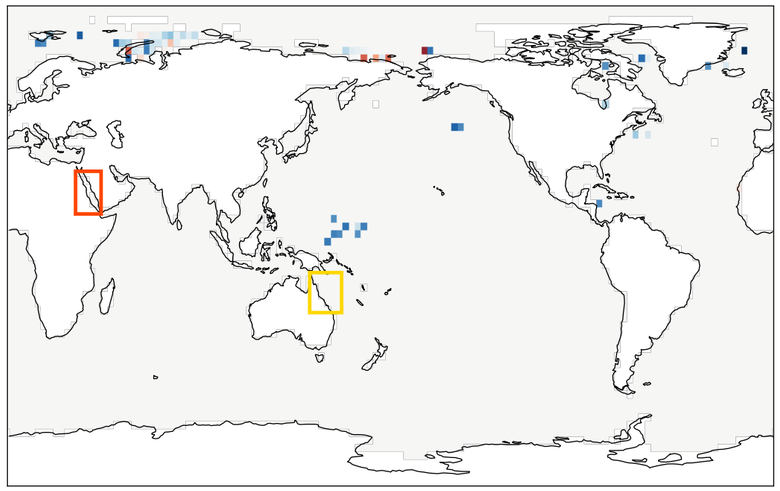
Actual DHW - Step 1



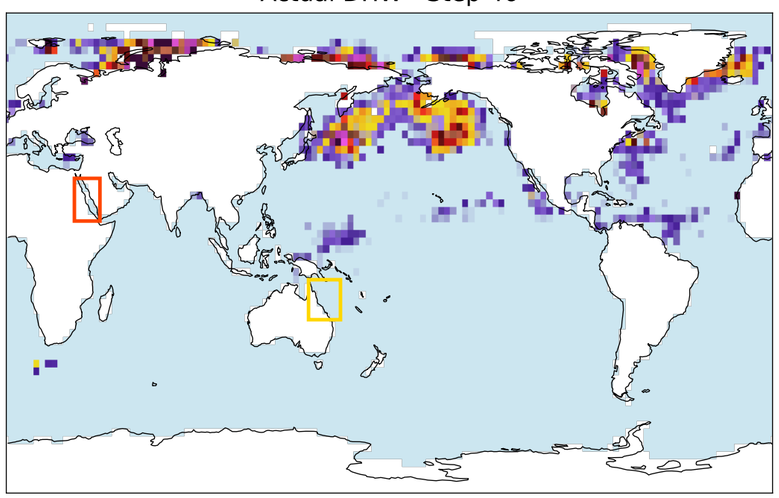
Predicted DHW - Step 1



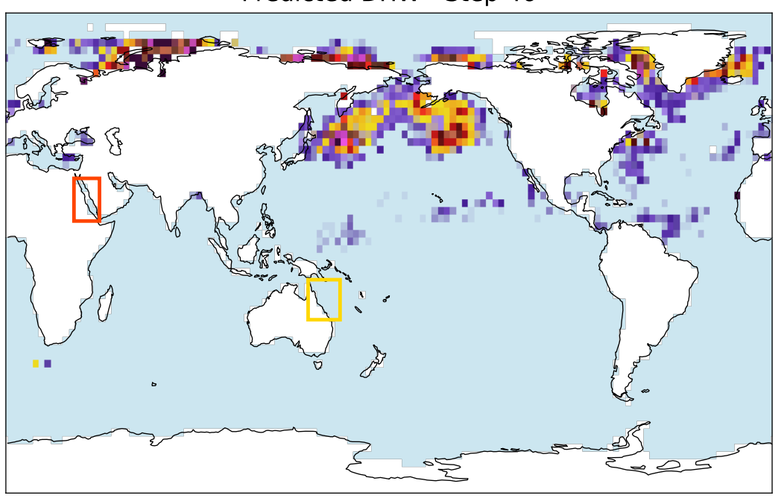
Error (Pred - Actual) - Step 1 [scale ±0.28]



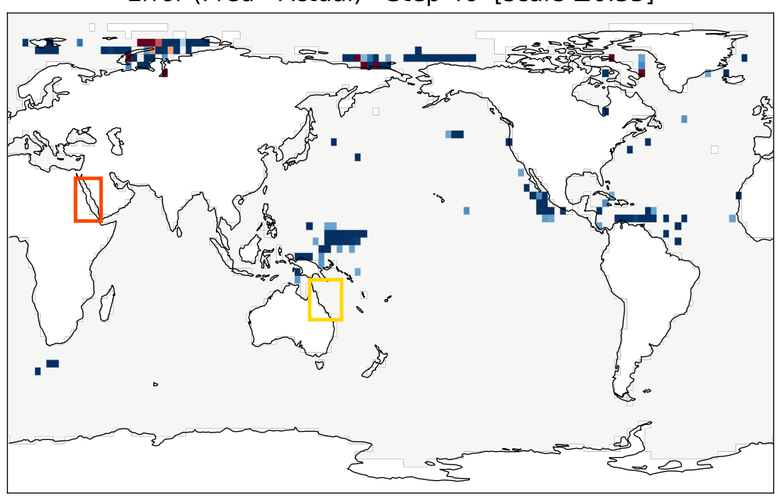
Actual DHW - Step 46



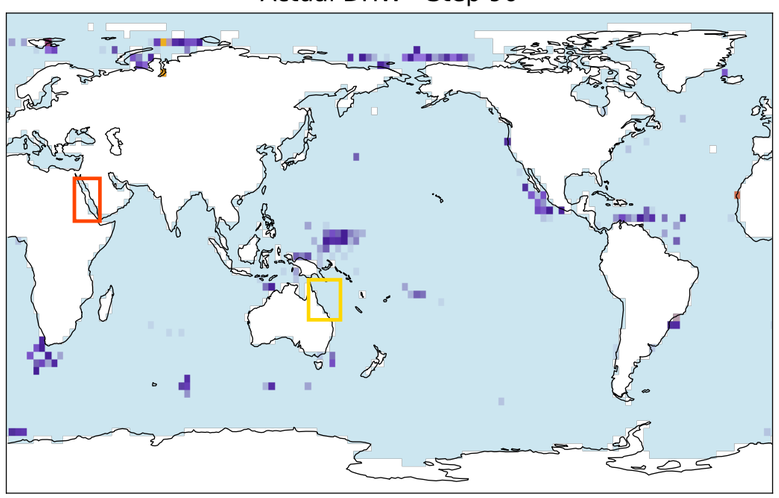
Predicted DHW - Step 46



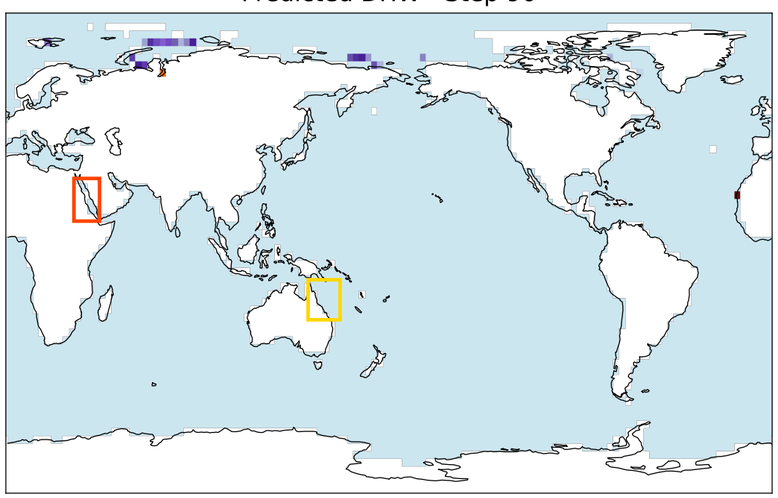
Error (Pred - Actual) - Step 46 [scale ±0.33]



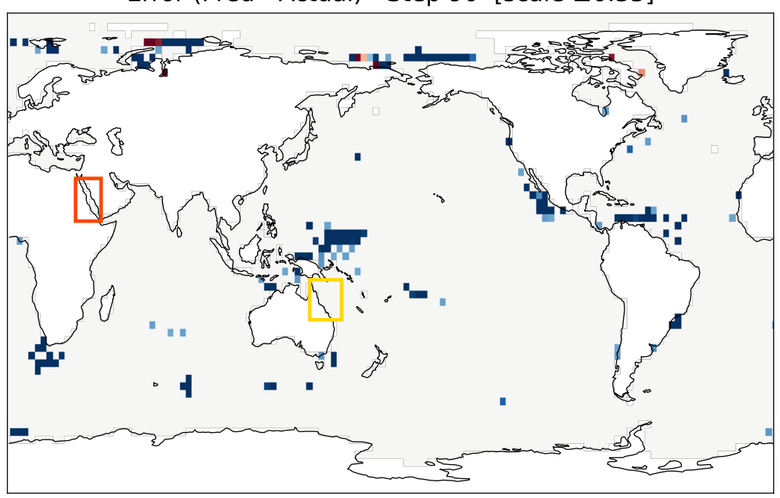
Actual DHW - Step 90

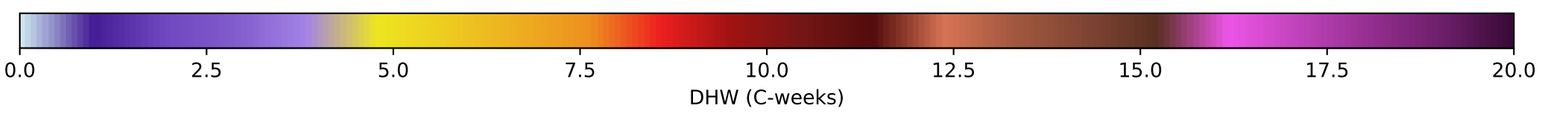


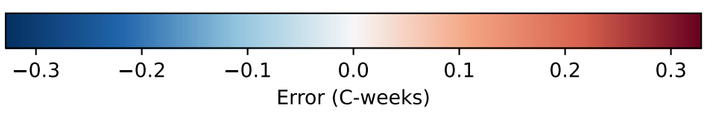
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.33]

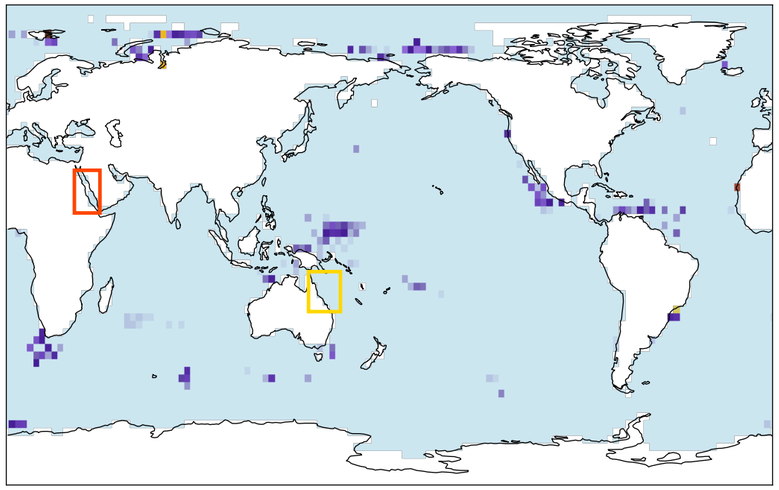




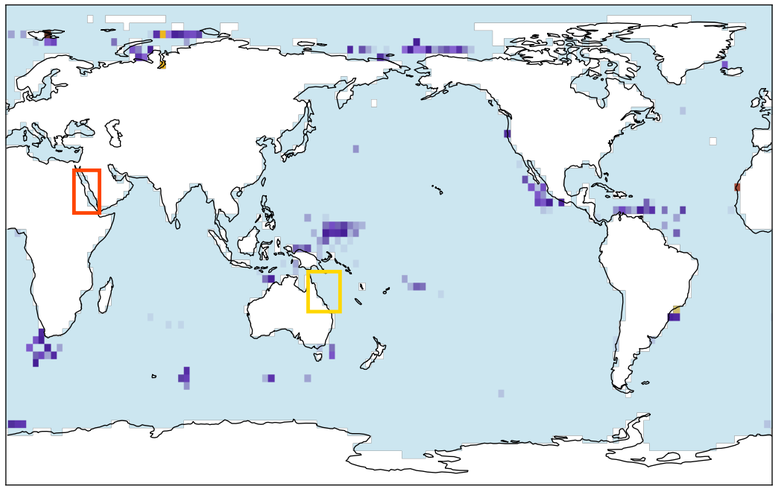


DHW Chain init: 2017-01-05 - Val (C-weeks)

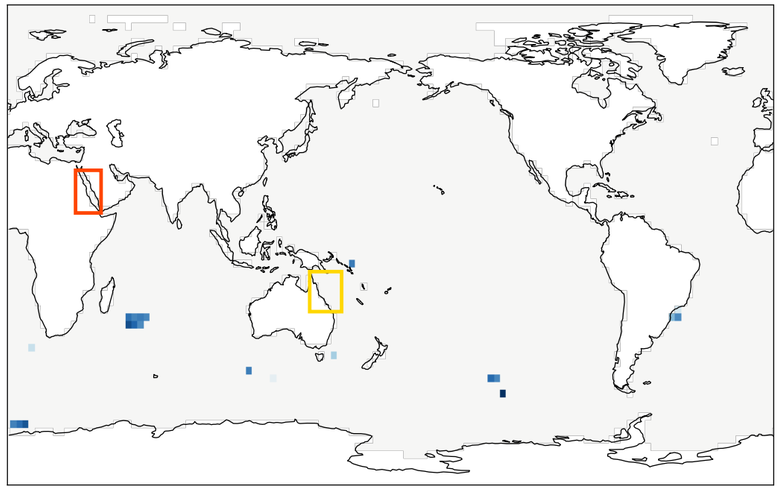
Actual DHW - Step 1



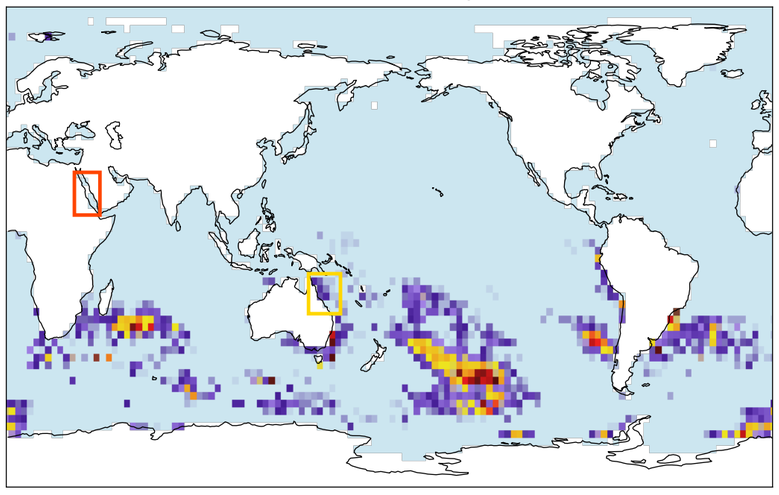
Predicted DHW - Step 1



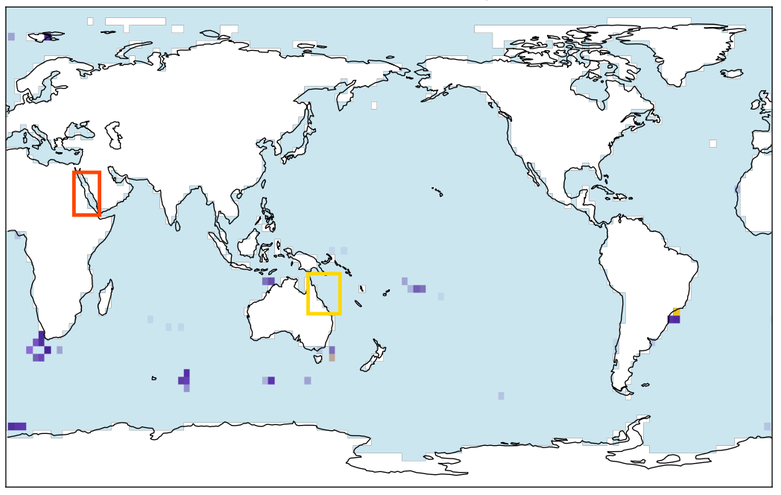
Error (Pred - Actual) - Step 1 [scale ±0.27]



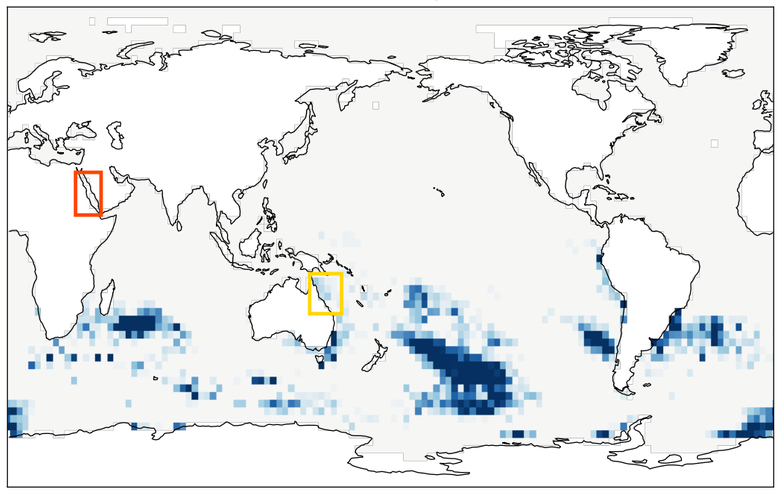
Actual DHW - Step 46



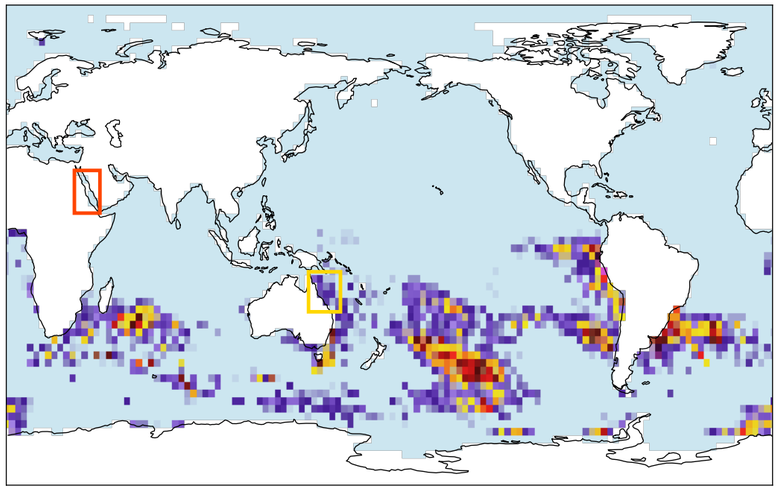
Predicted DHW - Step 46



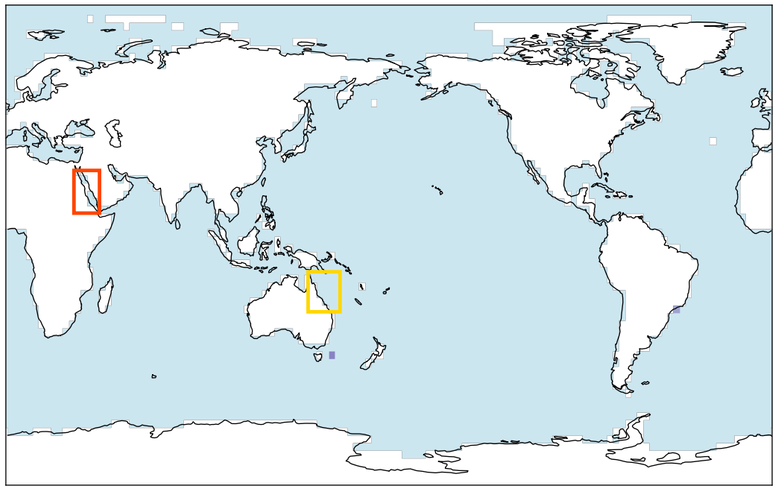
Error (Pred - Actual) - Step 46 [scale ±4.8]



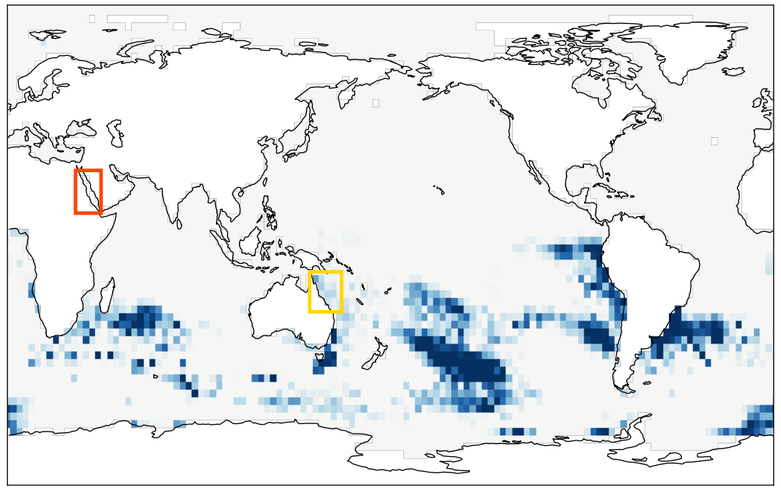
Actual DHW - Step 90

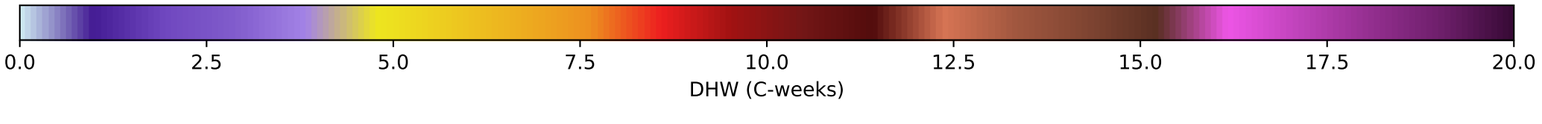


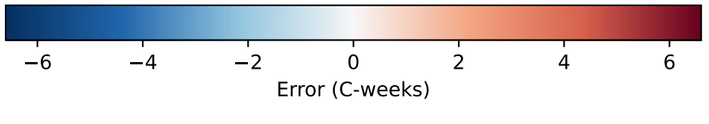
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±6.6]

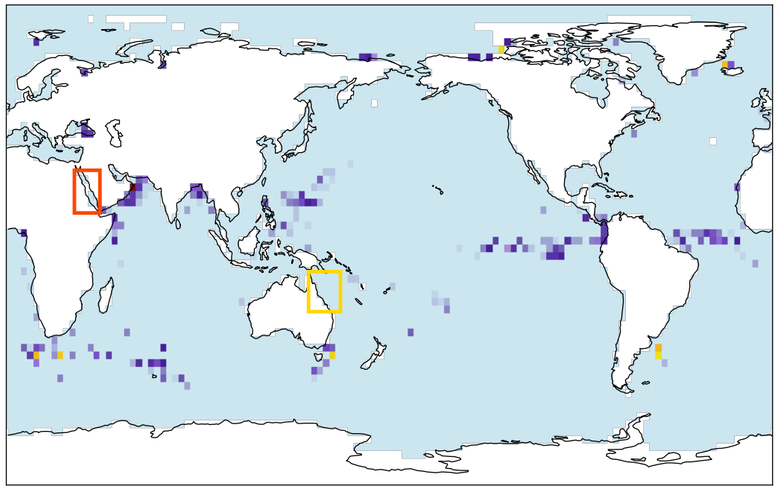




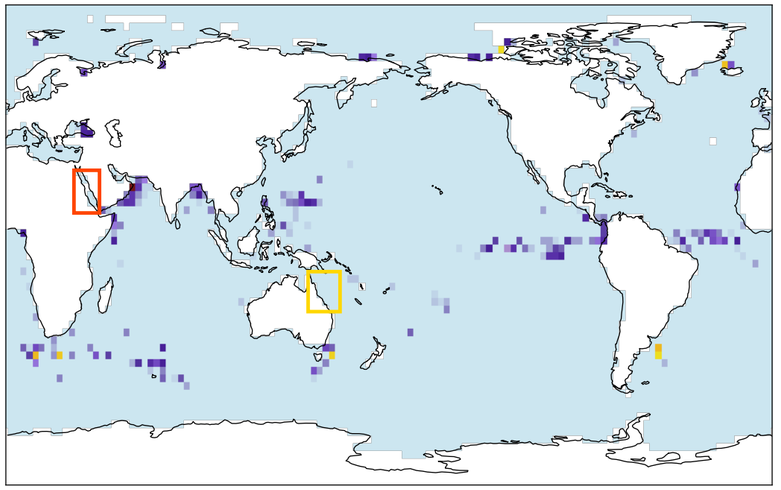


DHW Chain init: 2017-07-04 - Val (C-weeks)

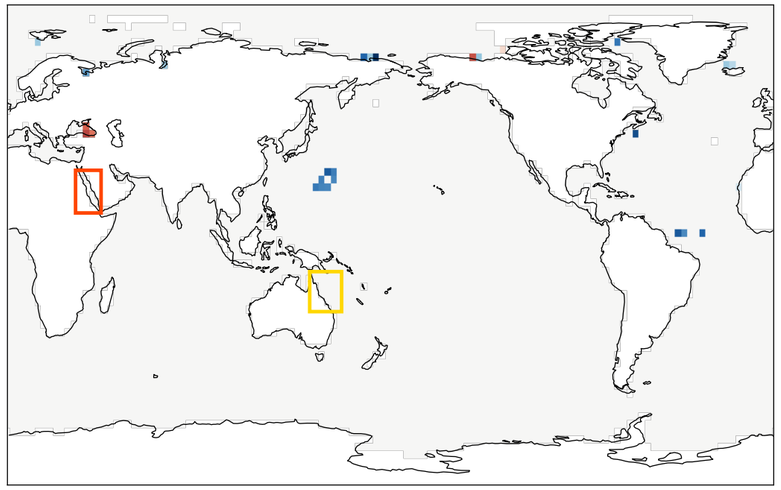
Actual DHW - Step 1



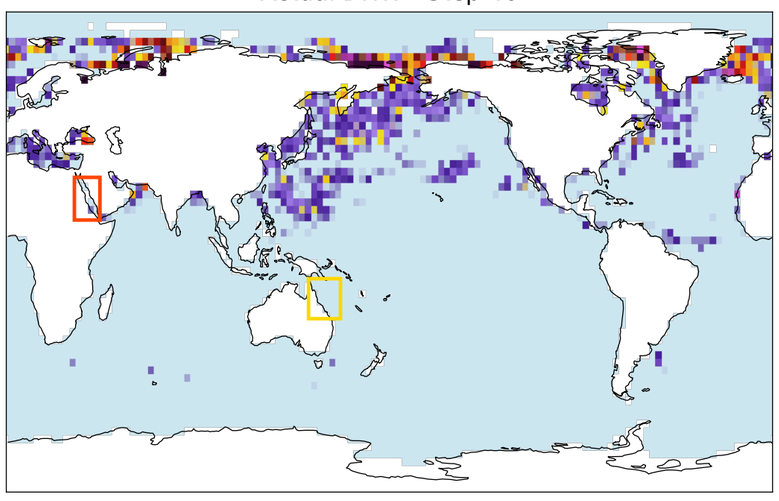
Predicted DHW - Step 1



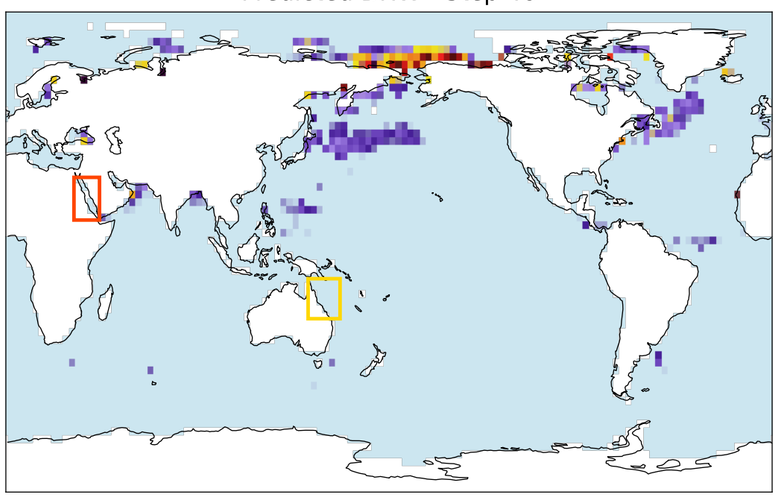
Error (Pred - Actual) - Step 1 [scale ±0.27]



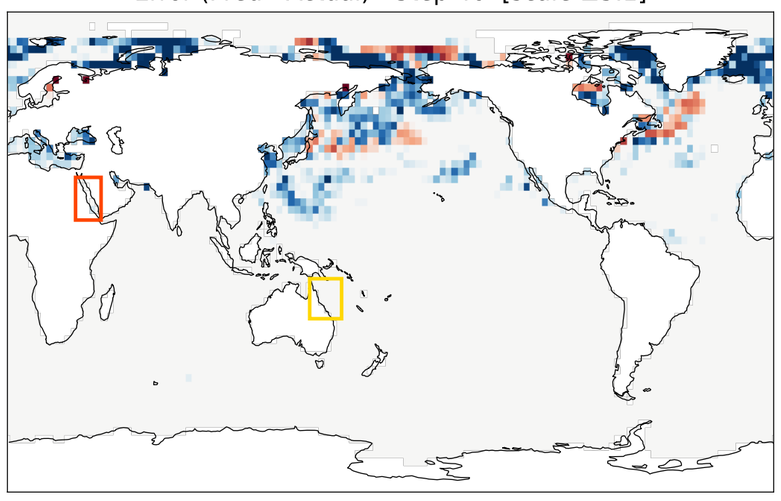
Actual DHW - Step 46



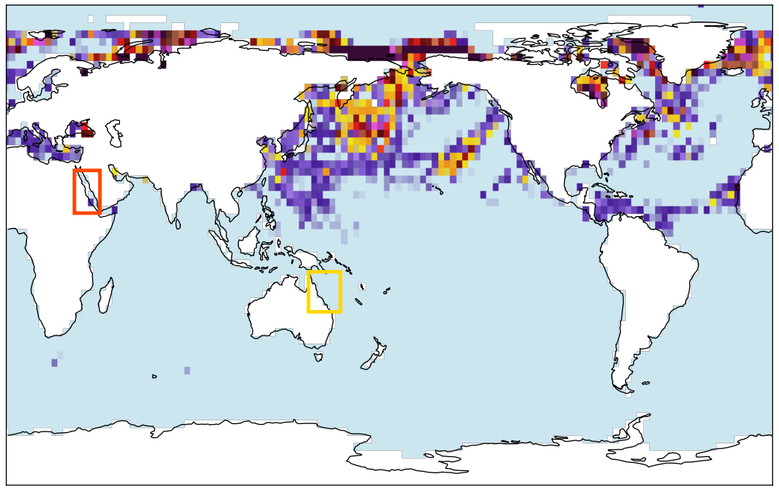
Predicted DHW - Step 46



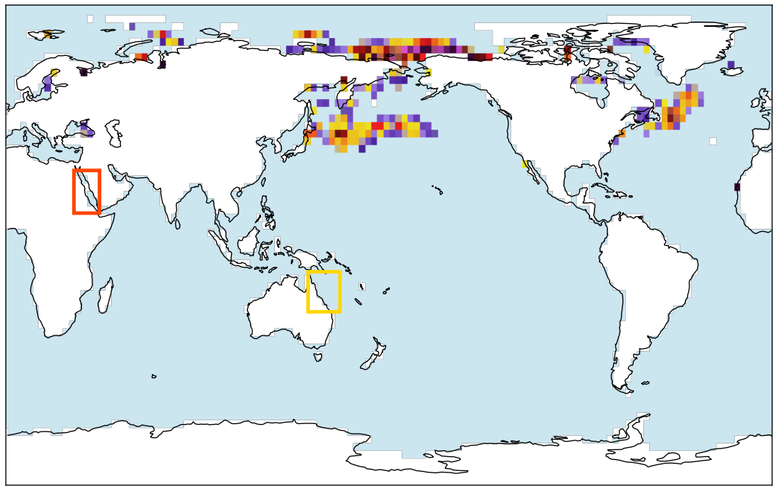
Error (Pred - Actual) - Step 46 [scale ±5.1]



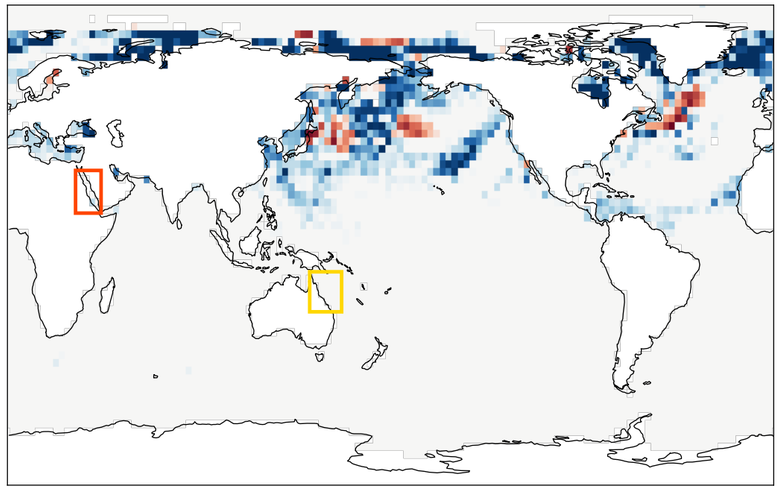
Actual DHW - Step 90

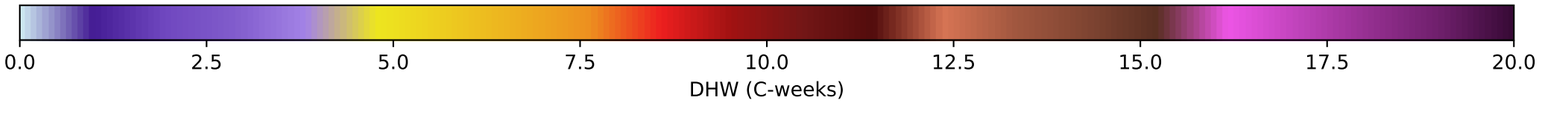


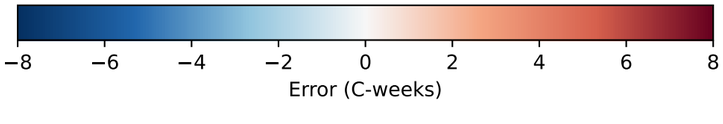
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

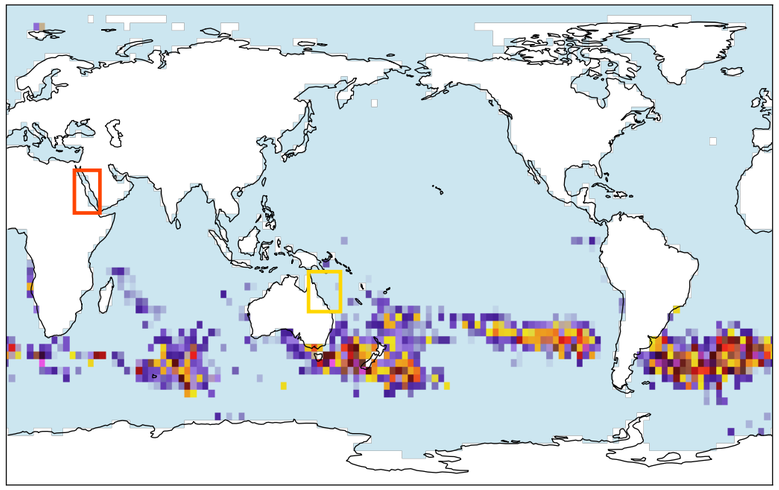




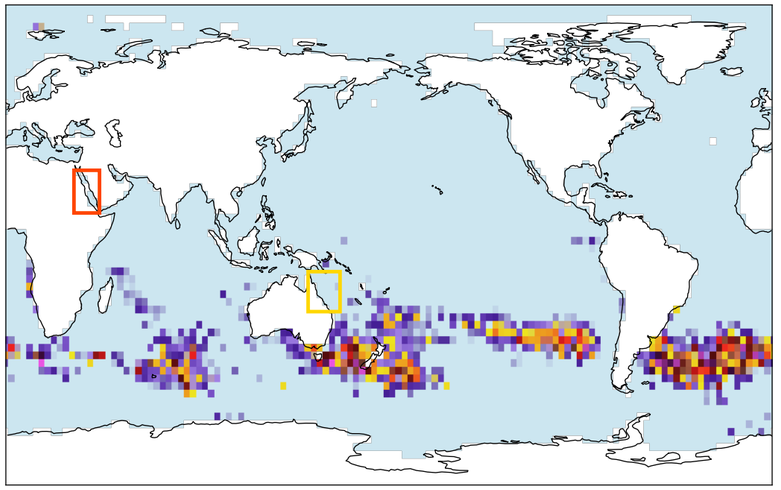


DHW Chain init: 2018-03-31 - Val (C-weeks)

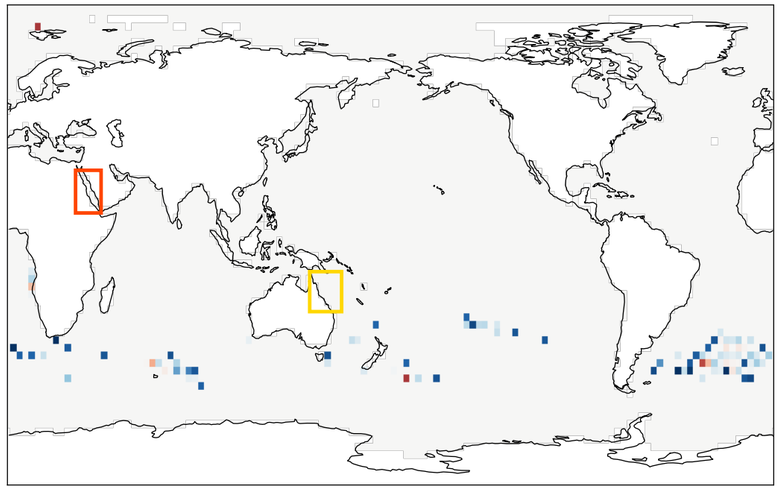
Actual DHW - Step 1



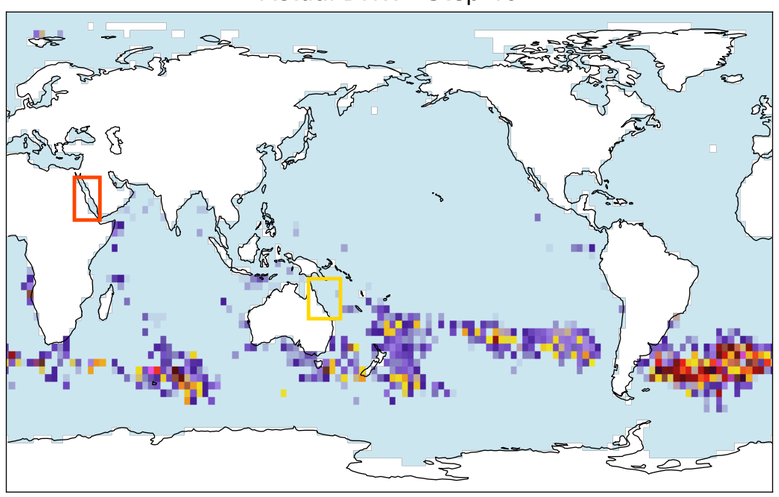
Predicted DHW - Step 1



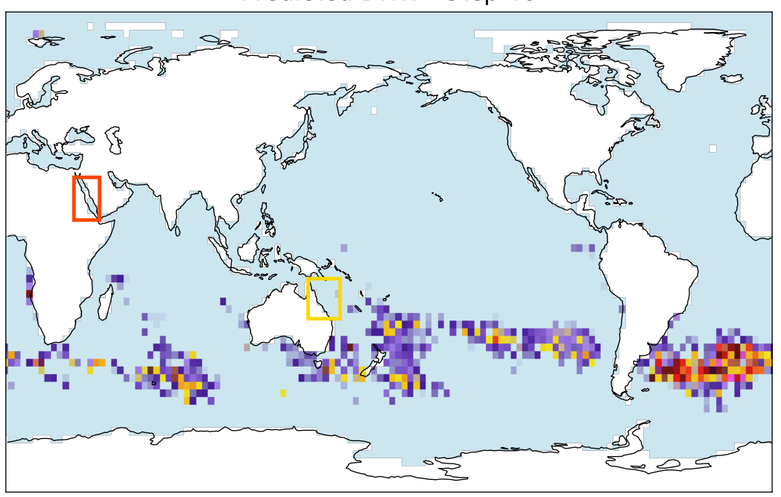
Error (Pred - Actual) - Step 1 [scale ±0.21]



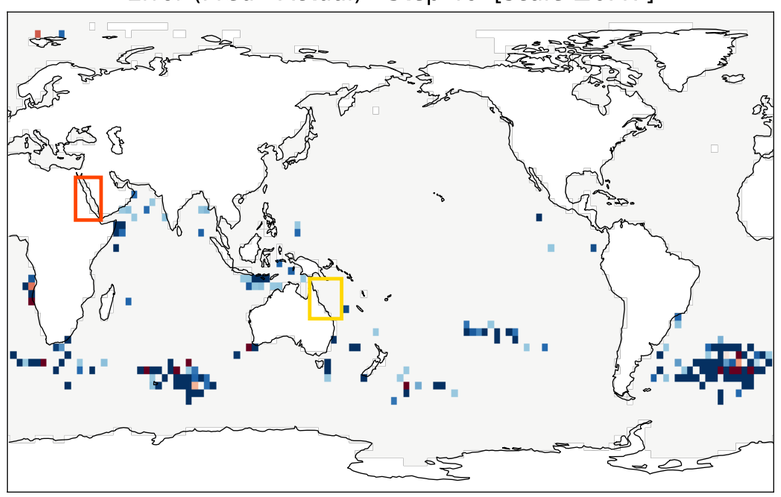
Actual DHW - Step 46



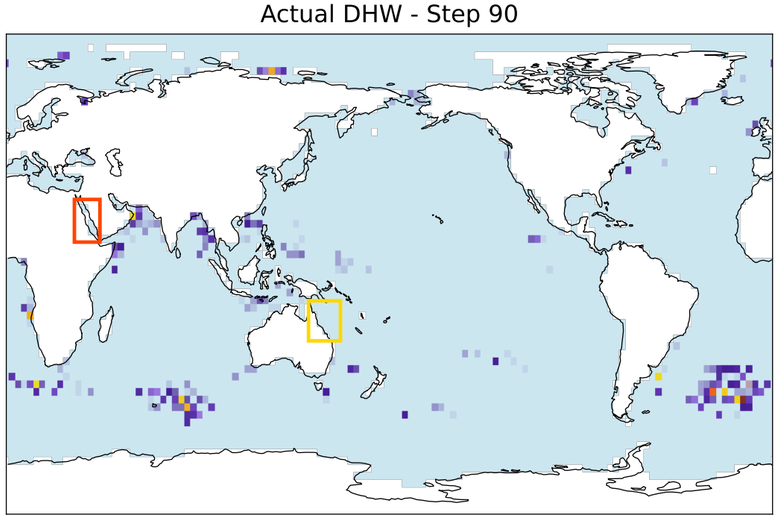
Predicted DHW - Step 46



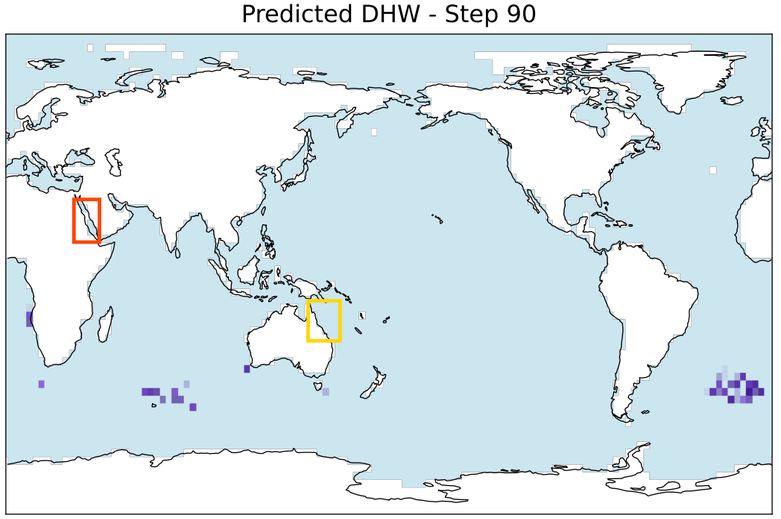
Error (Pred - Actual) - Step 46 [scale ±0.47]



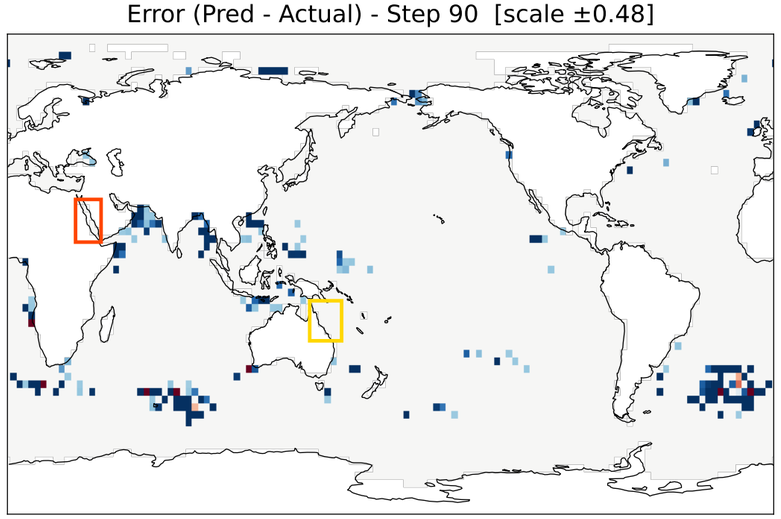
Actual DHW - Step 90

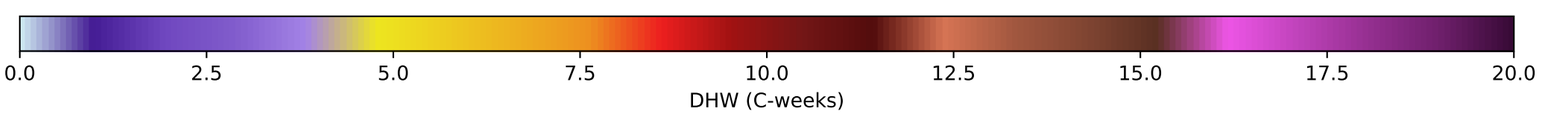


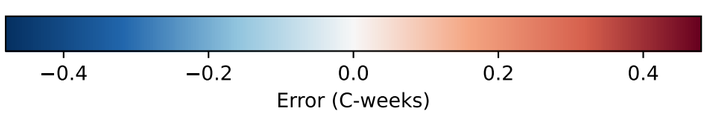
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±0.48]

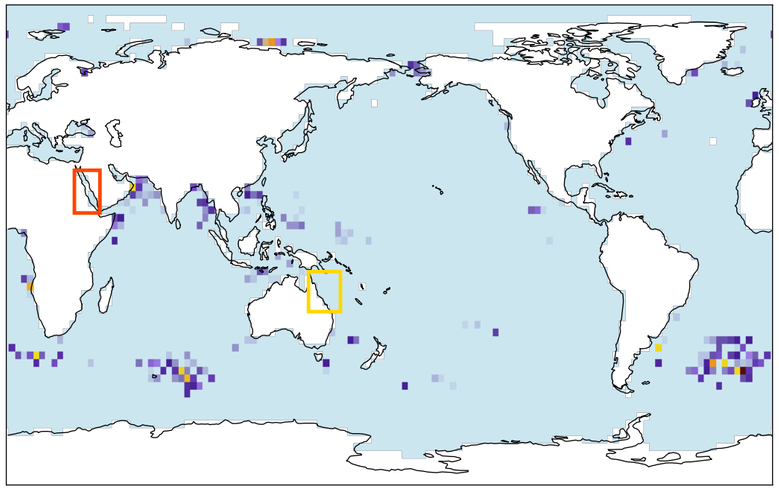




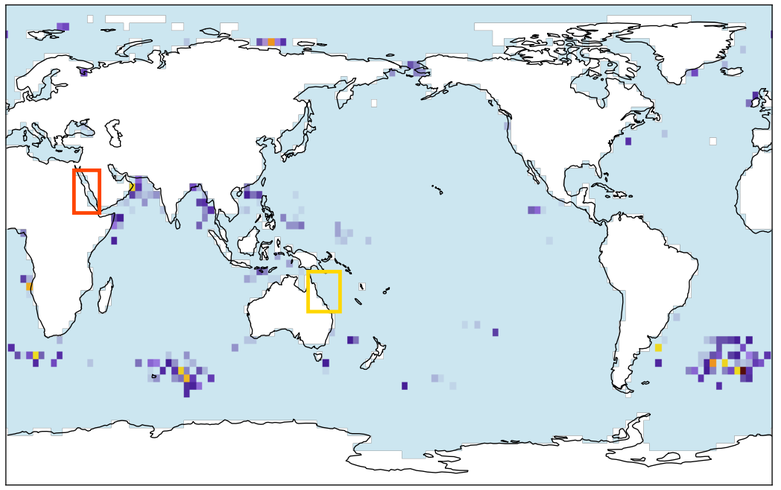


DHW Chain init: 2018-06-29 - Val (C-weeks)

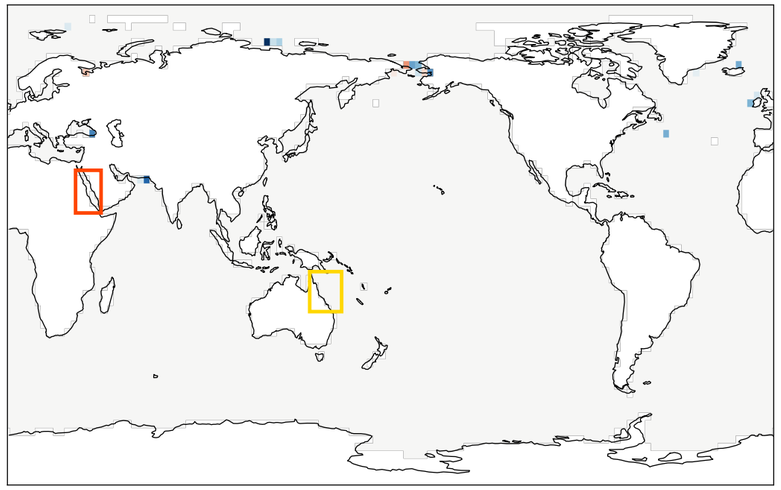
Actual DHW - Step 1



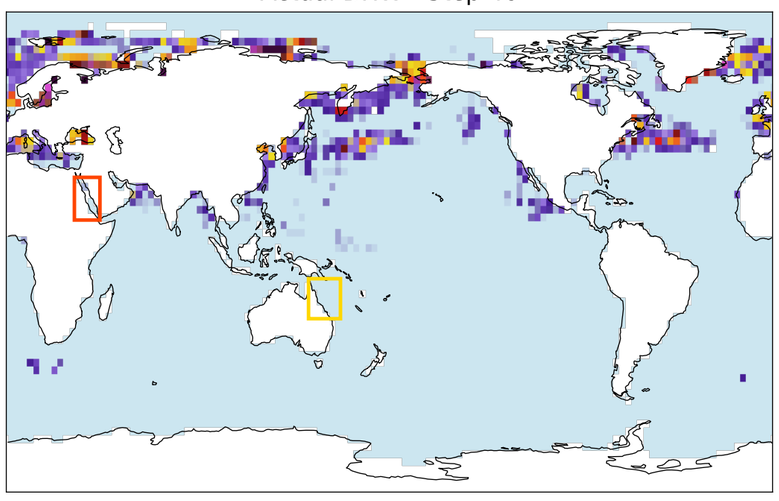
Predicted DHW - Step 1



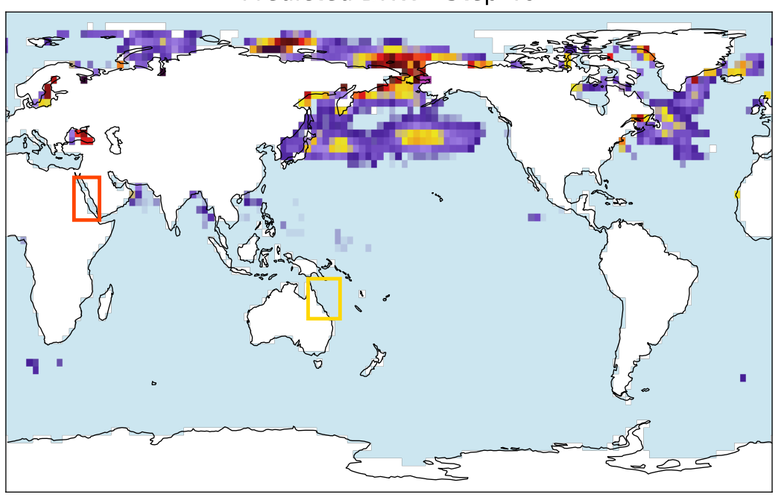
Error (Pred - Actual) - Step 1 [scale ±0.35]



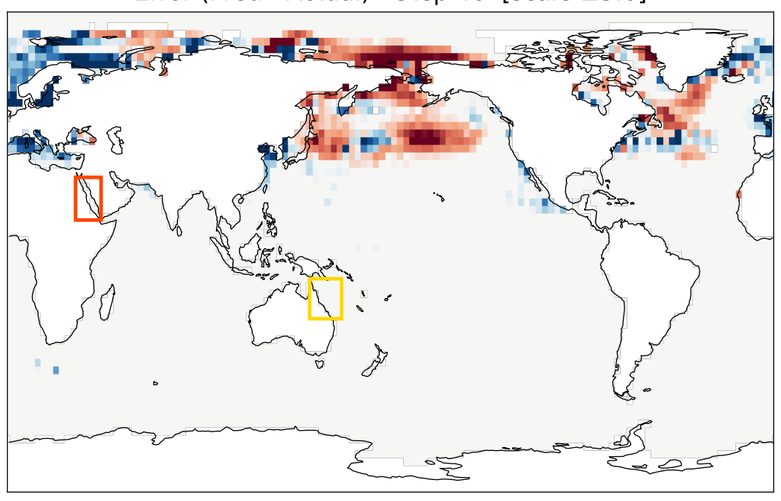
Actual DHW - Step 46



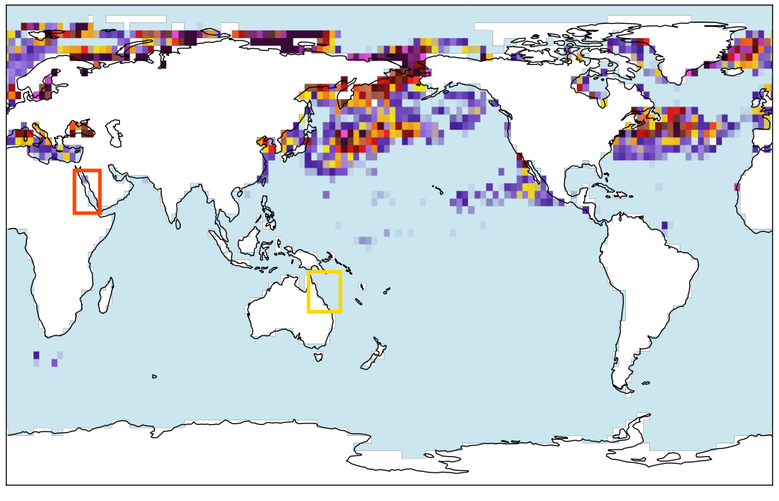
Predicted DHW - Step 46



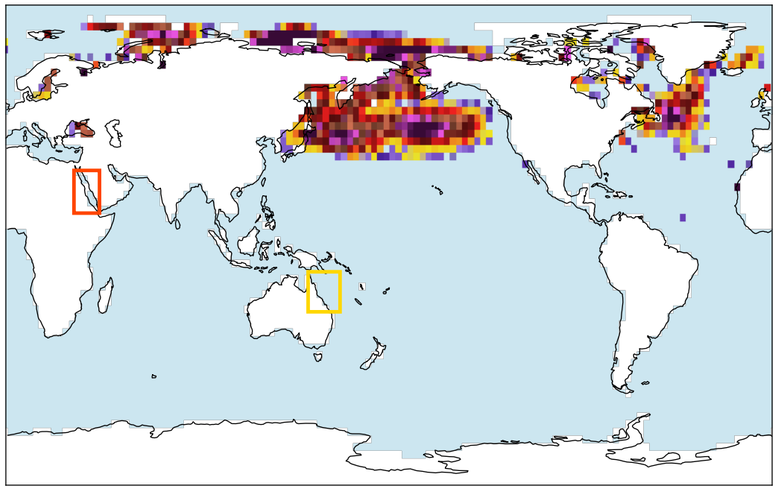
Error (Pred - Actual) - Step 46 [scale ±5.6]



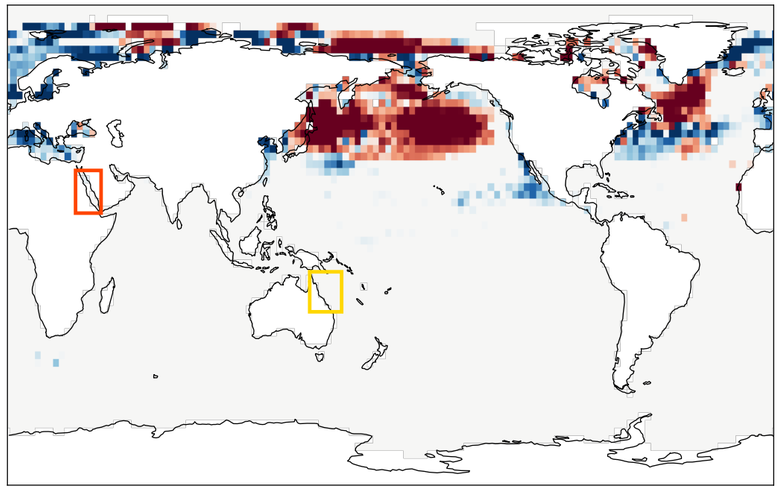
Actual DHW - Step 90

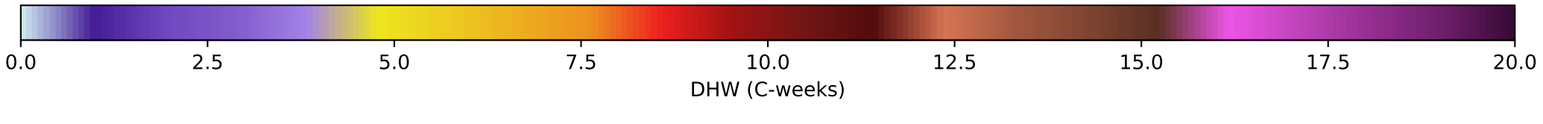


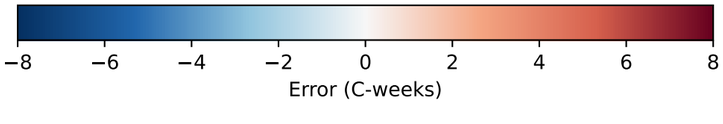
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

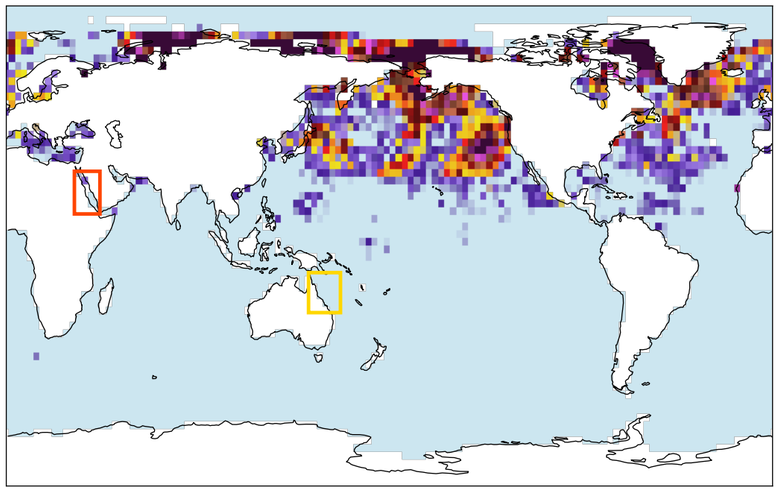




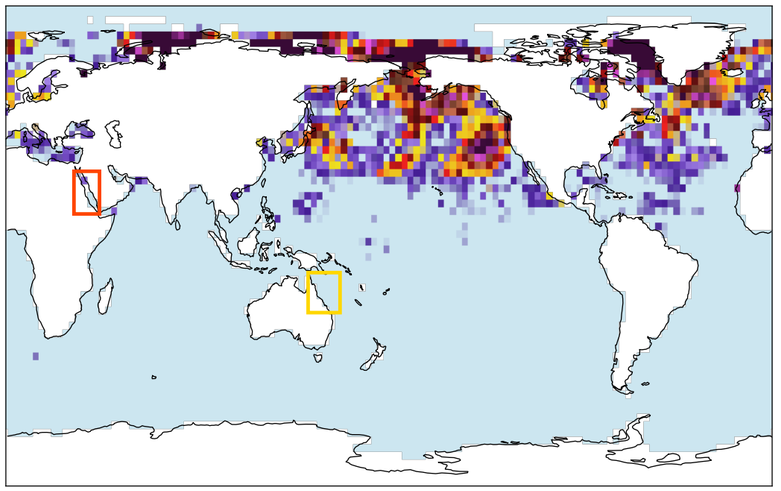


DHW Chain init: 2019-09-22 - Val (C-weeks)

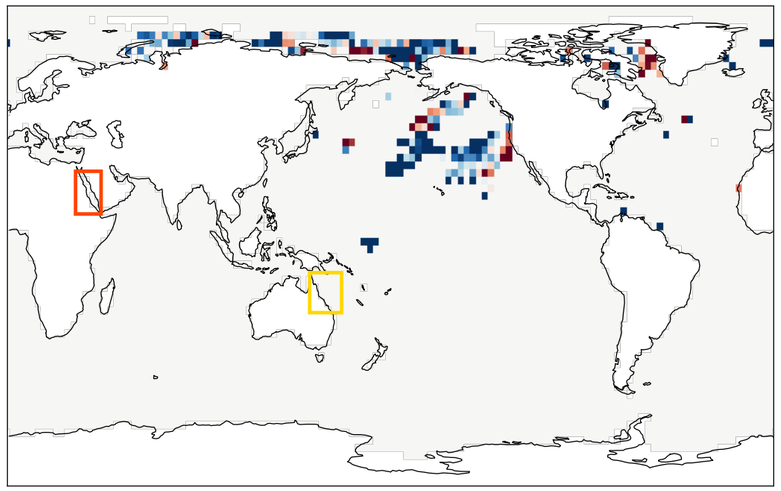
Actual DHW - Step 1



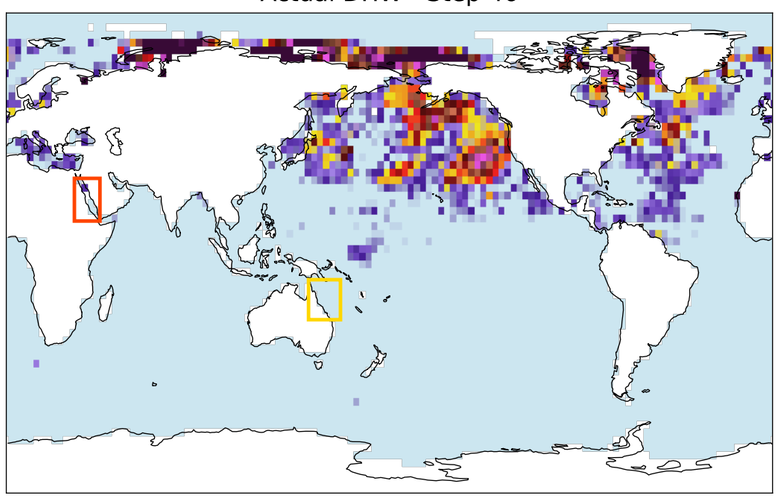
Predicted DHW - Step 1



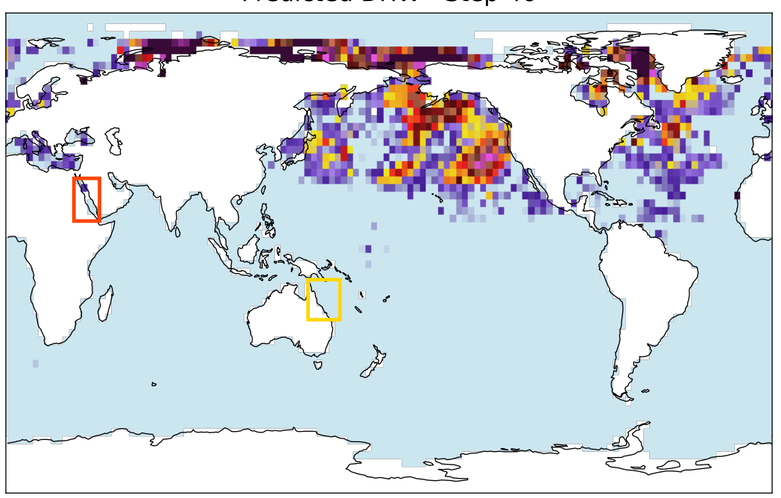
Error (Pred - Actual) - Step 1 [scale ±0.045]



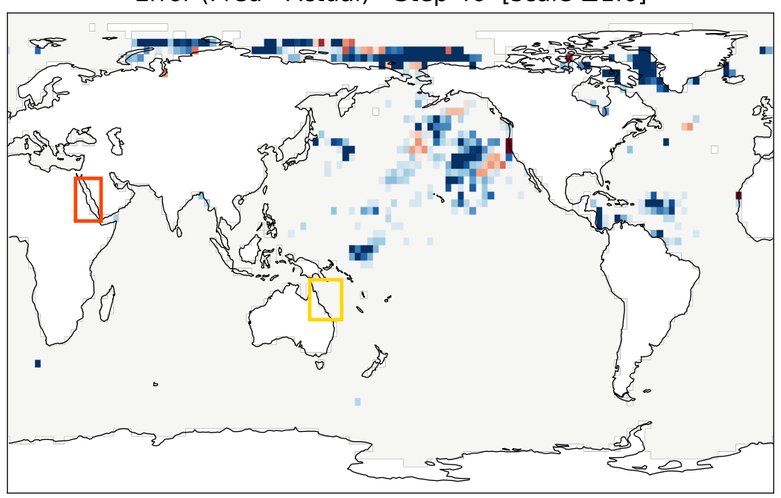
Actual DHW - Step 46



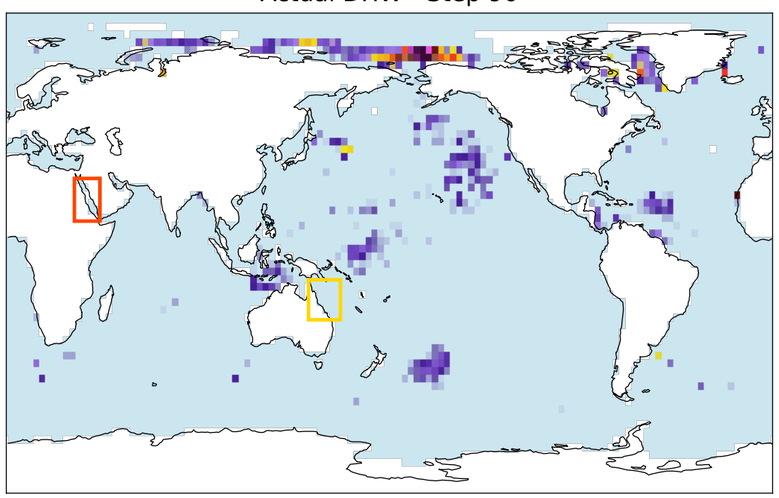
Predicted DHW - Step 46



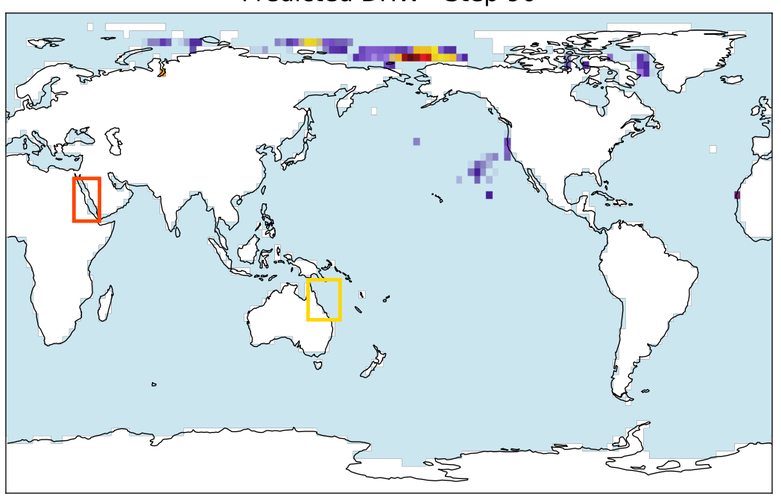
Error (Pred - Actual) - Step 46 [scale ±1.6]



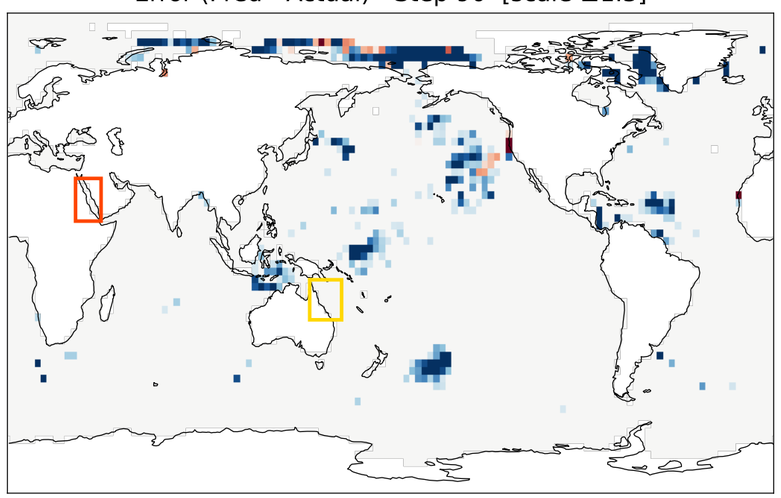
Actual DHW - Step 90

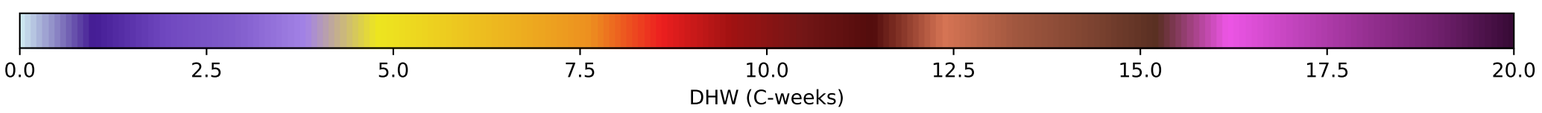


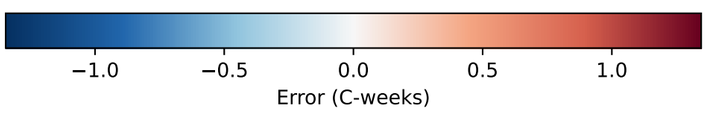
Predicted DHW - Step 90



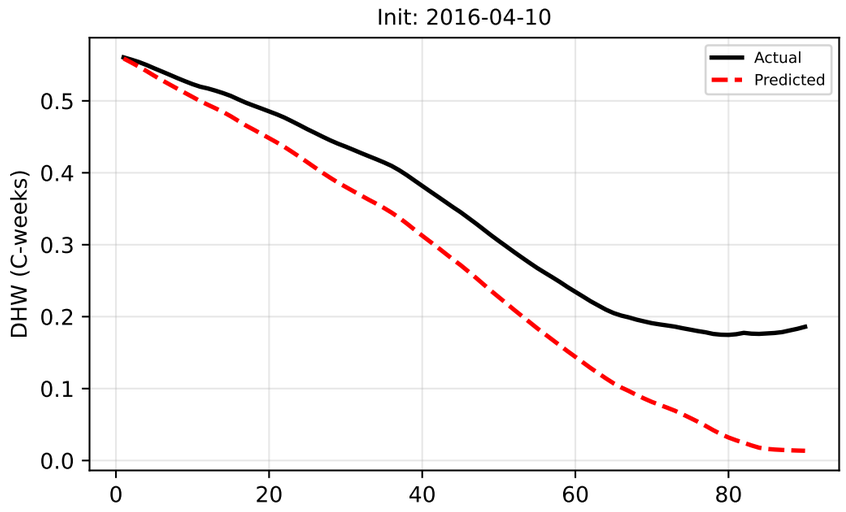
Error (Pred - Actual) - Step 90 [scale ±1.3]

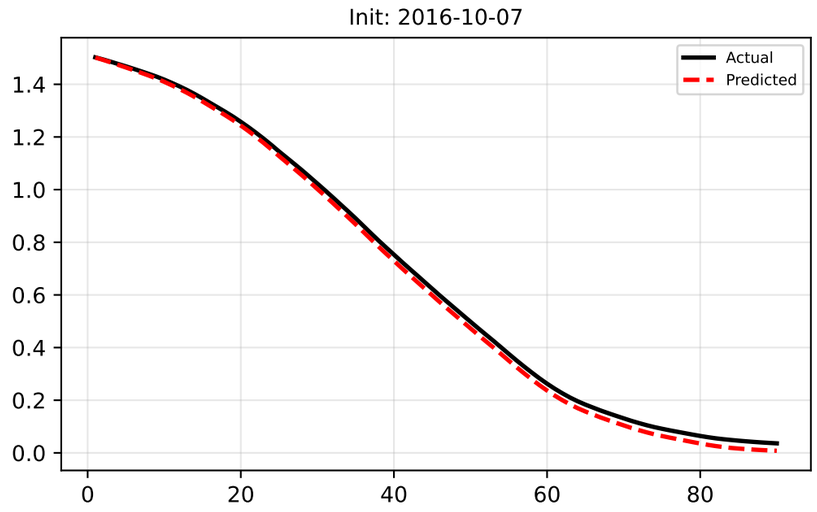


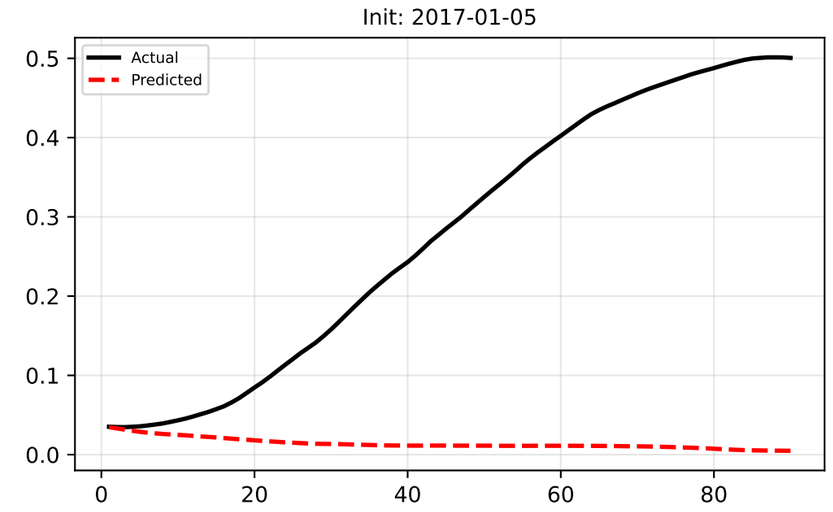


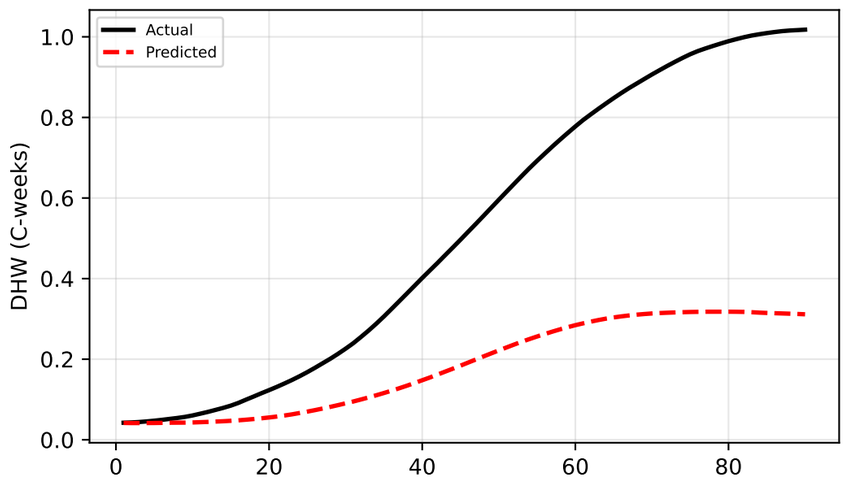


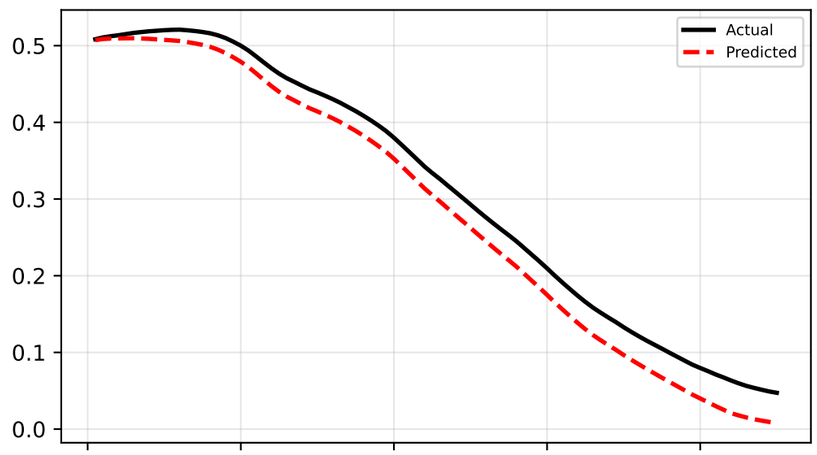
Global-Mean DHW - Val (C-weeks)

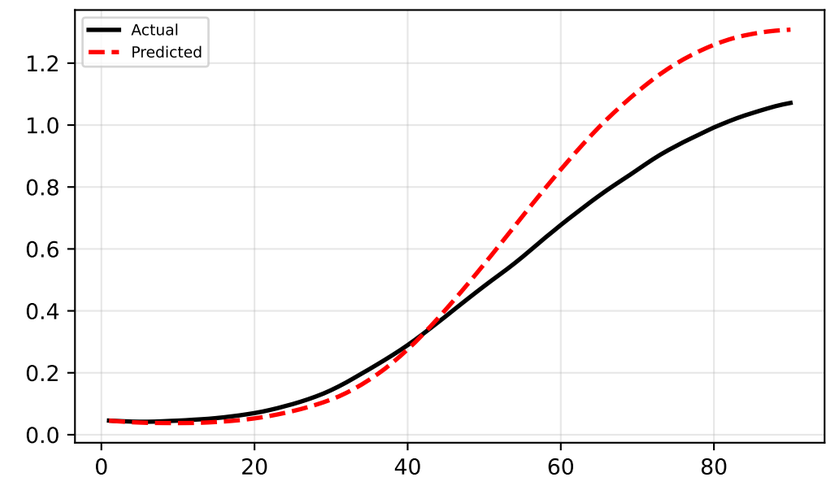


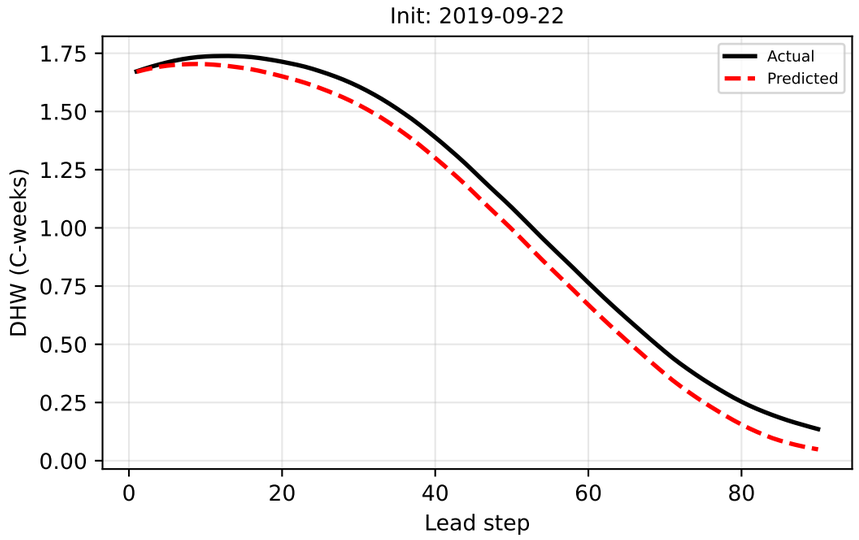




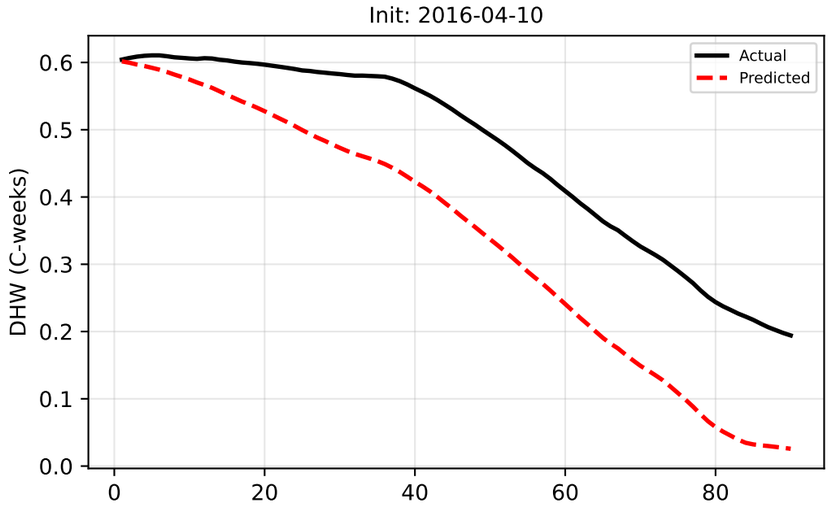


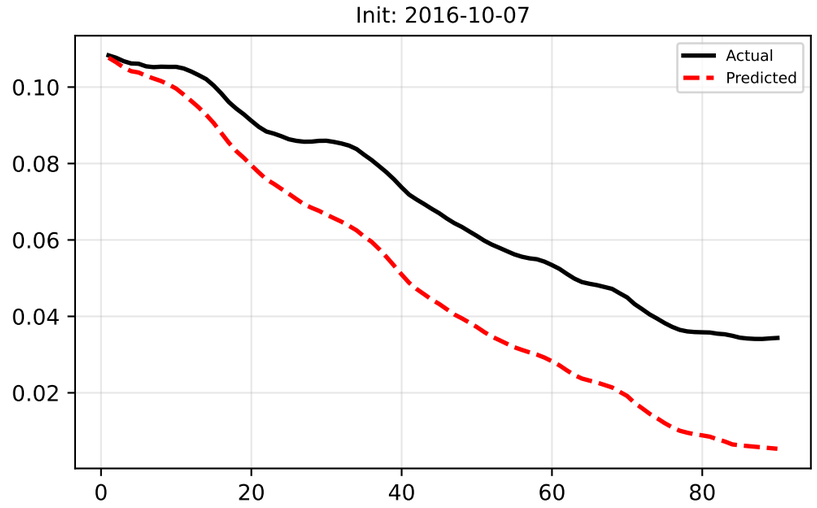


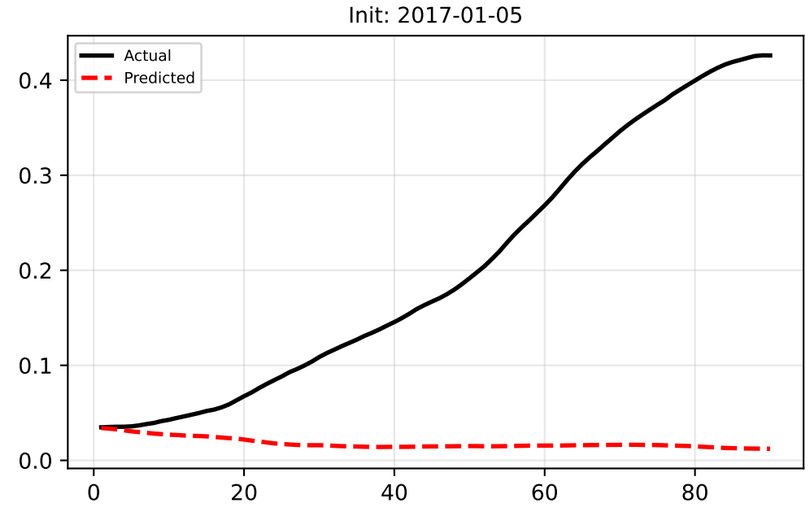


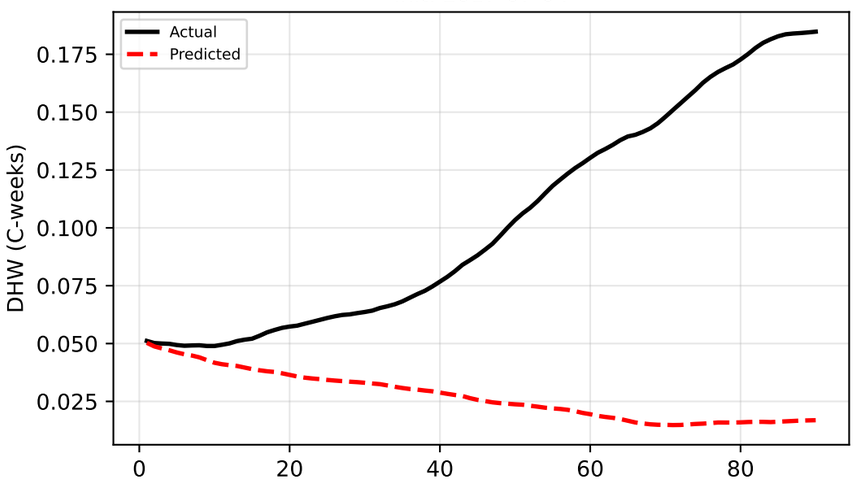


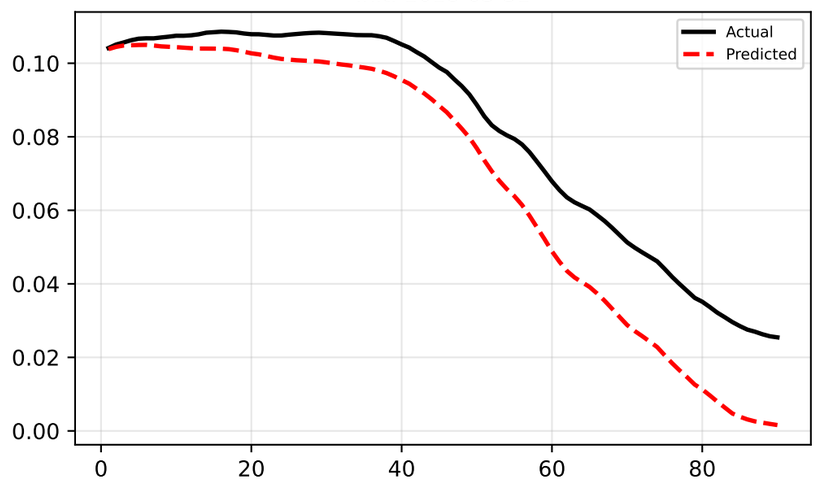
Tropical-Mean DHW (|lat|<=30) - Val (C-weeks)

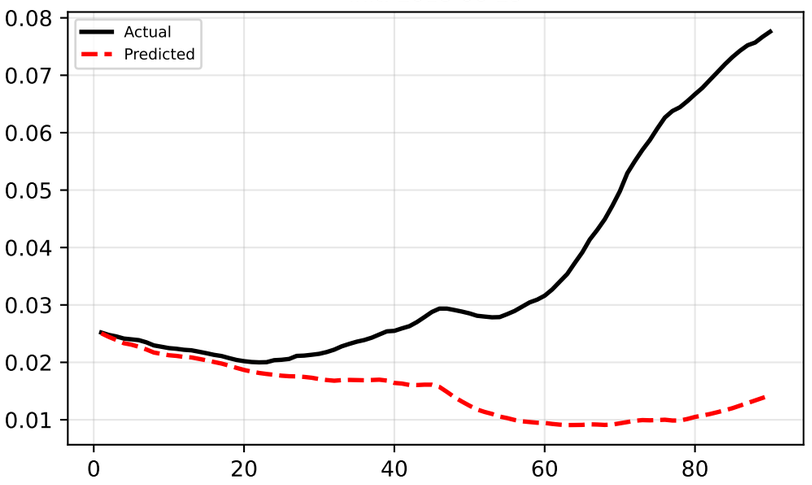


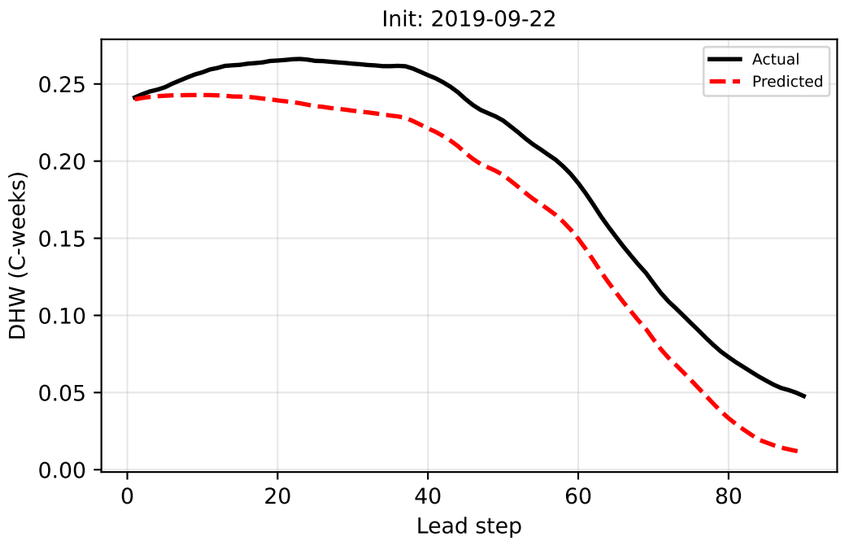




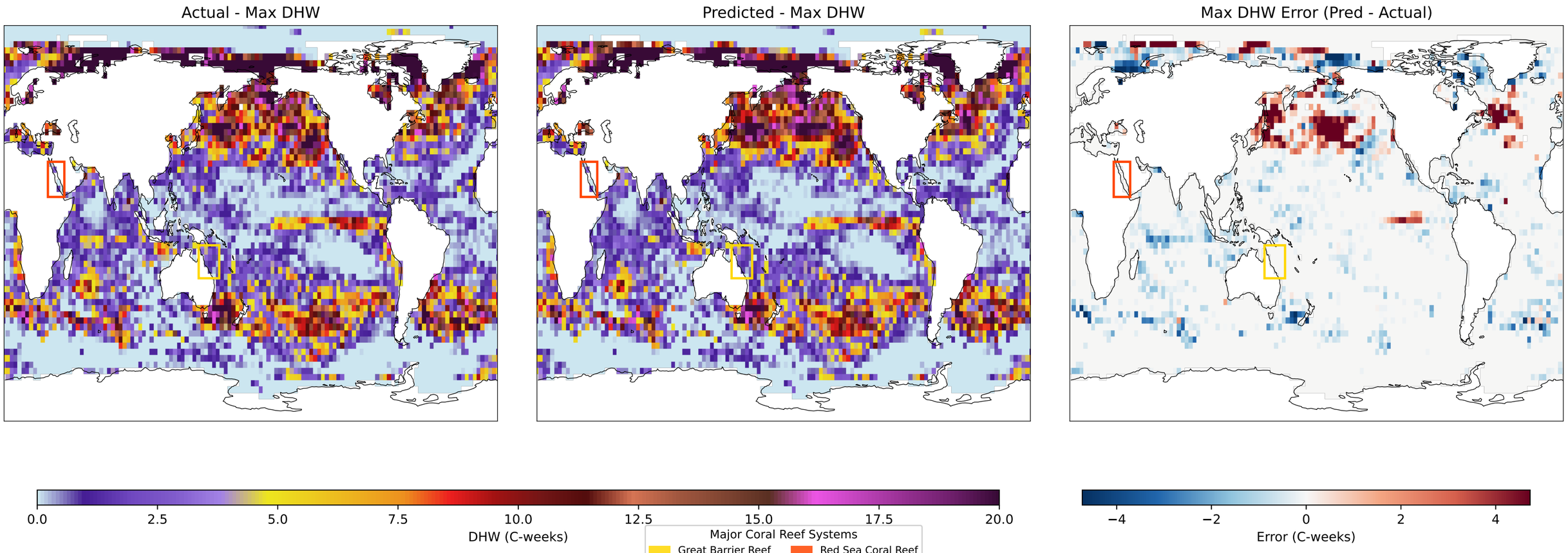


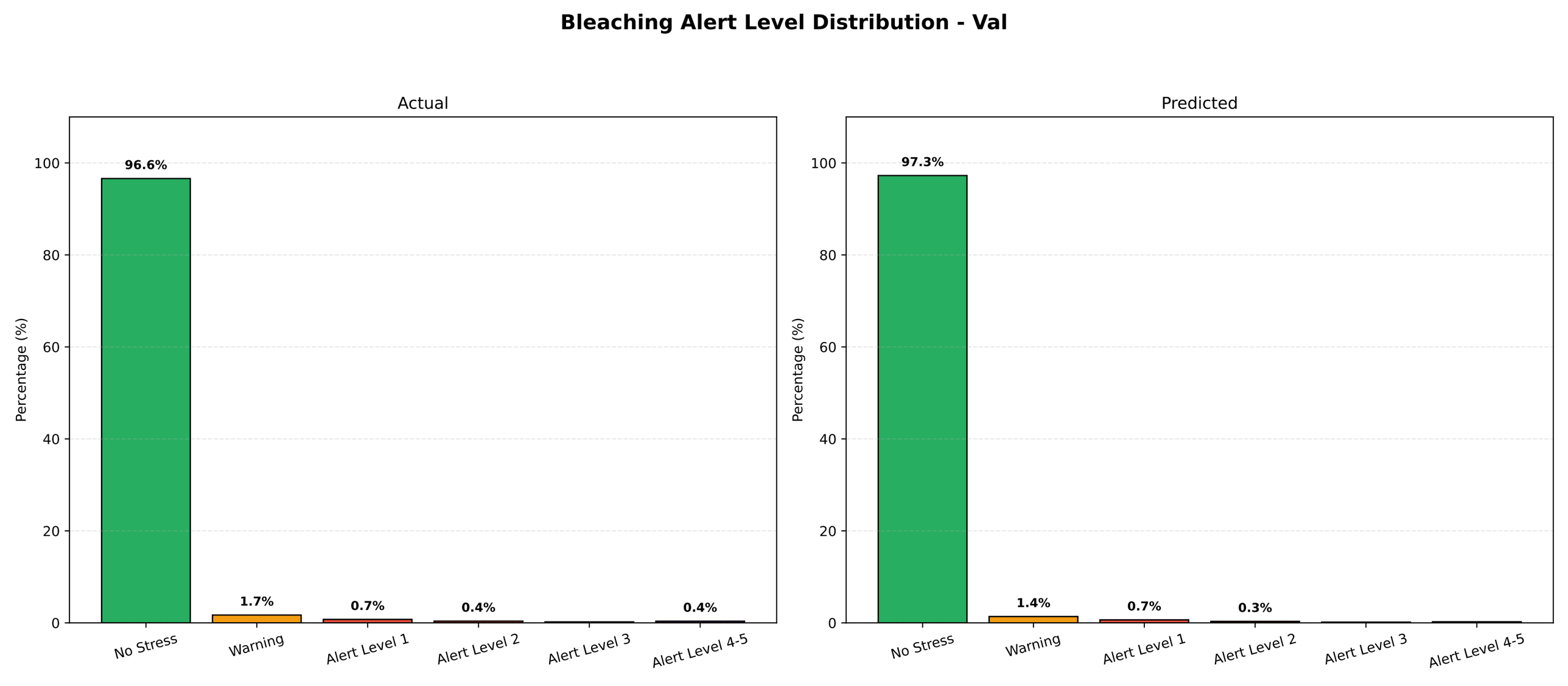






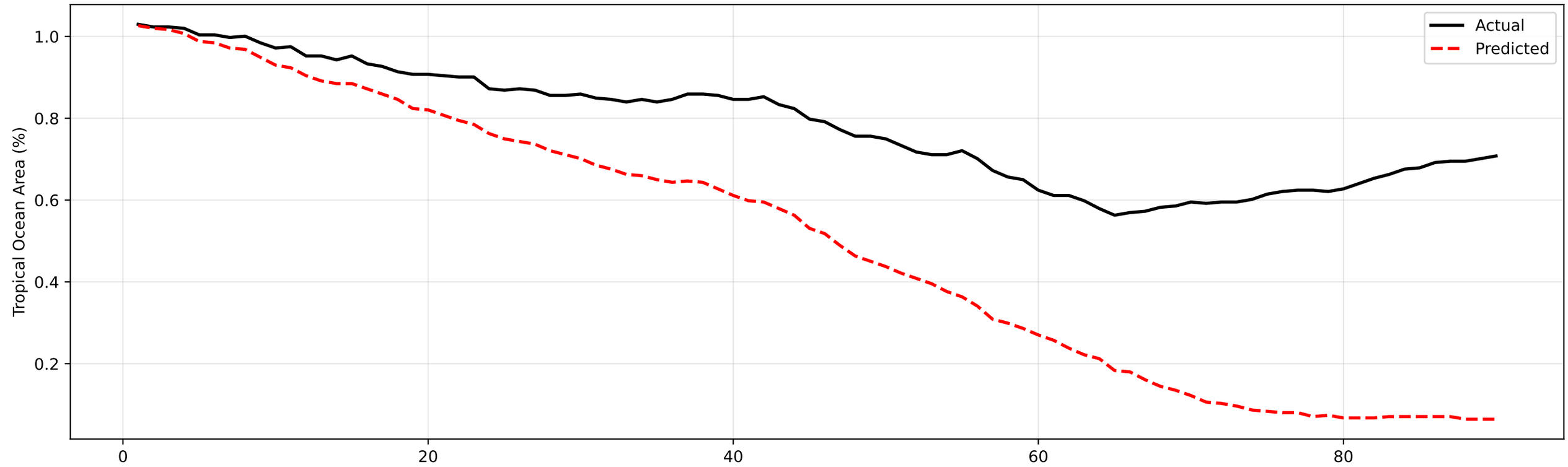
Peak DHW (max over all chains & leads) - Val (C-weeks)



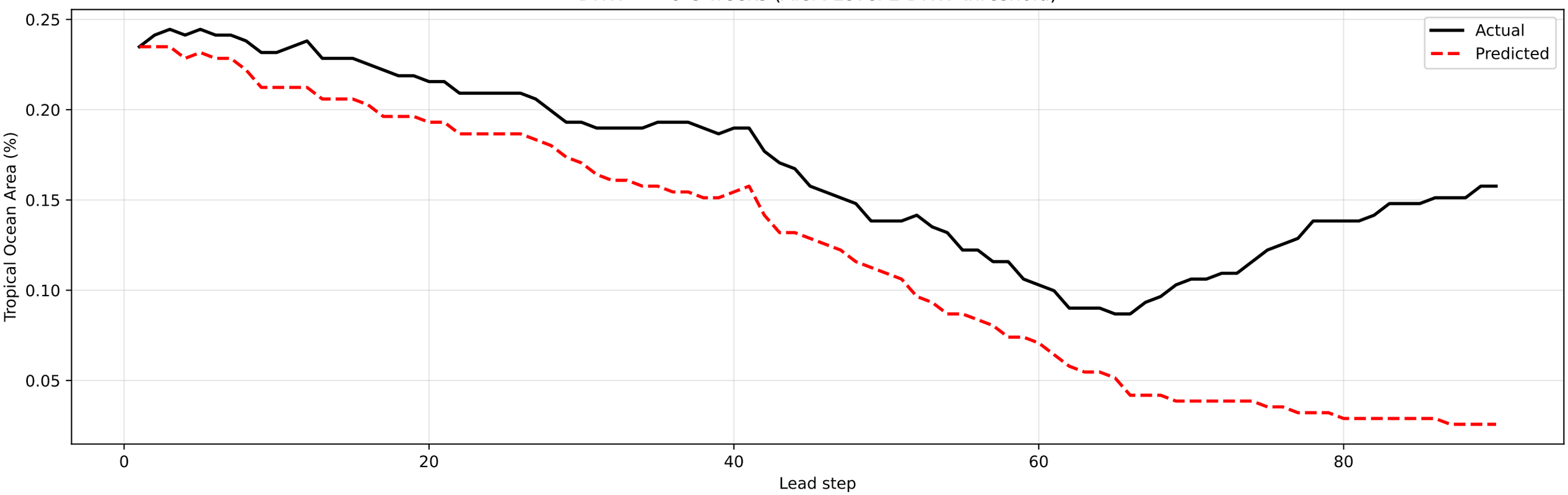

Bleaching Alert Level Distribution - Val
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
96.6%
1.7%
0.7%
0.4%
0.4%
97.3%
1.4%
0.7%
0.3%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)



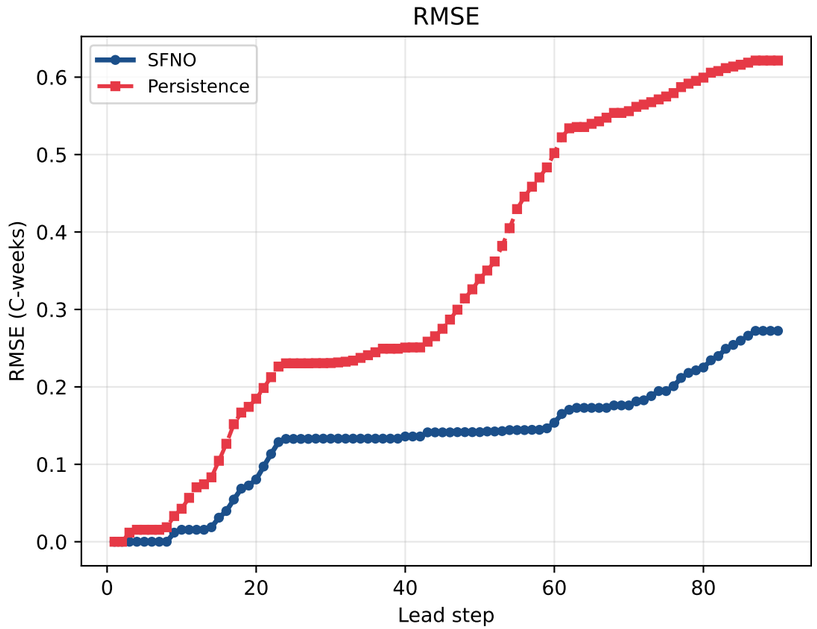
Great Barrier Reef Analysis - Val

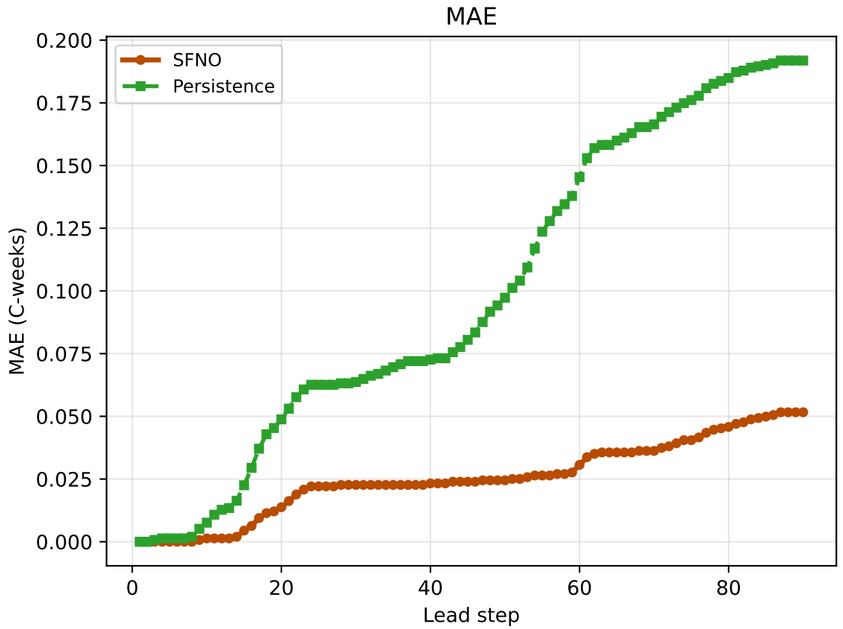
```
                    GREAT BARRIER REEF - VAL
Lat -25.0 deg to -10.0 deg  |  Lon 142.0 deg to 157.0 deg  (0-360)
                    Grid: 5 lat x 5 lon points
```

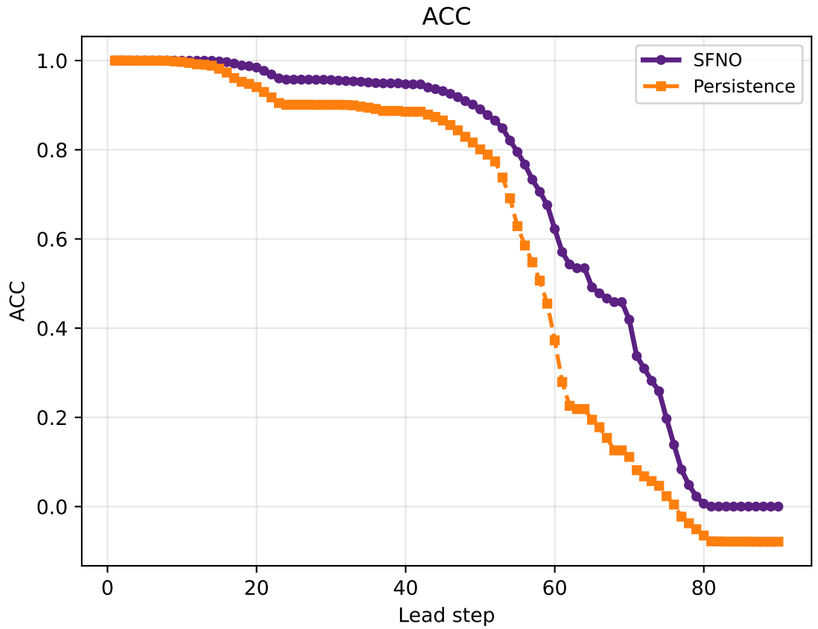
DHW Metrics by Lead Step - Val - GBR (15 chains scored at every lead)

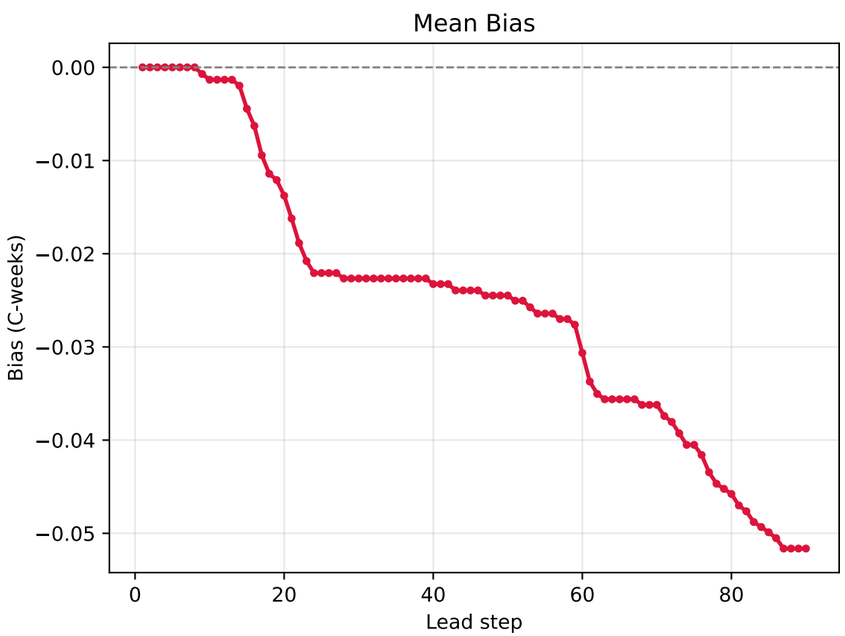
Lead Step	SFNO RMSE	Persist RMSE	Skill %	SFNO MAE	Persist MAE	SFNO Bias	SFNO ACC	Persist ACC	SFNO R²
1	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
2	0.0000	0.0000	n/a	0.0000	0.0000	0.0000	1.0000	1.0000	1.0000
3	0.0000	0.0117	100.0%	0.0000	0.0007	0.0000	1.0000	0.9998	1.0000
4	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
5	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
6	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
7	0.0000	0.0155	100.0%	0.0000	0.0013	0.0000	1.0000	0.9996	1.0000
8	0.0000	0.0187	100.0%	0.0000	0.0020	0.0000	1.0000	0.9994	1.0000
9	0.0117	0.0332	64.6%	0.0007	0.0052	-0.0007	0.9997	0.9981	0.9995
10	0.0155	0.0426	63.7%	0.0013	0.0076	-0.0013	0.9996	0.9969	0.9991
11	0.0155	0.0567	72.7%	0.0013	0.0108	-0.0013	0.9995	0.9947	0.9991
12	0.0155	0.0703	78.0%	0.0013	0.0127	-0.0013	0.9995	0.9917	0.9991
13	0.0155	0.0742	79.2%	0.0013	0.0134	-0.0013	0.9995	0.9907	0.9991
14	0.0187	0.0832	77.5%	0.0020	0.0163	-0.0020	0.9993	0.9884	0.9987
15	0.0311	0.1046	70.3%	0.0044	0.0225	-0.0044	0.9980	0.9817	0.9962
16	0.0397	0.1265	68.6%	0.0063	0.0295	-0.0063	0.9966	0.9732	0.9935
17	0.0545	0.1518	64.1%	0.0094	0.0371	-0.0094	0.9932	0.9608	0.9871
18	0.0686	0.1668	58.9%	0.0114	0.0429	-0.0114	0.9889	0.9523	0.9789
19	0.0726	0.1742	58.3%	0.0121	0.0453	-0.0121	0.9874	0.9478	0.9761
20	0.0804	0.1846	56.4%	0.0138	0.0488	-0.0138	0.9843	0.9404	0.9703
21	0.0971	0.1986	51.1%	0.0162	0.0531	-0.0162	0.9770	0.9296	0.9566
22	0.1133	0.2125	46.7%	0.0189	0.0576	-0.0189	0.9688	0.9174	0.9416
23	0.1287	0.2262	43.1%	0.0208	0.0607	-0.0208	0.9600	0.9047	0.9259
24	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
25	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
26	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
27	0.1328	0.2304	42.4%	0.0221	0.0626	-0.0221	0.9572	0.9009	0.9208
28	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
29	0.1332	0.2306	42.3%	0.0227	0.0632	-0.0227	0.9570	0.9007	0.9203
30	0.1332	0.2308	42.3%	0.0227	0.0637	-0.0227	0.9564	0.9008	0.9192
31	0.1332	0.2316	42.5%	0.0227	0.0649	-0.0227	0.9550	0.9008	0.9167
32	0.1332	0.2325	42.7%	0.0227	0.0662	-0.0227	0.9542	0.9004	0.9152
33	0.1332	0.2339	43.1%	0.0227	0.0669	-0.0227	0.9533	0.8996	0.9136
34	0.1332	0.2374	43.9%	0.0227	0.0683	-0.0227	0.9525	0.8967	0.9121
35	0.1332	0.2409	44.7%	0.0227	0.0696	-0.0227	0.9509	0.8944	0.9093
36	0.1332	0.2447	45.6%	0.0227	0.0708	-0.0227	0.9496	0.8915	0.9068
37	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
38	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
39	0.1332	0.2493	46.6%	0.0227	0.0720	-0.0227	0.9489	0.8871	0.9055
40	0.1359	0.2508	45.8%	0.0233	0.0726	-0.0233	0.9469	0.8852	0.9019
41	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
42	0.1359	0.2509	45.8%	0.0233	0.0732	-0.0233	0.9466	0.8853	0.9015
43	0.1413	0.2584	45.3%	0.0239	0.0756	-0.0239	0.9396	0.8788	0.8887
44	0.1413	0.2654	46.8%	0.0239	0.0776	-0.0239	0.9361	0.8735	0.8822
45	0.1413	0.2750	48.6%	0.0239	0.0805	-0.0239	0.9315	0.8656	0.8740
46	0.1413	0.2869	50.8%	0.0239	0.0834	-0.0239	0.9255	0.8554	0.8631
47	0.1415	0.2997	52.8%	0.0245	0.0876	-0.0245	0.9180	0.8435	0.8499
48	0.1415	0.3142	55.0%	0.0245	0.0917	-0.0245	0.9092	0.8291	0.8344
49	0.1415	0.3260	56.6%	0.0245	0.0942	-0.0245	0.9013	0.8163	0.8207
50	0.1415	0.3395	58.3%	0.0245	0.0973	-0.0245	0.8907	0.8009	0.8025
51	0.1424	0.3503	59.3%	0.0250	0.1012	-0.0250	0.8778	0.7892	0.7809
52	0.1424	0.3620	60.7%	0.0250	0.1041	-0.0250	0.8653	0.7741	0.7605
53	0.1429	0.3821	62.6%	0.0257	0.1093	-0.0257	0.8482	0.7377	0.7322
54	0.1442	0.4049	64.4%	0.0264	0.1169	-0.0264	0.8206	0.6912	0.6890
55	0.1442	0.4296	66.4%	0.0264	0.1236	-0.0264	0.7950	0.6287	0.6501
56	0.1442	0.4457	67.6%	0.0264	0.1278	-0.0264	0.7668	0.5854	0.6088
57	0.1445	0.4585	68.5%	0.0270	0.1318	-0.0270	0.7330	0.5477	0.5614
58	0.1445	0.4705	69.3%	0.0270	0.1346	-0.0270	0.7057	0.5066	0.5248
59	0.1465	0.4834	69.7%	0.0276	0.1379	-0.0276	0.6760	0.4551	0.4861
60	0.1537	0.5019	69.4%	0.0306	0.1454	-0.0306	0.6222	0.3728	0.4186
61	0.1651	0.5223	68.4%	0.0337	0.1529	-0.0337	0.5707	0.2794	0.3594
62	0.1704	0.5339	68.1%	0.0350	0.1569	-0.0350	0.5428	0.2259	0.3295
63	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
64	0.1729	0.5356	67.7%	0.0356	0.1582	-0.0356	0.5345	0.2187	0.3212
65	0.1729	0.5397	68.0%	0.0356	0.1599	-0.0356	0.4915	0.1947	0.2746
66	0.1729	0.5429	68.2%	0.0356	0.1611	-0.0356	0.4781	0.1779	0.2597
67	0.1729	0.5476	68.4%	0.0356	0.1629	-0.0356	0.4666	0.1539	0.2469
68	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
69	0.1760	0.5538	68.2%	0.0362	0.1653	-0.0362	0.4585	0.1260	0.2369
70	0.1760	0.5561	68.3%	0.0362	0.1664	-0.0362	0.4189	0.1113	0.2001
71	0.1812	0.5617	67.7%	0.0374	0.1695	-0.0374	0.3375	0.0816	0.1329
72	0.1827	0.5646	67.6%	0.0380	0.1713	-0.0380	0.3096	0.0678	0.1116
73	0.1881	0.5677	66.9%	0.0393	0.1731	-0.0393	0.2819	0.0571	0.0915
74	0.1946	0.5713	65.9%	0.0405	0.1749	-0.0405	0.2585	0.0465	0.0756
75	0.1946	0.5750	66.2%	0.0405	0.1761	-0.0405	0.1969	0.0233	0.0384
76	0.2009	0.5794	65.3%	0.0416	0.1778	-0.0416	0.1385	0.0044	0.0092
77	0.2116	0.5871	64.0%	0.0434	0.1808	-0.0434	0.0831	-0.0226	-0.0142
78	0.2180	0.5916	63.1%	0.0447	0.1826	-0.0447	0.0485	-0.0374	-0.0277
79	0.2214	0.5952	62.8%	0.0452	0.1837	-0.0452	0.0225	-0.0507	-0.0367
80	0.2250	0.5994	62.5%	0.0458	0.1849	-0.0458	0.0064	-0.0649	-0.0418
81	0.2344	0.6058	61.3%	0.0470	0.1872	-0.0470	0.0000	-0.0779	-0.0419
82	0.2398	0.6079	60.6%	0.0476	0.1879	-0.0476	0.0000	-0.0782	-0.0411
83	0.2492	0.6117	59.3%	0.0488	0.1890	-0.0488	0.0000	-0.0785	-0.0399
84	0.2542	0.6137	58.6%	0.0493	0.1896	-0.0493	0.0000	-0.0786	-0.0391
85	0.2597	0.6160	57.8%	0.0499	0.1901	-0.0499	0.0000	-0.0786	-0.0383
86	0.2661	0.6187	57.0%	0.0505	0.1907	-0.0505	0.0000	-0.0785	-0.0374
87	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
88	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
89	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373
90	0.2723	0.6214	56.2%	0.0516	0.1919	-0.0516	0.0000	-0.0788	-0.0373

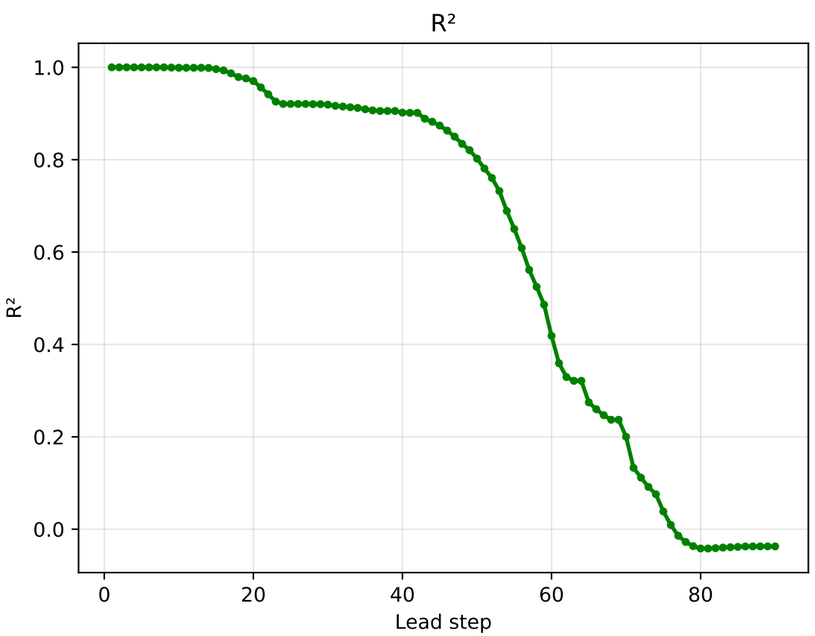
DHW Forecast Skill vs Lead Step - Val - GBR

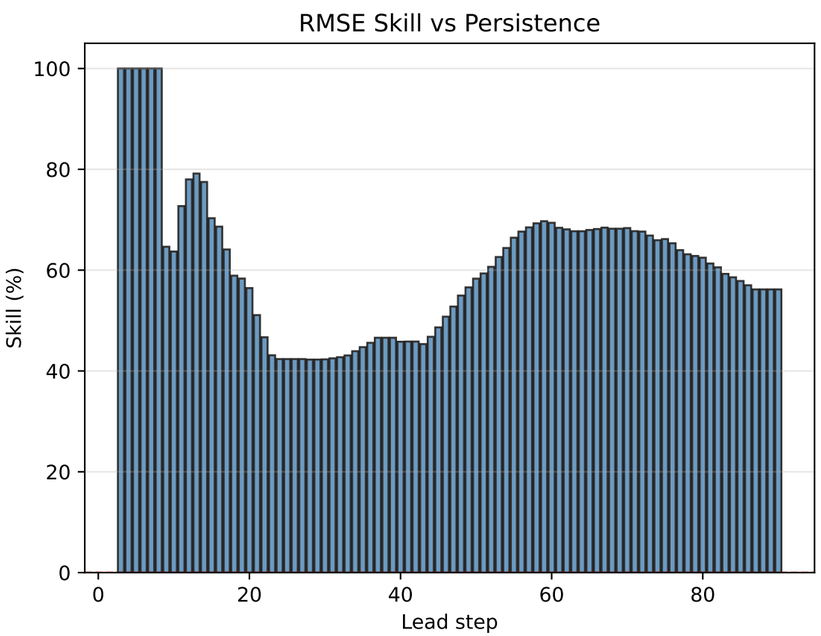


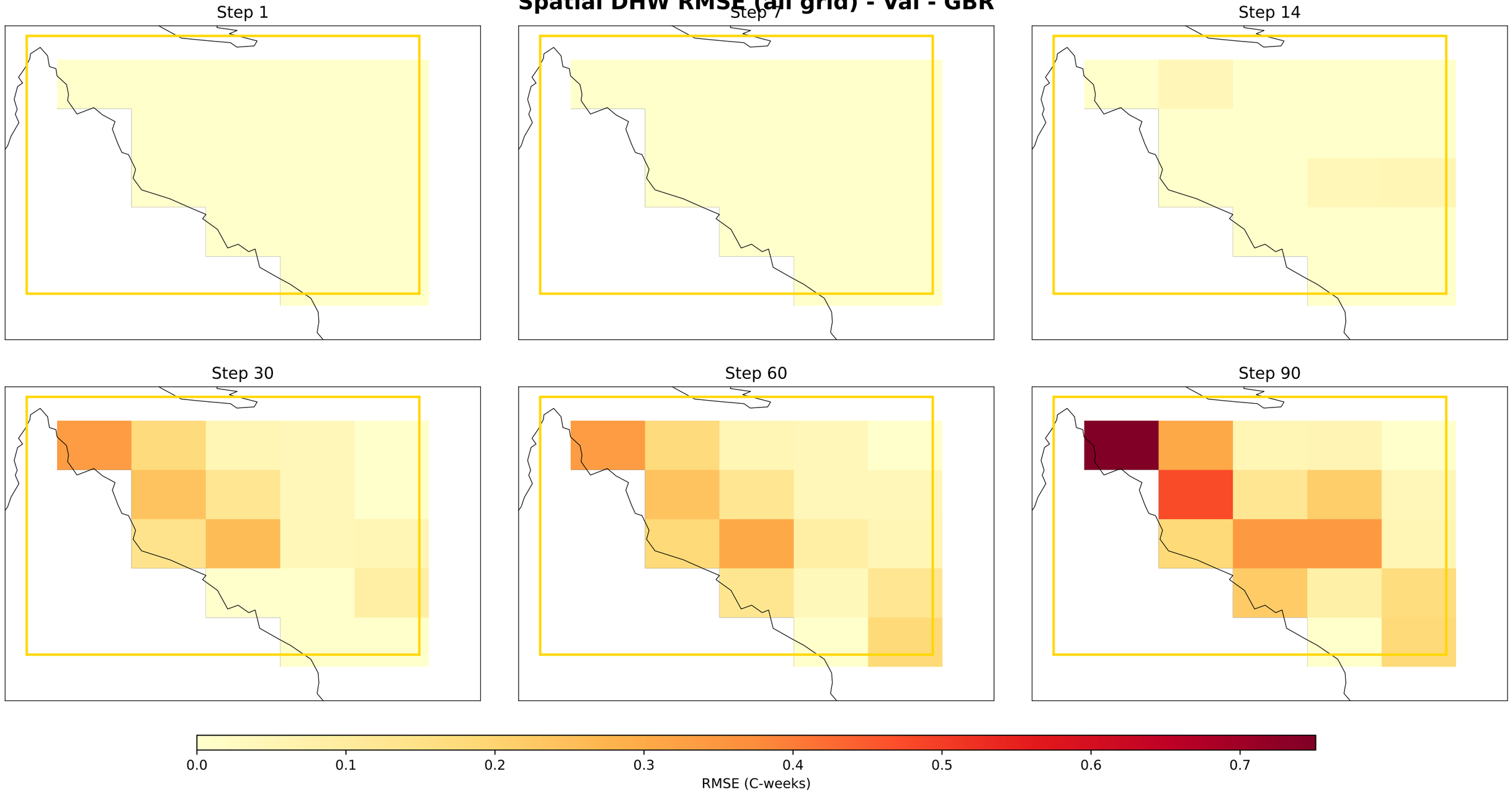


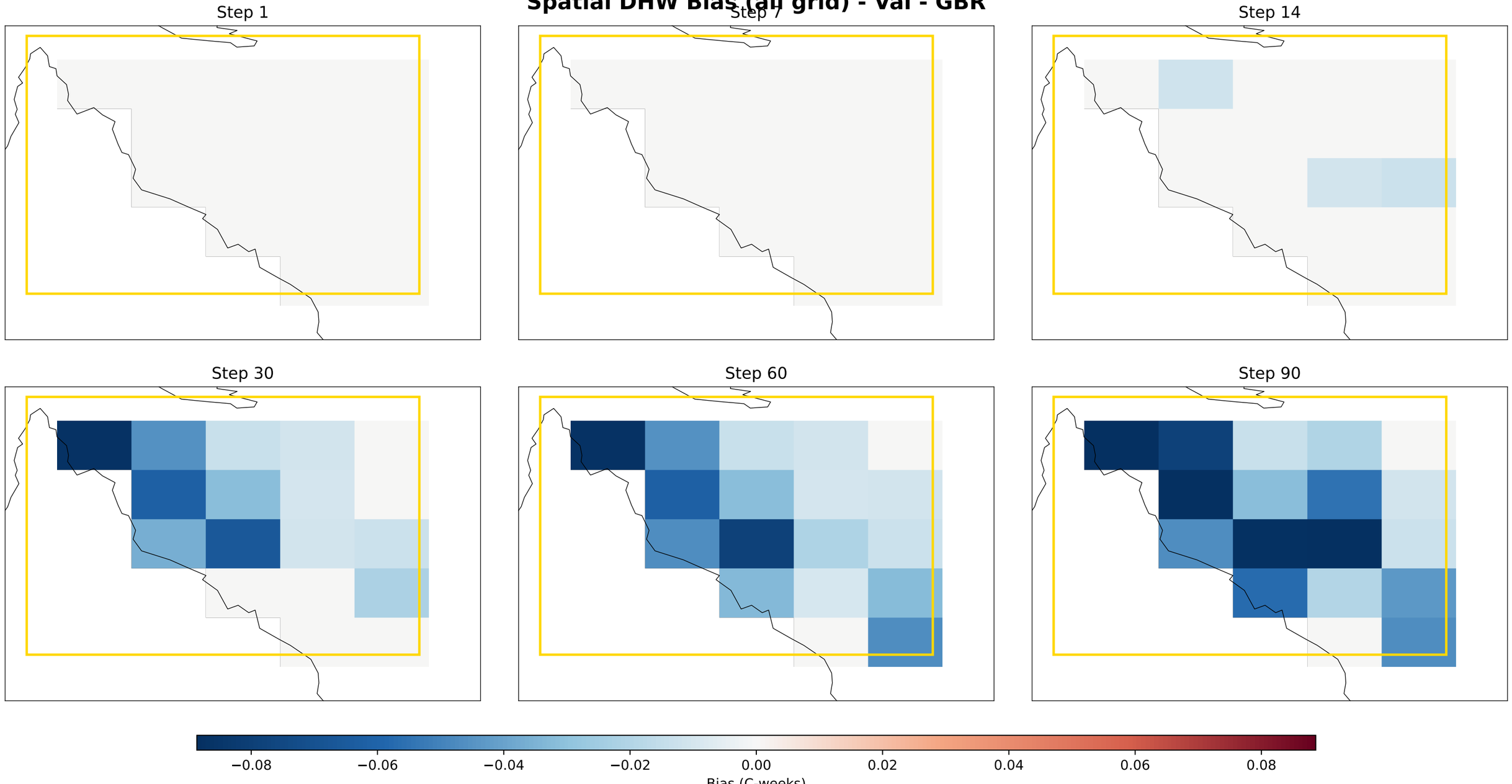


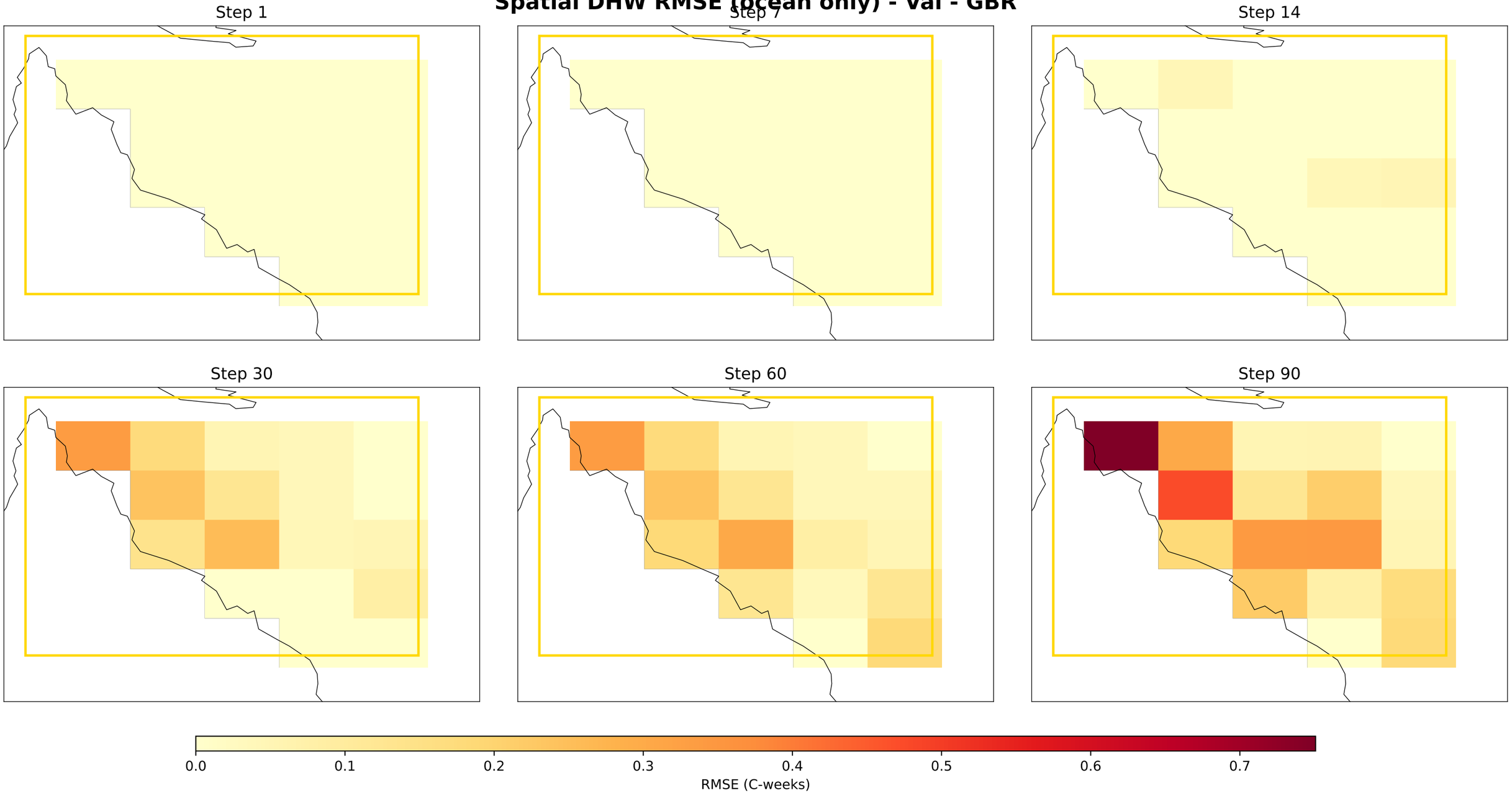


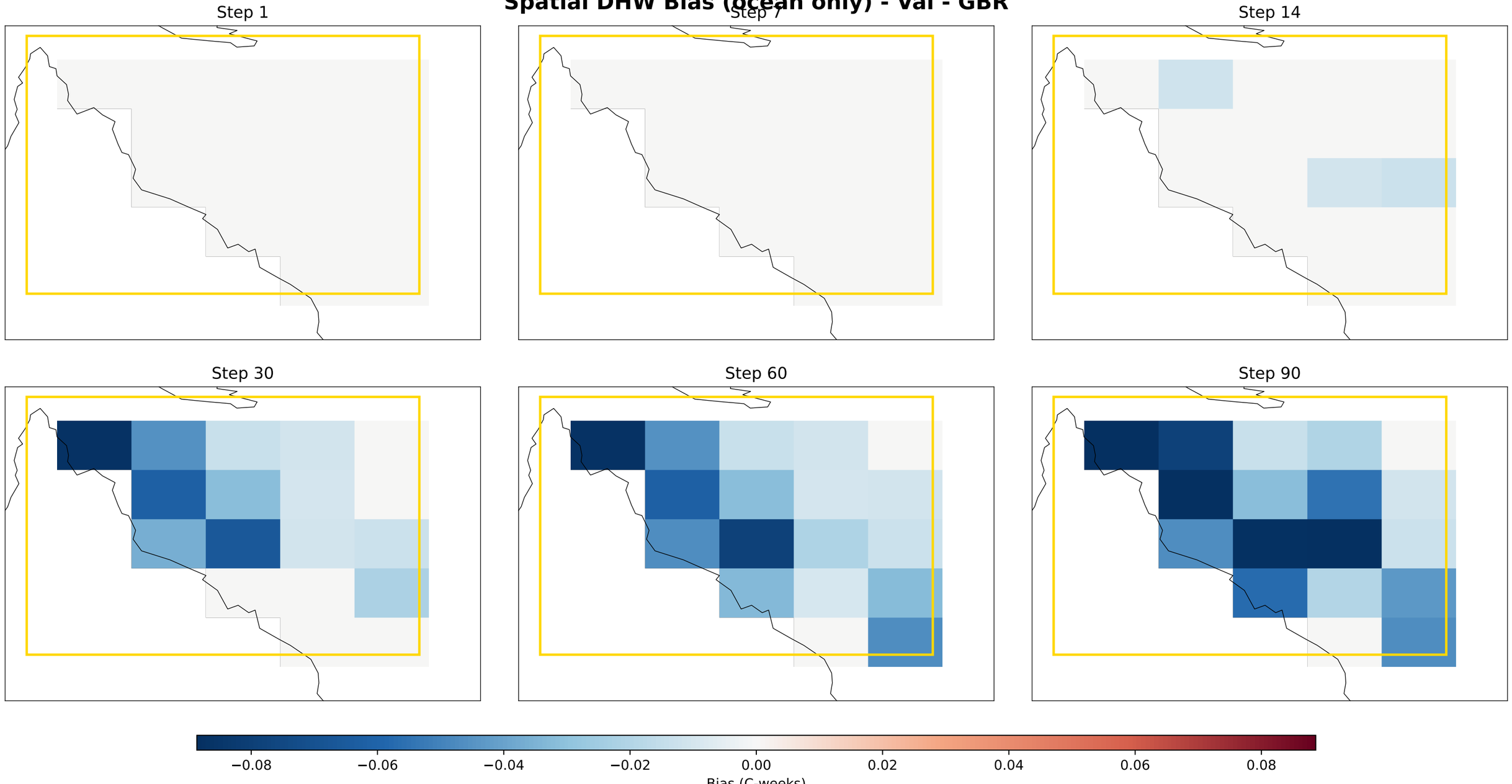





Spatial DHW RMSE (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

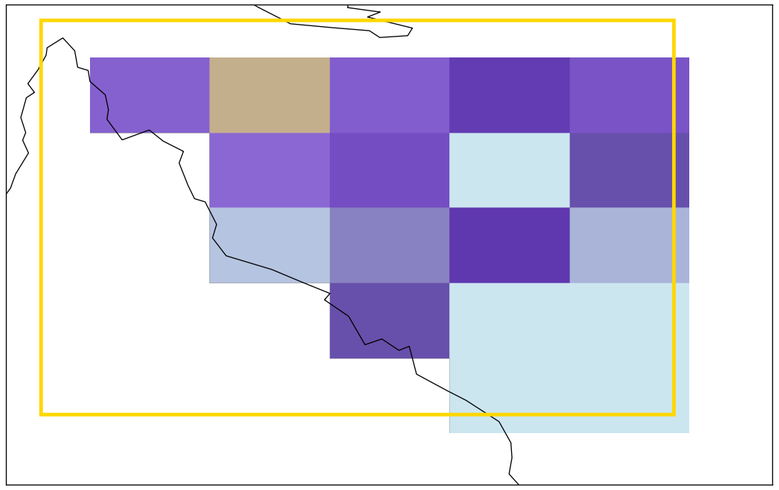

Spatial DHW Bias (all grid) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)


Spatial DHW RMSE (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
RMSE (C-weeks)

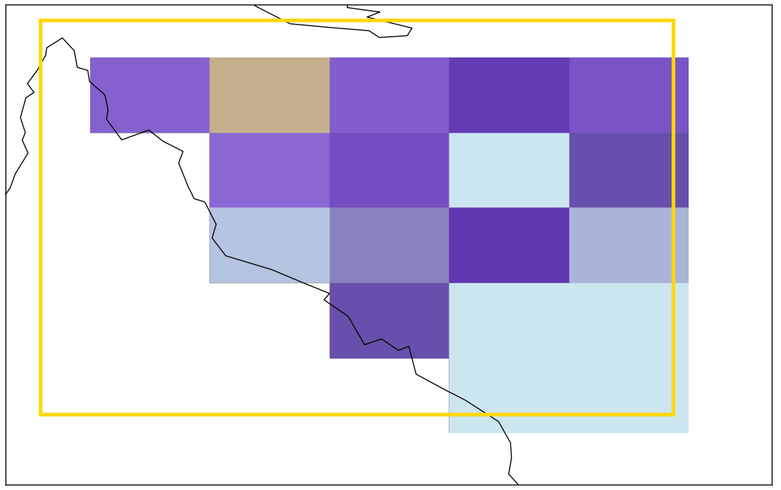

Spatial DHW Bias (ocean only) - Val - GBR
Step 1
Step 7
Step 14
Step 30
Step 60
Step 90
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
Bias (C-weeks)

DHW Chain init: 2016-04-10 - Val - GBR (C-weeks)

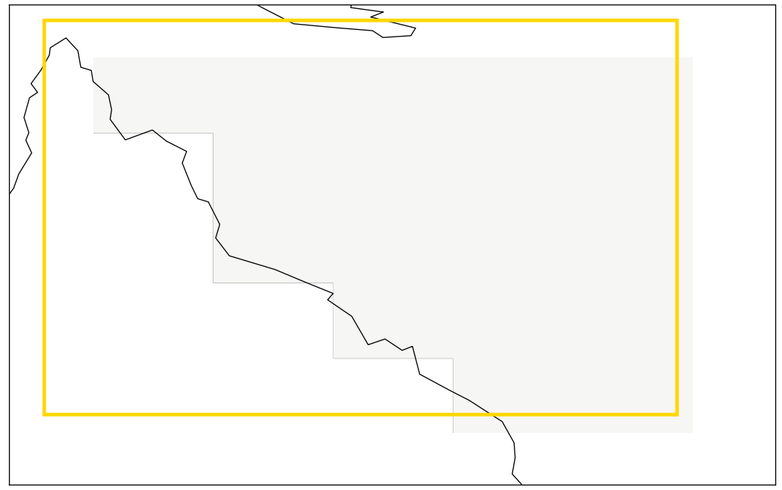
Actual DHW - Step 1



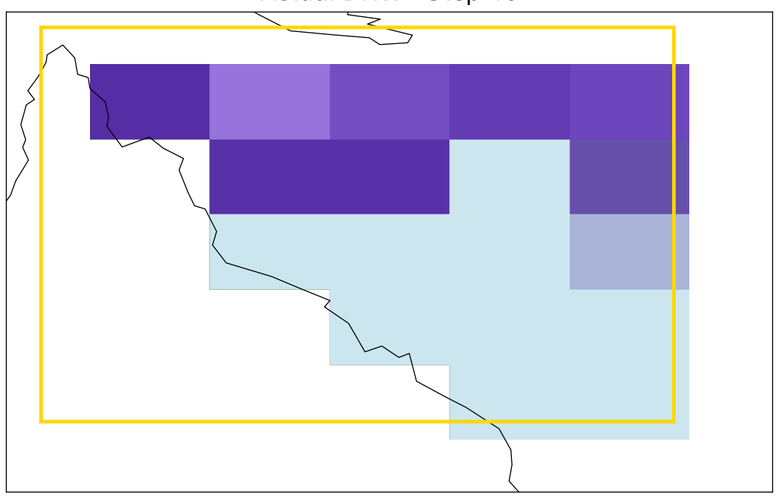
Predicted DHW - Step 1



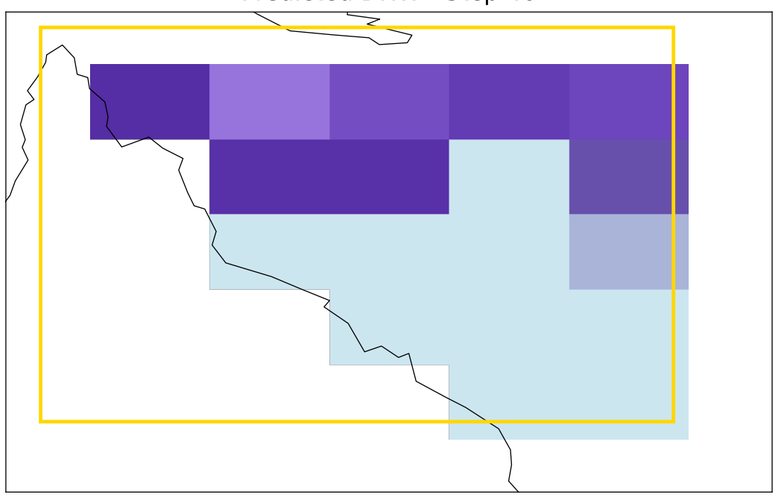
Error (Pred - Actual) - Step 1 [scale ±8]



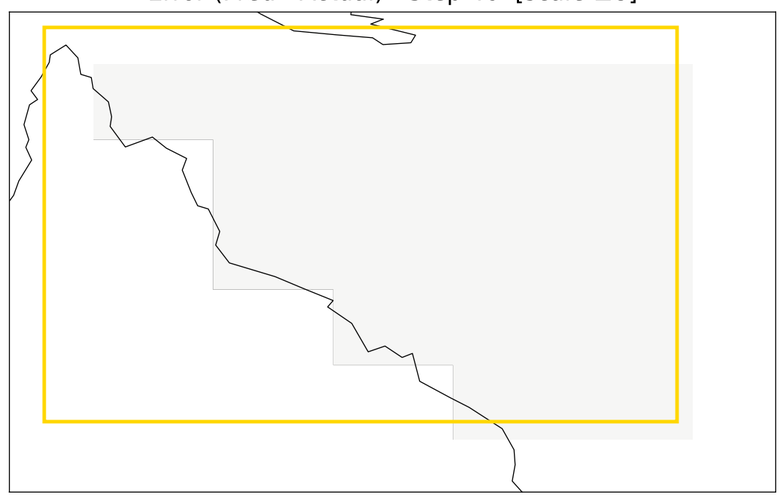
Actual DHW - Step 46



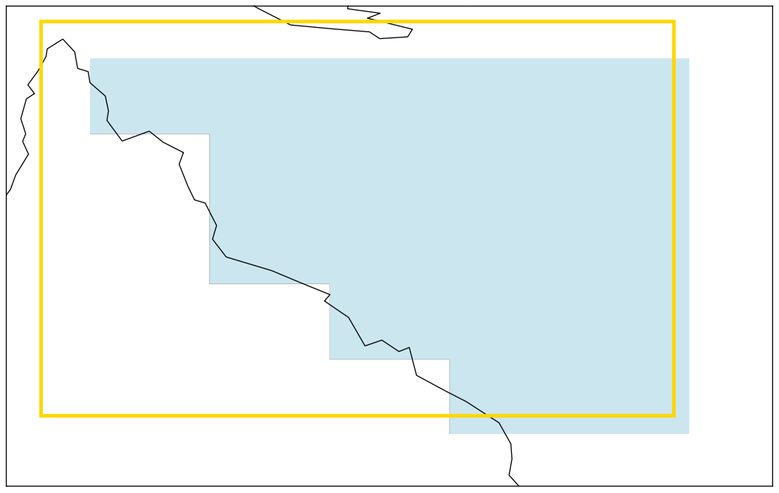
Predicted DHW - Step 46



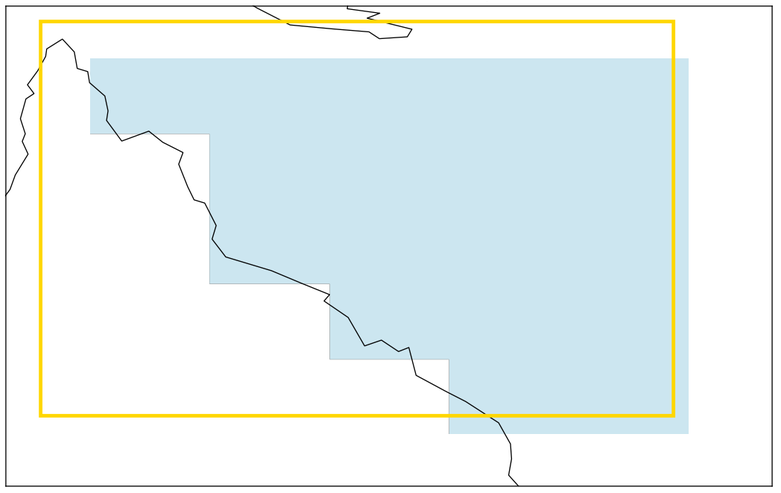
Error (Pred - Actual) - Step 46 [scale ±8]



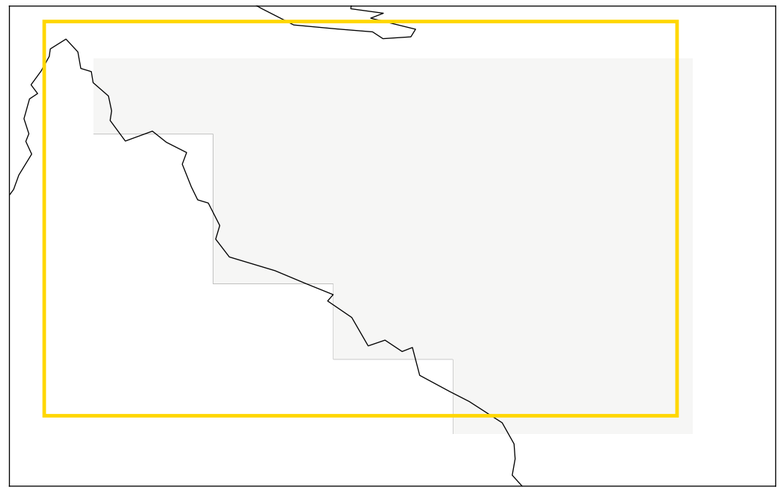
Actual DHW - Step 90

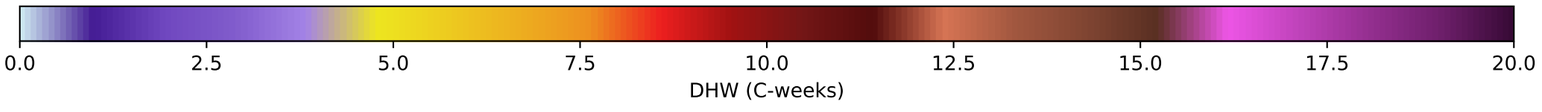


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

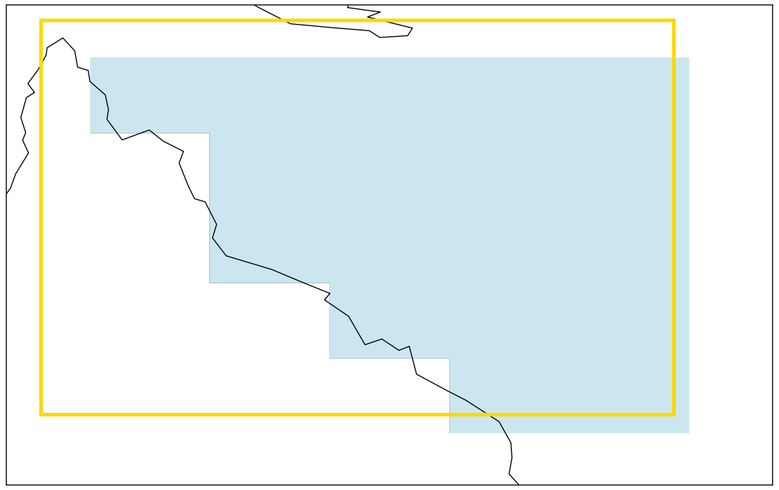




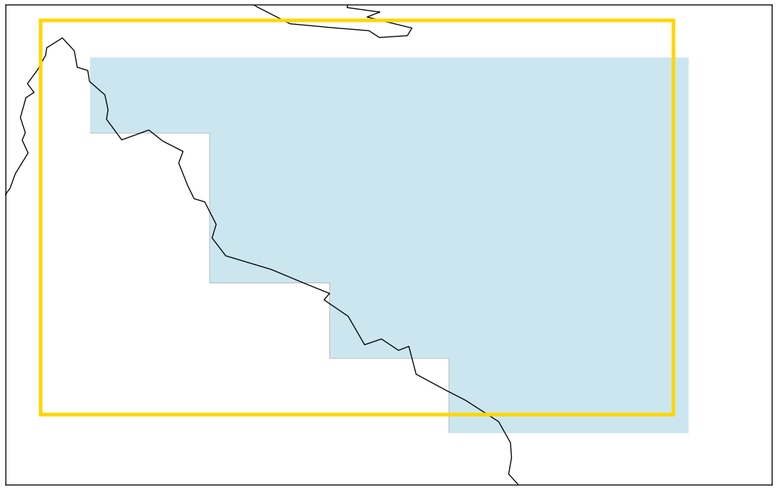


DHW Chain init: 2016-10-07 - Val - GBR (C-weeks)

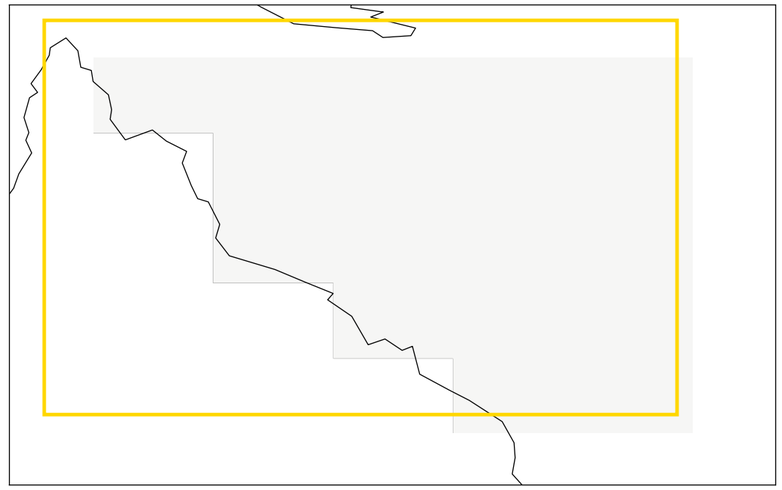
Actual DHW - Step 1



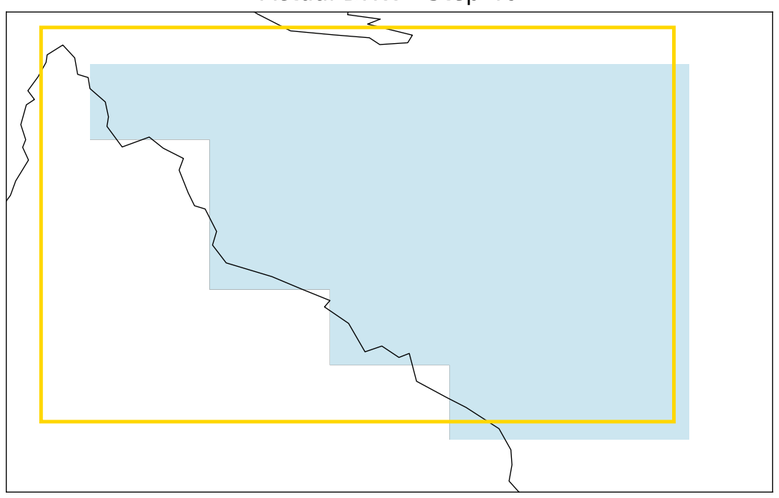
Predicted DHW - Step 1



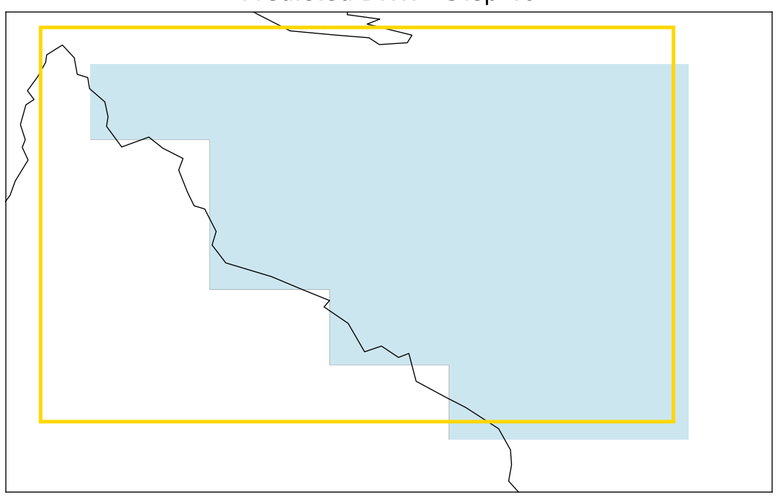
Error (Pred - Actual) - Step 1 [scale ±8]



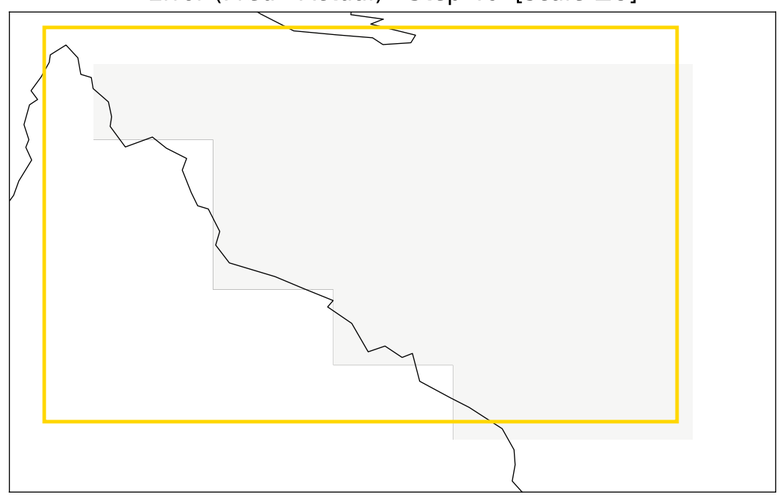
Actual DHW - Step 46



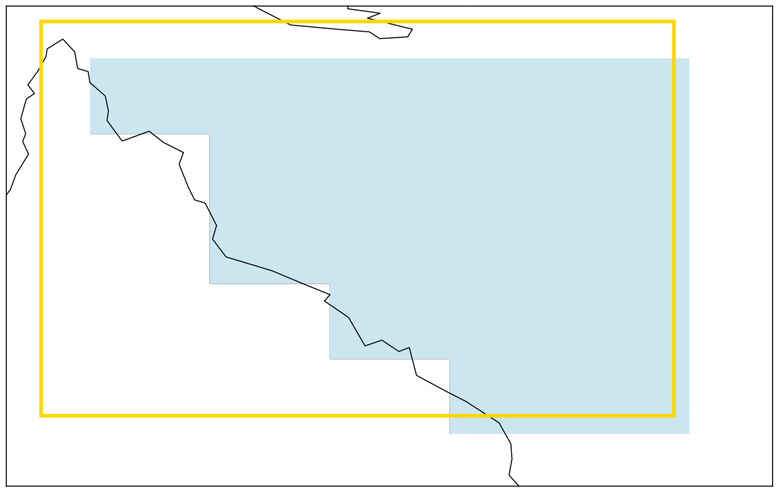
Predicted DHW - Step 46



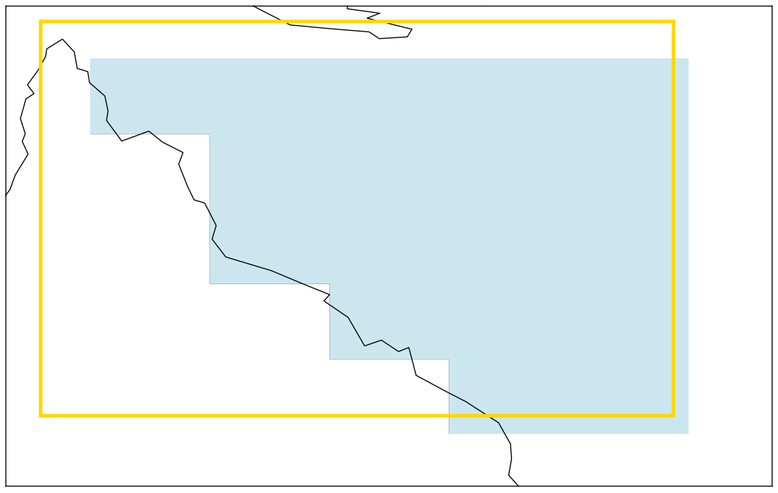
Error (Pred - Actual) - Step 46 [scale ±8]



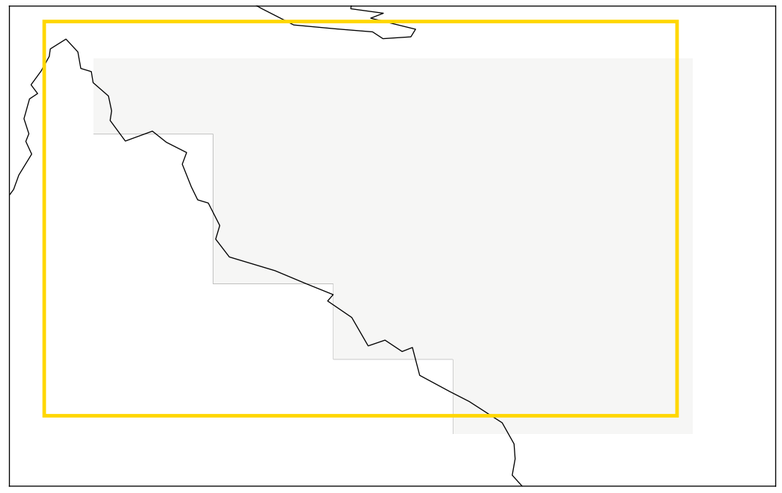
Actual DHW - Step 90

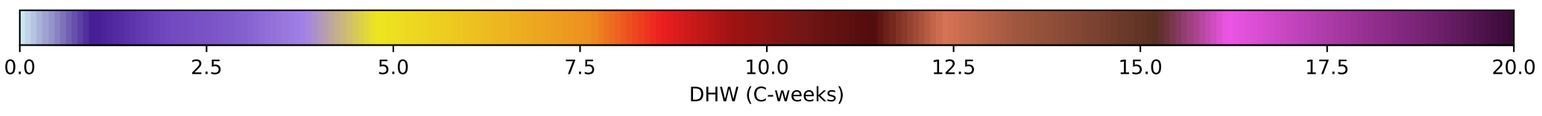


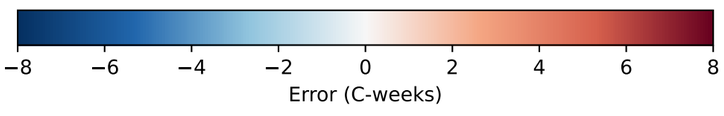
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

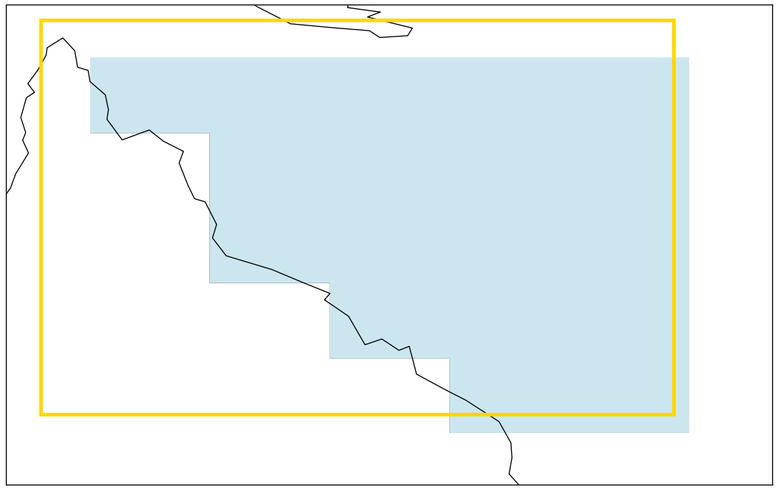




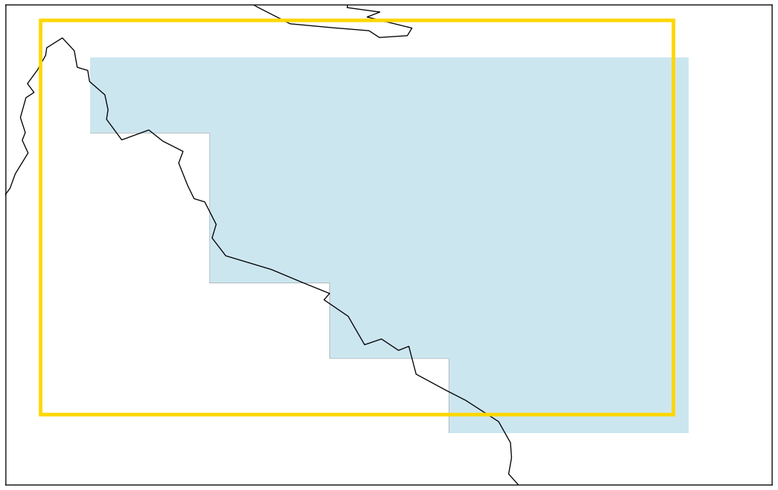


DHW Chain init: 2017-01-05 - Val - GBR (C-weeks)

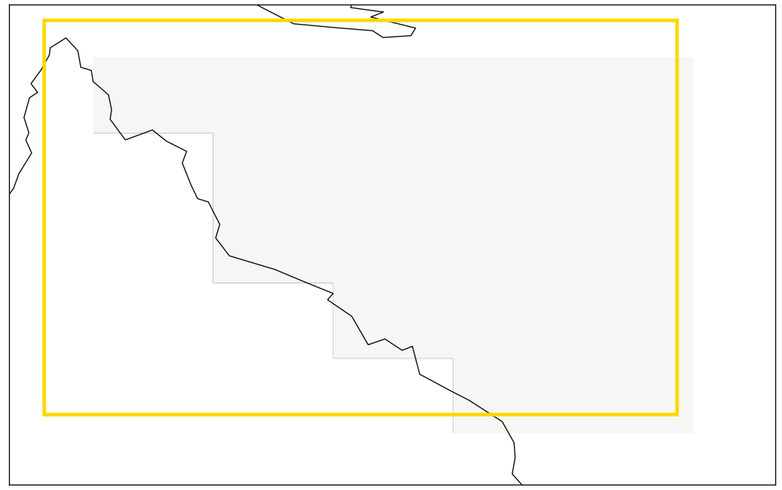
Actual DHW - Step 1



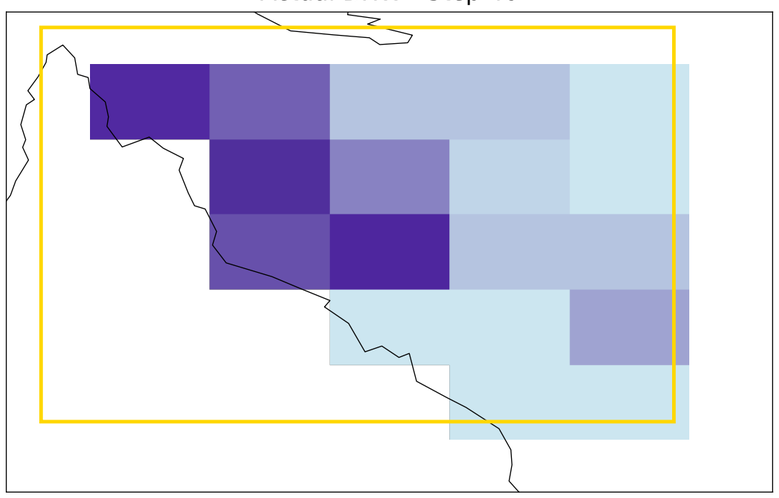
Predicted DHW - Step 1



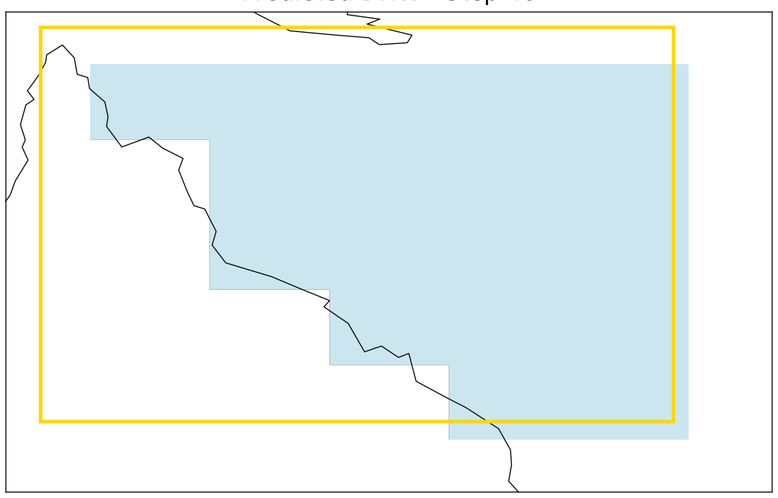
Error (Pred - Actual) - Step 1 [scale ±8]



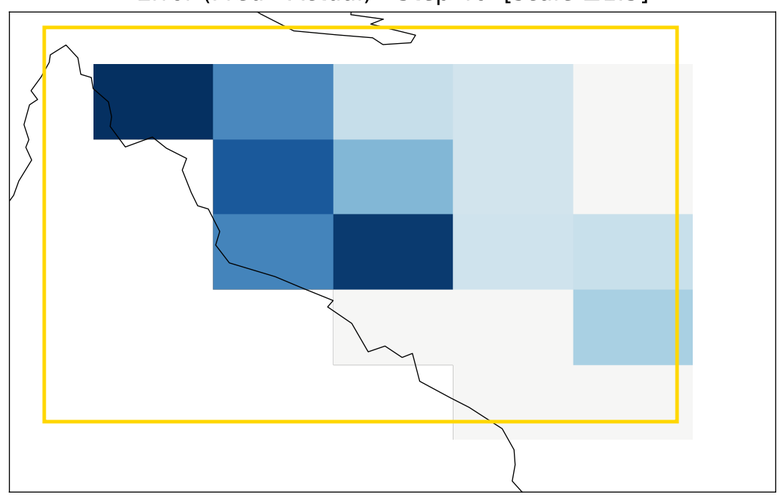
Actual DHW - Step 46



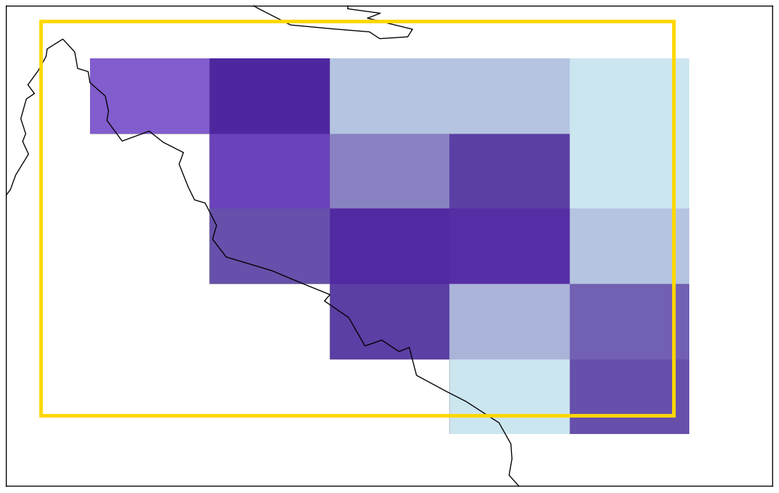
Predicted DHW - Step 46



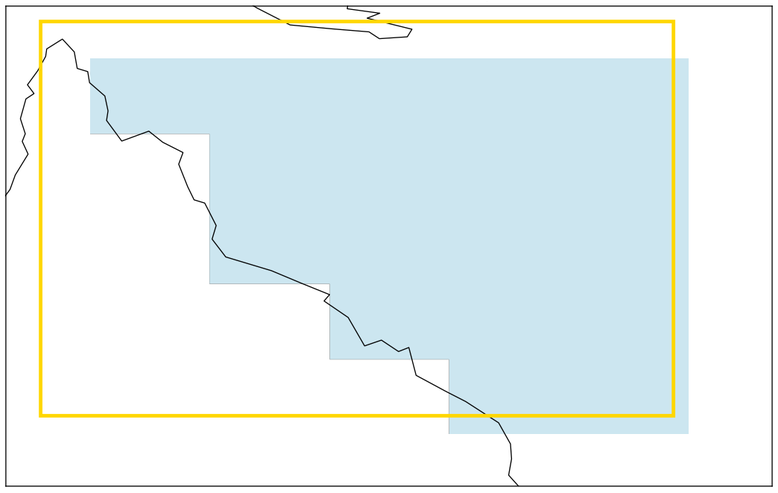
Error (Pred - Actual) - Step 46 [scale ±1.3]



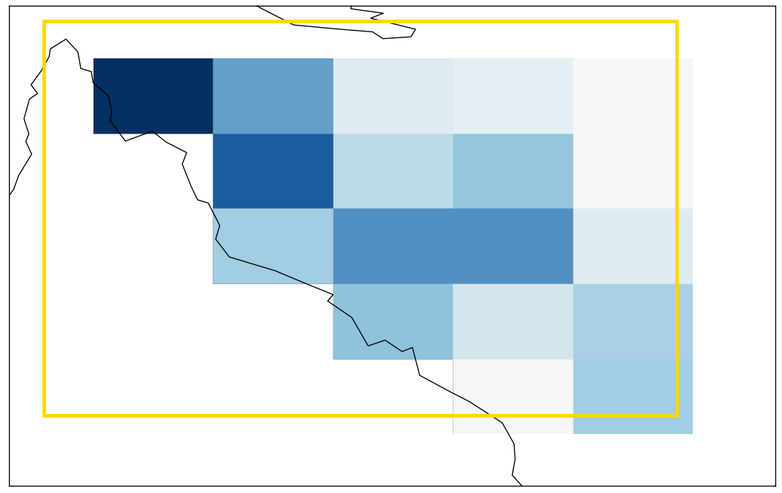
Actual DHW - Step 90

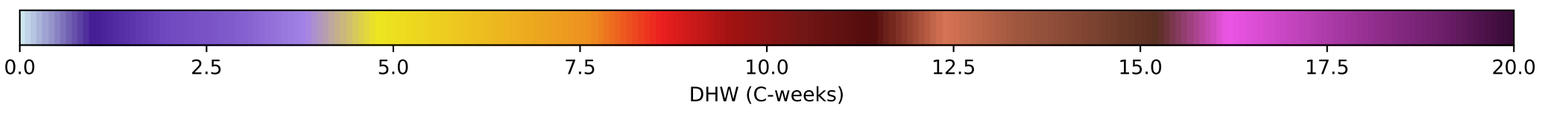


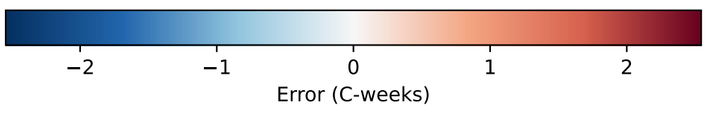
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±2.5]

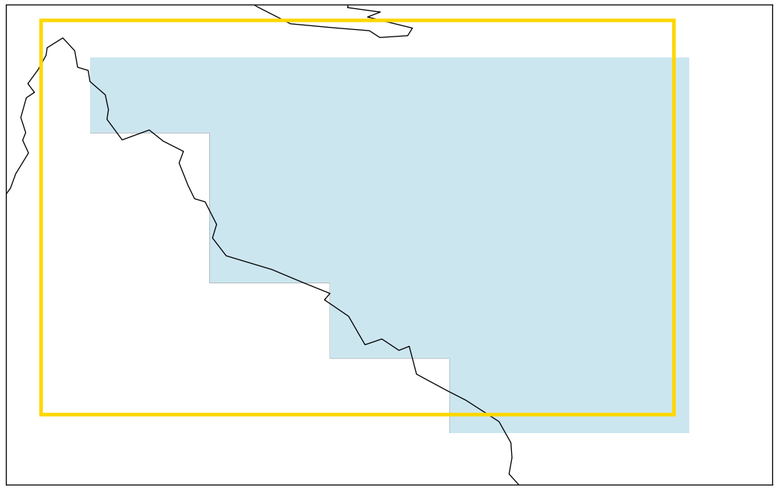




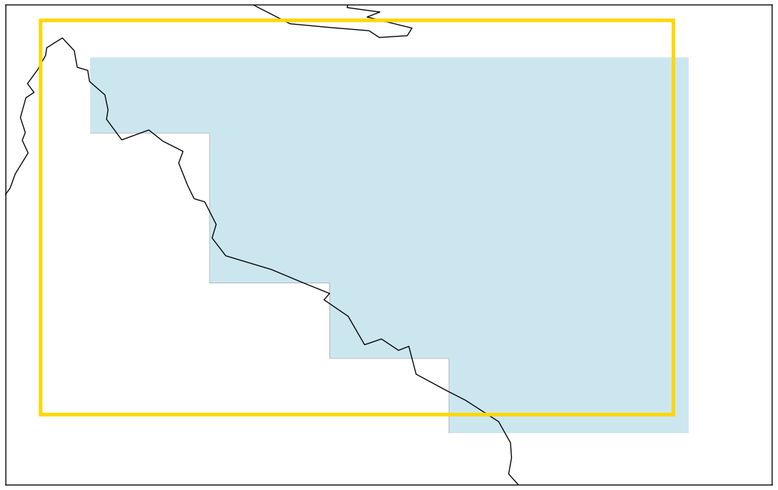


DHW Chain init: 2017-07-04 - Val - GBR (C-weeks)

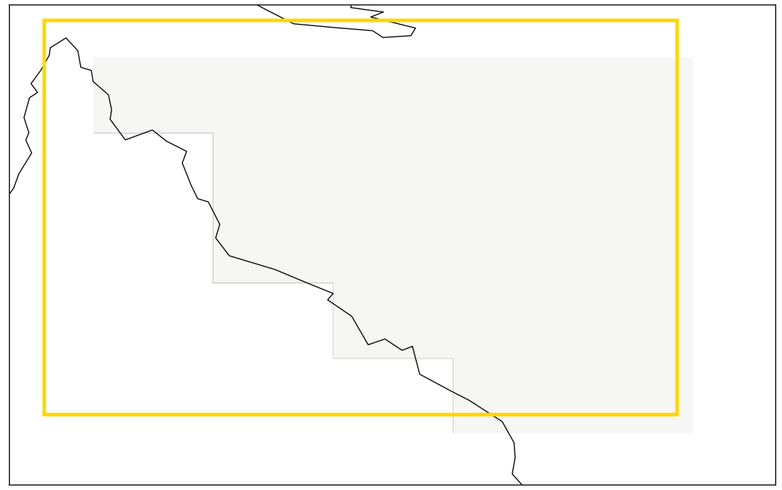
Actual DHW - Step 1



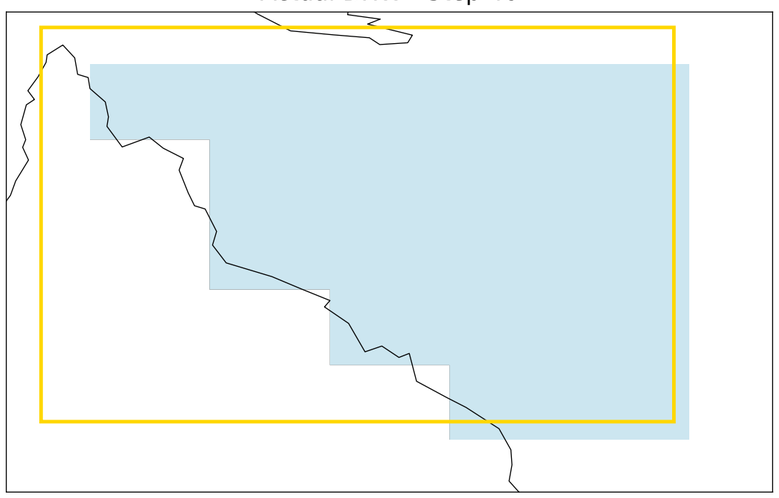
Predicted DHW - Step 1



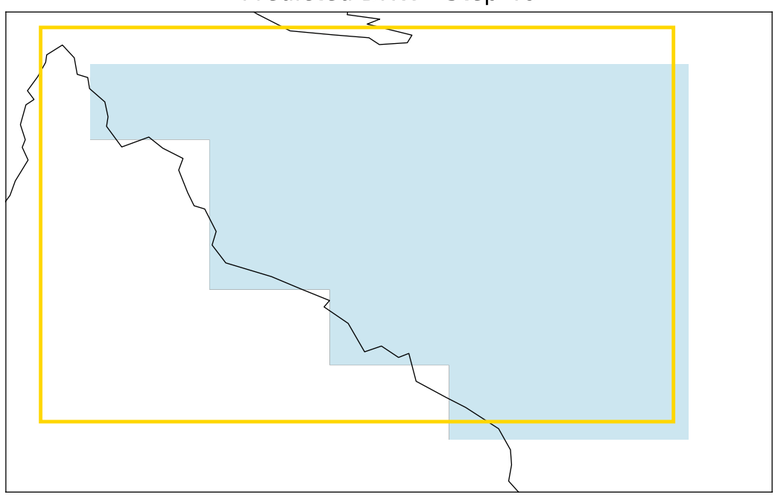
Error (Pred - Actual) - Step 1 [scale ±8]



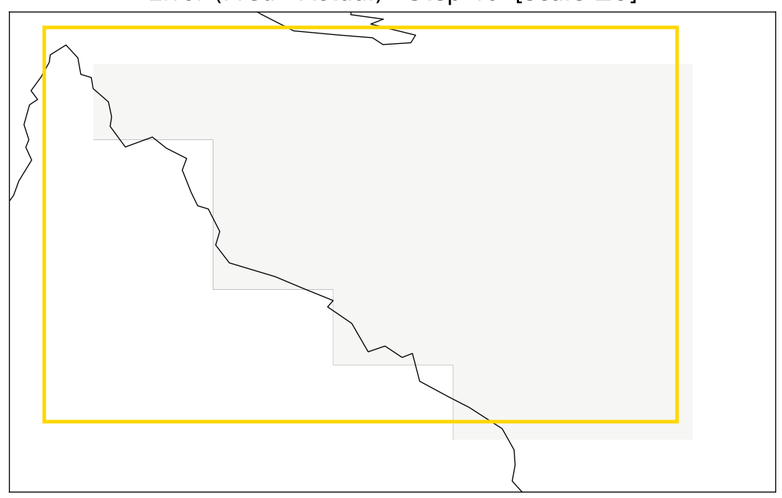
Actual DHW - Step 46



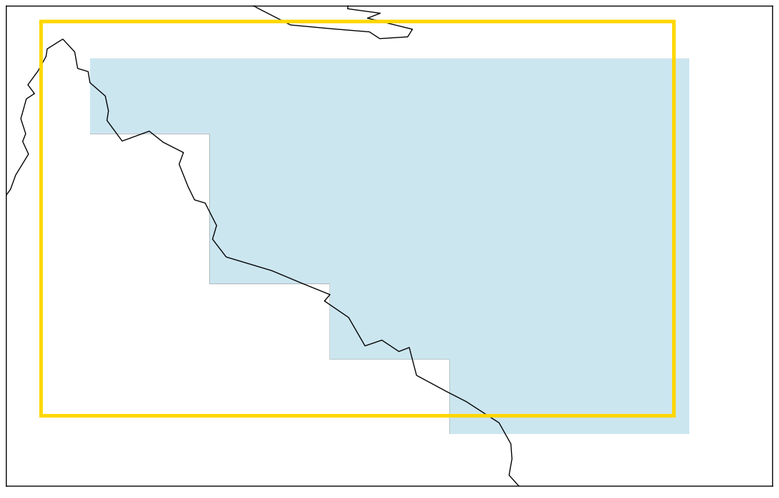
Predicted DHW - Step 46



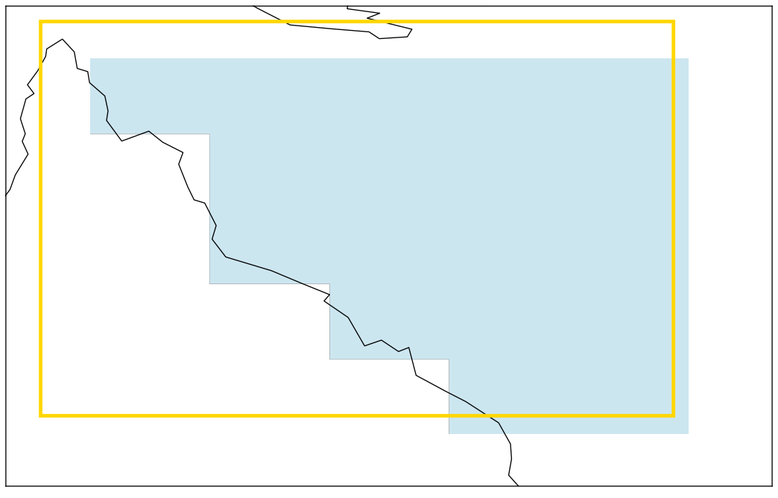
Error (Pred - Actual) - Step 46 [scale ±8]



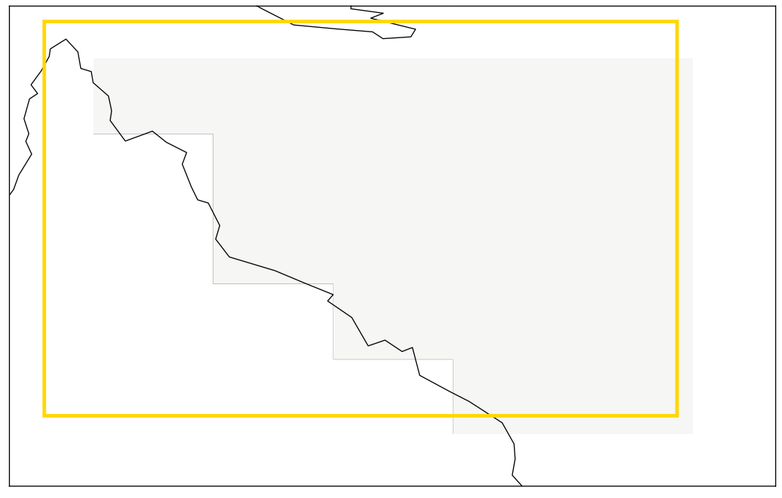
Actual DHW - Step 90

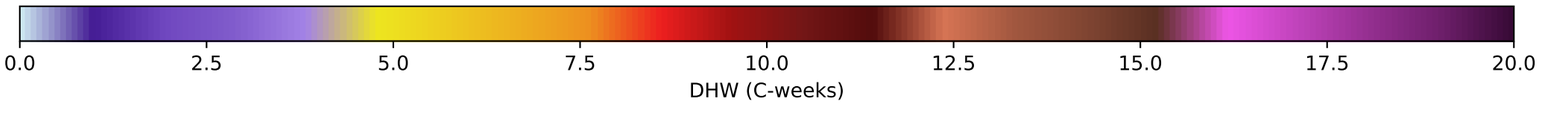


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

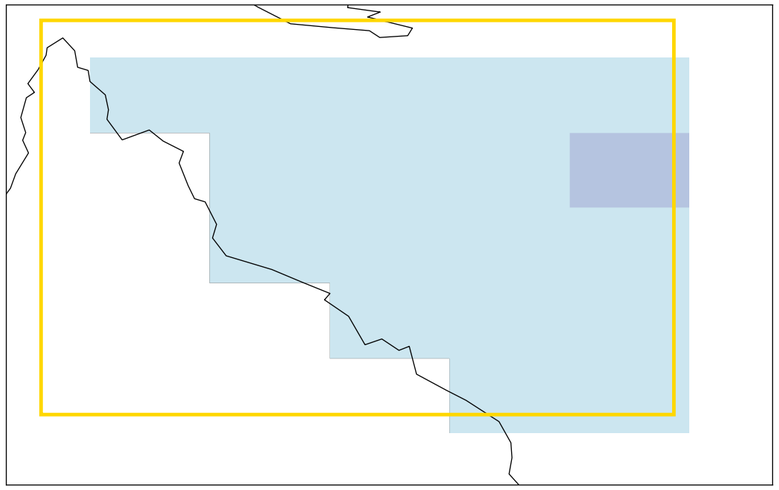




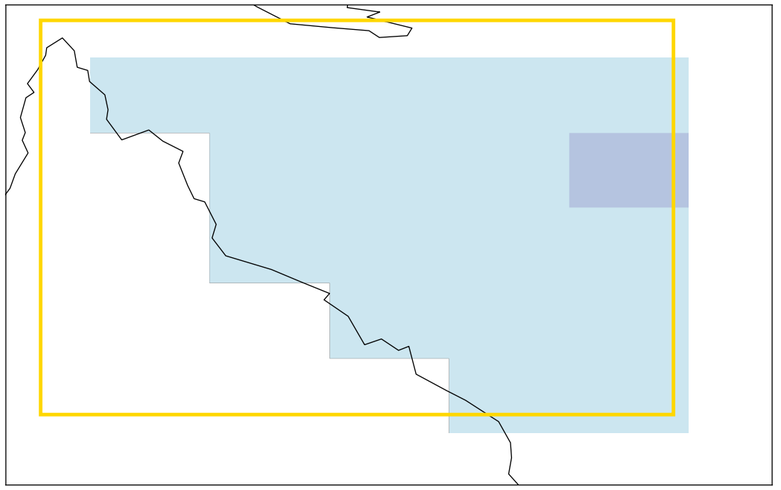


DHW Chain init: 2018-03-31 - Val - GBR (C-weeks)

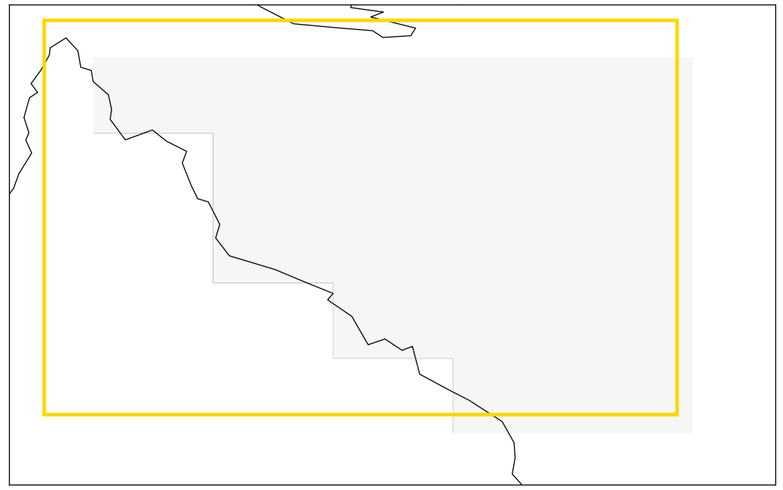
Actual DHW - Step 1



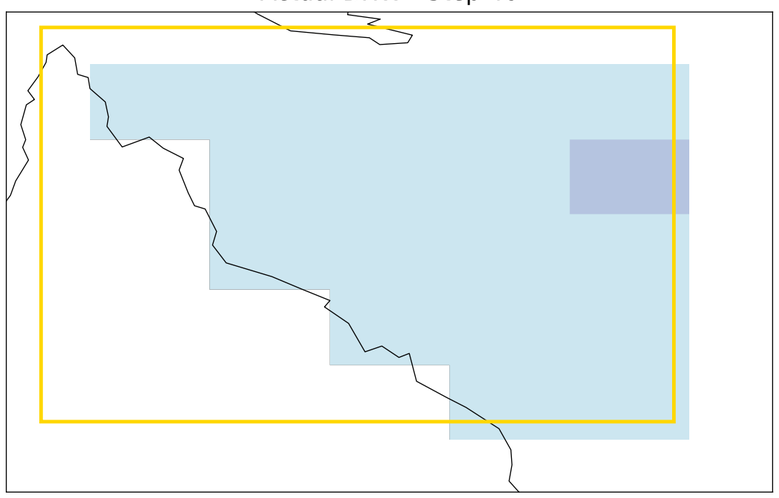
Predicted DHW - Step 1



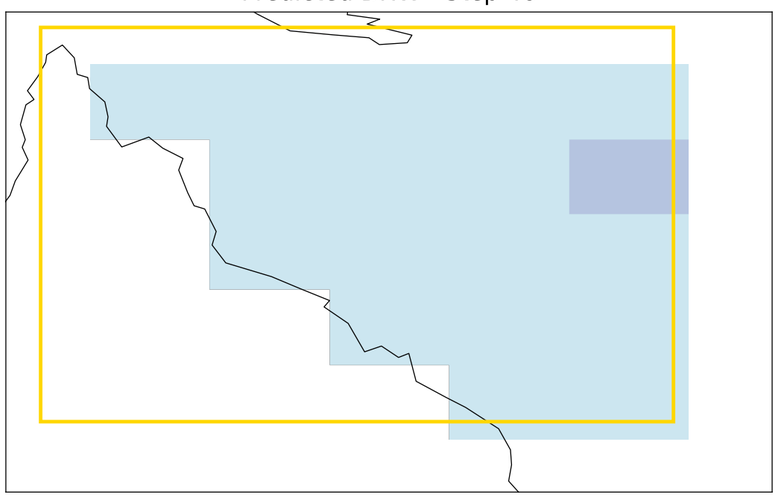
Error (Pred - Actual) - Step 1 [scale ±8]



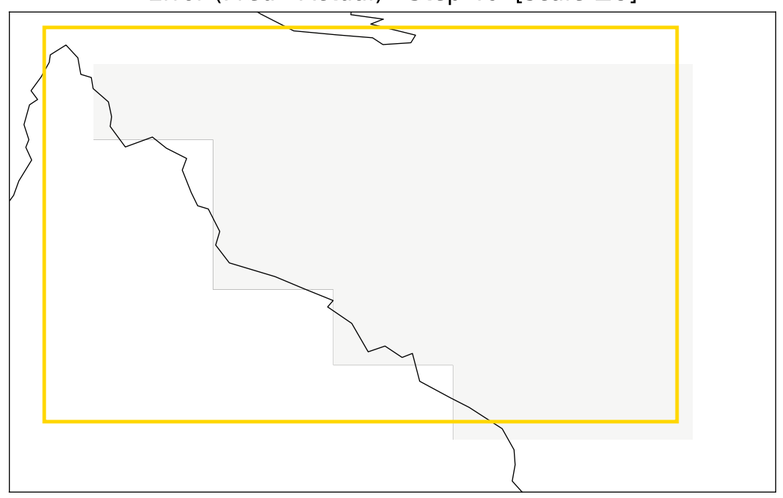
Actual DHW - Step 46



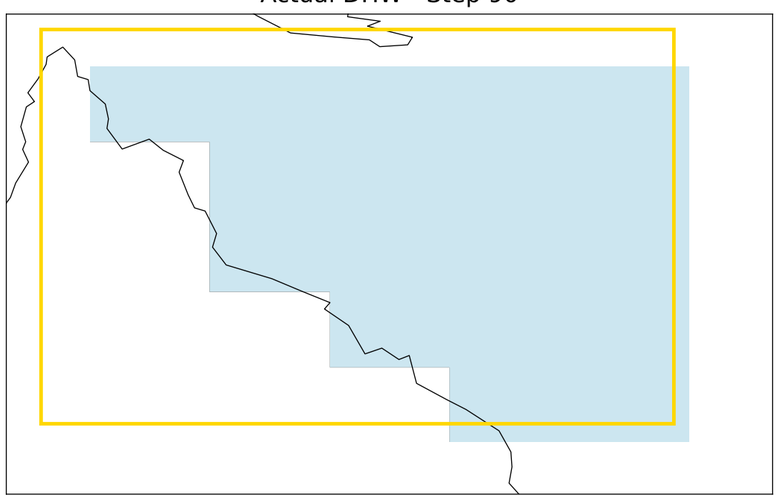
Predicted DHW - Step 46



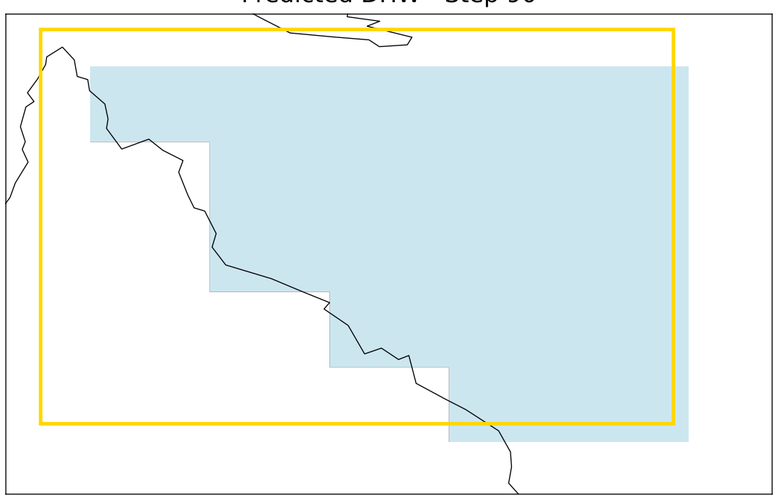
Error (Pred - Actual) - Step 46 [scale ±8]



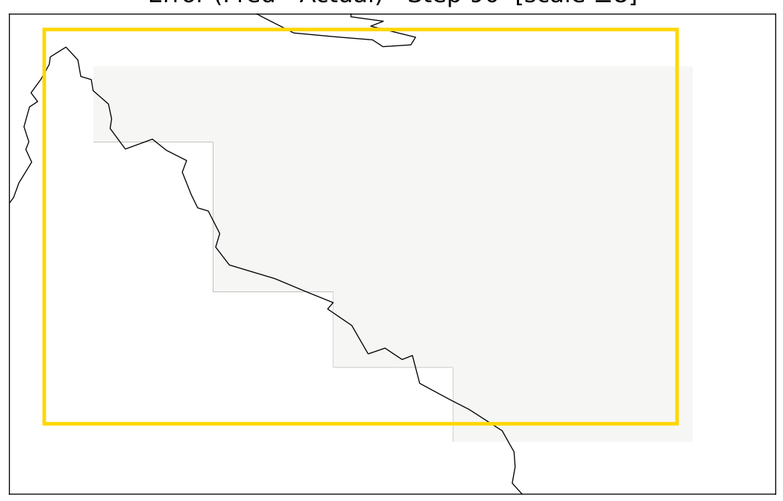
Actual DHW - Step 90

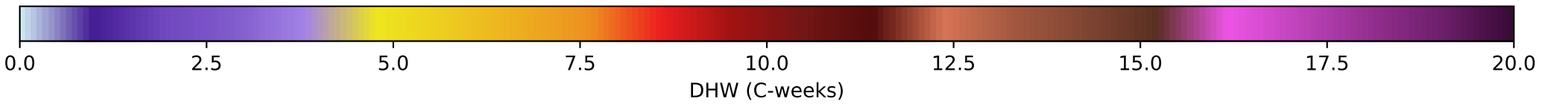


Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

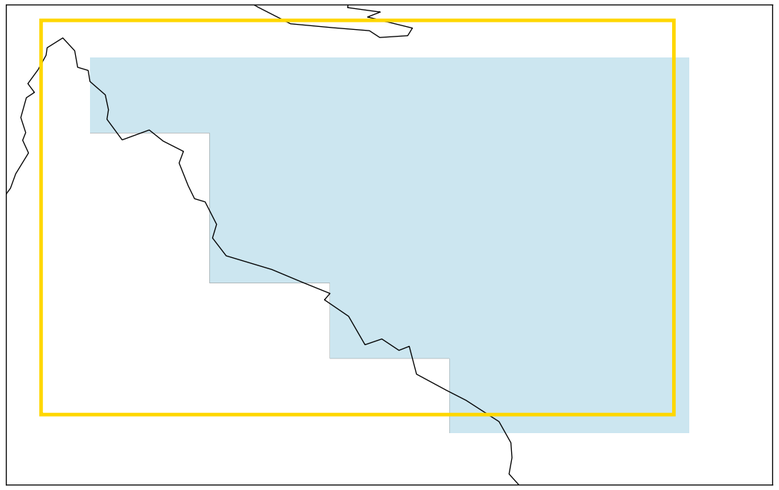




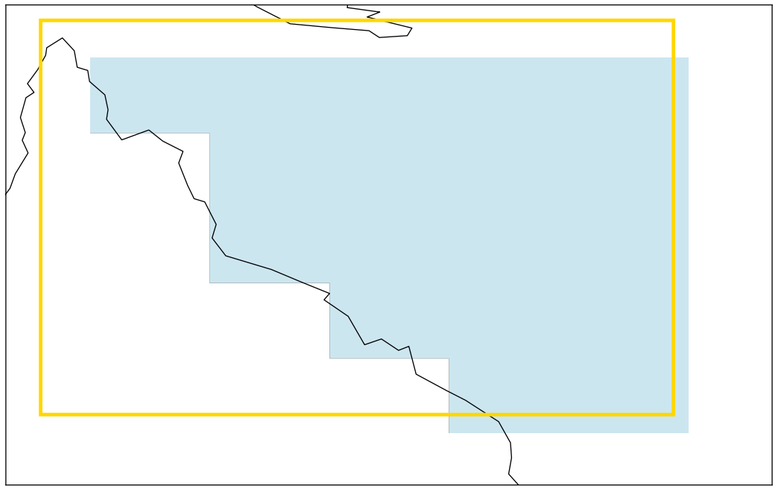


DHW Chain init: 2018-06-29 - Val - GBR (C-weeks)

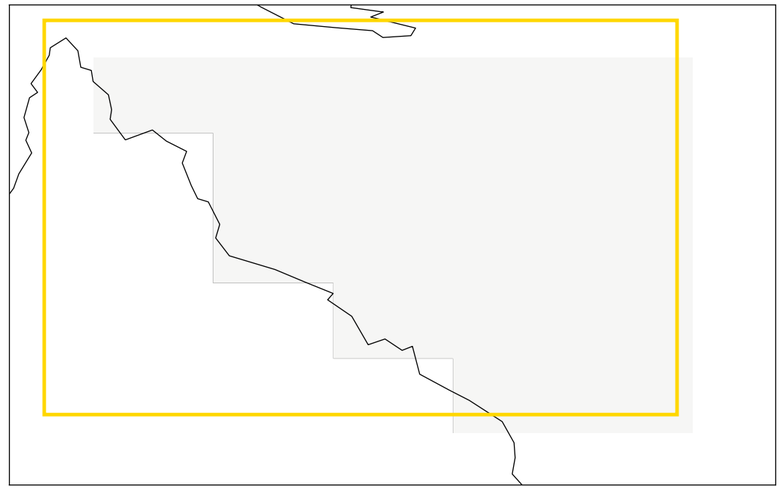
Actual DHW - Step 1



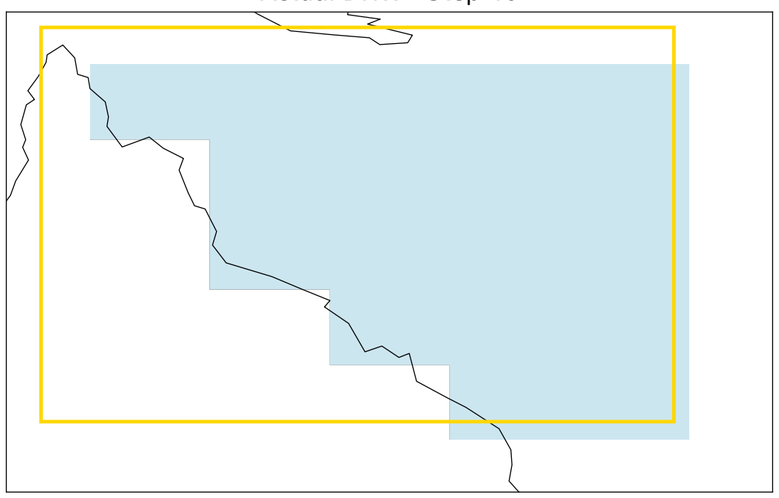
Predicted DHW - Step 1



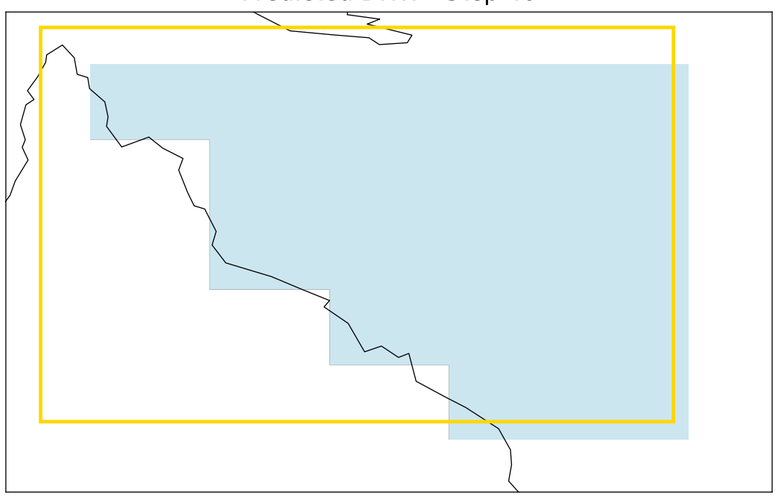
Error (Pred - Actual) - Step 1 [scale ±8]



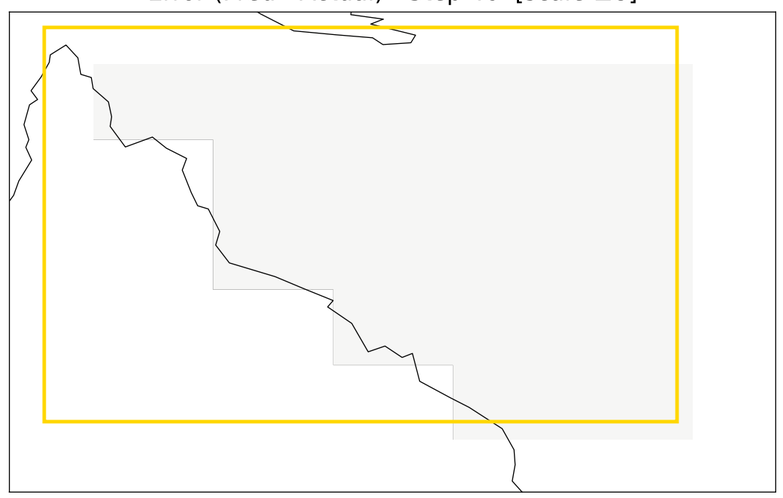
Actual DHW - Step 46



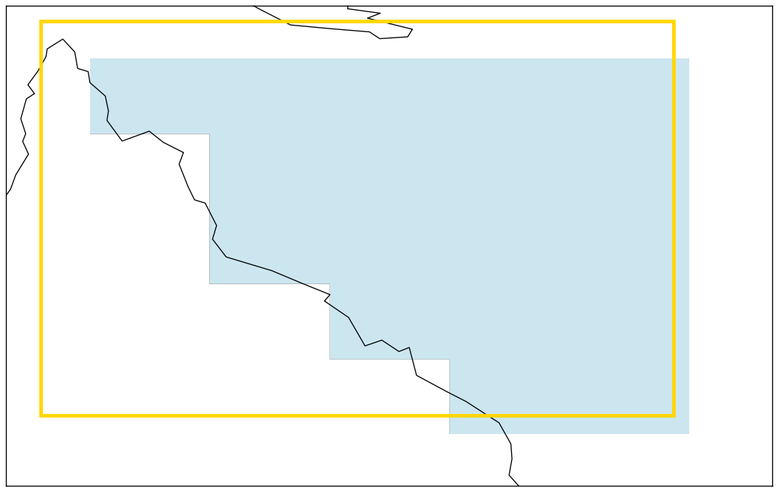
Predicted DHW - Step 46



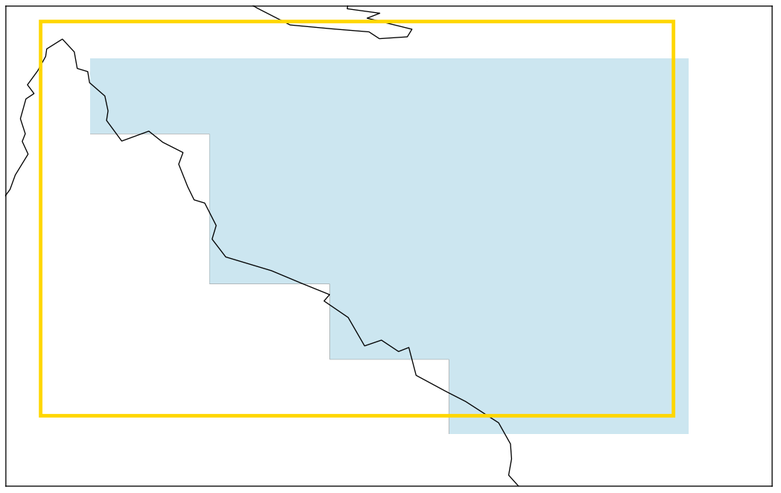
Error (Pred - Actual) - Step 46 [scale ±8]



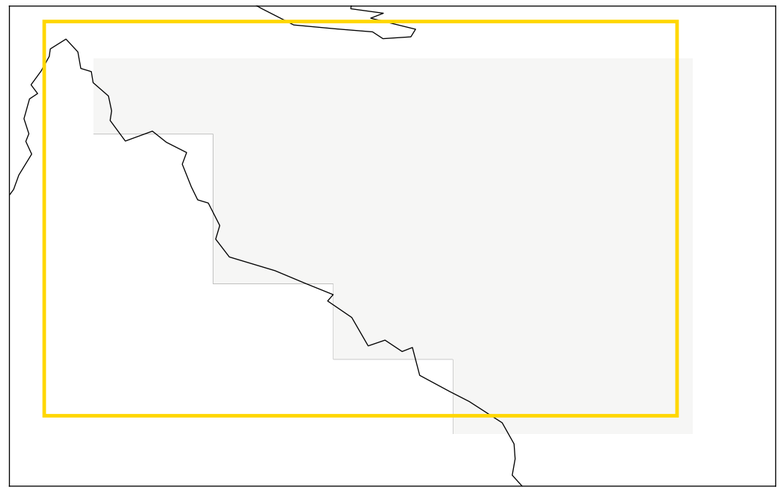
Actual DHW - Step 90

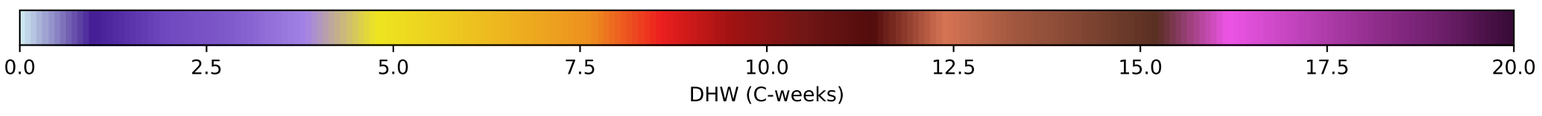


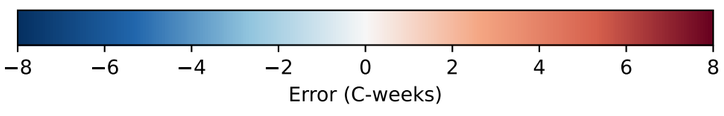
Predicted DHW - Step 90



Error (Pred - Actual) - Step 90 [scale ±8]

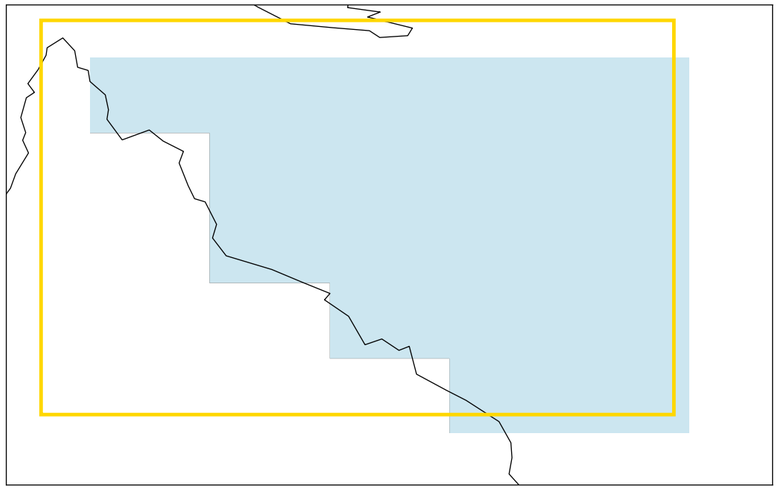




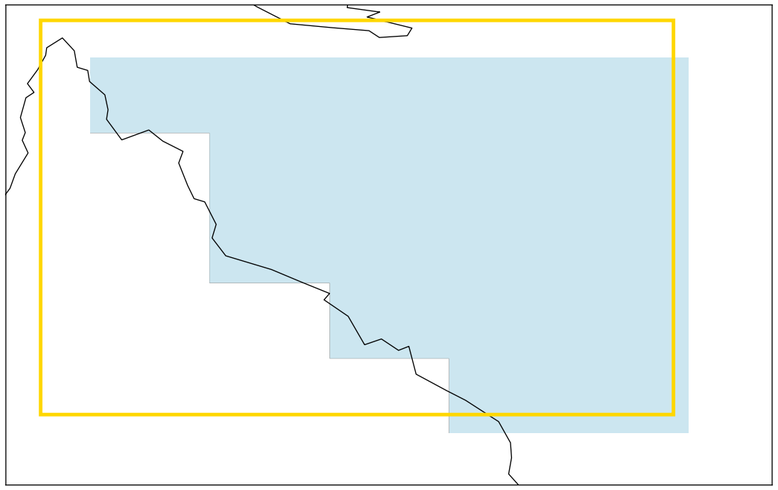


DHW Chain init: 2019-09-22 - Val - GBR (C-weeks)

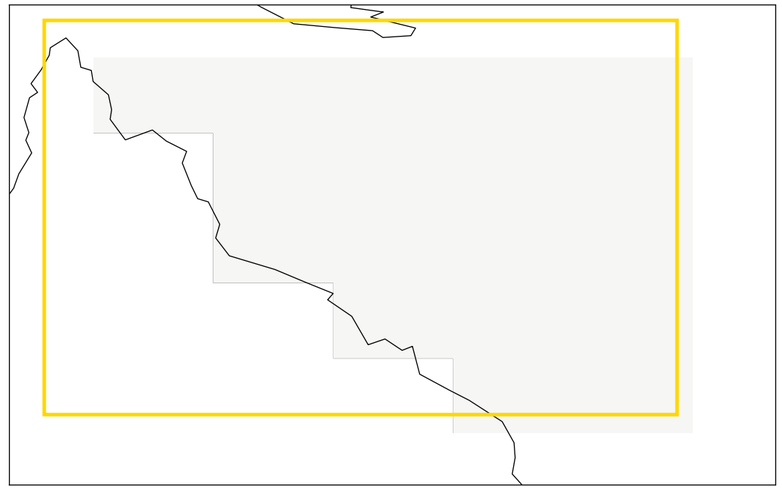
Actual DHW - Step 1



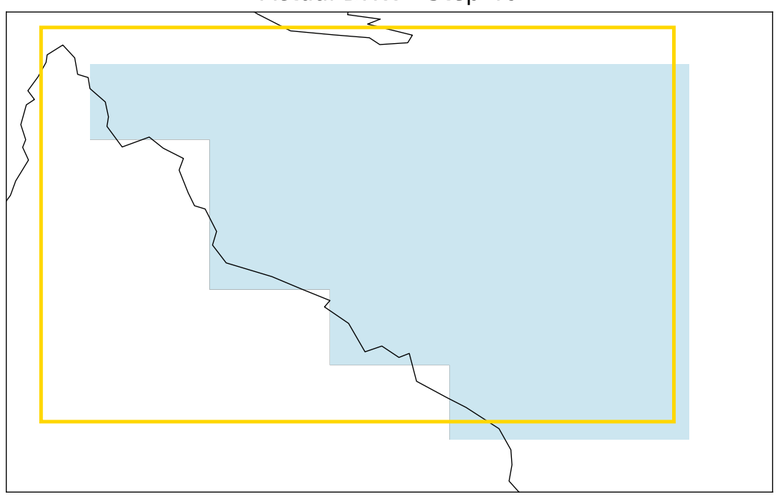
Predicted DHW - Step 1



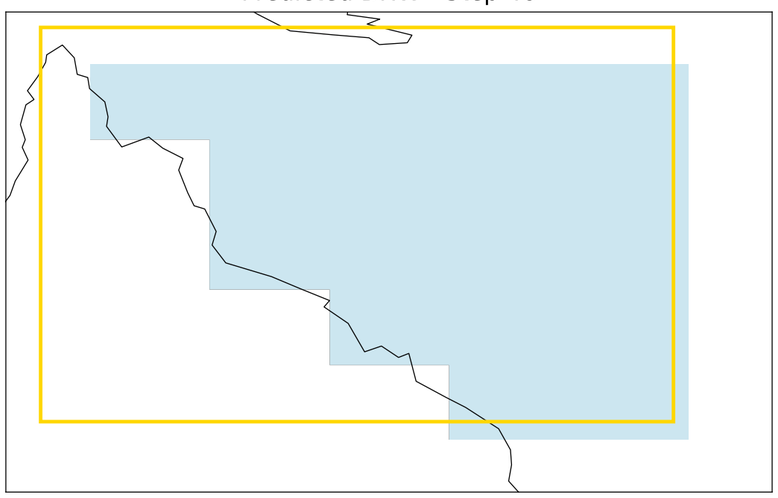
Error (Pred - Actual) - Step 1 [scale ±8]



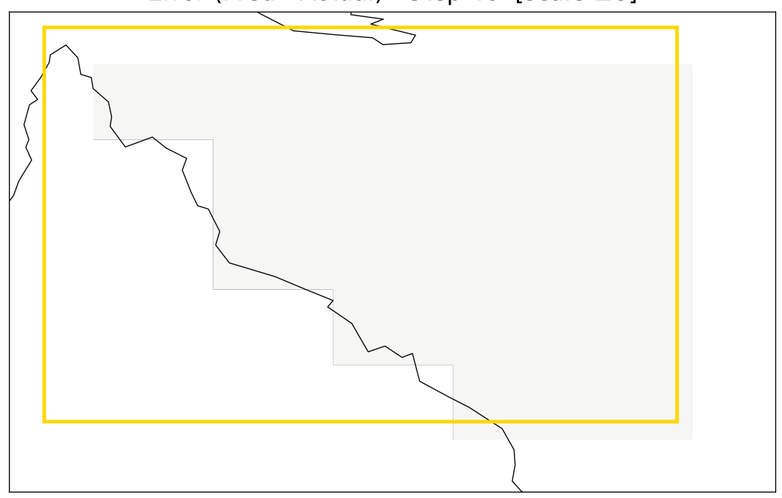
Actual DHW - Step 46



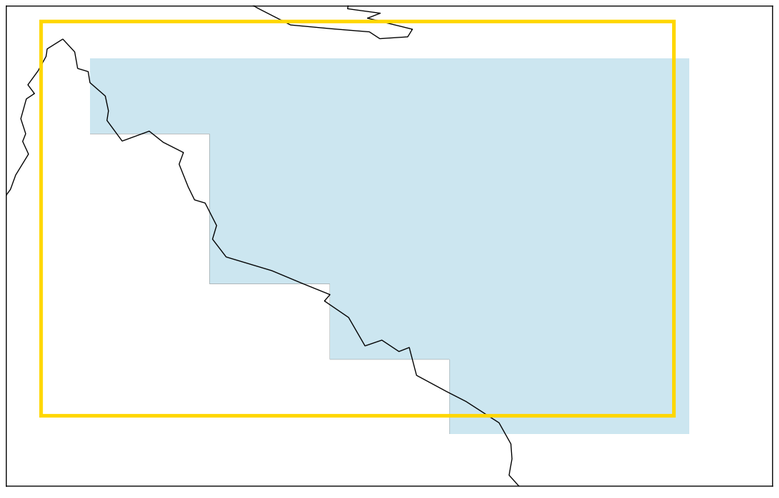
Predicted DHW - Step 46



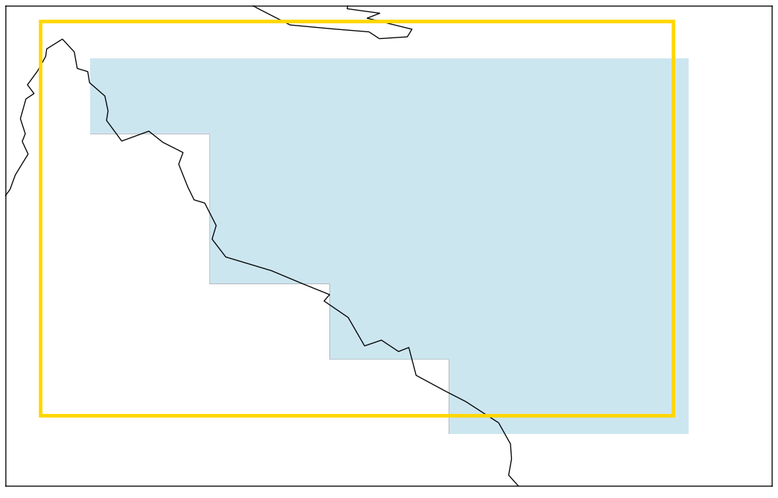
Error (Pred - Actual) - Step 46 [scale ±8]



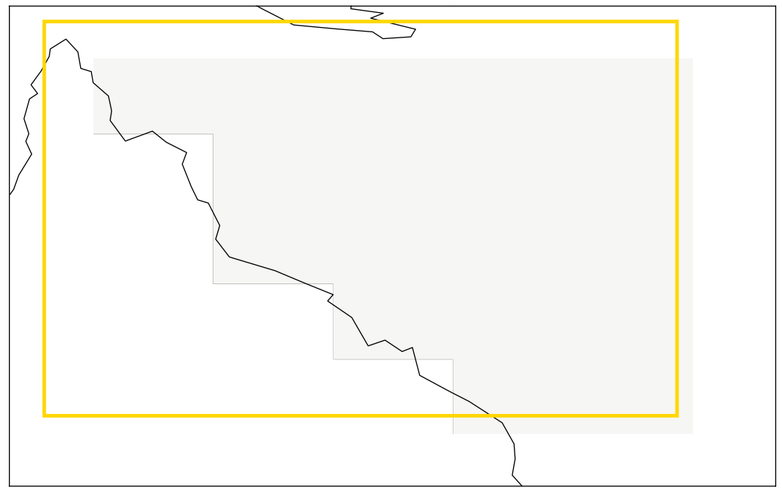
Actual DHW - Step 90

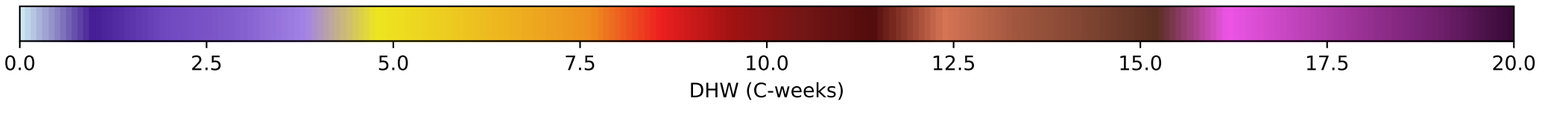


Predicted DHW - Step 90



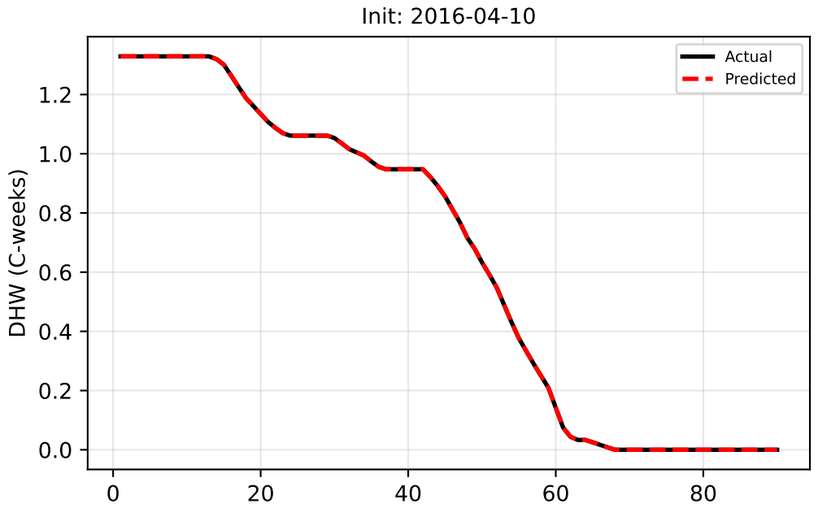
Error (Pred - Actual) - Step 90 [scale ±8]

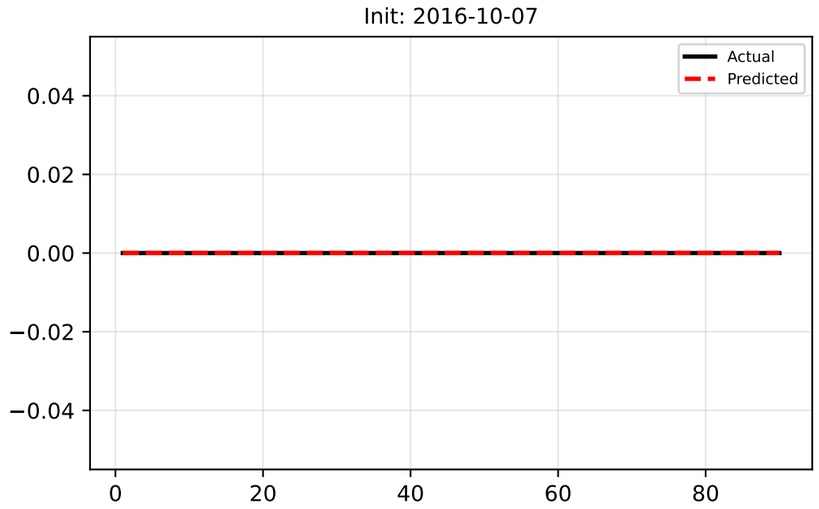


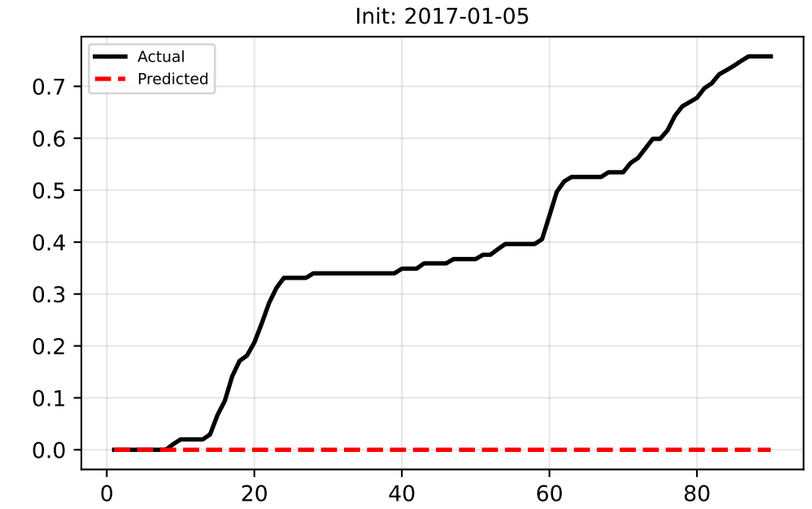


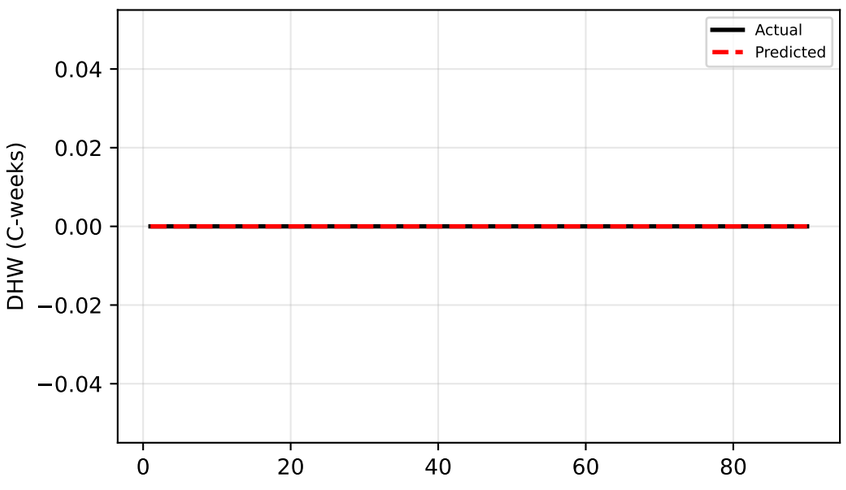


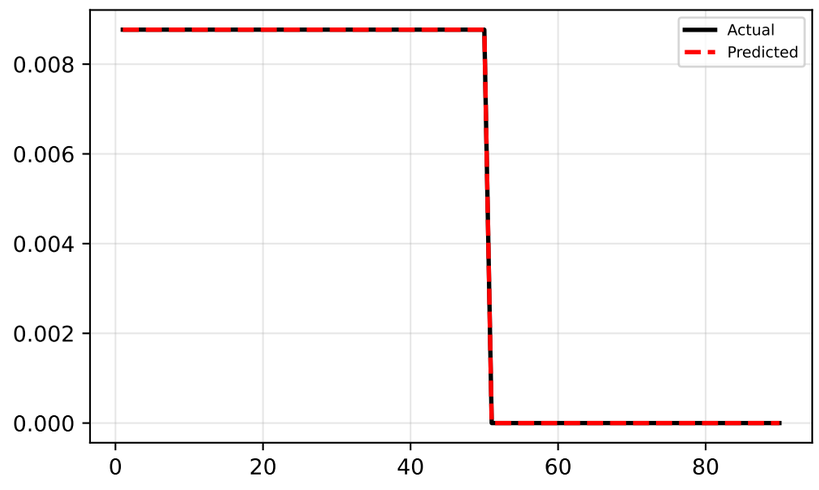
Global-Mean DHW - Val - GBR (C-weeks)

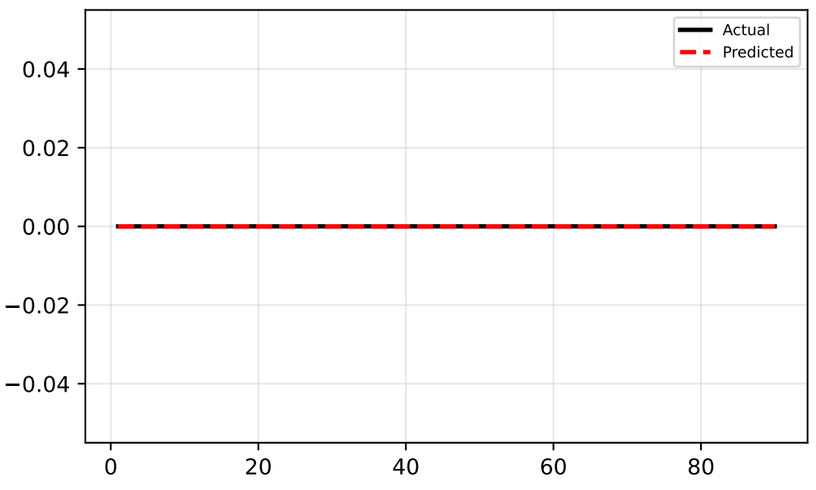


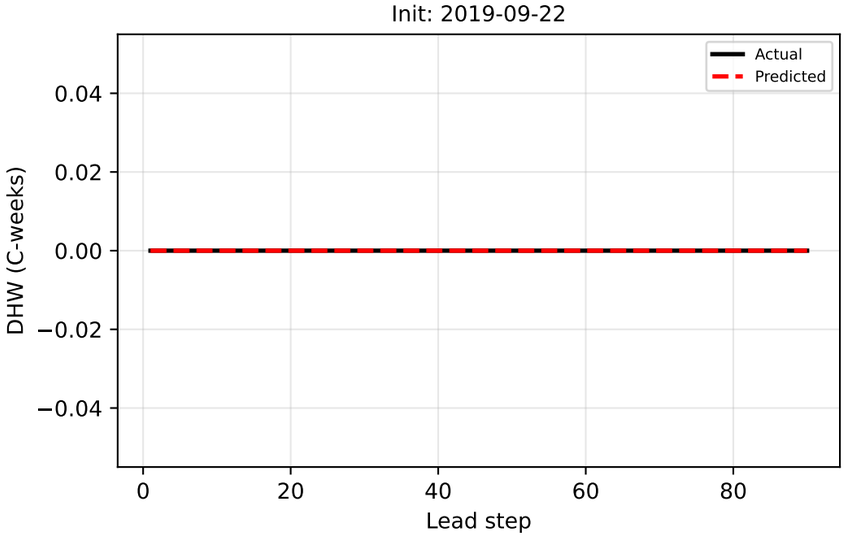


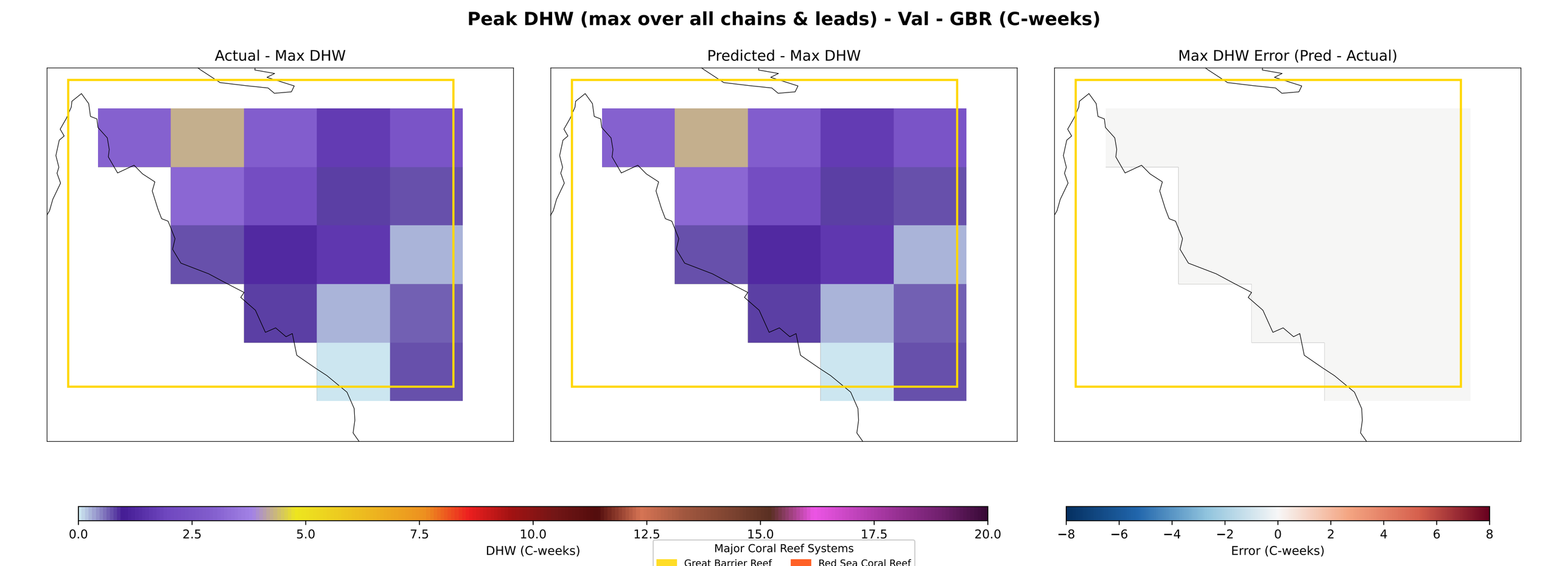


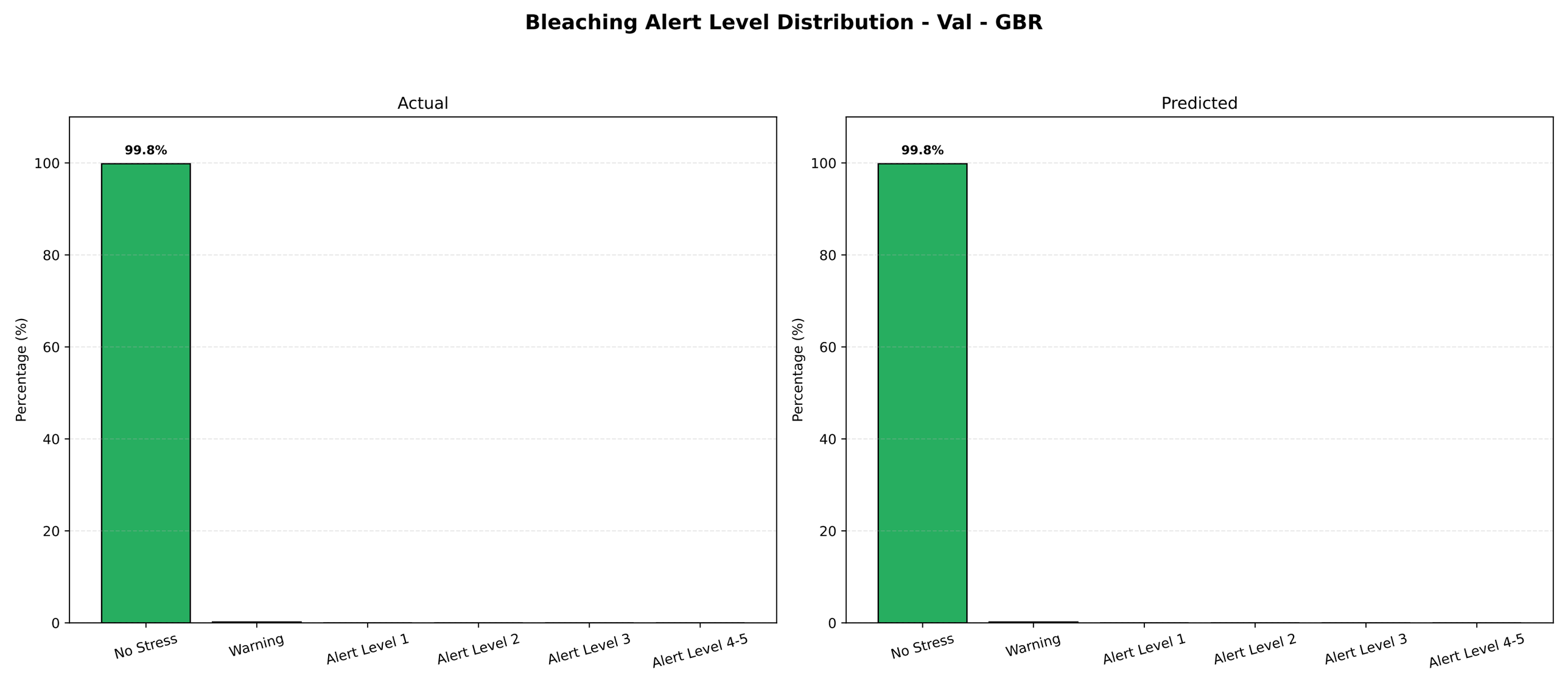






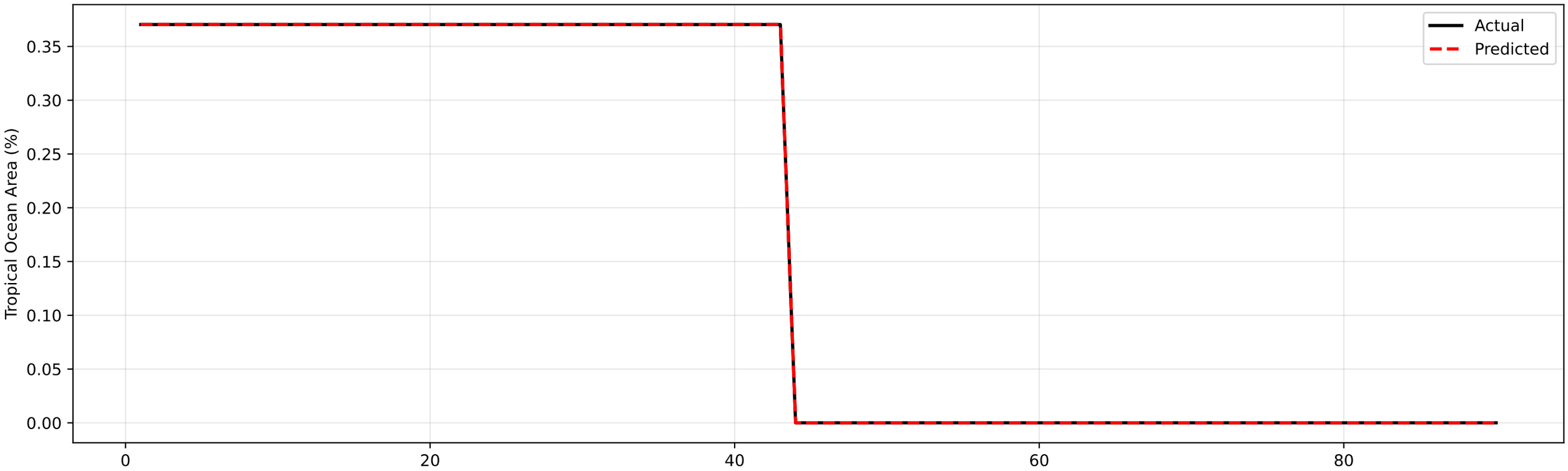



Peak DHW (max over all chains & leads) - Val - GBR (C-weeks)
Actual - Max DHW
Predicted - Max DHW
Max DHW Error (Pred - Actual)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
DHW (C-weeks)
Major Coral Reef Systems
Great Barrier Reef
Red Sea Coral Reef
−8
−6
−4
−2
0
2
4
6
8
Error (C-weeks)


Bleaching Alert Level Distribution - Val - GBR
Actual
Predicted
Percentage (%)
0
20
40
60
80
100
99.8%
No Stress
Warning
Alert Level 1
Alert Level 2
Alert Level 3
Alert Level 4-5

Tropical Ocean Area Under Bleaching Stress - Val - GBR

DHW >= 4 C-weeks (Alert Level 1 DHW threshold)



DHW >= 8 C-weeks (Alert Level 2 DHW threshold)

